

Optical Mater

5. Optical Materials and Their Properties

This chapter provides an extended overview on today's optical materials, which are commonly used for optical components and systems. In Sect. 5.1 the underlying physical background on light–matter interaction is presented, where the phenomena of refraction (linear and non-linear), reflection, absorption, emission and scattering are introduced. Sections 5.2 through 5.8 focus on the detailed properties of the most common types of optical materials, such as glass, glass ceramics, crystals, and plastics. In addition, special materials displaying “unusual nonlinear” or “quasi-nonreversible” optical behavior such as photorefractive or photorecording solids are described in Sect. 5.9. The reader could use this chapter as either a comprehensive introduction to the field of optical materials or as a reference text for the most relevant material information.

5.1 Interaction of Light with Optical Materials	250
5.1.1 Dielectric Function	250
5.1.2 Linear Refraction	255
5.1.3 Absorption	258
5.1.4 Optical Anisotropy	261
5.1.5 Nonlinear Optical Behavior and Optical Poling	265
5.1.6 Emission	269
5.1.7 Volume Scattering	271
5.1.8 Surface Scattering	275
5.1.9 Other Effects	278
5.2 Optical Glass	282
5.2.1 Chronological Development	282
5.2.2 Compositions of Modern Optical Glass	283
5.2.3 Environmentally Friendly Glasses ...	287
5.2.4 How to Choose Appropriate Optical Glasses	288
5.3 Colored Glasses	290
5.3.1 Basics	290
5.3.2 Color in Glass	292
5.4 Laser Glass	293
5.4.1 Common Laser Glasses and Properties	293
5.4.2 Laser Damage	297
5.4.3 Storage and Handling of Laser Glass	300
5.5 Glass–Ceramics for Optical Applications ..	300
5.5.1 Overview	300
5.5.2 Properties of Glass–Ceramics	301
5.5.3 Applications	306
5.6 Nonlinear Materials	307
5.6.1 Overview on Nonlinear Optical Materials	307
5.6.2 Application: All Optical Switching	312
5.6.3 Second Harmonic Generation in Glass	313
5.6.4 Glass Systems Investigated for Nonlinear Effects	313
5.6.5 NL–Effects in Doped Glasses	314
5.7 Plastic Optics	317
5.7.1 Moulding Materials	317
5.7.2 Manufacturing Methods	319
5.7.3 Manufacturing Process	320
5.7.4 Coating and Component Assembly	322
5.7.5 New Developments	322
5.8 Crystalline Optical Materials	323
5.8.1 Halides, CaF_2	323
5.8.2 Semiconductors	325
5.8.3 Sapphire	325
5.8.4 Optic Anisotropy in Cubic Crystals	326
5.9 Special Optical Materials	327
5.9.1 Tunable Liquid Crystal Electronic Lens	327
5.9.2 OLEDs	333
5.9.3 Photorefractive Crystals	339
5.9.4 Metal Mirrors	346
5.10 Selected Data	354
References	360



5.1 Interaction of Light with Optical Materials

In this section the general physics of the interaction of light with matter is briefly presented. A detailed insight into theoretical electrodynamics cannot be given here. The interested reader might refer to standard textbooks on electrodynamics e.g. [5.1, 2].

5.1.1 Dielectric Function

The starting point for an analysis of any interaction between electromagnetic waves with matter is Maxwell's equations. The static interaction for the dielectric displacement and the magnetic induction is described by

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \rho, \\ \nabla \cdot \mathbf{B} &= 0,\end{aligned}\quad (5.1)$$

whereas the dynamic interaction of the electric and magnetic fields is given by

$$\begin{aligned}\nabla \times \mathbf{E} &= -\dot{\mathbf{B}}, \\ \nabla \times \mathbf{H} &= \mathbf{j} + \dot{\mathbf{D}}.\end{aligned}\quad (5.2)$$

$\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields; $\mathbf{D}$ and $\mathbf{H}$ are the electric displacement and the auxiliary magnetic fields; ρ and $\mathbf{j}$ are the charge and the current density.

Material equations are needed to close Maxwell's equations:

$$\begin{aligned}\mathbf{D} &= \varepsilon_0 \mathbf{E} + \mathbf{P}, \\ \mathbf{B} &= \mu_0 \mathbf{H} + \mathbf{M},\end{aligned}\quad (5.3)$$

where $\mathbf{P}$ and $\mathbf{M}$ are the polarization and magnetization densities. The vacuum permittivity (in SI units) is $\varepsilon_0 = 8.854 \times 10^{-12} \text{ A s/V m}$ and the vacuum permeability is $\mu_0 = 4\pi \times 10^{-7} \text{ V s/A m}$.

The complete optical properties for any spatial combination of matter are included in the solution of (5.1), (5.2), which are closed by using the material equations (5.3) and by using appropriate boundary conditions. For only a few special cases such a solution can be written down directly. In the following we give a few examples.

Wave Equation in Vacuum

If we want to solve (5.1), (5.2) in infinite vacuum we have the following boundary conditions and material equations:

$$\mathbf{P}(\mathbf{r}) = 0, \quad \mathbf{M}(\mathbf{r}) = 0, \quad \rho(\mathbf{r}) = 0, \quad \mathbf{j}(\mathbf{r}) = 0, \quad (5.4)$$

where $\mathbf{r} = (x, y, z)$ are the three spatial coordinates. With these simplest possible boundary conditions the material equations (5.3) read

$$\begin{aligned}\mathbf{D} &= \varepsilon_0 \mathbf{E}, \\ \mathbf{B} &= \mu_0 \mathbf{H}.\end{aligned}\quad (5.5)$$

After applying a few vector operations, one gets the wave equation for the electromagnetic field $\mathbf{E}$ in vacuum:

$$\Delta \mathbf{E} - \mu_0 \varepsilon_0 \ddot{\mathbf{E}} = 0. \quad (5.6)$$

An identical wave equation can be derived for the magnetic field $\mathbf{B}$. (5.6) immediately defines the speed of light c (in vacuum).

$$c = \sqrt{\frac{1}{\mu_0 \varepsilon_0}}. \quad (5.7)$$

Equation (5.6) is generally solved by all fields which fulfill $\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \cdot f(\mathbf{k}\mathbf{r} \pm \omega t)$ involving any arbitrary scalar function f . The most common systems of function f are plane waves:

$$\mathbf{E}_s(\mathbf{r}, t) = \mathbf{E}_0 \text{Re} \left(e^{-i(\mathbf{k}\mathbf{r} - \omega t)} \right). \quad (5.8)$$

These plane waves, with a time and spatial dependent phase $\theta = \mathbf{k}\mathbf{r}$, are described by a wave vector $\mathbf{k}$, an angular frequency ω , and a corresponding wavelength $\lambda = 2\pi/k = 2\pi c/\omega$, where $k = |\mathbf{k}|$ is the absolute value of the wave vector. Describing an arbitrary field $\mathbf{E}$ in terms of plane waves is identical to decomposing this electrical field into its Fourier components.

Wave Propagation in an Ideal Transparent Medium

We can describe an ideal material by simply replacing the speed of light in vacuum by that of the medium.

$$c \rightarrow \frac{c}{n}, \quad (5.9)$$

where n is the (in this case only real) refractive index of the material. Wave propagation in a dispersing or weakly absorbing medium is considered at the end of the present section. In fact, most parts of an optical design can be done by treating optical glasses as such ideal transparent materials (see Sect. 5.1.2). Even though such an ideal material cannot exist in reality, optical glasses come very close to it (for electromagnetic radiation in the visible range). For such an ideal material the wave equation (5.6) reads:

$$\Delta \mathbf{E} - \frac{n^2}{c^2} \ddot{\mathbf{E}} = 0. \quad (5.10)$$

It is solved again by plane transverse waves. Where the speed of light is now reduced to the speed of light in the transparent medium $c_{\text{med}} = c/n$ and the wavelength of the lightwave is reduced to $\lambda_{\text{med}} = \lambda/n$.

Refraction and Reflection

We now derive the laws of refraction and reflection for the ideal transparent medium just described. They are obtained by solving Maxwell's equations at the (infinite) boundary between two materials of different refractive indices n_1 and n_2 (see Fig. 5.1). As boundary conditions one obtains that the normal component of the electric displacement (and magnetic induction) and the tangential component of the electric (and magnetic) field have to be continuous at the interface:

$$D_1^n = D_2^n, \quad E_1^t = E_2^t. \quad (5.11)$$

Further, a phase shift of an incoming wave occurs upon reflection

$$\theta_r = \pi - \theta_i, \quad (5.12)$$

where $\theta_{r,i}$ are the phases of the reflected and incident wave, respectively. If we solve Maxwell's equations for an incoming plane wave (applying the boundary conditions stated above), Snell's law of refraction is obtained

$$n_1 \sin \alpha_1 = n_2 \sin \alpha_2 \quad (5.13)$$

together with that of reflection

$$\alpha_r = \alpha_1. \quad (5.14)$$

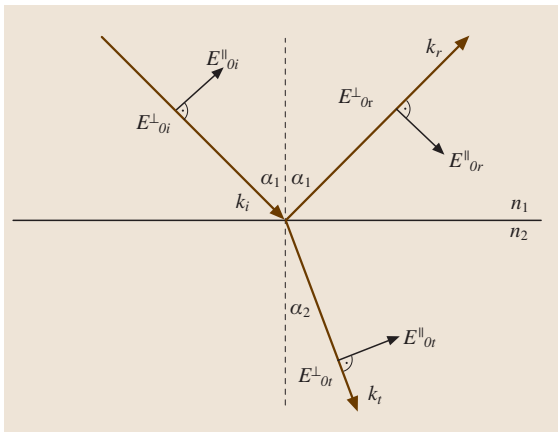


Fig. 5.1 The polarization directions of the E and B fields for reflection and refraction at an interface between two optical materials of different refractive indices are shown. A circle indicates that the vector is perpendicular to the plane shown

Now, the electric field E is decomposed into its components which are defined relative to the plane outlined by the three beams of incoming, transmitted and reflected light. This decomposition is shown in Fig. 5.1. The coefficients for reflection and transmission are defined as:

$$\begin{aligned} r_{\parallel} &= \frac{E_{0r}^{\parallel}}{E_{0i}^{\parallel}}; & t_{\parallel} &= \frac{E_{0t}^{\parallel}}{E_{0i}^{\parallel}}, \\ r_{\perp} &= \frac{E_{0r}^{\perp}}{E_{0i}^{\perp}}; & t_{\perp} &= \frac{E_{0t}^{\perp}}{E_{0i}^{\perp}}. \end{aligned} \quad (5.15)$$

The Fresnel formula for these coefficients can be derived as

$$\begin{aligned} r_{\perp} &= \frac{n_1 \cos(\alpha_1) - n_2 \cos(\alpha_2)}{n_1 \cos(\alpha_1) + n_2 \cos(\alpha_2)} = -\frac{\sin(\alpha_1 - \alpha_2)}{\sin(\alpha_1 + \alpha_2)}, \\ r_{\parallel} &= \frac{n_2 \cos(\alpha_1) - n_1 \cos(\alpha_2)}{n_1 \cos(\alpha_2) + n_2 \cos(\alpha_1)} = -\frac{\tan(\alpha_1 - \alpha_2)}{\tan(\alpha_1 + \alpha_2)}, \\ t_{\perp} &= \frac{2n_1 \cos(\alpha_1)}{n_1 \cos(\alpha_1) + n_2 \cos(\alpha_2)} \\ &= -\frac{2 \sin(\alpha_2) \cos(\alpha_1)}{\sin(\alpha_1 + \alpha_2)}, \\ t_{\parallel} &= \frac{2n_1 \cos(\alpha_1)}{n_1 \cos(\alpha_2) + n_2 \cos(\alpha_1)} \\ &= -\frac{2 \sin(\alpha_2) \cos(\alpha_1)}{\sin(\alpha_1 + \alpha_2) \cos(\alpha_1 - \alpha_2)}. \end{aligned} \quad (5.16)$$

Here the usual convention has been used that the coefficients of reflectivity obtain an additional minus sign in order to indicate backtraveling of light. The quantities that are measured in an experiment are intensities. The relationship between the intensities defines the reflectivity and transmissivity of a material

$$\begin{aligned} R_{\perp} &:= |r_{\perp}|^2; & R_{\parallel} &:= |r_{\parallel}|^2, \\ T_{\perp} &:= |t_{\perp}|^2; & T_{\parallel} &:= |t_{\parallel}|^2. \end{aligned} \quad (5.17)$$

The angular-dependent coefficients of reflection from (5.16) are displayed in Fig. 5.2. In Fig. 5.2a the case of light propagating from an optically thin medium with refractive index n_1 to an optically thicker medium with refractive index $n_2 > n_1$ is plotted. At the so called Brewster angle α_B the reflected light is completely polarized; α_B is given by the condition $\alpha_1 + \alpha_2 = \pi/2$. Therefore, the Brewster angle α_B results as a solution of

$$\alpha_1 = \frac{\pi}{2} - \arccos\left(\frac{n_2}{n_1} \cos \alpha_1\right) \quad (5.18)$$

which gives $\alpha_B = \arctan \frac{n_2}{n_1}$. In Fig. 5.2b the case of light propagating from an optically thick to an opti-

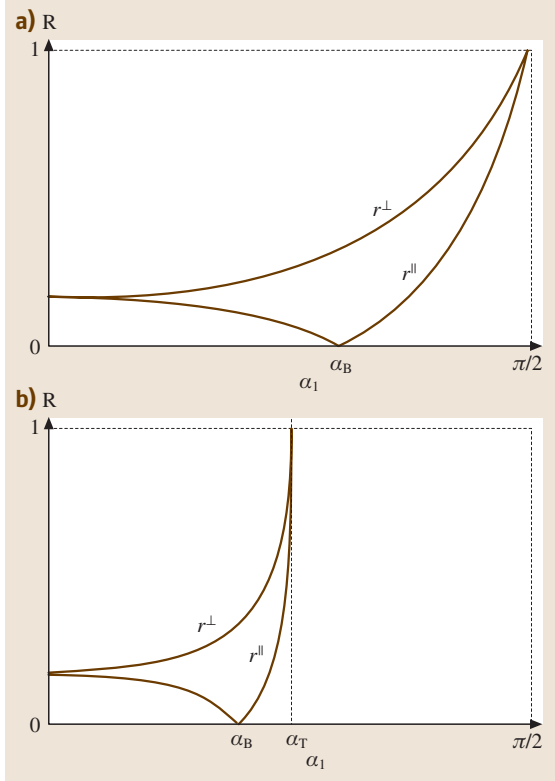


Fig. 5.2a,b The reflection coefficients are plotted as a function of incident scattering angle for light propagating from a medium of (a) smaller refractive index into a medium of larger index and (b) larger refractive index into a medium of smaller index. Here, total reflection occurs at an angle α_T and α_B is the Brewster angle

cally thin medium is plotted. Here an additional special angle occurs – the angle of total reflection, α_T . All light approaching the surface at an angle larger than α_T is totally reflected. At $\alpha_1 = \alpha_T$ the angle for refraction in the medium with refractive index n_2 is $\alpha_2 = \pi/2$. For α_T it follows that:

$$\alpha_T = \arcsin\left(\frac{n_2}{n_1}\right). \quad (5.19)$$

Evaluating (5.16) for the special case of incident light as $\lim_{\alpha \rightarrow 0}$ allows one to calculate the reflectivity for normal incidence

$$R_{\text{norm}} = \left| \frac{n_1 - n_2}{n_1 + n_2} \right|^2. \quad (5.20)$$

For practical applications it is important to note that the Fresnel equations remain valid in the case of weakly

absorbing media discussed at the end of the present section. Here, only the refractive indices have to be replaced by the complex quantities of (5.36). It is further helpful to define a transmittivity and reflectivity for unpolarized light

$$R^{\text{unpol}} = \frac{E_{0r}^{\parallel 2} + E_{0r}^{\perp 2}}{E_{0i}^{\parallel 2} + E_{0i}^{\perp 2}}; \quad T^{\text{unpol}} = \frac{E_{0t}^{\parallel 2} + E_{0t}^{\perp 2}}{E_{0i}^{\parallel 2} + E_{0i}^{\perp 2}}. \quad (5.21)$$

Inserting the expressions for the reflection coefficients we obtain e.g. the total reflectivity as a function of the incident and refracted angular

$$R_{\text{all}}^{\text{unpol}} = \frac{E_{0i}^{\parallel 2} \frac{\tan^2(\alpha_1 - \alpha_2)}{\tan^2(\alpha_1 + \alpha_2)} + E_{0i}^{\perp 2} \frac{\sin^2(\alpha_1 - \alpha_2)}{\sin^2(\alpha_1 + \alpha_2)}}{E_{0i}^{\parallel 2} + E_{0i}^{\perp 2}}$$

With the definitions from (5.21) the following sum rule must be fulfilled:

$$R^{\text{unpol}} + \frac{n_2 \cos \alpha_2}{n_1 \cos \alpha_1} T^{\text{unpol}} = 1. \quad (5.22)$$

The rule provides an easy check for transmitted and reflected total intensities, especially for normal incidence.

Wave Propagation in an Isotropic, Homogeneous Medium

We now consider wave propagation in an ideal optical material. This is a nonmagnetic, homogeneous, isotropic, perfectly insulating medium, which is further a perfectly linear optical material. Considering time dependence including retardation in the materials (5.2) leads to:

$$D(\mathbf{r}, t) = \varepsilon_0 \mathbf{E}(\mathbf{r}, t) + \mathbf{P}(\mathbf{r}, t). \quad (5.23)$$

The polarizability is related to the electric field via the susceptibility χ . In the case of a homogeneous isotropic material χ is a scalar function. In Sect. 5.1.4 we will consider the case of optically anisotropic media, where χ becomes a second-rank tensor.

$$\mathbf{P}(\mathbf{r}, t) = \int d\mathbf{r}' \int_{-\infty}^t dt' \chi(\mathbf{r} - \mathbf{r}', t - t') \mathbf{E}(\mathbf{r}, t'). \quad (5.24)$$

Fourier transformation in time and space deconvolutes the integral and leads to:

$$\mathbf{P}(\mathbf{k}, \omega) = \chi(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega), \quad (5.25)$$

where $\chi(\mathbf{k}, \omega)$ is, in general, a complex analytic function of the angular frequency ω . The complex function $\chi(\mathbf{k}, \omega)$ unifies the two concepts of a low-frequency

polarizability χ' and a low-frequency conductivity σ of mobile charges to a single complex quantity

$$\lim_{\omega \rightarrow 0} \chi(\omega) = \chi'(\omega) + 4\pi i \frac{\sigma(\omega)}{\omega}. \quad (5.26)$$

At larger frequencies the separation of the two concepts breaks down, since above the frequencies of optical phonon modes in the IR the bound charges are unable to follow the electric field, whereas below the phonon modes the charges can follow this motion (Sect. 5.1.3). The usual form in which the susceptibility enters the equations for optical purposes is via the dielectric function

$$\varepsilon(\mathbf{k}, \omega) = 1 + \chi(\mathbf{k}, \omega). \quad (5.27)$$

Inserting the dielectric function into the material equation (5.23) gives:

$$\mathbf{D}(\mathbf{k}, \omega) = \varepsilon_0 \varepsilon(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega). \quad (5.28)$$

Here, we restrict ourselves to ideal optically isotropic materials by neglecting the nature of the dielectric function as a second-rank tensor. In Sect. 5.1.4 we will extend our considerations to optically anisotropic materials. With the same steps as in Eqs. (5.5–5.6) a wave equation can be derived which has the following form in Fourier space:

$$\left[k^2 - \varepsilon(\mathbf{k}, \omega) \frac{\omega^2}{c^2} \right] \mathbf{E}_0 = 0, \quad (5.29)$$

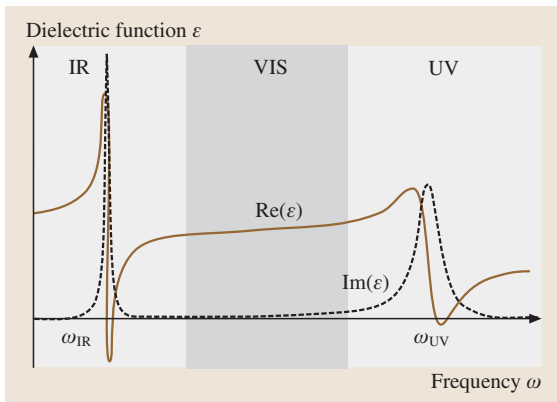


Fig. 5.3 Dielectric function $\varepsilon(\omega)$ for the “model optical solid” with one generic absorption in the infrared ω_{IR} and a second one in the ultraviolet ω_{UV} . The dielectric function is plotted on a logarithmic energy scale. The solid line is the real part and the dashed line is the imaginary part of $\varepsilon(\omega)$

where the relations $\partial^2/\partial t^2 \mathbf{E}_s(\mathbf{r}, t) \rightarrow -\omega^2 \mathbf{E}_s(\mathbf{r}, t)$ and $\Delta \mathbf{E}_s(\mathbf{r}, t) \rightarrow -k^2 \mathbf{E}_s(\mathbf{r}, t)$ have been used. The expression in brackets in (5.29) defines the dispersion relation for an optically linear, homogeneous, isotropic material.

Poynting Vector and Energy Transport

The energy flux density of the electric field is obtained via the Poynting vector, given by

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (5.30)$$

It gives the rate at which electromagnetic energy crosses a unit area and has the unit W/m^2 . It points in the direction of energy propagation. The time average of the absolute value of the Poynting vector $\langle |\mathbf{S}| \rangle$ is called the intensity I of the electromagnetic wave

$$I = \langle |\mathbf{S}| \rangle = \frac{1}{2} |\mathbf{E} \times \mathbf{H}| \quad (5.31)$$

and is the energy flux density of the electromagnetic radiation. In the special case of propagation of transverse plane waves [as given in (5.8)], it simplifies to

$$I = \frac{1}{2} \frac{n}{c\mu_0} |\mathbf{E}_0|^2, \quad (5.32)$$

where in vacuum $n = 1$ is valid.

General Form of the Dielectric Function

For most optical materials the dielectric function has a form in which a transparent frequency (or wavelength) window is bounded at the high energy side by electron–hole excitations (dominating the UV edge) and at the low energy side by IR absorptions given by optical phonon modes (lattice vibrations). The general form of the dielectric function is given by the Kramers–Heisenberg equation [5.3]

$$\varepsilon(\mathbf{k}, \omega) = 1 + \sum_j \frac{\alpha_{\mathbf{k},j}}{\omega^2 - \omega_{\mathbf{k},j}^2 - i\omega\eta_{\mathbf{k},j}}. \quad (5.33)$$

Here $\alpha_{\mathbf{k},j}$ is the amplitude, $\omega_{\mathbf{k},j}$ the frequency and $\eta_{\mathbf{k},j}$ the damping of the particular excitation j . A schematic view of the dielectric function is plotted in Fig. 5.3. Here we use a model for a transparent homogeneous, isotropic solid (such as glass) with one generic absorption at low energies (ω_{IR} in the infrared, IR) and another one at large phonon energies (ω_{UV} in the ultraviolet spectral range, UV). In the following this model solid is used to discuss optical material properties.

Dispersion Relation

Solving (5.29) gives two frequency-dependent solutions for the wave vector $\mathbf{k}$ as a function of ω since the left-hand side of (5.29) is quadratic in $\mathbf{k}$. Far away from

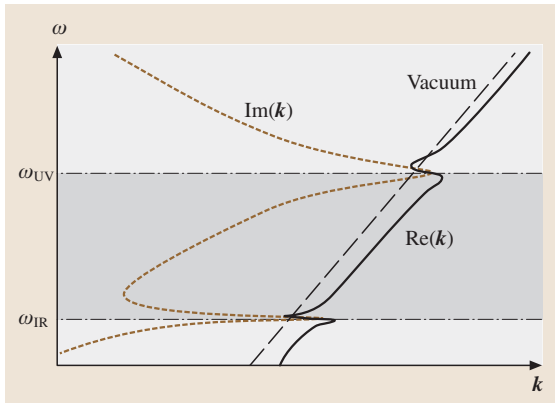


Fig. 5.4 Dispersion relation $\omega(k)$ for the “model optical solid” on a double logarithmic scale. The *solid line* is the real part of k and the *dashed line* the imaginary part of k . For comparison the simply linear dispersion relation for light propagating in vacuum $\omega = ck$ is shown with a *long-dashed line*

absorptions the dielectric function is real and positive. Here only one solution exists, which describes the wave propagating with the speed of light in the medium. Close to an absorption two solutions exist which are even more complicated. This means that near a resonance the dispersion of the light cannot be considered independently from the dispersion of the excitations in the material. They both form a composite “new” entity propagating in the medium. This is called the polariton [5.4]. For our model solid with two generic absorptions (ω_{IR} and ω_{UV}), the dispersion is plotted in Fig. 5.4.

Wave Propagation, Phase and Group Velocity

When an electromagnetic wave propagates through a medium one can define two velocities. The phase velocity is the speed with which a certain phase propagates. It is, for example, the velocity of the wavefront maxima moving through the medium. The phase velocity is given as

$$v_{\text{ph}} = \frac{\omega}{k}. \quad (5.34)$$

In Fig. 5.5 the phase velocity of the model solid is plotted on a logarithmic frequency scale. Close to the absorption edges of the material it loses its meaning because attenuation due to the absorption processes will dominate most processes. Far away from absorptions it reaches a nearly constant value. The second velocity is the group

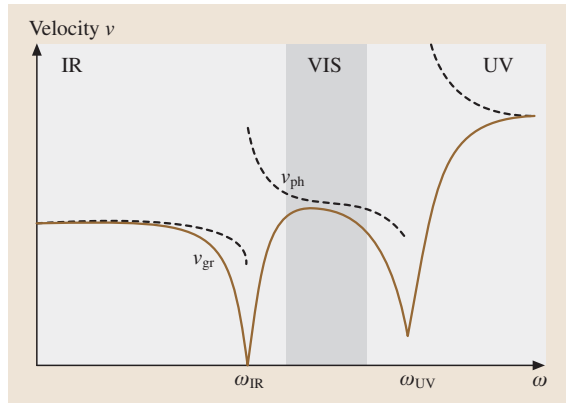


Fig. 5.5 The group velocity $v_{\text{gr}}(k) = \partial\omega/\partial k$ (*solid line*) and the phase velocity $v_{\text{ph}}(k) = \omega/k$ (*dashed line*) are plotted on a logarithmic frequency scale. Far away from absorptions both velocities approach each other while the group velocity is always smaller than the phase velocity

velocity

$$v_{\text{gr}}(k) = \frac{\partial\omega}{\partial k}. \quad (5.35)$$

This is the velocity at which a complete wave packet travels through the medium and is, hence, the speed with which information can travel through the system. It is plotted in Fig. 5.5 for the model solid. Reasonably far away from material absorptions the group and phase velocities approach each other. However, the group velocity is always smaller than the phase velocity.

Refractive Index

The refractive index n is the most widely used physical quantity in optical design. It is the square root of the dielectric function. The dynamic refractive index is generally a complex quantity

$$\tilde{n}(\omega) = n(\omega) + i\kappa(\omega) \quad (5.36)$$

and must fulfill the Kramers–Kronig relations [5.3]. The refractive index for our generic model solid is plotted as a function of logarithmic frequency in Fig. 5.6. In practical use, the wavelength dependence is often exploited

$$\tilde{n}(\mathbf{k}, \lambda) = \sqrt{\varepsilon(\mathbf{k}, 2\pi c/\lambda)}. \quad (5.37)$$

With a few basic steps, (5.33) can be rewritten as a function of wavelength alone. If one further restricts to wavelengths which are far away from absorptions ($\omega^2 - \omega_{k,j}^2 \ll \omega\eta_{k,j}$), the Sellmeier formula (see

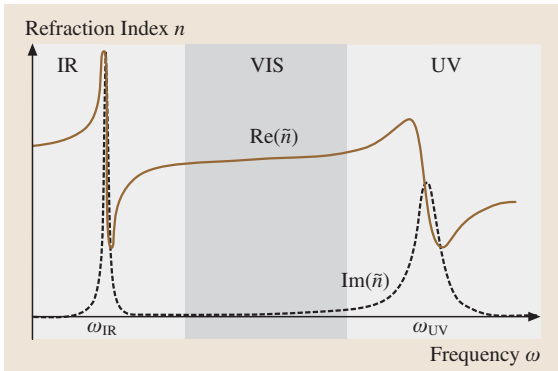


Fig. 5.6 The complex refractive index $n(\omega) = n^*(\omega) + i\kappa(\omega)$ is plotted on a logarithmic frequency scale. The real part is plotted with a *solid line* and the imaginary part with a *dashed line*

Sect. 5.1.2), which is widely used for characterizing optical materials, is obtained

$$n(\lambda)^2 \approx 1 + \sum_j \frac{B_j \lambda^2}{\lambda^2 - \lambda_j^2}, \quad (5.38)$$

where $\lambda_i = 2\pi c/\omega_i$ with $i \in \{(k, j)\}$ is used and $B_j = a_{k,j} \lambda_{k,j}/(2\pi c)^2$. Normally B_j and λ_j are just fitting constants to describe the dispersion of the refractive index over a certain wavelength range. They are, however, connected to the microscopic fundamental absorption behavior of the material. Sometimes also $n(\lambda)$ and not $n(\lambda)^2$ is approximated with a Sellmeier formula. Since as well $n(\lambda)$ as well as $n(\lambda)^2$ are complex differential (analytic) functions both formula give refractive indices and dispersions with the same accuracy. However care has to be taken, which quantity is expressed when using a Sellmeier formula.

Wave Propagation in Weakly Absorbing Medium

In this subsection the link between the attenuation of a wave and the imaginary part of the refractive index is given. A weakly absorbing medium is defined by the imaginary part of the refractive index (5.37) being much smaller than the real part

$$\kappa \ll n \quad (5.39)$$

(the coefficient κ/n is also called the attenuation index). In this case light propagates as transverse waves through the medium. We consider two points in our medium: P_1 and P_2 . Between these points the light travels the distance l . In the absence of absorption, the electric and

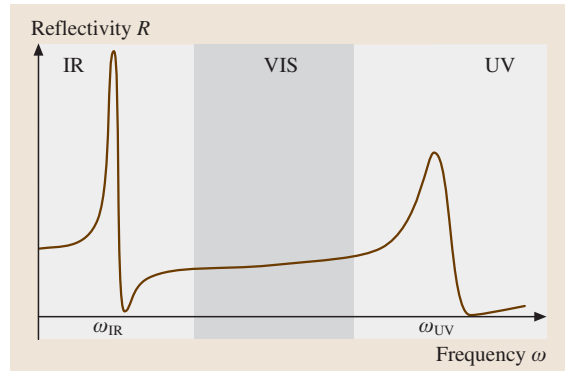


Fig. 5.7 Reflectivity R for normal incidence plotted on a logarithmic frequency scale

magnetic fields at point P_2 are given by

$$E_2 = E_1 e^{i\frac{\omega}{c}\tilde{n}l}, \quad H_2 = H_1 e^{i\frac{\omega}{c}\tilde{n}l}. \quad (5.40)$$

Using (5.31) we obtain for weak absorption the radiation intensity at point P_1 . The radiation intensity at P_2 is:

$$\begin{aligned} I_2 &= \frac{1}{2} |E_2 \times H_2|, \\ &= \frac{1}{2} |E_1 \times H_1| e^{-2\frac{\omega}{c}\kappa l}, \\ &= I_1 e^{-2\frac{\omega}{c}\kappa l} = I_1 e^{-\alpha l}. \end{aligned} \quad (5.41)$$

The absorption coefficient α is connected to the complex refractive index by

$$\alpha = 2\frac{\omega}{c}\kappa; \quad (5.42)$$

α can easily be measured and its importance for optical properties is discussed in Sect. 5.1.3.

It is also important to note that the Fresnel equations (5.16) remain valid in the case of a weakly absorbing medium if the complex refractive indices are used. As an example, we plot the reflectivity (at the interface air-model solid) near an absorption resulting from (5.20) for a complex refractive index $n_2 \rightarrow \tilde{n}_2$. In Fig. 5.7 the reflectivity is plotted on a logarithmic frequency scale. Note that the absorption seems to be shifted compared to the plots of the complex dielectric function or the complex refractive index. Measurement of the reflectivity is of importance for reflection spectroscopy.

5.1.2 Linear Refraction

As already introduced in Sect. 5.1.1, two phenomena occur when light impinges upon the surface of any optical material: reflection and refraction [5.5]. The reflected



light bounces off the glass surface, while the refracted light travels through the material. The amount of light that is reflected depends on the refractive index of the sample, which also affects the refractive behavior of the sample [5.6]. The refractive index of optical materials turns out to be one of the most important factors that must be considered when designing systems to transmit and modulate light [5.7]. The refractive index is a complex material property that depends on temperature and wavelength [5.8]. The wavelength dependence of the refractive index is the dispersion [5.5].

Law of Refraction

When a light ray impinges upon a glass surface, a portion is reflected and the rest is either transmitted or absorbed. The material modulates the light upon transmission. The light travels at a different velocity as it is transmitted through the glass as compared to through vacuum. As introduced in Sect. 5.1.1, the index of refraction (*n*), is defined as the ratio of the speed of light in vacuum (*c*) to that in the material (*c_m*) [5.11]:

$$n = \frac{c}{c_m} .$$

(5.43)

Most commonly, the reported refractive indices are relative to the speed of light in air, rather than in vacuum, no matter which technique is used to measure the refractive index [5.12]. The index of refraction for vacuum, by definition, must be exactly 1. The index of refraction of air is 1.00029 at standard temperature (25 °C) and pressure (1 atm) (STP). Therefore, the index of refraction of optical matter (*n_{rel}*) relative to air (*n_{air}*), rather than vacuum is [5.11]:

$$n_{rel} = \frac{n_m}{n_{air}} .$$

(5.44)

Table 5.1 Indices of common materials at standard temperature and pressure at 587.56 nm (helium d line) [5.9]

Material	<i>n_d</i>	Material	<i>n_d</i>
Vacuum	1	Crown glass	1.52
Air	1.00029	Sodium chloride	1.54
Water	1.33	Polystyrene	1.55
Acetone	1.36	Carbon disulfide	1.63
Ethanol	1.36	Flint glass	1.65
Sugar solution (30 wt %)	1.38	Sapphire	1.77
Fused silica	1.46	Heavy flint glass	1.89
Sugar solution (80 wt %)	1.49	Diamond	2.42

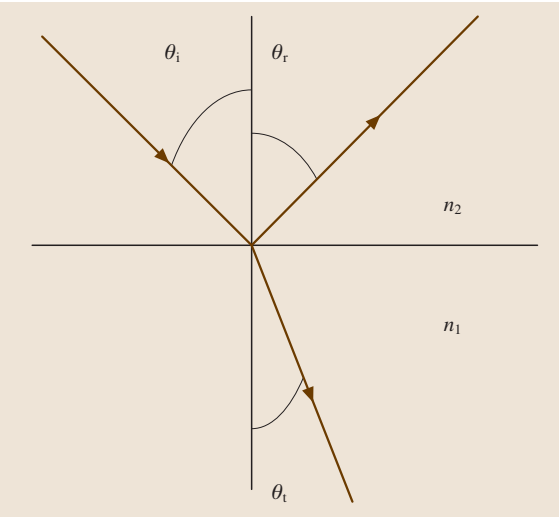


Fig. 5.8 Ray tracings of incident, reflected, and transmitted light from one medium to another representing the angles and indices necessary to apply Snell’s law (after [5.10])

The STP indices of some common compounds, and classes of compounds, are shown in Table 5.1 [5.9].

As discussed in Sect. 5.1.1, when light hits a glass surface at an angle α_i , it is Fresnel-reflected back at an angle α_t . The angle of incidence is equal to the angle of reflection ($\alpha_i = \alpha_r$), as shown in Fig. 5.8 [5.10]. The percentage of light reflected for $\alpha_i = 0$ at each interface (*R*) relative to the incident intensity (Sect. 5.1.1) is dependent on the index of refraction of the two media the light is passing through, typically air (*n₂*) and glass (*n₁*) Fig. 5.8 and (5.20) [5.10]

$$R_{norm} = 100 \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2 .$$

(5.45)

Fresnel’s formula (5.45) assumes smooth surfaces that produce only specular reflection. Diffuse reflection occurs when the surface is rough, so the incident light is reflected through a range of angles, thereby reducing the intensity of the reflected light at any given angle [5.10] (see Sect. 5.1.8). The specular reflection that is taken into consideration by Fresnel’s relationship can be monitored and used to estimate the refractive index of samples in situ [5.13].

The angle of the light transmitted within the material (α_t), relative to the incident light transmitted through air, is dependent on the refractive indices of the air (*n_{air}*) and solid (*n_m*) and the incident light angle (α_i) [5.10]:

$$n_{air} \sin \alpha_i = n_m \sin \alpha_t .$$

(5.46)

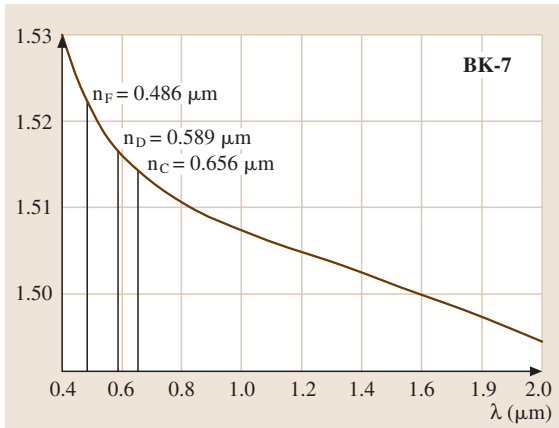


Fig. 5.9 Dispersion present in BK-7 optical glass. Common index of refraction measurement wavelengths are indicated

This is the general form of Snell's law of refraction to predict transmitted angles in media [5.14].

Dispersion Relationships in Glass

The refractive index of a medium is dependent upon the wavelength of the light being transmitted (see Sect. 5.1.1). This wavelength dependence is dispersion, which means that different wavelengths of light will be modulated differently by the same piece of matter [5.10]. Each wavelength of light will be subject to a different index of refraction. One ramification of dispersion is that white light can be separated into its principal visible components through a glass prism, or a simple raindrop. It is the dispersion of white light through raindrops that causes rainbows. The dispersion of light through optical materials results in the light being refracted at various angles because of Snell's law (5.46). The various components of white light experience different indices of refraction, which leads to different angles of exiting light. The difference in refractive index with wavelength is illustrated in Fig. 5.9 for BK-7 optical glass, which is a high dispersion material.

In normal dispersion, the index increases for shorter wavelengths of light [5.15]. Normal dispersion is valid only far away from absorption bands (see Fig. 5.6). Water has a normal dispersion response in the visible, so the red light is refracted by a lower index, and thus a greater angle, which is why red is on top in a rainbow. Discussion of rainbow formation is presented elsewhere in depth [5.10].

Anomalous dispersion is an increase in refractive index with an increase in wavelength. Anomalous dispersion typically occurs at frequencies that represent a crossover between the polarization mechanisms (dipolar, ionic, electronic etc.) that are responsible for absorption of electromagnetic radiation. This is further discussed in [5.11].

Due to dispersion, the index of refraction must be reported with the wavelength of measurement. The most common wavelengths at which n is measured are reported in Table 5.2. These wavelengths most often correspond to common sharp emission lines. The index can be determined most accurately ($\pm 1 \times 10^{-6}$) by measuring the angle of minimum deviation of light in a prism [5.15]. However, a Pulfrich refractometer ($\pm 1 \times 10^{-5}$) is most commonly used in industry. Details of measurement techniques are given in [5.16].

The index at various wavelengths is commonly referred to by the designations in Table 5.2: i. e. n_d is the refractive index measured at the yellow helium d line of 582.5618 nm. The dispersion is often given as a difference in n at two wavelengths. For instance, the primary dispersion is given by $n_F - n_C$ (hydrogen lines) and $n_F - n_{C'}$ (cadmium lines). The most commonly reported measure of dispersion is the Abbe number (ν), which is

Table 5.2 Wavelength of spectral lines used for measuring index of refraction, with the common designation and spectral line source [5.11]

Wavelength (nm)	Designation	Spectral line
2325.4		Hg IR line
1970.1		Hg IR line
1529.6		Hg IR line
1013.98	t	H IR line
852.1101	s	Cs IR line
706.5188	r	He red line
656.2725	C	H red line
643.8469	C'	Cd red line
632.8		He-Ne laser line
589.2938	D	Na yellow line (center of doublet)
587.5618	d	He yellow line
546.074	e	Hg green line
486.1327	F	H blue line
479.9914	F'	Cd blue line
435.8343	g	Hg blue line
404.6561	h	Hg violet line
365.0146	i	Hg UV line



commonly given for two sets of conditions

$$\nu_d = \frac{n_d - 1}{n_F - n_C}, \quad \nu_e = \frac{n_e - 1}{n_{F'} - n_{C'}}. \quad (5.47)$$

The Abbe number is a measure of the ratio of the refractive power to the dispersion. In most optical materials catalogs, a six-digit number is assigned to the solid that is dependent upon the index and the Abbe number: $1000(n_d - 1) + 10\nu_d$. Using this property, e.g. optical glasses are divided into two general categories: crowns and flints (see Sect. 5.2). Crown glasses typically have a low index of refraction and a high Abbe number ($n_d < 1.60$ and $\nu_d > 55$), whereas flint glasses have high indices and low Abbe numbers ($n_d > 1.60$ and $\nu_d > 50$) [5.15]. The terms crown and flint have historical significance in that flint glasses typically had lead oxide added to them to increase the refractive index (see Sect. 5.2) and crown glasses were typically blown and had curvature – a crown. Typically, to make an achromatic optical system crown and flint lenses are combined in series. The n_d and ν_d are plotted for various types of optical glasses in Fig. 5.87. The n_d and ν_d are listed with glass type and manufacture in Table 5.37.

Often it is desired to have a mathematical representation of the index as a function of wavelength. A considerable number of models exist for just this purpose. Perhaps the best known, and most widely used, is the Sellmeier form [5.15]

$$n^2(\lambda) - 1 = \frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{B_2\lambda^2}{\lambda^2 - C_2} + \frac{B_3\lambda^2}{\lambda^2 - C_3}. \quad (5.48)$$

Most major optical material manufacturers supply the Sellmeier coefficients for their glasses on the product data sheets. With the six Sellmeier coefficients it is possible to estimate the index of refraction at any wavelength, given that it is not near a strong absorption. Numerous other dispersion models have been developed and are presented elsewhere [5.10, 15].

The index of refraction is also dependent upon temperature, and similar formulas and tables of constants have been developed for the differential change in n with temperature [5.15]

$$\frac{dn_{\text{abs}}(\lambda, T)}{dT} = \frac{[n^2(\lambda, T_0) - 1]}{2n(\lambda, T_0)} \times \left(D_0 + 2D_1\Delta T + 3D_2\Delta T^2 + \frac{E_0 + 2E_1\Delta T}{\lambda^2 - \lambda_{\text{TK}}^2} \right), \quad (5.49)$$

where T_0 is 20 °C, T is the temperature in °C, ΔT is $T - T_0$, λ is the wavelength in microns, and λ_{TK} is the

average effective resonance wavelength for the temperature coefficients in microns. The constants E_0 , E_1 , D_0 , D_1 , and D_2 are provided on the manufacture product data sheets for each composition. The index of refraction must be measured.

5.1.3 Absorption

Based on the discussion in Sect. 5.1.1 the most common processes causing absorption in optical materials will be discussed here.

Introduction to Absorption

Absorption in glass is characterized by a decrease in transmitted light intensity through the sample that is not accounted for by reflection losses at the surface or scattering by inclusions [5.14]. As already introduced in Sect. 5.1.1 absorption is not uniform across all wavelengths of interest (UV–VIS–IR; ≈ 200 –2000 nm) and can be characterized by absorption bands [5.17]. The absorption bands are due to both intrinsic and extrinsic effects [5.14, 18, 19].

The quantity used to discuss absorption as a function of wavelength in glass is the transmittance (T), which is the ratio of the transmitted light intensity (I) to the initial light intensity (I_0) after passing through a glass plate of thickness L [5.20]:

$$T = I/I_0. \quad (5.50)$$

The transmittance is related to the more common percent transmittance (% T) through:

$$\%T = 100 T. \quad (5.51)$$

Sometimes it is more practical or convenient to look at the optical spectra in terms of absorbance rather than the transmittance. Absorbance (A) is defined as the $\log_{10}$ of ($1/T$) [5.20]

$$A = \log_{10} \frac{I_0}{I} = \log_{10} \frac{1}{T} = 2 - \log_{10} \%T, \quad (5.52)$$

$$\%T = 100/10^A. \quad (5.53)$$

Absorbance is often referred to as optical density (OD). When the absorbance changes linearly with concentration of the absorbing species (C), then Beer's law, which relates absorbance with concentration, path length (l) and extinction coefficient (ε), is applicable [5.20]

$$A = \varepsilon Cl. \quad (5.54)$$

The typical units of ε are liters mole⁻¹ cm⁻¹ and are often omitted.

From Beer's law we see that the absorbance changes with path length. To account for this, the spectra (in A or $\%T$) must list the sample thickness. Absorbance can also be normalized to path length in terms of an absorption coefficient (α). The connection of an optical absorption to a microscopic property of an underlying absorption process is straightforward. A light beam propagating through a medium is exponentially damped if absorption is present. This follows directly from the fact that the attenuation inside the optical material is proportional to the present (wavelength-dependent) intensity $I(\lambda)$ [5.21]

$$dI(\lambda) = N\sigma^{\text{abs}}(\lambda)I(\lambda)dz. \quad (5.55)$$

Where dz is an infinitesimally small part of the propagation part (unit meter) and N is the density of absorbing centers (in units of m^{-3}). The quantity $\sigma^{\text{abs}}(\lambda)$ is the absorption cross section. Since both sides of (5.55) must have the same units, it follows that the absorption cross section is measured in m^2 . It therefore has the meaning of an area and can be interpreted as the effective area perpendicular to the direction of an incoming beam that is seen by that beam and causes an absorption process. Figure 5.10 illustrates this. The solution of (5.55) for an optical plate of thickness L and initial intensity I_0 gives the intensity of the beam after traveling the distance L through the material [5.21]

$$I_{L,G}(\lambda) = I_{0,G}e^{-\alpha(\lambda)L} \quad (5.56)$$

with $I_{0,G}$ and $I_{L,G}$ being taken at both surfaces, but inside the solid. They are related to the measurable values I_0 and I_L outside the material by $I_{0,G} = (1 - R)I_0$ and $I_L = (1 - R)I_{L,G}$ with R being the Fresnel losses as described in Sect. 5.1.1. Equation (5.56) is known as

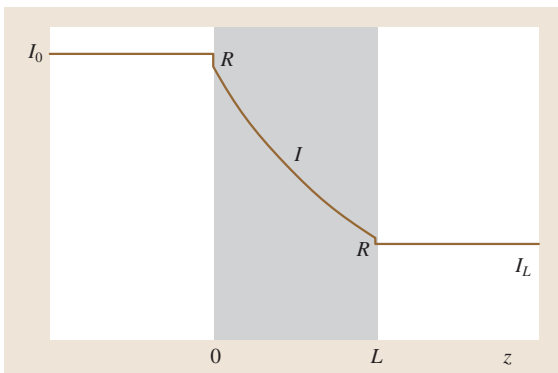


Fig. 5.10 Illustration of the Lambert–Beer law, displaying the bulk absorption inside the optical material and reflective Fresnel losses R at both surfaces

the law of Lambert–Beer, where the absorption coefficient $\alpha(\lambda)$ is defined as [5.21]

$$\alpha(\lambda) = N\sigma^{\text{abs}}(\lambda). \quad (5.57)$$

In contrast to the absorbance A , the absorption coefficient takes into account the reduction in transmission at both surfaces due to Fresnel-type losses [5.15]

$$\alpha = \frac{-\ln [T/(1 - R)^2]}{L}. \quad (5.58)$$

The absorption coefficient normalizes for path length, while the absorption cross section σ^{abs} normalizes for absorbing ion (AI) concentration $[AI]$ (in ions/m^3) [5.15]

$$\sigma(\lambda) = \frac{\alpha(\lambda)}{[AI]}. \quad (5.59)$$

The absorption cross section allows for a more fundamental comparison of various concentrations of absorbing ions (as long as they obey Beer's law) within different matrices.

Summary of Absorption Mechanisms in Glass

Glasses, as typical candidates for the simple optical solid model described in Sect. 5.1.1, are transparent from the near UV to the near IR. The regions of transparency and semi-transparency in the visible and near IR are shown for a typical optical glass (BK-7) in Fig. 5.11. The intrinsic absorption is dominated by electronic transitions in the UV–VIS range, while in the IR it is dominated by molecular vibrations [5.20]. There are multiple extrinsic mechanisms, which occur with the inclusion of ionic

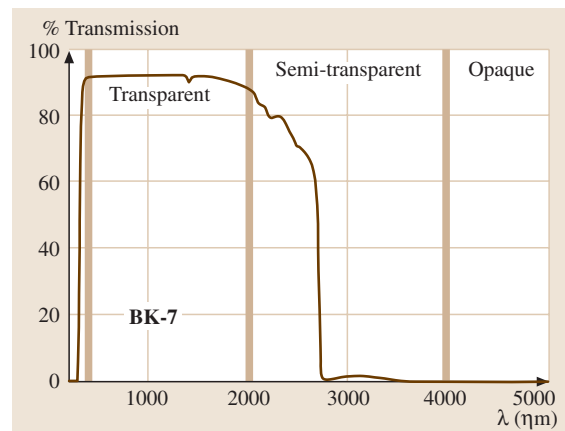


Fig. 5.11 Transmission through 10 mm thick BK-7 optical glass showing regions of transparency (visible), semi-transparency (near IR) and opacity (far IR)

complexes, insulators (crystals and phase separation), semiconductors, and conductors [5.19].

Intrinsic Absorption Mechanisms in Glass. The UV–VIS absorption is dominated by electronic transitions [5.22]. The electronic transitions that occur in the UV–VIS range involve a transition from the highest occupied molecular orbital (HOMO) to the lowest unoccupied molecular orbital (LUMO). The difference in energy (ΔE) between the LUMO (E_{LUMO}) and HOMO (E_{HOMO}) energy levels corresponds to the energy of the corresponding absorption maxima, which can be related to the wavelength, and frequency of the absorption through Planck's constant and the speed of light [5.22]

$$\Delta E = E_{\text{LUMO}} - E_{\text{HOMO}} = h\nu = hc/\lambda. \quad (5.60)$$

The bonds present dominate the energy gap between the HOMO and LUMO, and thus the energy of the transition [5.22]. For example, the UV absorption is very similar between crystalline and fused silica, indicating that the short-range structure dominates the electronic transitions [5.22]. It is in the long-range structure of silica that fused and crystalline differ due to the random connections of the SiO_4 tetrahedra in fused silica.

The absorption band present in Fig. 5.12 represents the intrinsic UV absorption in silica. The increase in peak width is due to the distribution of tetrahedra in the fused silica. The lowest energy peak is associated with a Wannier exciton, but the next three peaks are associated with electronic transitions between the oxygen 2p electrons with the silicon 3d electrons [5.15].

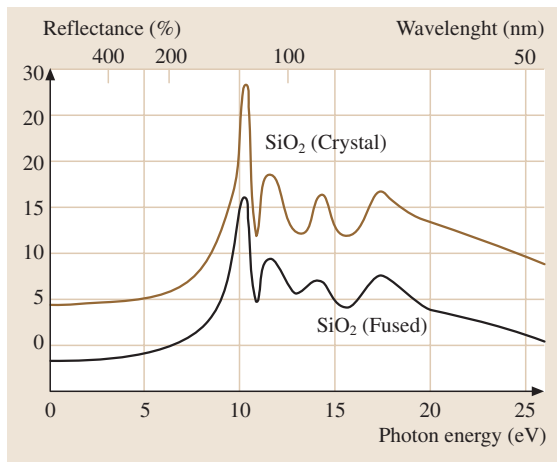


Fig. 5.12 Intrinsic absorption of glassy and crystalline SiO_2 (after [5.15]). The values for fused silica have been lowered by 5% for improved visibility

The intrinsic absorption of glass in the IR is due to molecular vibrations, which begin at the multiphonon edge [5.23]. These vibrations can be understood using a harmonic oscillator model. For a diatomic molecule, the frequency of vibration, ν_{ph} , is dependent upon the reduced mass (M) and the force constant (f) [5.15]:

$$\nu_{\text{ph}} = \frac{1}{2\pi} \sqrt{\frac{f}{M}}, \quad (5.61)$$

where the reduced mass between two atoms is

$$\frac{1}{M} = \frac{1}{m_1} + \frac{1}{m_2}. \quad (5.62)$$

However, in glasses we are not dealing with a simple diatomic system, which makes it really an *anharmonic* oscillator giving rise to overtone vibrations at higher energies (shorter wavelengths). However, the harmonic oscillator model can be used to gain a qualitative understanding of vibrational frequencies in various glasses. The most common extrinsic absorption centers in the IR are OH, CO_2 and iron [5.15]. The multiphonon edge is shifted to longer wavelengths (smaller frequencies) by heavier ions (greater M) and weaker bonds (smaller f), which is why chalcogenide glasses transmit further into the IR than more traditional oxide glasses [5.23].

In the visible wavelength region, glasses normally display no intrinsic absorption.

Ionic Absorption in Glass. Absorption in glass due to ionic impurities involves transition-metal (TM) and rare-earth (RE) ions that act as the central atom (CA) in ligand complexes absorbing light in different wavelength bands [5.20,22,24,25]. The absorption due to TM and RE ions results from electronic transitions within the d or f orbitals of the TM or RE elements [5.22]. The energy of the absorption band is dependent upon the coordination environment of the CA (coordination number and geometry), the chemical identity of the CA (Cr, Co, Cu ...) and the ligand (O versus F). The base composition of the glasses affects the nature of the ligand and bonds available for the CA. This change affects the energy of the absorption bands and thus the resulting color of the glass [5.19]. This effect is taken advantage of when designing filter glass (see Sect. 5.3), and can be used to explain the differences in color of the same TM or RE in different glass matrices. The absorptions result from a splitting of the d orbitals for TM elements and the 4f orbitals for the RE elements in the presence of a ligand field. The ligand field theory that governs these splittings and interactions are covered in detail for TM elements in *Bersuker* [5.22] and in general in

Douglas et al. [5.24]. The absorption cross section for TM elements is approx. 100 times that of the absorption cross section for RE elements [5.15]. This results in distinct coloration at the ppm concentration level for TM elements.

Figure 5.13 shows the variation in absorption due to various TM elements in BK 7 glass at the 1 ppm dopant level [5.15]. The intensity of cobalt and nickel absorption is clearly shown.

These intraband transitions due to electronic transitions within the ionic species d or f orbitals are formally Laporte-forbidden transitions by selection rules, so the intensity is low [5.22, 24]. Charge-transfer transitions involve an electron transition between the CA and the coordinating ligand (interband). These interband transitions are formally allowed, and thus have approx. 100–1000 times greater intensity than the intraband electronic transitions [5.24].

Absorption by Semiconductor Particles. Semiconductor particles in glass are typically too small (1–10 nm) to scatter visible light, however they absorb light over a continuum of wavelengths corresponding to energies greater than the band gap (E_g) of the semiconductor particles [5.10]. The band gap of the semiconductor particles is controlled by the size of the particle and the chemical composition. Typically, semiconductor glasses are melted with Zn, Cd, S, Se and Te raw materials in the batch and upon casting the glasses cool colorless [5.15]. Secondary heat treatment (striking) results in the crystallization of various semiconductor crystal phases in the glass, or a mixture thereof: ZnS, ZnSe, ZnTe, CdS, CdSe, and CdTe. The size and distribution of the semiconductor particles can be controlled by the heat treatment and thus so can the optical properties of the resulting glasses. With the proper heat treatment, one glass can be “struck” into multiple glasses with absorptions leading to red, yellow and orange coloration [5.15]. When the band gap of the particles is large enough, the absorption edge is shifted into the UV and the glass appears colorless. The opposite can also occur when the band gap energy is so low that it absorbs all visible light, and the sample appears black [5.15].

Semiconductor-doped glasses are often used as low-pass filter glasses because of their sharp absorption cutoff. Semiconductor-doped glasses are covered in more detail in Sect. 5.3.

Absorption by Conducting Particles. The coloration of glasses due to small conducting particles in the glass is

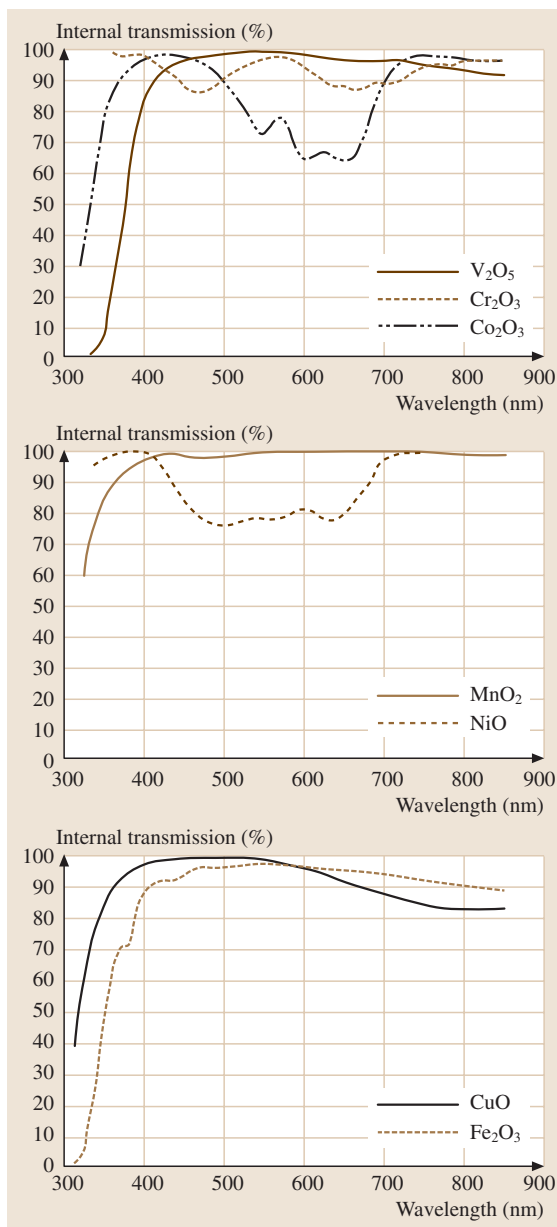


Fig. 5.13 Absorption due to 1 ppm of transition metal elements in BK 7 glass in 100 mm thick samples (after [5.15])

a combination of both absorption and scattering effects. For small metallic particles, the scattering is governed specifically by Mie's theory but can also be treated in general with the Rayleigh scattering theory [5.10]. Mie scattering is covered in depth in Sect. 5.1.7 and Rayleigh scattering in Sect. 5.1.8.



5.1.4 Optical Anisotropy

Maxwell's Equation in Dielectric Media

In the following section we will discuss the behavior of electromagnetic fields in media, especially in anisotropic materials. The basic interaction of light with matter consists of the interaction of the electric field with charges (electrons and ions) in a material (see Sects. 5.1–5.3). The propagation of light through matter is driven by the polarization of the material. Here we only consider linear effects in the material, i. e. the polarizability depends only linearly on the electric field strength. In a macroscopic formulation the material's response to the external field is given by the dielectric function (Sect. 5.1.1). From a microscopic point of view it is obvious that the polarization and thus the dielectric function

depend on the atomic structure of the material. Looking at the various crystal structures [5.26] shown in Fig. 5.14 one can easily understand that along different crystal orientations the dielectric function has different values since the atomic distances have different values. Thus the dielectric function of a crystalline material will not just be a number, but will be represented by a matrix (a two-dimensional tensor). The well-known occurrence of birefringence in certain materials is a consequence of the orientation-dependent dielectric function.

We start again with the Maxwell equation [5.27] for the solution of a plane wave. This is appropriate since any solution can be described as the superposition of plane waves. Inserting a plane wave ansatz

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} = \mathbf{E}_0 e^{i\omega \left(\frac{\mathbf{k}}{\omega} \cdot \mathbf{r} - t \right)} = \mathbf{E}_0 e^{i\omega \left(\frac{1}{v} \mathbf{l} \cdot \mathbf{r} - t \right)} \quad (5.63)$$

(with the phase velocity $v = \omega/k$ and $\mathbf{l} = \mathbf{k}/k$) into Maxwell's equations we find the following relations for the field components

$$\mathbf{D} = -\frac{1}{\mu\mu_0\omega} \mathbf{k} \times (\mathbf{k} \times \mathbf{E}_0) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}, \quad (5.64)$$

$$\mathbf{B} = \frac{1}{\omega} (\mathbf{k} \times \mathbf{E}_0) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}, \quad (5.65)$$

and

$$\mathbf{H} = \frac{1}{\mu\mu_0\omega} (\mathbf{k} \times \mathbf{E}_0) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (\mathbf{B} = \mu\mu_0\mathbf{H}). \quad (5.66)$$

The relation of the linear response of the material to the external electric field together with the Maxwell equation will give conditions for the existence of the above ansatz of a plane wave. Not taking into account dispersion, i. e. the frequency dependence of the dielectric function, the material's polarization is given by [5.28]

$$D_i = \varepsilon_{ij} \varepsilon_0 E_j \quad (5.67)$$

(where i and j are indices for the spatial components and the Einstein sum convention is used). Since the dielectric tensor is symmetric one can always find a coordinate system, where the tensor is diagonal, the system of principle axes [5.28, 29]. In the following we will always work in this reference system.

$$\begin{pmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{pmatrix}. \quad (5.68)$$

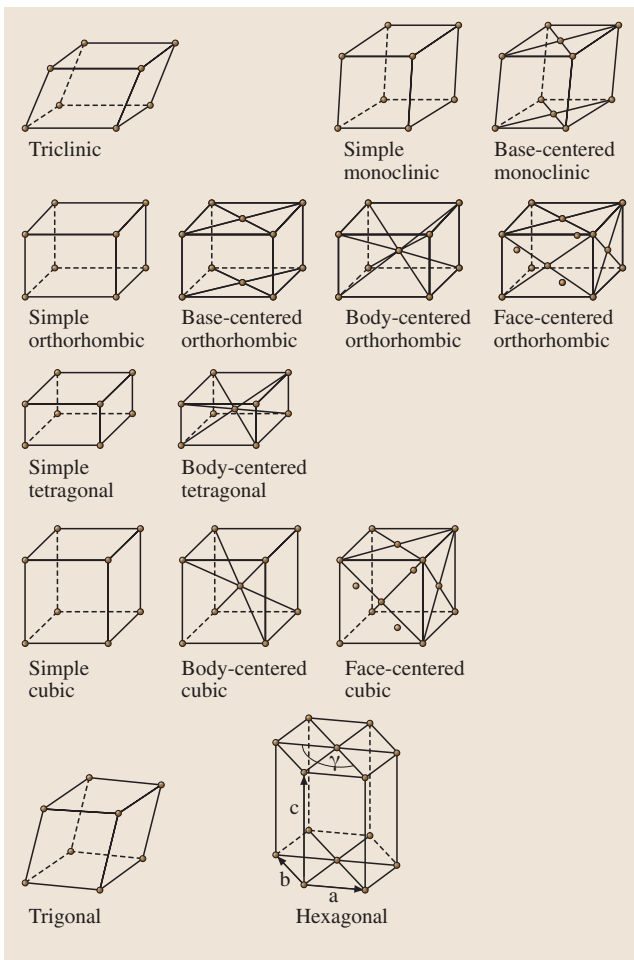


Fig. 5.14 The fourteen Bravais lattices [5.26]

The relation between D and E now gives $D_i = \varepsilon_i \varepsilon_0 E_i$, (no sum over i). Inserting this into (5.64) and using the expansion of two cross products one finds

$$\varepsilon_i \varepsilon_0 \mu \mu_0 \omega^2 E_i = k^2 E_i - k_i (\mathbf{k} \cdot \mathbf{E}). \quad (5.69)$$

This is the basic equation which determines the solution that can exist in the material. Introducing a unit vector in the direction of propagation $\mathbf{l}$ and defining a velocity v_i one gets:

$$\begin{aligned} \frac{\omega^2}{v_i} E_i &= k^2 [E_i - l_i (\mathbf{l} \cdot \mathbf{E})], \\ \left(\mathbf{k} &= k\mathbf{l}, \quad |\mathbf{l}| = 1, \quad \frac{1}{v_i} = \sqrt{\varepsilon_i \varepsilon_0 \mu \mu_0}, \right. \\ E_i &= \frac{l_i (\mathbf{l} \cdot \mathbf{E})}{1 - v^2/v_i^2}, \quad v = \frac{\omega}{k} \left. \right). \end{aligned} \quad (5.70)$$

Multiplying the last equation by l_i and summing over i results in a quadratic equation for v , the so called Fresnel equation of wave normals [5.28, 29]

$$\begin{aligned} \mathbf{l} \cdot \mathbf{E} &= \sum_i \frac{l_i^2}{1 - v^2/v_i^2} (\mathbf{l} \cdot \mathbf{E}) \\ \Leftrightarrow 1 &= \sum_i \frac{l_i^2}{1 - v^2/v_i^2} \left(1 = \sum_i l_i^2 \right) \\ 0 &= \sum_i \frac{l_i^2}{1 - v^2/v_i^2} - l_i^2 = v^2 \sum_i \frac{l_i^2}{v^2 - v_i^2} \\ \Leftrightarrow \sum_i \frac{l_i^2}{v^2 - v_i^2} &= 0. \end{aligned} \quad (5.71)$$

This is the equation usually discussed in crystal optics. Since it is a quadratic equation in the variable v , there are at most two solutions for the phase velocity. This is the origin of birefringence which we will discuss in detail in the section on uniaxial materials.

Classes of Optical Anisotropy Materials

From the representation of the dielectric tensor in the system of principle axes we see that there are at most three different values possible for the dielectric function. Before discussing in detail birefringence in matter, we discuss first the different classes of optical anisotropic materials resulting from the dielectric tensor [5.29].

Group I. Crystals in which all three values in the dielectric tensor in (5.68) are equal, $\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = \varepsilon$, belongs to the so called cubic class. The material is said to be *optical isotropic*. The relation between D and E reduces to the well-known form $\mathbf{D} = \varepsilon \mathbf{E}$.

Group II. The next possible case is that in which two of the three axes are equivalent. The corresponding dielectric tensor has the property $\varepsilon_1 = \varepsilon_2 \neq \varepsilon_3$. Such a crystal is said to be *uniaxial*. This group of crystals belong to the trigonal, tetragonal and hexagonal system (Fig. 5.14).

Group III. The last group of crystals has three different values in the dielectric tensor $\varepsilon_1 \neq \varepsilon_2 \neq \varepsilon_3$. All three axes are different and the crystals belong to the orthorhombic, monoclinic and triclinic class (Fig. 5.14).

Uniaxial Materials:

Ordinary and Extraordinary Rays

As discussed above a material is said to be uniaxial if two of the three values of the dielectric tensor are equal $\varepsilon_1 = \varepsilon_2 \neq \varepsilon_3$. As a result the velocities v_1 and v_2 are also equal ($v_1 = v_2$). The Fresnel equation of the wave normals (5.71) for this case have the following form

$$\begin{aligned} 0 &= (v^2 - v_1^2)(v^2 - v_3^2)(l_1^2 + l_2^2) + (v^2 - v_1^2)^2 l_3^2 \Leftrightarrow \\ 0 &= (v^2 - v_1^2) \left[(v^2 - v_3^2)(l_1^2 + l_2^2) + (v^2 - v_1^2) l_3^2 \right]. \end{aligned} \quad (5.72)$$

This equation has two solutions for the phase velocity

$$\begin{aligned} 1: v &= v_1 \\ 2: (v^2 - v_3^2)(l_1^2 + l_2^2) + (v^2 - v_1^2) l_3^2 &= 0 \\ \Rightarrow v^2 &= v_3^2 (l_1^2 + l_2^2) + v_1^2 l_3^2. \end{aligned} \quad (5.73)$$

Expressing the unit vector $\mathbf{l}$ in spherical coordinates, $l_3^2 = \cos^2(\vartheta)$, $l_1^2 + l_2^2 = \sin^2(\vartheta)$, one gets

$$v^2 = v_3^2 \sin^2(\vartheta) + v_1^2 \cos^2(\vartheta). \quad (5.74)$$

Usually one uses the notation $v_o = v_1$ (index o: ordinary) and $v_e = v_3$ (index e: extraordinary) (Fig. 5.16). Finally we arrive at the two solutions for the phase velocity of a plane wave propagating through the medium

$$\begin{aligned} 1. v &= v_o, \\ 2. v &= \sqrt{v_e^2 \sin^2(\vartheta) + v_o^2 \cos^2(\vartheta)}. \end{aligned} \quad (5.75)$$

The z -axis is called the optical axis. For the refractive index one obtains

$$\begin{aligned} n &= \frac{c}{v} \Rightarrow \\ n(\vartheta) &= c \frac{1}{\sqrt{v_o^2 \cos^2(\vartheta) + v_e^2 \sin^2(\vartheta)}} \\ &= \frac{n_o n_e}{\sqrt{n_o^2 + (n_e^2 - n_o^2) \cos^2(\vartheta)}}. \end{aligned} \quad (5.76)$$



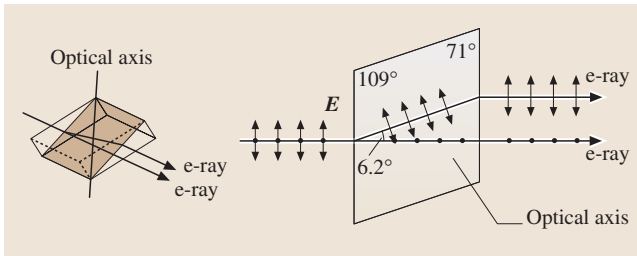


Fig. 5.15 Birefringence in calcite (CaCO_3), ordinary (o) and extraordinary ray (e) observed by perpendicular incidence to a surface that has an angle of approximately 45° to the optical axis [5.30]

The case $v_e > v_o$, corresponding to $n_o > n_e$, is called a positive crystal; the other case $v_e < v_o$ corresponding to $n_o < n_e$ is called a negative crystal.

Thus we have derived that in a uniaxial crystal there are always two solutions – the ordinary and extraordinary beam – with different phase velocities; one of them, the extraordinary beam, depends on the direction. To be able to understand the picture of these two rays traveling through for example a calcite crystal one must consider another quantity used in the theory of electro-magnetic radiation – namely the Poynting vector $\mathbf{S}$, which describes the energy flux of electromagnetic radiation (Sect. 5.1.1). For a plane wave with propagation vector $\mathbf{k}$ it follows from the Maxwell equation that the field vectors $\mathbf{E}, \mathbf{H}, \mathbf{S}$ and $\mathbf{D}, \mathbf{H}, \mathbf{k}$ form a set of orthogonal vectors:

$$\begin{aligned} \mathbf{S} &= \mathbf{E} \times \mathbf{H} \rightarrow \mathbf{S} \perp \mathbf{H} \text{ and } \mathbf{E}, \\ \mathbf{k} \cdot \mathbf{D} &= 0 \rightarrow \mathbf{k} \perp \mathbf{D}, \\ \mathbf{k} \cdot \mathbf{H} &= 0 \rightarrow \mathbf{k} \perp \mathbf{H} \\ \mathbf{D} &= -\frac{1}{\omega}(\mathbf{k} \times \mathbf{H}). \end{aligned} \quad (5.77)$$

The angle α between the vectors $\mathbf{k}$ and $\mathbf{S}$ is the same as that between $\mathbf{D}$ and $\mathbf{E}$ (Fig. 5.17). In an isotropic medium the angle α is zero, since $\mathbf{D}$ is proportional to $\mathbf{E}$. As a consequence, $\mathbf{k}$ and $\mathbf{S}$ point in the same direction. In an anisotropic medium this is generally no longer true. The light traveling through a uniaxial material and in particular the two rays shown in the calcite example are in fact the $\mathbf{S}$ vectors of the ordinary and extraordinary rays [5.30].

The angle between the ordinary and extraordinary ray is given by α , since the ordinary ray travels along $\mathbf{k}$ (remember, $\mathbf{k}$ and $\mathbf{S}$ point in the same direction). Thus the angle α gives the splitting between the ordinary

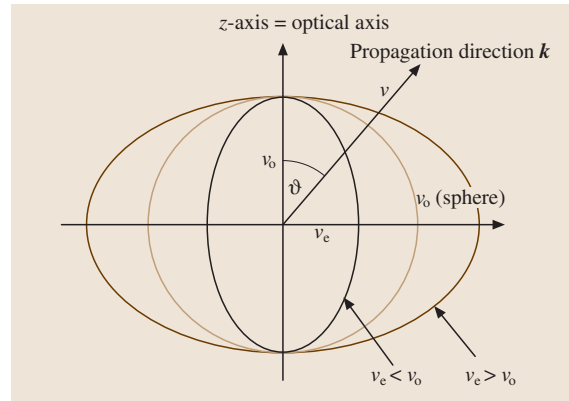


Fig. 5.16 Graphical representation of the solution to Fresnel's equation (Indikatrix)

and extraordinary ray in a uniaxial crystal. In the following derivation we want to obtain a rather simple expression for the angle α determined by the refractive indices n_e and n_o . Since we are only interested in the angle between two vectors we can consider unit vectors. Taking $\mathbf{D}$ as a unit vector, $\mathbf{E}$ is given for the material equation. In the framework of the principle axis

$$\mathbf{k} = \begin{pmatrix} \sin(\vartheta) \cos(\varphi) \\ \sin(\vartheta) \sin(\varphi) \\ \cos(\vartheta) \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \sin(\vartheta') \cos(\varphi') \\ \sin(\vartheta') \sin(\varphi') \\ \cos(\vartheta') \end{pmatrix}, \quad (5.78)$$

since $\mathbf{k} \cdot \mathbf{D} = 0$, it follows that

$$\begin{aligned} \cos(\vartheta) \cos(\vartheta') + \sin(\vartheta) \sin(\vartheta') \cos(\varphi - \varphi') &= 0 \Leftrightarrow \\ \cot(\vartheta') &= -\tan(\vartheta) \cos(\varphi - \varphi'). \end{aligned}$$

For the angle-independent solution v_o it follows that

$$\begin{aligned} \cos(\varphi - \varphi') &= 0 \Rightarrow \varphi' = \varphi - \frac{\pi}{2} \\ \text{and } \cot(\vartheta') &= 0 \Rightarrow \vartheta' = \frac{\pi}{2}. \end{aligned} \quad (5.79)$$

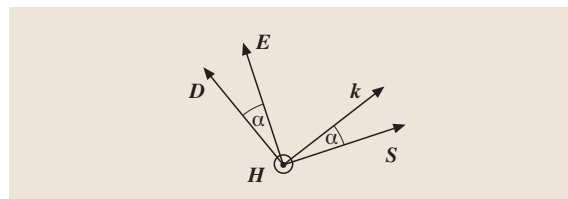


Fig. 5.17 Angle α between $\mathbf{k}$, $\mathbf{S}$, $\mathbf{D}$ and $\mathbf{E}$

For the second solution, which must be orthogonal to the first, it follows that: $\varphi' = \varphi$, $\vartheta' = \vartheta - \frac{\pi}{2}$.

Together with the relation between D and E one has

$$D^{(1)} = \begin{pmatrix} -\cos(\vartheta) \cos(\varphi) \\ -\cos(\vartheta) \sin(\varphi) \\ \sin(\vartheta) \end{pmatrix}, \quad D^{(2)} = \begin{pmatrix} \sin(\varphi) \\ -\cos(\varphi) \\ 0 \end{pmatrix}$$

$$\text{and } E^{(1)} = \begin{pmatrix} -\cos(\vartheta) \cos(\varphi) \frac{1}{\varepsilon_1} \\ -\cos(\vartheta) \sin(\varphi) \frac{1}{\varepsilon_1} \\ \sin(\vartheta) \frac{1}{\varepsilon_3} \end{pmatrix}, \quad E^{(2)} = \frac{1}{\varepsilon_1} D^{(2)}.$$

(5.80)

In the second case $D \propto E$ and, hence, $\alpha = 0$, which is obviously the ordinary ray. For the first solution, the extraordinary ray, we calculate the angle α from the scalar product $D \cdot E = |D| |E| \cos(\alpha)$, substituting D and E it follows that

$$\begin{aligned} & \frac{1}{\varepsilon_1} \cos^2(\vartheta) \cos^2(\varphi) \\ & + \frac{1}{\varepsilon_1} \cos^2(\vartheta) \sin^2(\varphi) + \frac{1}{\varepsilon_3} \sin^2(\vartheta) = \\ & 1 \left(\frac{1}{\varepsilon_1^2} \cos^2(\vartheta) \cos^2(\varphi) + \frac{1}{\varepsilon_1^2} \cos^2(\vartheta) \sin^2(\varphi) \right. \\ & \quad \left. + \frac{1}{\varepsilon_3^2} \sin^2(\vartheta) \right)^{1/2} \cos(\alpha) \Leftrightarrow \\ & \frac{1}{\varepsilon_1} \cos^2(\vartheta) + \frac{1}{\varepsilon_3} \sin^2(\vartheta) = \\ & \left(\frac{1}{\varepsilon_1^2} \cos^2(\vartheta) + \frac{1}{\varepsilon_3^2} \sin^2(\vartheta) \right)^{1/2} \cos(\alpha) \Leftrightarrow \\ & \cos(\alpha) = \frac{\varepsilon_3 \cos^2(\vartheta) + \varepsilon_1 \sin^2(\vartheta)}{(\varepsilon_3^2 \cos^2(\vartheta) + \varepsilon_1^2 \sin^2(\vartheta))^{1/2}}, \\ & \cos(\alpha) = \frac{n_e^2 \cos^2(\vartheta) + n_o^2 \sin^2(\vartheta)}{(n_e^4 \cos^2(\vartheta) + n_o^4 \sin^2(\vartheta))^{1/2}}, \end{aligned} \quad (5.81)$$

with the relation $\varepsilon = n^2$ and the notation $n_3 = n_e$, $n_1 = n_o$. As an example let us consider the aforementioned calcite. The surface of the calcite that one gets by cleaving the crystal has an angle of approximately 45° . Light incident perpendicular to the surface thus gives $\cos^2(\vartheta) = \sin^2(\vartheta) = 0.5$. The refractive indices for the ordinary and extraordinary rays in calcite are: $n_e = 1.4864$, $n_o = 1.6584$. From these values one finds the angle between the ordinary and extraordinary

ray as (Fig. 5.15):

$$\cos(\alpha) = \frac{1}{\sqrt{2}} \frac{n_e^2 + n_o^2}{\sqrt{n_e^4 + n_o^4}} \Rightarrow \alpha = 6.22^\circ. \quad (5.82)$$

5.1.5 Nonlinear Optical Behavior and Optical Poling

Since the first observation of nonlinear (NL) effects, many materials have been investigated concerning their nonlinear optical properties [5.31, 32]. Nonlinear materials can be found among gases and vapors, polymers, liquid crystals, organic solvents, or crystals – in short, in nearly every material system.

A linear dielectric medium is characterized by a linear relation between the polarization density P and the electric field E , $P = \varepsilon_0 \chi E$ (5.25), where ε_0 is the vacuum permittivity and χ is the electric susceptibility of the medium (see Sect. 5.1.1). A nonlinear dielectric medium, on the other hand, is characterized by a nonlinear relation between P and E , as illustrated in Fig. 5.18.

The nonlinearity may be of microscopic or macroscopic origin. The polarization density $P = Np$ is a product of the individual dipole moment p (which is induced by the applied electric field E) and the number density of dipole moments N . The nonlinear behavior may have its origin in either p or N .

The relation between p and E is linear when E is small, but becomes nonlinear as E acquires values comparable with interatomic electric fields (typically 10^5 – 10^8 V/m). This may be explained in terms of the simple Lorentz model in which the dipole moment is $p = -ex$, where x is the displacement of a mass with charge $-e$ to an applied electric force $-eE$. If the restraining elastic force is proportional to the displacement (i. e., if Hooke's law is satisfied), the equilibrium displacement x is proportional to E . Hence, P is proportional to E , and the medium is linear. However, if the restraining force is a nonlinear function of the

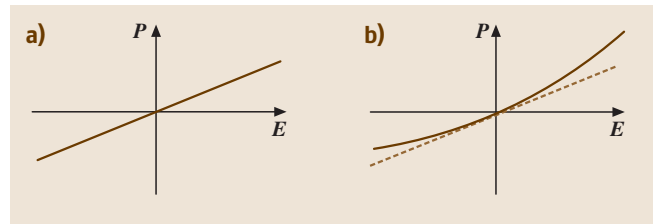


Fig. 5.18a,b The P – E relation for (a) a linear dielectric medium, and (b) a nonlinear medium

displacement, the equilibrium displacement $\mathbf{x}$ and the polarization density $\mathbf{P}$ are nonlinear functions of $\mathbf{E}$ and, consequently, the medium is nonlinear.

Another possible origin of the nonlinear response of an optical material to light is the dependence of the number density N on the optical field. An example is a laser medium for which the number of atoms occupying the energy levels involved in the absorption and emission of light is dependent upon the light intensity itself (see Sect. 5.1.7).

Since externally applied optical fields are typically small in comparison with characteristic interatomic or crystalline fields, even when focused laser light is used the nonlinearity is usually weak. The relation between $\mathbf{P}$ and $\mathbf{E}$ is then approximately linear for small $\mathbf{E}$, deviating only slightly from linearity as $\mathbf{E}$ increases (Fig. 5.18). Under these circumstances, it is possible to expand the function that relates $\mathbf{P}$ to $\mathbf{E}$ in a Taylor series about $\mathbf{E} = 0$

$$\mathbf{P}(\mathbf{E}) = \varepsilon_0(\chi^{(0)}\mathbf{E} + \chi^{(2)}\mathbf{E}^2 + \chi^{(3)}\mathbf{E}^3 + \dots), \tag{5.83}$$

where $\chi^{(0)}$ is the linear susceptibility and $\chi^{(n>0)}$ are the higher-order coefficients.

For the nonlinear refractive index n , we get

$$n(\mathbf{E}) = n_0 + n_1\mathbf{E} + n_2\mathbf{E}^2 + \dots \tag{5.84}$$

Equations (5.83) and (5.84) provide the basic description for a nonlinear optical medium. Anisotropy (Sect. 5.1.4), dispersion (Sect. 5.1.2), and inhomogeneities (Sect. 5.1.7) have been ignored both for simplicity and to enable us to focus here on the basic nonlinear effect.

In centrosymmetric media with inversion symmetry (like glass), in which the properties of the medium are not altered by the transformation $\mathbf{r} \rightarrow -\mathbf{r}$, the $\mathbf{P}(\mathbf{E})$ function must have odd symmetry so that the reversal of $\mathbf{E}$ results in the reversal of $\mathbf{P}$ without any other change. The second-order nonlinear coefficient $\chi^{(2)}$ must then vanish, and the lowest nonlinear term is of third order. Typical values of the second-order nonlinear coefficient $\chi^{(2)}$ for dielectric crystals, semiconductors, and organic materials lie in the range $\chi^{(2)} = 10^{-24}$ to 10^{-21} m/V. Typical values of the third-order nonlinear coefficient $\chi^{(3)}$ for glasses, crystals, semiconductors, semiconductor-doped glasses, and organic materials are $\chi^{(3)} = 10^{-34}$ to 10^{-29} (m/V)².

Nonlinear Wave Equation

The propagation of light in a nonlinear medium is governed by the wave equation (5.29), which can be written

in the form [5.34]:

$$\Delta \mathbf{E} - \frac{n^2}{c_0^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = -\mu_0 \frac{\partial^2 \mathbf{P}_{NL}}{\partial t^2} \tag{5.85}$$

where $\mathbf{P}_{NL}$ is the nonlinear part of the polarization density:

$$\mathbf{P}_{NL} = \varepsilon_0(\chi^{(2)}\mathbf{E}^2 + \chi^{(3)}\mathbf{E}^3 + \dots). \tag{5.86}$$

It is useful to regard (5.85) as a wave equation in which the term $-\mu_0 \partial^2 \mathbf{P}_{NL} / \partial t^2$ acts as a source radiating in a linear medium of refractive index n . Because $\mathbf{P}_{NL}$ is a nonlinear function of $\mathbf{E}$, (5.85) is a nonlinear partial differential equation in $\mathbf{E}$. This is the basic equation that underlies the theory of nonlinear optics. There are two approximate approaches to solve the nonlinear wave equation: the Born approximation and the coupled wave theory.

Optical Classification of Nonlinear Effects

Generally, the term “nonlinear optics” describes that part of the interaction between light and matter in which the optical material properties of the light beam passing through a nonlinear medium are changed by the beam itself or another electromagnetic field. Table 5.3 gives a short overview over the various nonlinear effects and their order of nonlinearity.

Here, we will focus our discussion on one of the most prominent effects: second-harmonic generation.

Second-Harmonic Generation (SHG)

In the case of the main second-order application [5.31] – second-harmonic (SH) generation – which e.g. can

Table 5.3 Nonlinear processes [5.33]

Nonlinear optical effect	Order
n th harmonic generation (e.g. SHG, rectification)	n
n -wave mixing	$n - 1$
n th photon absorption	n
Phase conjugation	3
Intensity-dependent refraction (e.g. Kerr effect, self phase modulation, self modulation)	2, 3
Optical parametric oscillation	2
Induced opacity	2
Induced reflectivity	2
Raman and Brillouin scattering	2

tive indices below 2 but some glasses which fulfill these requirements are the chalcogenide glasses, chalcogenides, heavy metal oxide (HMO)- and RE-doped oxide glasses [5.37, 38, 41–43].

Coherent Nonlinear effects are characterized by an instantaneous response of the material to the incident fields. This condition is fulfilled if the corresponding electronic excitation in the dielectric material is created only virtually – using the language of coupled wave theory, see p. 266 – or, within the quantum mechanical treatment, if we treat only interactions of coherent polaritons [5.31].

Incoherent Nonlinear effects involve modifications of the optical properties by really or incoherently excited species, e.g. electron–hole pairs, excitons, or phonons. These species have finite lifetimes τ which can be on the order of ns to ms. Due to this finite lifetime, their density does not instantaneously follow the incident light field, but depends on the generation and depopulation rate, which are characterized by an instantaneous response of the material to the incident field. Table 5.4 lists some effects in optical materials and typical values.

Creation of Artificial Second-Order Nonlinearity: Optical Poling

As mentioned before, the SHG process is related to the second-order susceptibility and thus allowed only in non-centrosymmetric materials. On the other hand, third-order nonlinear materials such as glass offer advantages such as easier and cheaper fabrication with high homogeneity. Thus, attempts have been made either to increase the field strength (electric poling) or create artificial symmetry (thermal or optical poling [5.37, 38, 44, 45]), or combine the two methods.

There exist two types of electric field poling: electrode poling (in which a DC field is applied to a pair

of electrodes) and corona discharge poling (where the surface of a film is charged under a strong potential). Annealing in vacuum is effective in enhancing the SHG intensity, whereas addition of hydrogen at room temperature increases the lifetime of the optical nonlinearity. Further methods are poling by mechanical stress or with a chemical dilutant [5.33, 40, 42, 46–48].

Higher SHG intensities than those obtained with thermal or electric poling can usually be derived by optical poling. This is due to the fact that an artificial phase matching (quasi phase matching) can be achieved by creating a stable $\chi^{(2)}$ -grating in the glass (see Sect. 5.6). Thus, with optical poling phase matching over a sample length of several cm can be achieved, whereas the active region for thermal or electric poling is only between 5–15 μm .

Optical poling is achieved by intensive irradiation by, e.g. 1064 nm light and irradiation by additional light of the SH wavelength during the poling procedure (seeding light). After this procedure intensive SHG (e.g. in Pb–silica, Ge–silica, metal- or SC-doped, or ZnO–TeO₂ glasses) has been observed. Fundamental and SH cause a semipermanent DC electric field in the glass (by photoinduced current or structural orientation) which cancels the inversion symmetry and allows the periodic phase matching of the SHG (see also Table 5.22).

Optical poling can be also used for organic materials, e.g. dyes, which have asymmetric structure and exhibit polarization. Dispersed in a glass matrix, they have random orientation and thus are isotropic. Even growth of crystals in the glass does not result in the desired second-order nonlinearity because of the usually centrosymmetric (antiparallel) arrangement of the polar molecules. By irradiation with two laser beams or by applying an electric field, the random orientation can be changed to an ordered configuration.

In detail, poling is a process which can be found especially in the fabrication of photonic devices. Most

Table 5.4 Third-order susceptibility, response time τ , and magnitude of n_2 for various sources of n_2 (NRI) [5.43] (SC = semiconductor, MQW = multi quantum well)

Mechanism	Typical materials	$\chi^{(3)}$ (esu)	Response time τ (s)	n_2 (esu)	Type
Semiconductor bandfilling, exciton effects	GaAs, MQW, SC-doped glasses	$10^{-2} - 10^{-8}$	$\approx 10^{-8}$		Resonant
Nonlinear electronic polarizability	PTS, glass	$10^{-11} - 10^{-14}$	$10^{-14} - 10^{-15}$	$10^{-8} - 10^{-14}$	Non-resonant
Optical Kerr effect			$10^{-11} - 10^{-12}$	$10^{-11} - 10^{-12}$	
Electrostriction			$10^{-7} - 10^{-8}$	$10^{-7} - 10^{-8}$	
Photorefractive effect	BaTiO ₃ , CdTe	$10^{-4} - 10^{-5}$	$10^{-3} - 1$		Resonant
Molecular reorientation	CS ₂	$10^{-12} - 10^{-14}$	$\approx 10^{-12}$		Non-resonant
Thermal effects	ZnS, ZnSe	$10^{-4} - 10^{-6}$	$10^{-3} - 1$	$10^{-4} - 10^{-5}$	Resonant

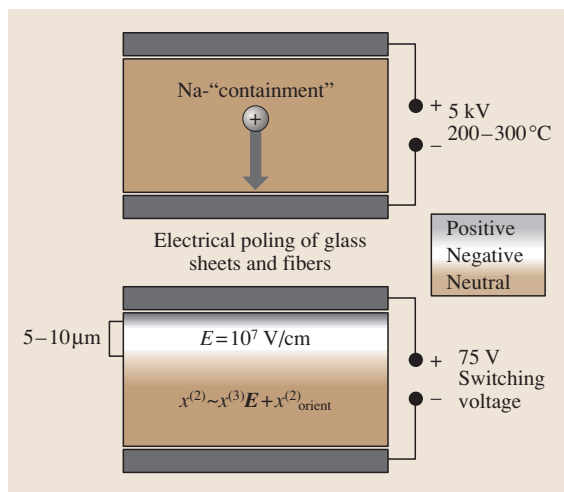


Fig. 5.20 Electrical poling of glass sheets

of the optical poling experiments to date have been performed in fibers. But additional **SHG** in waveguide geometries is also possible. In the case of waveguides, the poling was performed near the glass transition point of a nonlinear core polymer in order to establish a permanent orientation of the nonlinear chromophore dipoles. In the case of e.g. glasses containing Na^+ and H^+ a charge field is established by wandering of these mobile ions in a strong electric field (thermo-electric poling).

5.1.6 Emission

For numerous applications for active optical materials the interrelation between specific absorption and emission processes [5.49] is of crucial importance. One example is a laser or an optical amplifier, where a specific absorption on a well-defined level in the optically active material is needed for pumping. This leads to the building of an inversion in a lower lying state. The inversion can either be emptied by spontaneous emission, which causes noise, or by stimulated emission which amplifies a laser (or signal) mode.

The microscopic processes between these absorption and emission levels can have different origins, for example several levels of 4f electron systems of a rare earth. These levels only interact weakly with lattice vibrations and therefore barely show thermal broadening (zero phonon lines). Examples for laser materials are Nd^{3+} -doped **YAG** (yttrium aluminate garnet) single crystals and Nd^{3+} -doped phosphate glasses. Examples of optical amplifier materials are

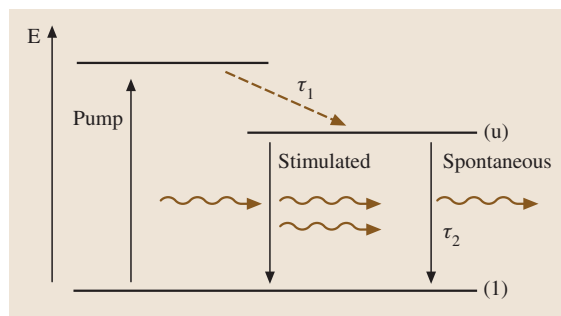


Fig. 5.21 Schematic plot of a three level laser system Via a pump level with a very short lifetime $\tau_1 \ll \tau_2$ the upper laser level (u) is populated. Depopulation can either occur via stimulated emission, where copies of photons are made or via spontaneous emission where the probability is determined by the lifetime τ_2

Er^{3+} -doped fused silica and $\text{Er}^{3+}/\text{Yb}^{3+}$ -doped multicomponent glasses (see Fig. 5.23). Here the optical emission of Er^{3+} in the wavelength range of optical telecommunication is used. Further laser systems are e.g. ruby lasers (Cr^{3+} + doped Al_2O_3); semiconductor lasers, where the pumping is done electronically; gas lasers such as the helium–neon laser, where the energy levels of He are used for pumping and the inversion is built up in the energy levels of Ne; or dye lasers where broad π -electron systems of organic molecules are used to provide the necessary tunable energy levels. For more details on laser physics see also [5.49]. The absorption and the absorption cross sections can be directly measured and are described in Sect. 5.1.3. As we will see, the approach to treating spectral emission of an excited material is more sophisticated.

Emission Cross Section

Although absorption has a well-defined meaning with respect to attenuation of light and can be measured quantitatively via the absorption coefficient α it has a quantummechanical origin. Especially for making absorption (and emission) quantitative the quantummechanical concept of transition probabilities is needed. It is also usually expressed via oscillator strengths of certain transitions. The fact that light is absorbed at certain frequencies ω (or energies $\hbar\omega$) means that light is portioned in packages (photons) of energy given by

$$\hbar\omega = \frac{hc}{\lambda}, \quad (5.87)$$

where h is Planck's quantum and $\hbar = h/2\pi$. The absorption cross section is connected to the transition probability R_{ij} , which is the probability for absorbing a photon to induce a transition in the medium from state i to state j . In general the transition probability depends on the detailed wavelength dependence of the transition and on the wavelength dependence of the incoming light. Therefore the transition probability is related to the cross section as follows:

$$R_{ij} = \int \sigma(\omega) \Phi(\omega) d\omega = \int \frac{-2\pi c}{\lambda^2} \sigma(\lambda) \Phi(\lambda) d\lambda, \quad (5.88)$$

where Φ is the photon flux measured in units of m^{-2} , which is related to the light intensity $I = \hbar\omega\Phi$.

Quantum Emissions

Furthermore, emission and emission cross sections have a quantum-mechanical nature. Historically the work of Einstein did not only bring its author a Nobel prize but also laid the foundation of laser physics. Einstein introduced the fundamental concepts of stimulated and spontaneous emission. The stimulated emission process makes copies of photons with identical energy and – more importantly – phase. This is the process behind laser activity and amplification. While stimulated emission is just the opposite process of absorption, spontaneous emission is caused by the finite lifetime of an excited state. It is responsible for noise in a laser or optical amplifier. The lifetime τ_2 of the upper, excited level with only one transition $2 \rightarrow 1$ is the inverse of the transition probability A_{21}

$$\tau_2 = \frac{1}{A_{21}}. \quad (5.89)$$

In the case of different decay channels the lifetime is the inverse of a sum over all transition probabilities

$$\tau_2 = \frac{1}{\sum_m A_{2m}}. \quad (5.90)$$

We now want to define an emission cross section. If we consider an ideal system with a single nondegenerate transition and no nonradiative decay channels, the emission and absorption cross sections are identical $\sigma^{\text{abs}} = \sigma^{\text{emis}}$.

Real systems deviate from this in the following ways:

- The upper as well as the lower laser level are composed of an ensemble of sublevels. Especially in disordered systems such as glasses an inhomogeneous broadening of the energy levels can become

important. Therefore an effective density of states for each level has to be taken into account.

- The spacing of the sublevels is such that thermal occupation has to be considered, i. e. where $\Delta E = \hbar\Delta\omega < k_B T$ where k_B is the Boltzmann constant.
- Different degeneracies of the energy levels are involved. If the upper level has a degeneracy of g_2 and the lower level a degeneracy of g_1 , the emission and absorption probabilities have to be multiplied by these degeneracy factors. If an energy state is connected with a magnetic moment, a canceling of degeneracies due to crystal field splitting can occur, which leads to an ensemble of sublevels (see above).
- There exist nonradiative processes that empty the upper state and decrease the radiative emission cross section. At high enough energies lattice vibrations (phonons) can empty an excited state in a nonradiative way.

For these reasons the effective emission cross section for a real system deviates from the absorption cross section in a nontrivial way

$$\sigma^{\text{abs}}(\lambda) \neq \sigma^{\text{emis}}(\lambda). \quad (5.91)$$

An ideal two-level system and a real system are sketched schematically in Fig. 5.22a,b. The emission cross section is related to the transition probability by

$$\sigma^{\text{emis}}(\lambda) = \frac{\lambda^2 A_{21}}{8\pi} S(c/\lambda), \quad (5.92)$$

where $S(c/\lambda) = S(\nu)$ is the lineshape function. The lineshape function is normalized to unity $\int S(\nu) d\nu = 1$. The

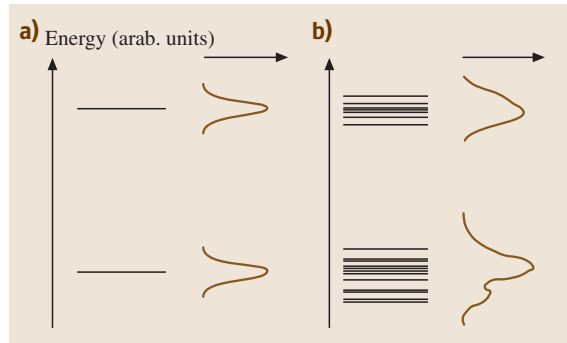


Fig. 5.22a,b In (a) the schematic terms for an ideal, only lifetime-broadened emission process from an excited state to the ground state is plotted. Many real systems sketched in (b) show a splitting into different sublevels, different degeneracies and nonradiative transitions

question remains if and how it is possible to measure the emission cross section. While for the absorption cross section such a measurement is straightforward, it is far more complicated to complete this process for an emission cross section. The lineshape can be determined by saturating a transition with a light source, then switching off the light and measuring the resulting fluorescence, which is decaying exponentially in time, with a spectrometer. The wavelength dependent signal of the spectrometer is proportional to the emission cross section. If a sample of volume V is irradiated with light until a stationary state is reached, the inversion level $0 < N_2(x) < 1$ is a measure of the excited optical centers. The probability for an emission process to occur in the wavelength range between λ and $\lambda + \Delta\lambda$ is

$$W(\lambda) = \frac{\sigma^{\text{emis}}(\lambda)\Delta\lambda}{\int \sigma^{\text{emis}}(\lambda)d\lambda}. \quad (5.93)$$

Therefore the light intensity emitted by an infinitesimal volume element $dV(x)$ is given by

$$dI(\lambda, x) = NN_2\tau^{-1}dV(x)h\nu W(\lambda). \quad (5.94)$$

Here again N is the concentration of optical centers and $h\nu = hc/\lambda$ is the energy of each emitted phonon with Planck's constant h . For a measurement of $W(\lambda)$, this intensity, which is emitted with equal probability in all spatial directions, has to be collected in a spectrometer which normally covers only a small part of the spatial angle in which the emission takes place. On the way to the spectrometer the light may also travel through other volume elements causing absorption or stimulated emission. Therefore the absolute signal measured in the spectrometer is related in a nontrivial way to the emission cross section. For measuring cross sections the following special conditions are thus used:

- Excitation and measurement are done close to the surface of a sample to avoid the emitted light traveling through large parts of the sample. This is done either in a backscattering geometry or at a corner of the material.
- The excitation is so strong (e.g., with a laser) that saturation is reached $N_2 \approx 1$. In this way absorption of emitted light traveling through the sample is suppressed.
- Standards are used for a comparison. These are e.g. rare-earth ions of well-defined concentration in single crystals, glasses with well-controlled compositions or fluid organic dyes.

Even if it is a nontrivial task to measure the emission cross sections, it is the most important microscopic quantity for the laser activity of a transition.

Example for Er Ions

As an example we show here the emission and absorption cross sections of Er^{3+} for the transition around 1550 nm. This transition is important for telecommunication applications since it takes place in the wavelength range where silica fibers exhibit minimum light attenuation. The states involved in this transition have the spectroscopic symbols $E_1 \rightarrow {}^4I_{15/2}$ and $E_2 \rightarrow {}^4I_{13/2}$ which stand for spin, angular and total moment of the correlated eleven 4f electron system which forms the Er^{3+} system. In Fig. 5.23 we plot an example of this transition of Er^{3+} , which is of high technological importance. The value of the emission cross section is influenced by the fact that the ground state is eightfold degenerate whereas the degeneracy of the excited state is sevenfold. It is further influenced by nonradiative decay channels due to lattice vibrations. The spectral shape of the cross sections is formed by the level splitting in the local crystal field formed by the glass environment and the thermal occupation of the different sublevels. In addition there is inhomogeneous broadening due to the disordered structure of the glass.

5.1.7 Volume Scattering

This section deals with the interaction of light with particles inside a certain volume leading to light scattering. Before going into details the term scattering must be defined and explained. Afterwards scattering is subdivided

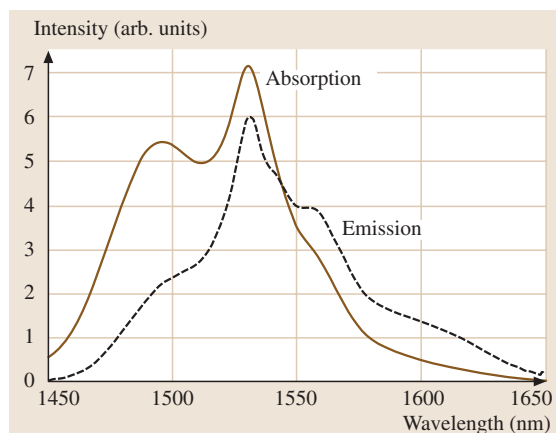


Fig. 5.23 Plot of the emission and absorption cross section for an Er^{3+} -doped glass

into single scattering, multiple scattering, and coherent scattering.

Definition and Basics

Scattering is defined here as energy absorption of incident light followed by re-emission of part of this light at the same frequency. Thus inelastic effects such as Raman scattering or Brillouin scattering are not discussed in this section.

The *origins* of light scattering are surface irregularities. From this point, volume scattering is scattering at the surface of particles in a certain volume. Surface scattering is discussed in Sect. 5.1.8.

Obviously scattering is strongly related to diffraction, as seen from the definition of scattering. Indeed, diffraction is scattering by a flat particle [5.50].

In the following, volume scattering is subdivided into (Fig. 5.24)

- single scattering,
- multiple scattering,
- coherent scattering.

Volume scattering can be regarded as the sum of single scattering events as long as the density of scattering particles is not too high. Mathematically this is expressed with the packing fraction η defined by

$$\eta := \frac{NV_{\text{scat}}}{V_{\text{vol}}}, \quad (5.95)$$

where N is the number (integer) of the single scatterer, V_{scat} is the volume of a single scatterer, and V_{vol} is the

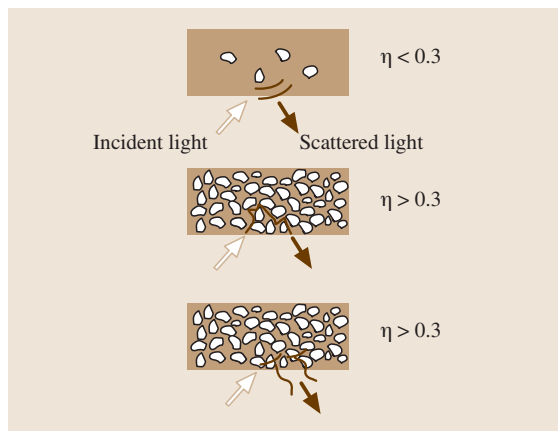


Fig. 5.24 Illustration of the subdivision of volume scattering: single scattering (*top*), multiple scattering (*middle*), and coherent scattering (*bottom*)

whole volume in which the N identical scatterers are located.

Before now discussing these three different subdivisions of volume scattering a few basic items must be described.

Scattering attenuates light, as does absorption by the medium itself. Both effects together are referred to as *extinction*, where: extinction = absorption + scattering [5.50].

The total (over all directions) scattered power P_{scat} (in units of W) can be calculated from the scattering cross section C_{scat} (in units of m^2) by

$$P_{\text{scat}} = C_{\text{scat}} I_{\text{in}}, \quad (5.96)$$

where I_{in} is the incident light intensity (in units of W/m^2). Typically one wants to know the power scattered in a certain direction (Fig. 5.25) and this is described by the differential scattering cross section $dC_{\text{scat}}/d\Omega$ (Ω : spatial angular)

$$\frac{dC_{\text{scat}}}{d\Omega} = R^2 \frac{I_{\text{scat}}}{I_{\text{in}}} \quad (5.97)$$

where R is the distance from scatterer to observer and I_{scat} is the scattering intensity. The scattering cross section can be obtained from the differential scattering cross section by:

$$C_{\text{scat}} = \int_{4\pi} \frac{dC_{\text{scat}}}{d\Omega} d\Omega. \quad (5.98)$$

Unfortunately the scattering cross section and the differential scattering cross section can only be calculated for a few geometries, such as spheres (Mie scattering).

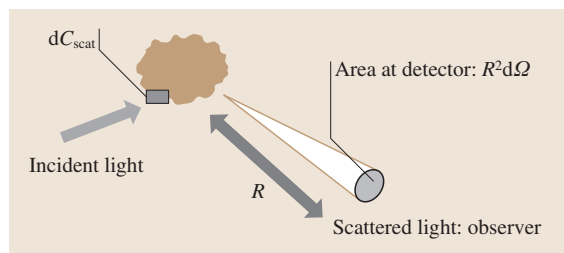


Fig. 5.25 Scattering in a certain direction described by the scattering cross section

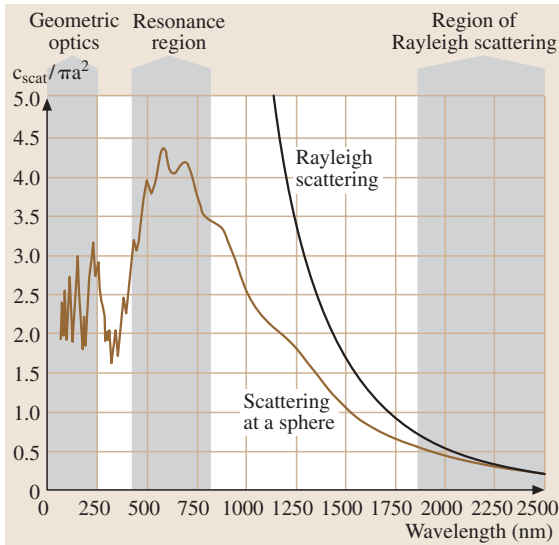


Fig. 5.26 Scattering cross section per sphere area (cross section) versus wavelength for a glass sphere ($n_{\text{sphere}} = 1.50$) of radius $a = 400$ nm in air ($n_{\text{medium}} = 1.0$)

Single Scattering (Mie Scattering)

A very important geometry of single scatterers is the sphere. In 1908 *Mie* [5.51] derived an analytic theory that completely describes the scattering from a (conducting or nonconducting) sphere embedded in a nonconducting medium. A compact survey of the formulas can be found in [5.52]. Many basic characteristics of scattered power can be obtained by studying scattering by a sphere. The calculated scattering cross section normalized to the geometric cross section of a glass sphere ($= \pi a^2$ where a is the sphere radius) can be seen in Fig. 5.26.

Figure 5.26 demonstrates the different wavelength dependence of the scattering cross section and thus of the scattered power according to (5.96). Depending on the geometric sphere size (characterized by the diameter $2a$) relative to the wavelength, three important regions can be identified:

- $\lambda \ll 2a/n_{\text{medium}}$, region of *geometric optics*. Surprisingly the term $C_{\text{scat}}/\pi a^2 = 2$. This effect is called the *extinction paradox* and the factor 2 is due to diffraction effects [5.53, 54].
- $\lambda \approx 2a/n_{\text{medium}}$, *resonance effect*, the term $C_{\text{scat}}/\pi a^2$ has its maximum. Therefore the highest scattered power can be measured if the geometric size of an object is of the order of the wavelength.
- $\lambda \gg 2a/n_{\text{medium}}$, *Rayleigh scattering*. The scattered power is proportional to $1/\lambda^4$. This effect is respon-

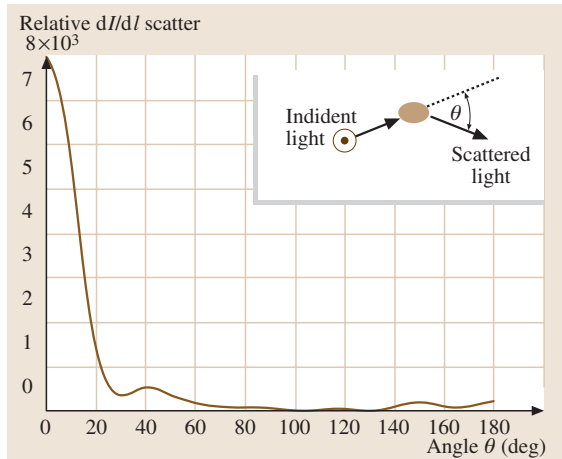


Fig. 5.27 Differential scattering cross section multiplied by 1.5×10^9 particles/mm³ for a glass sphere ($n_{\text{sphere}} = 1.51$) of radius $a = 400$ nm embedded in a medium with $n_{\text{medium}} = 1.50$ at a wavelength of 550 nm

sible for the blue sky, where small particles inside the atmosphere scatter blue light (≈ 400 nm) more than red light (≈ 800 nm).

These basic characteristics of a single scatterer – here shown for the special case of a sphere – are typical for all single scatterers of arbitrary geometry. As discussed and shown in Fig. 5.26 different behavior of the scattered power can be observed based on the size of the scatterer relative to the wavelength.

Multiple Scattering

In the preceding section scattering at a single isolated particle has been described. Now, multiple scattering is discussed, which is the weighted superposition of many single scatterers without interference effects. Multiple scattering with interference effects is called coherent scattering and is discussed below.

Multiple scattering can only be described numerically and must be used whenever the packing fraction η is larger than about 0.3. Based on the knowledge of the scattering function (described by the differential scattering cross section) of a single scatterer, the overall scattering function is calculated by tracing rays throughout the volume.

For example, multiple scattering of 1.5×10^9 particles/mm³ resulting from a packing fraction of $\eta = 0.8$ is calculated here (this can be the case for a glass ceramic, Sect. 5.5). The single scatterer is assumed to be a sphere of radius 400 nm with a refractive

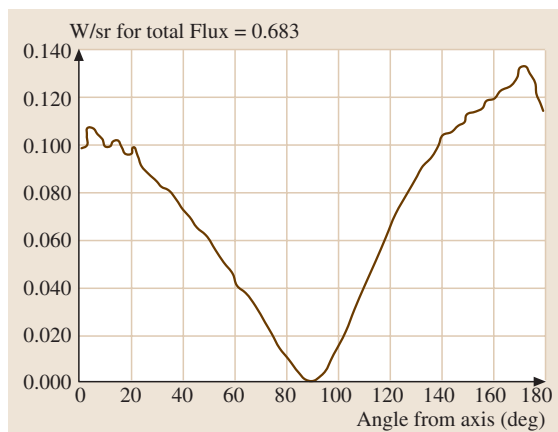


Fig. 5.28 Multiple scattering with $\eta = 0.8$ based on the data from Fig. 5.27. The differential scattering cross section due to multiple scattering inside the volume is totally different from that shown in Fig. 5.27 (the angle is defined in the inset of Fig. 5.27)

index of $n_{\text{sphere}} = 1.51$ embedded in a glass with refractive index $n_{\text{medium}} = 1.50$ at a wavelength of 550 nm. The differential scattering cross section of this single scatterer multiplied by 1.5×10^9 particles/mm³ is shown in Fig. 5.27.

The differential scattering cross section of a single scatterer (as seen in Fig. 5.27) is used to calculate the differential scattering cross section of the whole volume with a packing fraction of $\eta = 0.8$. The result, shown in Fig. 5.28, has been obtained by calculating the multiple scattering numerically.

As shown in Fig. 5.28 most of the power will be scattered in the backward direction (180°). Less power will be scattered into the forward direction (0°) and no power will be scattered in the sideways direction (90°).

By comparing Fig. 5.27 with Fig. 5.28 it can be seen that the differential scattering cross sections are totally different. The reason for that is multiple scattering inside the volume.

Coherent Scattering

Coherent scattering – the phenomenon in which a series of independent single-particle scattering events is replaced by collective light manipulation from an ensemble of different scattering centers – is the focus of this section. When dealing with materials containing scatterers separated by distances greater than the coherence length (the distance necessary for propagating waves to lose their coherence), the scattering events can be treated as independent occurrences even under

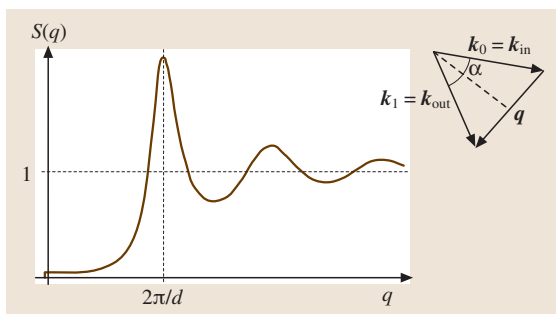


Fig. 5.29 (a) An example of the static structure factor, for the case of hard spheres. (b) The scattering vector q

multiple scattering conditions. Such scattering is called incoherent, and the resulting intensity of such radiation is simply a summation of the intensity contributions of all the independent scattering centers.

However, when the distances between scatterers are on the order of or less than the coherence length, coherent scattering effects must be considered [5.55]. In this case, wave package interference takes place as the package, is capable of interacting with more than one scattering center at a time; several scatterers distort the photon package simultaneously. Thus, a relationship exists between the phases of the light signals arising from the different scatterers. The events are no longer separate; they are correlated, and the resulting intensity of transmitted light is no longer a simple sum. Thus, in coherent scattering, wave packages experience a combined interaction with several scattering centers that affects their transport through the medium. In photon group interference (a quantum effect), the wave package comes apart upon scattering but the different sinusoidal components meet again and interfere. This interference affects the intensity of light transmitted through the sample; thus, coherent scattering studies can yield valuable structural information.

As discussed, scattering is coherent when the phases of the light signals arising from different scattering centers are correlated and incoherent when the phases are not. Hence, the propagation of coherently scattered light depends strongly on the direction of the scattering vector q – the difference between the incident and outgoing wave vectors, $k_{\text{in}} - k_{\text{out}}$ (Fig. 5.29b) – whereas incoherently scattered light can propagate in any direction regardless of the phase relation between waves from different scattering centers [5.56]. When the average distance between scattering centers (d) is on the order of the coherence length or less,

the interference effects are significant and quantitatively describable by the static structure factor $S(q)$, which gives correlations in the positions of particle centers (Fig. 5.29a). $S(q)$ is the link between the theoretical description of structural inhomogeneities and the actual experimental scattering measurements [5.57]. An extreme case of coherent scattering is a strictly periodic configuration of scattering centers. The scattering intensity shows maxima at Bragg peaks and the structure factor consists of sharp δ peaks. Note that the relation between incoherent and coherent scattering terms is usually expressed via the Debye-Waller factor.

The plot of the structure factor in real space is the pair correlation function, $g(r)$, which gives the probability of finding a pair of centers at a specific distance r apart in a sample. The relationship between $g(r)$ and $S(q)$ is

$$S(q) = 1 + 4\pi\rho \int_0^\infty r^2 [g(r) - 1] \frac{\sin(kr)}{kr} dr, \quad (5.99)$$

where ρ is the density of scattering centers. There is a mathematical connection between the cross section of the scatterers and the structure factor. For the general case of scattering from a correlated group of particles the scattering cross section is

$$C_{\text{scat}} = \frac{1}{k_0^2} \int_0^{2k_0a} F(y, k_0a) S(y) y dy, \quad (5.100)$$

where $y = qa$, $F(y)$ is the form factor, a is the radius of the scatterer, and k_0 is the incident wave vector [5.58].

Coherently scattered photons have a phase relationship, consequently exhibiting more wavelike behavior. In experiments in which coherent scattering is a prominent effect, the interference between scattering paths must be considered – and can be exploited. For example, light from a monochromatic coherent source scatters from a sample and exhibits a characteristic speckle pattern – an array of bright, non-overlapping spots due to interference effects – based on the composition and structure of the sample, as long as single scattering is the dominant effect [5.56]. Thus, such patterns contain structural information. An additional field is the analysis of backscattering cones. Another example of coherent scattering effects is evidenced by the unexpectedly high transparency of a glass ceramic – a phenomenon that cannot be explained by Rayleigh or Mie scattering theories (Sect. 5.5).

5.1.8 Surface Scattering

This section discusses in more detail light scattering due to surface irregularities. The Rayleigh criterion assesses when scattering is important and must be included. How roughness is related to the scattered light can be estimated by the so called total integrated scatter (TIS). TIS is an integrated (over all directions) value, so the angular dependence of scattering is not covered. The angular dependence of surface scattering can be estimated by assuming a sinusoidal surface roughness. A general (scalar) scattering theory is also briefly discussed.

More details on surface scattering can be found in [5.59, 60].

Basics and Rayleigh Criterion

Perfectly smooth surfaces only reflect light specularly. As stated in Sect. 5.1.7, due to surface irregularities light is also scattered in directions other than the specular one. This is illustrated in Fig. 5.30.

As can be seen from Fig. 5.30 in addition to the specular reflected light, scattered light is also generated due to the rough surface. This stray light is scattered over a wide angle around the specular direction. The rougher the surface the more light is scattered away from the specular direction.

But when is a surface smooth? This is answered by the so called *Rayleigh criterion* [5.60].

As can be seen from Fig. 5.31, a plane wave incident on a rough surface experiences a phase difference between the two outer “rays”. If this phase difference Φ

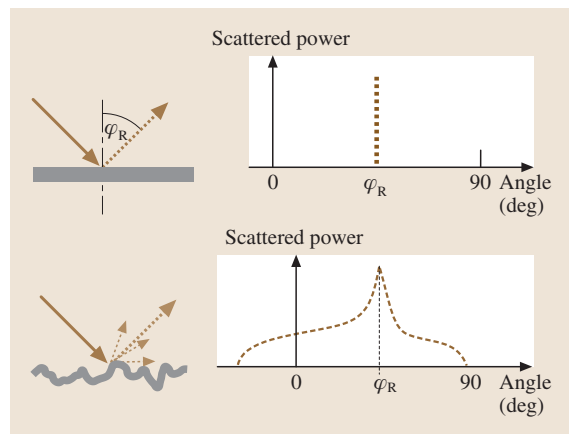


Fig. 5.30 Specular light reflection at a perfectly smooth surface (*top*) and light scattering at a rough surface (*bottom*). Dotted line = scattered (reflected) light, solid line = incident light



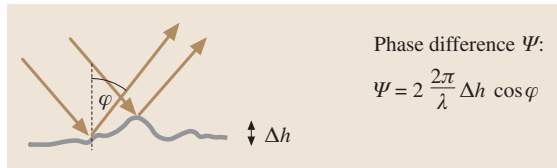


Fig. 5.31 A plane wave incident on a rough surface and its phase difference due to different optical paths

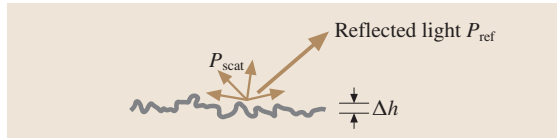


Fig. 5.32 Surface roughness and TIS

is smaller than $\pi/2$, the surface is called smooth (i. e. Rayleigh criterion)

$$\Delta h < \frac{\lambda}{8 \cos \alpha}, \quad (5.101)$$

where Δh is the **RMS** roughness (standard deviation).

Scattering (TIS) and Surface Roughness

The way in which surface roughness is related to scattered light can be seen in the so called total integrated scatter (TIS). The TIS is the ratio of scattered power P_{scat} (without specularly reflected light) in one hemisphere to the specularly reflected power P_{ref} . If the surface roughness is smaller than the wavelength, then TIS is given

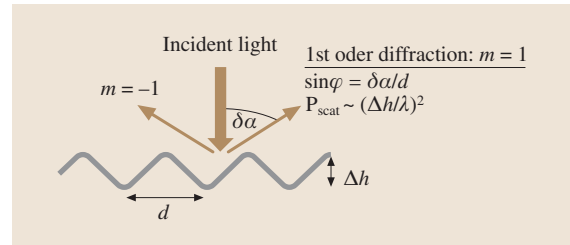


Fig. 5.34 Sinusoidal surface roughness and its parameters for the 1st diffraction order

by [5.59]:

$$TIS = \frac{P_{\text{scat}}}{P_{\text{ref}}} = \left[\frac{4\pi \cos \alpha_R}{\lambda} \right]^2 (\Delta h)^2, \quad (5.102)$$

where α_R is the angle of incidence (reflectance) as shown in Fig. 5.30 and Δh is the **RMS** roughness of the surface (Fig. 5.32).

From (5.102) it can be seen that the scattered power is proportional to the square of the surface roughness and depends as $1/\lambda^2$ on the wavelength. The surface roughness vs. wavelength for various TIS values is plotted in Fig. 5.33. Consider a specific glass surface and incident light at 500 nm. For this wavelength, only a thousandth part of the reflected light should be allowed to be scattered ($TIS = 10^{-3}$). For this wavelength and TIS value the **RMS** roughness is just 1.3 nm. This value is between normal and super-polished glass surface. Hence, normal polishing is *not* sufficient here. Therefore, a more expensive super-polishing must be used.

The TIS can also be used for estimating the surface roughness. By measuring the TIS at a determined wavelength, as described in [5.59, 61], the surface roughness can be calculated with the help of (5.102).

Angular Scattering from a Sinusoidal Surface Roughness

As already mentioned TIS is an integrated value, and thus angular dependencies of the scattered can not be described by TIS. But in many cases the direction of scattered light must be known, e.g. to determine the scattered contribution perpendicular (90°) to the incident light. To get a feeling for the angular dependence the easiest form of surface roughness – a sinusoidal surface – is considered (Fig. 5.34). An arbitrary surface roughness can be regarded as the superposition of sinusoidal rough surfaces. This approach is used in [5.60]. As already mentioned, scattering is strongly related to diffraction.

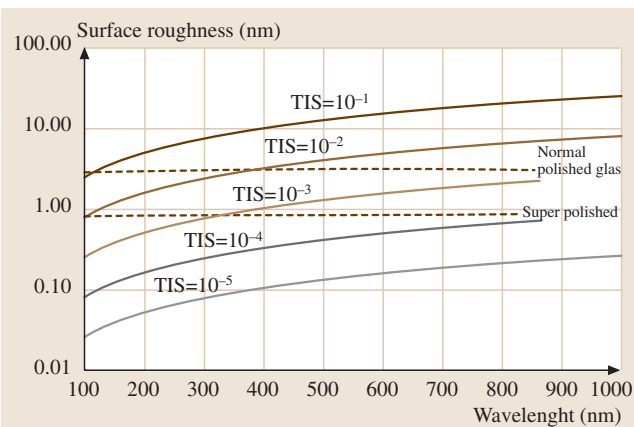


Fig. 5.33 Surface roughness plotted against wavelength for various TIS values. Typical roughness values for super and normal polished glass are also plotted. Note the logarithmic scale of the roughness axis

Such a sinusoidal grating diffracts (scatters) light into many orders. Light scattered into the m -th diffraction order is given by [5.59]

$$P_{\text{scat}}^m \propto \left(\frac{\Delta h}{\lambda} \right)^{2|m|}, \quad (5.103)$$

where the scattering angle belonging to this m -th order can be calculated from the grating equation

$$\sin \Delta \alpha_m = m \frac{\lambda}{d}. \quad (5.104)$$

$\Delta \alpha_m$ is the diffraction angle measured relatively to the direction of specular reflection. Here we will restrict the following considerations to the case of normal incidence, i. e. $\Delta \alpha = \alpha$. As an example, the case of a smooth surface is considered.

Smooth surface, i. e. $\Delta h/\lambda \ll 1$ and thus only the first-order diffraction need to be considered, as can be seen in Fig. 5.34. For that case the angle of the first diffraction order is only determined by the periodicity d and wavelength of the grating – *not* by the roughness Δh . The scattered power is proportional to $(\Delta h/\lambda)^2$, as it was obtained from TIS (which is valid only for smooth surfaces). The scattering angle $\Delta \alpha$ depends only on the periodicity d and the wavelength λ . Here three cases are of special importance:

- $\lambda/d \ll 1$, the periodicity is much larger than the wavelength and thus according to (5.104) with $m = 1$, the angle $\Delta(\alpha)$ equals 0. This means that the light will mainly scatter in the specular direction.
- $\lambda/d \approx 1$, the periodicity is of the order of the wavelength and so the scattering angle equals 90° .
- $\lambda/d < 1$, the periodicity is larger than the wavelength and light will be scattered between 0° and 90° .

All three cases for a smooth surface are illustrated in Fig. 5.35, which clearly shows the angular light scattering difference for different periodicities d (compared to the wavelength).

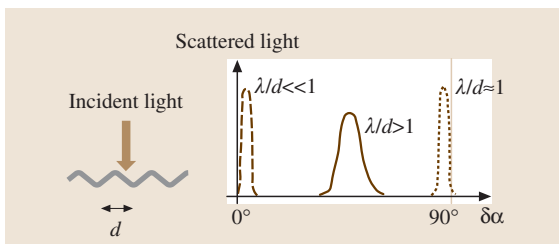


Fig. 5.35 Angular scattered light of a smooth surface for different periodicities d

Unfortunately, reality is much more complicated and a sinusoidal surface roughness cannot be found. But, as previously explained, a random surface roughness can be decomposed into a superposition of sinusoidal surfaces. This is done in much more detail in [5.60]. The angular scattering behavior of a surface is normally measured by the so called angular-resolved scatter (ARS). These measurements can be used to extract the angular scattering characteristics (the bidirectional scattering distribution function, BSDF) used in simulation tools for predicting scattering characteristics.

General Scalar Scattering Theory (First Born Approximation)

If the geometry of the roughness and the different refractive indices are known, then a general theory for calculating the scattered electric field can be developed. This theory is accurate for small refractive index changes (the so called first Born approximation), but after iterative use of the result stronger changes in the refractive index can also be calculated.

The total electric field $U(\mathbf{r}, \omega)$, i. e. the incident field U^i and the scattered field U^s ($U = U^i + U^s$), obeys the following differential equation [5.62] (in SI units):

$$\Delta U + \left(\frac{2\pi}{\lambda_0} \right)^2 U = -4\pi F(\mathbf{r}, \omega) U, \quad (5.105)$$

where λ_0 is the free space wavelength, $\mathbf{r}$ is the arbitrary point of observation, and the scattering potential F is

$$F(\mathbf{r}, \omega) = \frac{1}{4\pi} k^2 \left[n^2(\mathbf{r}, \omega) - 1 \right]. \quad (5.106)$$

The right-hand side of (5.105) describes the source of the scattering. Thus, as can be seen from (5.106), this source is only determined by the spatially and frequency-dependent refractive index n . Here it is clearly shown that the source of light scattering is the spatially dependent refractive index (rough surface).

As shown in [5.62] the solution of (5.105) for the scattered electric field is given by

$$U^s(\mathbf{r}, \omega) = \int_V F(\mathbf{x}, \omega) U(\mathbf{x}, \omega) \frac{e^{ik|\mathbf{r}-\mathbf{x}|}}{|\mathbf{r}-\mathbf{x}|} d^3x. \quad (5.107)$$

This form has the drawback that the total electric field inside the integral (5.107) must be known. But U also contains the scattered electric field. This problem can be overcome by using the so called first Born approximation (or Rayleigh–Gans theory). The first Born approximation assumes a smooth surface (weak change



in the refractive index) and also a far field [5.62]:

$$U \approx U_1 = e^{ik_i r} + \frac{e^{ikr}}{r} \int_V F(\mathbf{x}) e^{i(k_i - \mathbf{k})\mathbf{r}} d^3x, \quad (5.108)$$

where $\mathbf{k}_i$ is the wave vector of the incident wave and $\mathbf{k}$ is the wave vector of the origin and point of observation.

Equation (5.108) is the first Born approximation and the scattered electric field is given by

$$U^s = \frac{e^{ikr}}{r} \int_V F(\mathbf{x}) e^{i(\mathbf{k}_i - \mathbf{k})\mathbf{r}} d^3x. \quad (5.109)$$

The theory is limited to smooth surfaces, but iterative use of (5.109), as described in [5.54], allows its extension to rougher surfaces.

Unfortunately, the spatial distribution of the refractive index must be known. This is a problem and thus the elegant (5.109) has its limits. Therefore, TIS and ARS measurements are mainly used for describing the scattering of rough surfaces.

5.1.9 Other Effects

In addition to the previously discussed “classical” optical material properties:

- (non-)linear refractive index
- absorption coefficient
- emission cross section
- scattering cross section

there exist numerous effects which occur in optical materials when additional physical fields such as:

- quasi-DC electric field
- quasi-DC magnetic field
- mechanical stress

additionally influence the optical properties of the solid. From these phenomena, the resultant electro-optical and piezo-optical effects are described below.

For some optical materials, incoming light of certain intensity or energy causes (quasi-)permanent changes in optical properties. The unwanted effects of solarization and laser damage are described in Sect. 5.4. On the other hand, the desired permanent property changes one can encounter are:

- photochromism
- photorecordability
- photorefractivity.

Photochromism is briefly introduced here while photorecordability and photorefractivity are extensively discussed in Sect. 5.9.3.

The Electro-Optic Effect

The electro-optic effect [5.63] (Pockels effect) in special crystalline optical materials is the change in the indices of refraction of the ordinary and extraordinary rays (Sect. 5.1.4) that is caused by and is proportional to an applied electric field. The electro-optic effect was discovered by Roentgen and was extensively investigated by Pockels around the turn of the nineteenth century. In principle, the electro-optic effect is a special case of nonlinear optical behavior, where the perturbing electromagnetic field is of quasi-DC type.

The electro-optic effect provides a convenient means of controlling the intensity or phase of the propagating radiation. The linear effect can be characterized by a perturbative expansion of the dielectric tensor (see Sect. 5.1.4)

$$\varepsilon_{ij} = \varepsilon_{ij}^0 + \Delta\varepsilon_{ijk}^E E_k, \quad (5.110)$$

where ε_{ij}^0 describes the electric-field-independent contribution to the dielectric tensor and $\Delta\varepsilon_{ijk}^E$ is the first derivative of ε_{ij} with respect to each component k of the applied electric field E .

From symmetry arguments similar to the consideration in Sect. 5.1.4, the 27 different coefficients $\Delta\varepsilon_{ijk}^E$ collapse to only 18 parameters, because $\Delta\varepsilon_{ijk}^E = \Delta\varepsilon_{jik}^E$.

Table 5.5 Properties of most important electrooptical materials (after [5.63])

Material	Electrooptic constant r_{63} (pm/V)	Typical half-wave voltage at 546 nm (kV)	Approximate n_o
Ammonium dihydrogen phosphate (ADP)	8.5	9.2	1.526
Potassium dihydrogen phosphate (KDP)	10.5	7.5	1.51
Potassium dideuterium phosphate (KD*P)	26.4	2.6–3.4	1.52
Potassium dihydrogen arsenate (KDA)	10.9	6.4	1.57
Rubidium dihydrogen phosphate (RDP)	11.0	7.3	
Ammonium dihydrogen arsenate (ADA)	5.5	13	1.58

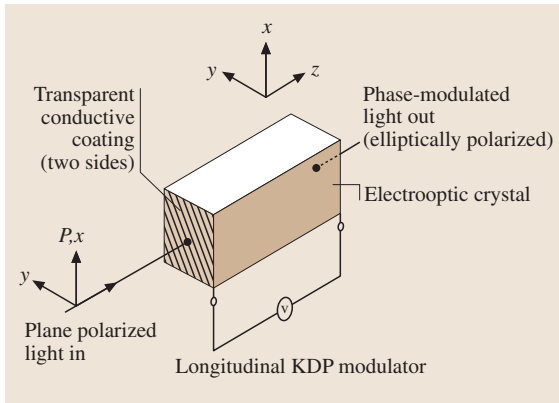


Fig. 5.36 Longitudinal electrooptical modulator (after [5.63])

These coefficients can be measured and provide the information necessary to compute the electrically induced birefringence. Some of the materials which exhibit the electro-optic effect and are used in electro-optic devices are listed in Table 5.5.

A potassium dihydrogen phosphate (KDP) longitudinal modulator is illustrated in Fig. 5.36. When no voltage is applied, the indices of refraction for the ordinary (o) and extraordinary (e) rays are identical along the optic axis. The incoming beam is plane-polarized with the plane of polarization aligned to the x - or y -axis. When a voltage is applied parallel to the light beam the crystal becomes birefringent. The extraordinary ray is retarded with respect to the ordinary ray and, upon emerging from the output face of the electro-optic crystal, the two polarized beams have acquired a relative phase shift due to the retardation of the light in the plane of the slow axis – resulting in an elliptically polarized beam. The retardation is given by

$$\Phi_{eo} = \frac{n_o^3 \Delta \epsilon_{123}^E U}{\lambda}, \quad (5.111)$$

where Φ_{eo} is the number of wavelengths of retardation, n_o is the ordinary index of refraction, $\Delta \epsilon_{123}^E$ is the electro-optic coefficient ($\mu\text{m}/\text{V}$), U is the applied voltage (V), and λ is the wavelength of incident light (μm). For the value $\Phi_{eo} = 1/2$, $U = U_{1/2}$ is defined as the half-voltage (see Table 5.5).

A transverse electro-optic effect is also exhibited by some crystals. In this case the birefringence changes are induced by a voltage applied transversely to the beam propagation.

Materials of prime importance concerning electro-optic properties are ADP, KDP, KD*P, KTN, LiNbO₃,

LiTaO₂, Ba₃NaNb₅O₁₅, Sr_xBa_{1-x}Nb₂O₆, CuCl, ZnSe, hexamethylenetetramine, and Se. Some materials have a quadratic rather than linear electro-optic effect. That is, the retardation is proportional to the square of the applied voltage. Among these are strontium titanate, potassium tantalate, and KTN.

Piezo-Optic Effect

The Piezo-optic effect [5.63] is analogous to the electro-optic effect. When pressure is applied to a piezo-optic crystal, the birefringence of the material changes. Some of these piezo-optic crystals are originally isotropic and cubic, but when stressed become birefringent. If such a crystal is stressed in a direction perpendicular to a light beam, the phase retardation is proportional to the stress. This effect is applied in acousto-optic modulators. These devices function based on the fact that refraction and diffraction effects occur when light passes through a solid medium transversely to a high-frequency (ultrasonic) acoustic field propagating in the same medium. The interaction can affect the deflection, polarization, phase, frequency or amplitude of the optical beam. Ultrasonic waves range from 10^5 Hz (above the upper range, approximately 2×10^4 Hz, of the human ear) to 10^9 Hz. Light passing through such a crystal in a direction transverse to the acoustic-wave direction is modulated by the wave.

Table 5.6 lists some of the more important acousto-optic materials and their characteristics. There are many figures of merit used in comparing these materials, two that are commonly employed are

$$F_1 = \frac{n^6 (\Delta \epsilon^P)^2}{\rho v^3}, \quad F_2 = \frac{n^7 (\Delta \epsilon^P)^2}{\rho v}, \quad (5.112)$$

where n is the optical index of refraction, $\Delta \epsilon^P$ is the appropriate component of the photoelastic tensor (analogous to $\Delta \epsilon^E$ in the electro-optic effect), ρ is the mass density, and v is the acoustic phase velocity. Since $\Delta \epsilon^P$, v , and ρ exhibit no extreme variations in the various acousto-optic materials, the candidates of most interest are those with good optical and acoustic qualities with large n , although optical and acoustic parameters are not included in the figures of merit.

In Table 5.6, the longitudinal (L) wave is one where the displacement u is parallel ($\parallel$) to the acoustic wave propagation direction ($\mathbf{k}$ vector), whereas the transverse (T) wave is one where the displacement u is normal ($\perp$) to the acoustic-wave propagation direction ($\mathbf{k}$ vector).

The polarization of the optical wave is defined as parallel (||) or perpendicular (⊥) to the plane formed by the acoustic and optic *k* vectors.

Photochromism
As discussed in the previous subsections, the interaction of electromagnetic waves with matter is generally

Table 5.6 Properties of most important piezooptical materials (after [5.63])

Material	λ (μm)	<i>n</i>	Polarization and direction ^a		Figure of merit	
			Acoustic wave	Optical wave ^b	$\frac{n^6 p^2}{p v^3} \times 10^{-18}$	$\frac{n^7 p^2}{p v} \times 10^{-7}$
Fused Quarz	0.63	1.46	L	⊥	1.51	7.89
			T	or ⊥	0.467	0.963
GaP	0.63	3.31	L, [110]		44.6	590
			T, [110]	or ⊥, [010]	24.1	137
GaAs	1.15	3.37	L, [110]		104	925
			T, [100]	or ⊥, [010]	46.3	155
TiO ₂	0.63	2.58	L, [11̄20]	⊥, [001]	3.93	62.5
LiNbO ₃	0.63	2.20	L, [11̄20]		6.99	66.5
YAG	0.63	1.83	L, [100]		0.012	0.16
			L, [110]	⊥	0.073	0.98
YIG	1.15	2.22	L, [100]	⊥	0.33	3.94
LiTaO ₃	0.63	2.18	L, [001]		1.37	11.4
As ₂ S ₃	0.63	2.61	L	⊥	433	762
	1.15	2.46	L		347	619
SF-4	0.63	1.616	L	⊥	4.51	1.83
β-ZnS	0.63	2.53	L, [110]	, [001]	3.41	24.3
			T, [110]	or ⊥, [001]	0.57	10.6
α-Al ₂ O ₃	0.63	1.76	L, [001]	, [11̄20]	0.34	7.32
CdS	0.63	2.44	L, [11̄20]		12.1	51.8
ADP	0.63	1.58	L, [100]	, [010]	2.78	16.0
			T, [100]	or ⊥, [001]	6.43	3.34
KDP	0.63	1.51	L, [100]	, [010]	1.91	8.72
			T, [100]	or ⊥, [001]	3.83	1.57
H ₂ O	0.63	1.33	L		160	43.6
Te	10.6	4.8	L, [11̄20]	, [0001]	4400	10 200
α-HIO ₃ ^c	0.63		L-a	a-c	48.2	
				b-c	20.8	
				c-b	46.0	
			L-b	a-c	41.6	
				b-c	58.9	
				c-a	32.8	
			L-c	a-b	83.5	
				b-a	77.5	
				c-a	63.0	
			Shear a-b	a-c	17.1	

^a L = longitudinal, T = transverse (shear)
^b Polarization is defined parallel (||) or perpendicular (⊥) to the plane formed by the acoustic and optic propagation directions (*k*-vectors)
^c Lattice constants: *a* = 5.888 Å, *b* = 7.733 Å, *c* = 5.538 Å

characterized by terms such as dispersion, absorption, scattering, refraction etc. Here we deal with absorptive effects which are induced in optical solids by the electromagnetic waves themselves.

Electromagnetic waves can cause electronic transitions from one state to another, create new particles or quasi-particles, or excite vibrations of ions and ensembles of charged particles. These effects are well-known absorption mechanisms. Obviously, the absorption spectrum depends on the electronic and ionic configuration. Consequently, the transition from one electronic or ionic configuration into a different configuration necessarily changes the absorption spectrum of the system under consideration. The control of the absorbance spectrum of any material by photons is called photochromism [5.64], regardless of the spectral range in which the absorbance is changed and regardless of the energy of the photons that induce the photochromism. Following this definition, a rich variety of possible effects exists that can be exploited for photochromism: the famous transition of F-centers into Q- and F'-centers or, more generally, the transformation from one to another color centre by illumination, the photon-induced orientation of color centers (photodichromism), the spectral hole burning as well as photon-induced polymerization or decomposition of compounds and photon-induced bleaching of absorption bands for Q-switching of a laser, just to name a few.

The change in the absorption spectrum, however, is hard to see if it occurs in a spectral region with a large absorption constant. Therefore, one must focus on materials and spectral regions where both the absolute and relative change of the absorbance is large if one wants to exploit the corresponding effect of photochromism

for technical applications. Challenging mass applications of photochromic effects are optical data storage and “smart” windows (see Sect. 5.8.3). However, several qualitative and quantitative requirements must be fulfilled; for instances the requirement of special spectral ranges optimized both for the inducing photons and for the induced absorbance is very important in many applications. It is equally important to discern whether the effect is reversible spontaneously or under irradiation, whether it shows fatigue, and how efficiently the absorption spectrum changes per incident photon. As a consequence, the photon-induced control of the absorbance – which is obviously very intriguing from a theoretical point of view – is reduced to only a few applications in practice.

The most prominent and widespread application of a photochromic effect seems to have occurred in photochromic eye-glasses (see Fig. 5.37 left). Although photochromic effects have long been known to occur in glasses, the basis of a material which is useful for photochromic sunglasses was developed by *Armistead* and *Stookey* [5.65] in the 1950s and 1960s. These scientists introduced silver halides into suitable glass melts and succeeded in forming small silver halide precipitates or particles in the glass. The basic effect, namely the photolytical decomposition of silver halides, is very well known in chemistry. In a test tube, the silver halide is decomposed into Ag clusters, which are responsible for the absorbance in the visible spectral region, and volatile halogens. The photolytical decomposition is not reversible in a test tube, however, if the halogens can escape. In a glassy matrix, on the other hand, the photolytic decomposition products remain close together and can recombine into silver halides. The chemical and phys-

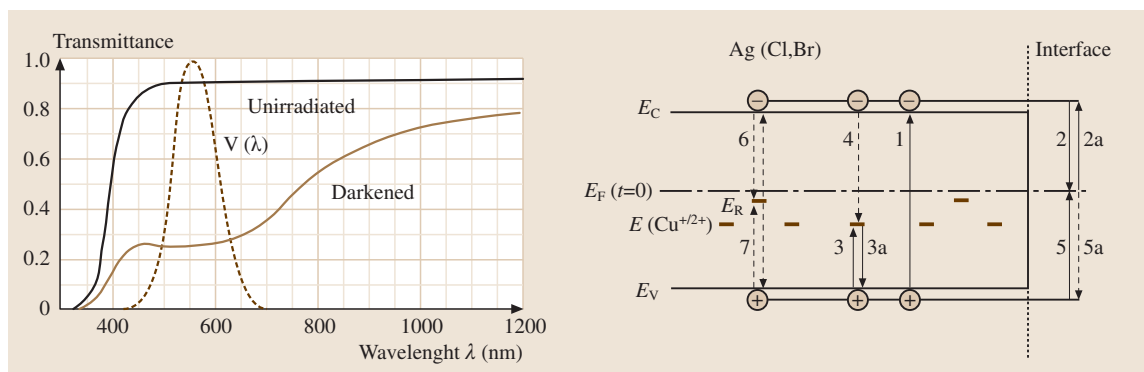


Fig. 5.37 Left: Transmittance of a commercially available photochromic glass as a function of wavelength before and under solar irradiation after about 15 min. Right: Energy band diagram of different photoelectronic processes which can occur in the silver halide particles at the interface between halide particle and vitreous glass matrix. (after [5.64])

ical details of the photochromism of glasses doped with silver halides (Fig. 5.37 right) have been discussed e.g. in [5.64].

Besides silver halides, many other different compounds can be decomposed due to light changing their absorption spectra under irradiation. To list a small ar-

bitrary selection of inorganic systems, we mention the Pb-, Cu-, and Cd-halides, the transition metal oxides, and some other compounds. However, most systems are interesting only from a scientific point of view and have little technical potential – or other practical disadvantages reduce their applicability.

5.2 Optical Glass

Every discussion about optical glass, especially an intensive one such as that following, has to start with the definition of the term *optical glass* and its delimitation from other possible types of glasses, e.g. filter glass, technical glass and others.

(*Classic*) *optical glass* is the traditional generic term for all glass types which possess the ability to influence the path of transmitting electromagnetic rays in the region between 200 nm to 1500 nm, e.g. to guide light [5.66]. This optical effect, and therefore the optical glass type itself, is usually fixed by the two main optical values: the refractive index (n_d) and the Abbe number (ν_d , $1/\text{dispersion}$) at the wavelength of the spectral d-line (587.6 nm, yellow helium line) [5.66, 67]. The relevant optical range was traditionally seen between $n_d = 1.38\text{--}2.20$ and $\nu_d = 18\text{--}100$. However, actual application trends lead to less extreme regions, so a reduction to $n_d = 1.40\text{--}2.00$ and $\nu_d = 20\text{--}90$ seems to be as suggestive as the prefix “classic”. This prefix preserves the delimitation from other types, nowadays often declared as subgroups of optical glass, e.g. color and filter glass and active optical glass. Here this prefix will be neglected, the term optical glass always refers to the aforementioned traditional definition.

Accordingly, optical glass is mainly used for components (lenses, prisms) in light-guiding application fields where there are imaging optics (microscopy, photo objectives, etc.), (digital) projection (color management systems, beamers, cinema objectives), telecommunications (hosting for active elements, ball, and rod lenses, graded index lenses, lens arrays), optical communications engineering (light and image fiber guides, data pick-up systems, read–write systems) and laser technology (laser optics, hosting for active elements).

An overview of available optical glass types is usually given in optical-glass diagrams, wherein they are specified by their two main optical values n_d/ν_d only. Those diagrams, provided (currently even by internet) by all well-known glass manufacturers (Schott [5.66], Ohara, Hoya, Sumita, Pilkington, in former times Corning), itemize the glasses independently of possi-

ble applications and other optical and physico-chemical properties. Up to now they are divided into historically developed sections, see for example Fig. 5.87.

5.2.1 Chronological Development

The first known optics shops and the early optical industry (ophthalmic, astronomy and microscopy) suffered from the lack of well-defined, reproducible optical glass. To cut a long story short, they had to take and to work with what they obtained from more or less irreproducible pot melts which suffered from poor raw material purity and availability. In addition to the poor optical quality of materials produced in this way, there did not exist any assured knowledge about the relation between composition and properties. At most, raw materials and their effects could be classified by supplier [5.68, 69].

Increasing chemical knowledge allowed the identification of raw material additives and impurities. This enabled, in combination with increasing abilities in chemical separation, the first relations between composition/compounds and properties/defects to be developed. This in turn allowed for the defined adjustment of optical properties, whereas until that time attention was tuned only by the optical position, i. e. the explicit values of n_d/ν_d as mentioned above. Otto Schott, a German chemist from a family of glass makers, was the first to investigate the composition/property relation with scientific methods and achieved a fundamental knowledge about melting consistent predeterminable glasses. Ernst Abbe and Carl Zeiss, owners of a firm for high-quality optical instruments, recognized the advantages for their business and, by association with Schott, easily emerged to be the most successful in their profession [5.69].

Prior to 1880, only simple crown and flint glasses were known (approximately 30 types) [5.70]. Crown glasses, consisting of soda–lime silicates, exhibit low refractive indices and high Abbe numbers, although from a current point of view the Abbe numbers would be better judged as moderate. Flint glasses [5.71], with high refractive indices and low Abbe numbers, were

represented solely by lead alkali silicates. The modern borderline between flint and crown is defined by the Abbe number, with flint glasses having $\nu_d < 50$ at $n_d > 1.60$ and $\nu_d < 55$ at $n_d < 1.60$. At that time, the distinction resulted naturally via composition.

From 1880, Otto Schott enlarged the glass map by establishing two new glass formers besides silicate: fluorine (very high ν_d and low n_d) and boron (moderate n_d/ν_d). Additionally he started to work with network modifiers by using BaO as a compound (moderate ν_d , high n_d). So, a variety of new glasses with a broader range of optical positions was generated, which led to the additional classifications: heavy (high refractive index) and light (low refractive index) besides crown and flint (the indication of dispersion/Abbe number) [5.72].

Around 1930 a new era in glass development started, when compounds such as rare earths (especially lanthanum), titania, zirconia and the additional network former phosphorous could be obtained and used in sufficient quantity and quality [5.73, 74]. These glasses led to further new optical positions, whose regions in the Abbe diagram were named by adding the new compound's chemical symbol to the traditional heavy/light crown/flint nomenclature, e.g. PSK (phosphorous heavy crown, the German *phosphor schwerkron*) or LaSF (lanthanum heavy flint, the German *Lanthanschwerflint*) [5.72].

In this context, it is easy to understand that a strong correlation between optical position (n_d/ν_d) and composition evolved, which led to the historical borders in the Abbe diagram and optically defined glass families. Hence, a glass's name, only defined by its optical position followed by a consecutive number, gave strong hints towards its composition.

Even though these boundaries have been valid until now, this strong correlation no longer exists because of the availability of highly pure (classical) raw materials and the accessibility of other, new raw materials obtained through new ways of mining, cleaning, purification etc.. Therefore, it is now possible to reach one optical position by several base-glass compositions, which leads to an increased number of compositional glass families with strong overlaps that cross the traditional boundaries. As a result, the information given by the glass type name (e.g. PSK 54) no longer provides reliable information about the components and glass family and thus about their properties. In this case, one cannot be sure about the existence of phosphorous, because, for example, PSK types can be obtained with and without phosphorous. And though both of these glass families occupy the same optical position

(e.g. Schott's N-LaSF 31A, Hikari's E-LaSF 08 and Hoya's TaFD 30), their physico-chemical properties are quite different (chemical resistance, Knoop hardness, temperature–viscosity profile, etc.).

It remains to remark that mainly the Japanese glass manufacturers Ohara and Hoya recently diverged from this traditional nomenclature and implemented their own representative system of glass naming. Nevertheless they maintained the traditional borderlines in the Abbe diagram.

5.2.2 Compositions of Modern Optical Glass

Today we know of more than 300 different types of optical glasses that can be classified by their chemical composition (Figs. 5.38 and 5.87, and Table 5.37). Because of better raw material availability, there exist more than several compositional methods to reach an optical position. Deviation from the traditional base systems and known compositions is often useful because of current enhanced glass-development specifications. Those specifications comprise customer- and process-referred profiles in which it is not unusual to find up to 15 properties, besides the well-known pure optical-position

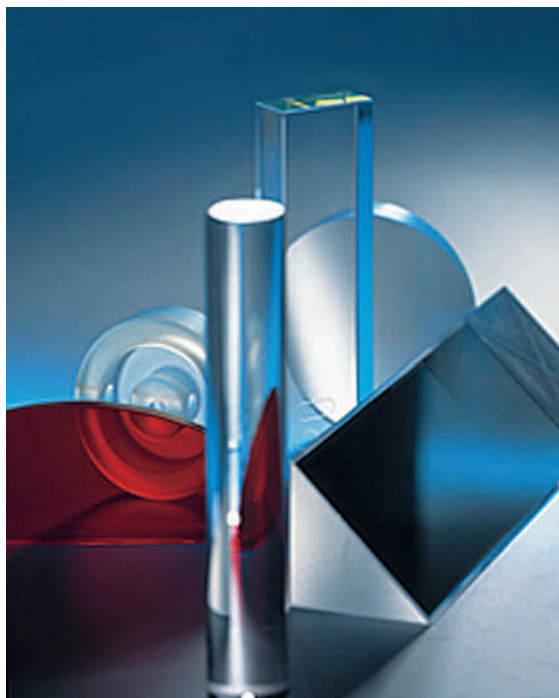


Fig. 5.38 Today, optical glasses with high quality exist in a broad range of shapes and products



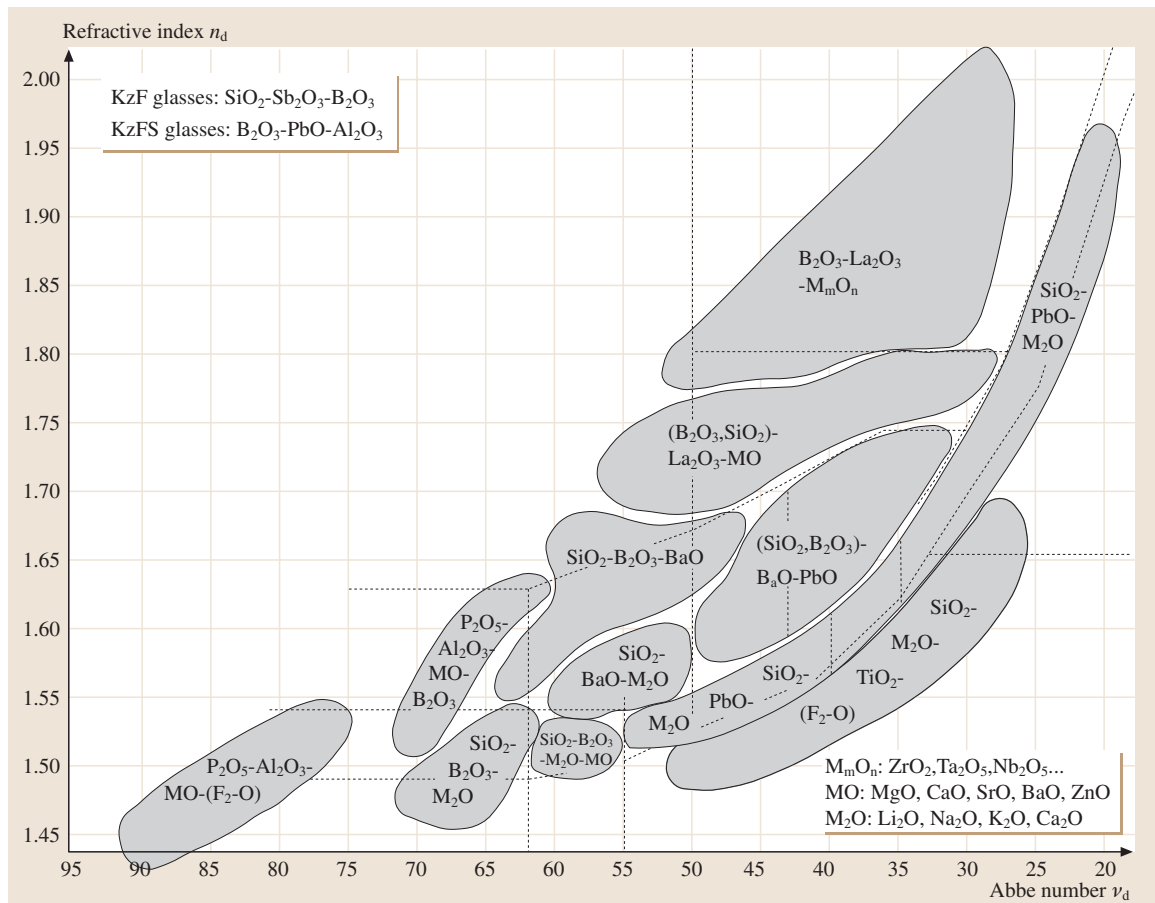


Fig. 5.39 Glass systems of optical glass in the n_d/v_d plot

specification. Commonly specified properties belong to the following fields:

- *application*: T_g , CTE, specific density, spectral transmission, chemical resistance, solarization, compaction, birefringence coefficient, fracture behavior;
- *pollution control and operational safety*: chemical and thermal reactivity of batch and raw materials, environmental impact of raw materials (mining, cleaning, use, application);
- *melting and hot forming*: viscosity–temperature profile, CTE, adjustment to standard processes, crystallization, aggressiveness against fireproof material;
- *cold working*: CTE, stiffness, hardness, fracture behavior, abrasion rate, chemical resistance;
- *patent law* (very active international patent awareness): adaptation of compositions to relevant legal surrounding, including variations, admixtures, etc.

Increasingly a transition has occurred from developing a) as far as possible an optimized glass out of the traditional optical glass families to b) specially designed single glasses for specific application or process fields arranged on the Abbe diagram only for traditional reasons. Therefore only the most important base systems, as shown in Fig. 5.39, will be discussed in detail.

1. $\text{SiO}_2\text{-B}_2\text{O}_3\text{-M}_2\text{O}$

This system is formed by the network formers SiO_2 and B_2O_3 . Alkaline oxides (M_2O) have been introduced as network modifiers. The presence of B_2O_3 and alkaline oxides is essential, as the glass-forming oxide SiO_2 has melting temperatures that are too high for conventional melting techniques. The addition of B_2O_3 and alkaline oxides facilitates the meltability of these glasses. The optical glass BK 7 is one of the main representative glasses of this system. It is the optical glass

most frequently used and can be produced with extreme homogeneity.

2. $\text{SiO}_2\text{--B}_2\text{O}_3\text{--BaO}$ and 3. $\text{SiO}_2\text{--BaO--M}_2\text{O}$

If BaO is used as the main network modifier instead of the alkaline oxide or B_2O_3 , this results in two large glass systems. In contrast to other earth-alkaline network modifiers, the introduction of BaO has various advantages. Besides PbO there is no other divalent oxide that increases the refractive index as strongly as BaO. Additionally, BaO neither decreases the Abbe number nor shifts the UV transmission edge significantly to greater wavelengths as PbO does. BaO-containing glasses normally show good abrasion hardness. In some glasses, BaO is partly replaced by ZnO.

4. $(\text{SiO}_2, \text{B}_2\text{O}_3)\text{--M}_2\text{O--MO}$

To obtain glasses of this glass system, divalent oxides (MO) are exchanged for B_2O_3 . From the divalent oxides only CaO, ZnO, and PbO are used in this system. For the production of glasses with high chemical resistance and good crystallization stability, ZnO is particularly important since it increases durability with respect to water and acids much more than CaO, provided that the ZnO content exceeds approximately 10 wt %. It is also possible to reduce the melting temperatures by a simultaneous exchange of ZnO for SiO_2 and to increase the alkaline content without harming durability against weathering. In terms of these characteristics, ZnO is much more effective than B_2O_3 . The use of CaO leads to the production of glasses with good mechanical properties and increased abrasion hardness. The chemical durability of these glasses increases with increasing CaO content.

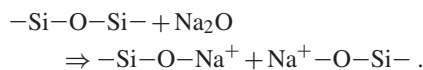
5. $(\text{SiO}_2, \text{B}_2\text{O}_3)\text{--BaO--PbO}$

This glass system is a mixture of the glass systems $\text{SiO}_2\text{--BaO--M}_2\text{O}$ and $\text{SiO}_2\text{--PbO--M}_2\text{O}$. Thus these glasses represent a transition from one glass system to the other. In the system $\text{SiO}_2\text{--BaO--M}_2\text{O}$, the exchange of PbO for BaO leads to the production of glasses with a lower tendency to crystallize. Compared with similar BaO-containing glasses, PbO-containing glasses are characterized by lower softening points and higher expansion coefficients. In some glasses of this system, BaO is partly replaced by ZnO.

6. $\text{SiO}_2\text{--PbO--M}_2\text{O}$

Glasses of this system have been known for a very long time. They are widely used for optical glasses and crystal glasses. As a glass component, PbO plays a vital role because on the one hand it increases the refractive in-

dex and on the other it decreases the Abbe number very strongly. Furthermore, PbO significantly affects the partial dispersion of glasses since a high PbO content leads to a positive deviation from the “normal line”. High amounts of PbO lead to a high glass density and a rather low chemical resistance, but effects an interesting birefringence coefficient near zero. PbO can be introduced into glasses at more than 80 wt %, which indicates that it does not only have network modifying properties. Normally, if a network modifier is introduced into a glass, e.g. Na₂O into a SiO_2 glass, bridging oxygen atoms are converted into non-bridging oxygen atoms:



With an increasing content of network modifiers the network progressively weakens and the viscosity of the glass melt decreases. If the content of the network modifiers is increased further, more and more bridging oxygen atoms are converted into non-bridging atoms. For a ratio of network modifiers to network formers of about 1 : 1, a three-dimensional network can no longer be formed and glass formation becomes impossible. However, the glass-forming tendency of binary PbO– SiO_2 systems leads to a PbO content of more than 70 mol %. According to Zachariasen [5.75, 76] and Warren and Loring [5.77] these glasses would normally not exist if PbO was only a network modifier. Fajans and Kreidl [5.78] and Stanworth [5.79] explained how the introduction of such a high amount of PbO is possible. In small concentrations, PbO has the same function as any other network modifier. High concentrations of PbO, however, cause an increase in the formation of $[\text{PbO}_4]$ tetrahedra. These $[\text{PbO}_4]$ tetrahedra can act as network formers and are incorporated into the three-dimensional network. Glasses that contain large amounts of PbO are usually more or less yellow in color. Impurities, such as Fe_2O_3 or Cr_2O_3 , have very strong effects on the color of PbO-containing glasses, even stronger than in soda–potash–lime glasses. It was further found that, first, besides the different structures of the PbO-containing glasses, there exists an equilibrium between Pb^{2+} and Pb^{4+} and, second, brown PbO_2 is formed if the melt is supersaturated with oxygen. Unlike the yellow color that is due to impurities, the yellow tint caused by the equilibrium between Pb^{2+} and Pb^{4+} can be reversed by a variation of the redox condition of the melt. For a long time, attempts have been made to increase the chemical stability of optical glasses with high PbO contents. Replacing parts of the PbO content with TiO_2 (in the range of some wt%) has been a common method



to increase the chemical durability. A sufficient TiO_2 content not only improves the chemical durability but, unfortunately, also leads to a transmission loss in the UV range. A solution to this problem is reached by replacing parts of the PbO content with both TiO_2 and ZrO_2 . The chemical durability is improved and the optical properties of the original glass ($\text{SiO}_2\text{--PbO--M}_2\text{O}$) are again obtained.

7. $\text{P}_2\text{O}_5\text{--MO--B}_2\text{O}_3\text{--Al}_2\text{O}_3$

Glasses in which P_2O_5 acts as a network former have been known for a long time. P_2O_5 forms glasses with rather low dispersions (high Abbe numbers) and high refractive indices, compared with the classic glasses based on B_2O_3 or SiO_2 . Glasses with a high P_2O_5 content have long dispersion spectra in the blue range. These glasses are normally used to correct chromatic aberration. The second and very important property of glasses containing P_2O_5 is their low birefringence coefficient and excellent UV transparency. A disadvantage of phosphate glasses is that their chemical durability is poor compared with the durability of SiO_2 -based glasses.

8. $\text{P}_2\text{O}_5\text{--Al}_2\text{O}_3\text{--MO--(F}_2\text{--O)}$

In order to obtain glasses with lower dispersions than those of phosphate glasses, one can introduce fluorine. Most of these glasses are based on a $\text{P}_2\text{O}_5\text{--Al}_2\text{O}_3\text{--MO}$ glass system which contains high amounts of fluorine. This glass system shows a relatively high chemical durability and a relatively low devitrification tendency. Fluorine takes the places of oxygen atoms so that one oxygen is replaced by two fluorine atoms. The formation of the fluoro-phosphate glasses of this system is rather complex and can be described with amorphous fluoro-alumino-phosphates. As fluorine has a smaller ionic radius and a higher electronegativity than oxygen, the polarization of the cations in these glasses is increased so that the refractive index and the dispersion are decreased. The replacement of oxygen by fluorine leads to glasses that have distinct LgK characteristics, which leads to a long spectrum in the blue spectral region, a significantly positive $\Delta P_{\text{g,F}}$ (partial dispersion deviation coefficient) and hence the aptitude for correcting achromatic aberration in combination with a KzF(S) type (short flint) with a short spectrum in the blue spectral region (negative $\Delta P_{\text{g,F}}$). The term LgK was neglected for a long time in favor of the systematic terms FK (fluoro crown) and PK (phosphate crown) which both indicate the optical position of fluoro-phosphate types (see Abbe diagram).

9. $(\text{B}_2\text{O}_3, \text{SiO}_2)\text{--La}_2\text{O}_3\text{--MO}$

In connection with B_2O_3 , glasses containing large amounts of La_2O_3 form a very extensive glass system. Glasses belonging to this system normally have a high refractive index with a relatively low dispersion. As their chemical resistance is rather poor, their durability can be increased by replacing SiO_2 with B_2O_3 and by the introduction of divalent oxides. In addition, the introduction of ZrO_2 and TiO_2 is another possibility for achieving better chemical properties. In the past, ThO_2 and CdO have also been used to achieve this aim, but for environmental protection these oxides have been eliminated from these glasses. The large disadvantage of these glasses is their poor crystallization stability, especially if SiO_2 is used.

10. $\text{B}_2\text{O}_3\text{--La}_2\text{O}_3$

These glasses are based on the $\text{B}_2\text{O}_3\text{--La}_2\text{O}_3$ glass system. They all have high refractive indices and high dispersion. In order to obtain high refractive indices, large numbers of oxides that are known to cause a high refractive index, such as Gd_2O_3 , Y_2O_3 , TiO_2 , Nb_2O_5 , Ta_2O_5 , WO_3 , and ZrO_2 are incorporated in the glasses of this system. The large number of network modifiers leads to glasses whose viscosities depend to a high degree on temperature (i. e., they are called short) and which show a very strong tendency to crystallize. On the other hand, these glasses have a distinct hardness and chemical resistance, which increase further with higher refractive indices. In this respect, the glasses differ from those of the system $\text{SiO}_2\text{--PbO--M}_2\text{O}$, which also have a high refractive index.

11. $\text{SiO}_2\text{--Sb}_2\text{O}_3\text{--B}_2\text{O}_3$

The replacement of PbO by Sb_2O_3 leads to an extensive glass system. Sb_2O_3 reduces the dispersion of the blue part of the visible spectrum. Consequently, short-flint glasses are obtained. The introduction of Sb_2O_3 into a glass gives results similar to those in the case of PbO , although Sb_2O_3 glasses seem to form long chains of --Sb--O--Sb--O-- units [5.80].

12. $\text{B}_2\text{O}_3\text{--PbO--Al}_2\text{O}_3$

If SiO_2 is replaced by B_2O_3 , glasses with primarily very poor chemical properties are obtained. Glasses of this system have an intensive short-flint characteristic, since the PbO content induces absorption in the blue range of the visible spectrum, and B_2O_3 causes strong absorption in the infrared range of the spectrum. Al_2O_3 is added to improve the chemical properties to an acceptable level.

13. $\text{SiO}_2\text{--M}_2\text{O--TiO}_2\text{--(F}_2\text{--O)}$

The PbO content in glasses of the system $\text{SiO}_2\text{--M}_2\text{O--PbO}$ can be replaced by TiO_2 . As TiO_2 causes an increase in the refractive index, glasses of the system $\text{SiO}_2\text{--M}_2\text{O--TiO}_2\text{--(F}_2\text{--O)}$ are characterized by high refractive indices. The dispersion of these glasses is increased to a higher degree than that of glasses containing the same amount of PbO. The introduction of TiO_2 into glasses is limited, because when the TiO_2 content is increased the glasses tend to crystallize strongly. This should be avoided by the introduction of alkaline oxides into the glass. In spite of the large content of alkaline oxides, these glasses are highly resistant to the influences of water and acids. Glasses with high TiO_2 contents show strong absorption in the UV range. This absorption is stronger than that of glasses containing PbO or Sb_2O_3 . Both glass types are characterized by a positive deviation of the partial dispersion from the normal line.

The break up of traditional glass families into ambitious, multi-property specified designer glasses led to rethinking, and thereby to new methods, in glass development. The well-known stepwise successive method has (where suggestive) now been replaced by an intensive field-screening method supported by computer-assisted statistical experimental design (SED; multi-linear regression-driven analysis).

This breaking of traditional borders and innovation concerning methods and materials is the reason for drastic changes in the assortments of optical glasses during the last five to ten years. First, the environmentally driven conversion to lead- and arsenic-free optical glasses took place [5.81] (see next section). In combination, a simplification of the assortments has been carried out. Old glass types with small market volume have been eliminated and high-performance glasses which were offered by several glass manufacturers were slightly redesigned to be optical interoperable. This led to a remarkable shrinking of the available glass assortment. In contrast the development of new glass types with opportunity for less crystallizing types and better application- and production-fitted glasses expanded the list of available glasses. Presumably, this will proceed in cycles in the future.

5.2.3 Environmentally Friendly Glasses

New recipes for many traditional optical glasses have recently been developed by all the major glass suppliers to make them ecologically friendly. The primary elements of attention in the development of ecologically friendly

optical glasses are arsenic and lead, although other toxic elements have also required attention. For example, in 1980, the use of ThO_2 was stopped completely and the use of CdO was restricted to colored optical glass.

In the late 1980s, when the public became increasingly sensitive to health problems caused by Pb, the whole glass-producing industry started developing PbO-free glasses as a first step towards the production of environmentally friendly glass. Apart from the health aspect, the reduction and replacement of PbO in the glass composition led to a considerable weight reduction, which is advantageous in several respects for outdoor applications. Since 1995 the use of another substance, As_2O_3 , has been restricted; it is used as a refining agent in trace amounts only. With the intention of replacing PbO and As_2O_3 , the glass manufacturers initiated a comprehensive program that has not yet been completed [5.81].

The main target has been to redesign all glass types so as to preserve n_d and ν_d while gaining a Pb- and As-free composition. Moreover, the partial-dispersion values $P_{g,F}$ and $\Delta P_{g,F}$ have been preserved for several special glasses. There are, however, some special glasses whose optical properties cannot be preserved without the addition of PbO. Of course, these glass types will be available in the future in their known quality and composition. They will be clearly marked as Pb-containing products. The bio-availability of the pure toxic substances in the products that are finally sold is negligibly low. For example, even high amounts of PbO in the composition of a glass would not cause any health problems. But although there is no objective reason, the customers' negative associations regarding Pb or As lead to a strong rejection of products containing these substances. There are, however, serious problems caused by hazardous compounds that make expensive and time-consuming development necessary. High amounts of fine dust are produced during acquisition of the raw material and during batch preparation. Of course, the dust does not contain a higher percentage of toxic substances than the base material, but it has a very much larger specific surface. This large surface in combination with the fact that dust tends to deposit in the lung tissue in large amounts causes a bio-availability of the toxic compounds that is in the latent-toxic range. High yields of grinding waste result from finishing processes, particularly from grinding and polishing. These must be disposed of at high costs on hazardous-waste disposal sites. Experience in this field shows that there is a high risk of incorrect handling and a tendency towards illegal disposal be-



cause of the high costs. Improving technical measures to protect directly affected persons from the unhealthy effects of dust and waste is theoretically possible, but hard to control. In particular in foreign production, for example in developing countries, safety could not be guaranteed [5.81].

For these reasons, there has been a tendency to dispense with these substances and to use TiO_2 , Nb_2O_5 , ZrO_2 , WO_3 and others instead of PbO and As_2O_3 as a refining agent is replaced by Sb_2O_3 . All glass compositions reformulated in this way are designated by a prefix to the original name, e.g. “N-” with Schott, “E-” with Hikari, “S-” with Ohara. For 50% of all glass compositions originally containing As_2O_3 but no PbO , replacing As_2O_3 by Sb_2O_3 does not change their properties.

For the compositions originally containing As_2O_3 and PbO (up to 80%), some physical and optical properties are significantly changed upon the replacement of both elements. Besides improvements, these changes also include some disadvantages. Improvements are: better chemical resistance; higher Knoop hardness; higher T_g point; lower density. Disadvantages are: higher tendency of crystallization; higher T_g , reduced transmission in the UV region; changed values for $P_{g,F}$ and $\Delta P_{g,F}$ (except for the N-KzFS series).

5.2.4 How to Choose Appropriate Optical Glasses

In this section practical guidelines for choosing appropriate glasses for optical applications are given. As stated above, optical glasses are ranked by refractive index and dispersion. However, during this selection process, not only the optical properties but also mechanical and chemical properties should be considered. We present each mechanical and chemical property individually along with a brief explanation of its importance.

Transmission and Absorption

Absorption in optical glass can be expressed in a number of ways and the reader is referred to Sect. 5.1.3 of this text for a more rigorous treatment. We simply state here that most optical glass manufacturers list for their glasses internal transmission values that neglect the losses from reflection at air–glass interfaces, and absorption values are thus conveniently calculated according to $\alpha = -\ln(\tau_1/\tau_2)/\Delta x$, where τ_1 and τ_2 are the internal transmission values for samples of thickness x_1 and x_2 ($x_1 > x_2$), Δx is the difference in thickness between the two samples, and α is the absorption coefficient (expressed in units of cm^{-1} when x is in cm).

Partial Dispersion

Since the refractive index varies with wavelength, not all colors are focused at the same point with a simple single-component lens. This effect is referred to as chromatic aberration. For the fabrication of optical systems, achromatic and apochromatic systems are often employed. In an achromatic system, chromatic aberration is minimized for two colors, e.g., red and blue. The other part of the visible spectrum remains uncorrected and is referred to as the “secondary spectrum”. The chromatic aberration in an apochromatic system, however, is corrected through the whole visible region. Most optical glasses show normal dispersion where the partial dispersion of refractive index varies nearly linearly with the Abbe number. These glasses are not suitable for the construction of apochromatic systems. Therefore, glasses that deviate from this normal dispersion behavior are required for the construction of apochromatic systems. As a result, glasses with a high dispersion (short) in the blue region of the visible spectrum have been developed. Deviation of partial dispersion from the normal, or linear, behavior is described by a series of partial dispersion deviation coefficients, $\Delta P_{X,Y}$, where X and Y indicate the wavelengths of interest.

Photoelastic Constants

The refractive index of glasses is altered by the application of stress. The change is generally different for light linearly polarized in a plane parallel to the direction of stress than for light linearly polarized perpendicular to the stress. The magnitude of the change is given by

$$n_{\parallel} = n_0 + K_{\parallel} S \quad \text{and} \quad n_{\perp} = n_0 + K_{\perp} S \quad (5.113)$$

where n_0 is the refractive index in the absence of applied stress, S , and $K_{\parallel}$ and $K_{\perp}$ are the photoelastic constants and $n_{\parallel}$ and $n_{\perp}$ are the refractive indices for light linearly polarized parallel and perpendicular to the direction of stress, respectively. The commonly used stress optic coefficient (K) is the difference between the two photoelastic constants.

Thermal Expansion

Inorganic glasses expand upon heating and shrink upon cooling. The coefficient of linear thermal expansion α is defined as $\alpha = \Delta L/L_0\Delta T$, where ΔL is the change in length of a material of length L_0 that experiences a temperature change of ΔT . The value of α is generally a function of temperature. The coefficient of thermal expansion values found in optical glass catalogs represent only a linear fit and care must be taken when comparing

two glasses that the expansion information is taken over the same temperature range.

Thermal expansion is an important property since it is one indicator of the extent to which a particular glass will be prone to thermal fracture. Higher expansion values are generally an indicator that a glass is relatively weak against thermal shock.

Transformation Range

The thermal expansion curve for glasses begins to deviate significantly from a roughly linear shape over a temperature interval referred to as the transformation range. Other temperature-dependent glass properties also change rapidly with temperature over this same range. In addition, if cooling through this range takes place rapidly or unevenly, permanent residual stresses can be imparted to the glass due to unequal amounts of thermal contraction.

The transformation temperature, T_g , is found in glass catalogs and can be used as an indication of where the transformation range exists and what processing or environmental temperatures need to be avoided to avoid changes in surface figure or the creation of residual stresses that can reduce the quality of a transmitted optical wavefront through stress birefringence effects. In this regards, as a rule of thumb, parts should not be heated above a value of $T_g - 200^\circ\text{C}$.

Softening Point

Similar to the transformation temperature discussed above, another characteristic temperature for glass is the temperature at which the glass begins to deform under its own weight. The corresponding material viscosity is $10^{7.6}\text{dPa}$.

Thermal Properties

Additional thermal properties include heat capacity, c_p , and thermal conductivity, λ . The former provides information on the quantity of heat, dQ , required to raise the material temperature by an amount dT : $c_p = dQ/dT$; and the latter on the heat flux, Q , that will pass through a slab of material of thickness L , in a time period t , when placed across a thermal gradient generated by a temperature difference ΔT ; $\lambda = QLt/\Delta T$. Another thermal property found in some optical glass catalogs is the thermal diffusivity K , which is related to the thermal conductivity, heat capacity and glass density ρ by $K = \lambda/\rho c_p$.

All three thermal properties discussed above have a weak dependence on temperature up to values just below the transformation range. As with thermal expansion

information, comparison of glasses in critical situations requires that property information be stated at the same temperature.

Elastic Properties

When a glass bar is placed under tension it stretches. The amount of stress, S , and strain, ϵ , experienced by the bar are related by $S = E\epsilon$ where E is the Young's modulus of the material (i. e. Hooke's law). The ratio of lateral to longitudinal stress in the bar is Poisson's ratio, μ . In the case where a compressive stress is applied, the proportionality constant between stress and strain is now the bulk modulus, K . Finally in the case of an applied shear stress, the proportionality constant is the shear modulus, G .

Young's modulus and Poisson's ratio values are typically provided in optical glass catalogs. These elastic constants are interrelated such that if any two are known the remaining pair can be readily calculated. We have for example, $K = E/[3(1 - 2\mu)]$ and $G = E/[2(1 + \mu)]$.

Hardness

The classic hardness scale (Moh scale) is based on a series of minerals and their ability to scratch one another. Modern hardness values are collected based on indentation methods using points of prescribed geometry applied at specific load levels and times. Two common hardness measurement methods, with different indenter geometries and load condition, lead to values of Knoop and Vickers hardness. Because test conditions vary between different techniques it is important to make comparisons only between measurements employing the same standard.

Chemical Resistance

Selection of glasses based on chemical properties, commonly chemical resistance test results in optical glass catalogs, is difficult at best since a wide range of different test standards exist among the various suppliers. Consequently, only a brief introduction is provided here, based on some of the more common types of chemical durability tests with some examples of specific test recipes.

Please note that there are some optical glasses that show no staining or other evidence of chemical attack in one test type, but have a low chemical resistance in one or more other tests. Consequently it is important when evaluating the chemical behavior of optical glasses to consider all resistance test results.

Climate Resistance Test. Climate resistance is intended to give guidance concerning the sensitivity of a glass to



water vapor in ambient air. Typically, but not always, a slow process, this form of chemical attack gradually leads to the formation of a cloudy surface film that cannot be wiped off. An accelerated procedure is thus typically used for testing the climatic resistance of glasses.

In one possible test technique, polished uncoated glass plates are exposed to a water-vapor-saturated atmosphere the temperatures of which are alternated between 40 °C and 50 °C on an hourly basis. Since the temperature increase in the glass plates follows that of the atmosphere, water condenses on the glasses during the warming phase. In the cooling phase the temperature of the atmosphere initially falls faster than that of the glass plates causing a drying of the glass surface. After an exposure time of, for example, 30 h the glasses are taken out of the climatic chamber. The difference, pH , between the transmission haze before and after testing is used as a measure of the resulting surface change. Classifications can then be done based on the transmission haze increase, pH , after the test period has been completed.

Resistance to Staining. The test procedure gives information on possible changes in the glass surface (stain formation) under the influence of lightly acidic water (for example perspiration, acidic condensates) without vaporization. The class of stain resistance is typically determined by placing a plane-polished glass sample in contact with a test solution (for example a standard acetate at $pH = 4.6$ or a sodium acetate buffer at $pH = 5.6$, depending on the relative durability of the glass) using a prescribed temperature and exposure time.

Interference color stains develop as a result of decomposition of the surface of the glass by the test solution. One possible measure for classifying glasses is the time which elapses before the first stain occurs. This change in color indicates a chemical change in the previously defined surface layer of 0.1 mm thickness.

Resistance to Acids. When an acidic aqueous medium reacts with the surface of a glass stains can form, the glass can be decomposed, or both can occur. In contrast to the staining resistance tests discussed above, acid resistance testing provides information concerning dissolution since testing is done with larger quantities of solutions, often at lower pH .

ISO 8424 is a standard for evaluating acid resistance of glass. Here, the time required to dissolve a layer with a thickness of 0.1 mm serves as a measure of the resistance to acids. Two aggressive solutions are used in determining the resistance to acids: a strong acid, nitric acid, $pH 0.3$ is used for the more resistant glass types and a weak acidic solution with a pH value of 4.6 (standard acetate) is used for glasses with lower acid resistance.

Resistance to Alkali. These tests concern resistance to alkalis and are provided because many fabrication processes occur in water-based media (for example, grinding and polishing compounds) that usually become increasingly alkaline through the chemical reactions of the water and the abraded glass particles over time. This particularly applies when such solutions are recycled.

ISO 10629 is a standard for evaluating alkali resistance of glass. This test also takes into consideration the fact that higher temperatures can occur as a result of the polishing process through abrasion. The alkali resistance classes indicate the resistance of glasses to alkaline solutions and therefore as a warning signal to finishers of glass of possible problematic materials. As with the acid resistance testing discussed above, alkali resistance is based on the time required to remove a layer thickness of glass of 0.1 mm in an alkaline solution (sodium hydroxide, $c = 0.01 \text{ mol/l}$, $pH = 12$). In ISO 106529, the layer thickness is calculated from the weight loss per surface area and the density of the glass.

5.3 Colored Glasses

5.3.1 Basics

Color in glass arises from the selective attenuation or amplification of the incident light within special regions of wavelength in the visible part of the spectrum, i. e. in the range from about 380 nm to 780 nm [5.82]. Generally, the shape of the transmission curve versus the wavelength λ is described by (5.114) (see

also Sect. 5.1.3)

$$\tau_i(\lambda) = \exp \sum_m (-\varepsilon_{\lambda,c,m} c_m d) , \quad (5.114)$$

$$\tau_i(\lambda) = \frac{(\Phi_{e\lambda})_{\text{ex}}}{(\Phi_{e\lambda})_{\text{in}}} , \quad (5.115)$$

$$\tau(\lambda) = \frac{(\Phi_{e\lambda})_{\lambda}}{\Phi_{e\lambda}} = P \tau_i(\lambda) , \quad (5.116)$$

where ε and c are the extinction coefficient and the concentration of the coloring species m , respectively, d is the thickness of the probe and τ_i the internal transmission. The internal spectral transmittance $\tau_i(\lambda)$ in (5.115) is the ratio of the emerging spectral radiant flux $(\Phi_{e\lambda})_{\text{ex}}$ to the penetrating radiant flux $(\Phi_{e\lambda})_{\text{in}}$. It describes the transmission of a homogeneous absorbing filter material without allowing for reflection losses; $\tau(\lambda)$ characterizes virtually every optical filter, regardless of its structure or mode of action. The extinction coefficient for a given base glass is a function of the wavelength, the coloring species m and, in some cases, the concentration of this species. Keep in mind that one coloring ion in two different oxidation states acts like two different species (e.g. Fe^{2+} , Fe^{3+}).

Unfortunately, there is no way to calculate ε from theoretical considerations so it must be taken from experiments for every coloring species. Some data have been compiled by Bamford [5.83], but not for all elements and combinations. Additionally, all data taken from the literature depend on the special base-glass composition. So, for a given new problem, the extinction coefficients must first be established experimentally.

In many cases (5.114) can be simplified. Very often, glasses are colored by only one species in one oxidation state and the extinction coefficient is not a function of concentration. This is especially true if the concentration of the species is low. Then (5.114) can be written as (Lambert–Beer law)

$$\tau_i(\lambda) = \exp(-\varepsilon_\lambda cd). \quad (5.117)$$

From this equation, the transmission plots for different concentrations of the coloring species and different probe thicknesses can easily be calculated from one experiment, since ε is then a constant for a given wavelength.

As already mentioned, the color of a glass can be taken from a plot of its internal transmission against wavelength. A translucent body showing an internal transmission of $\tau_i = 1$ at every wavelength in the visible range is called colorless (see Fig. 5.40); the usual optical glasses are examples of these. When the internal

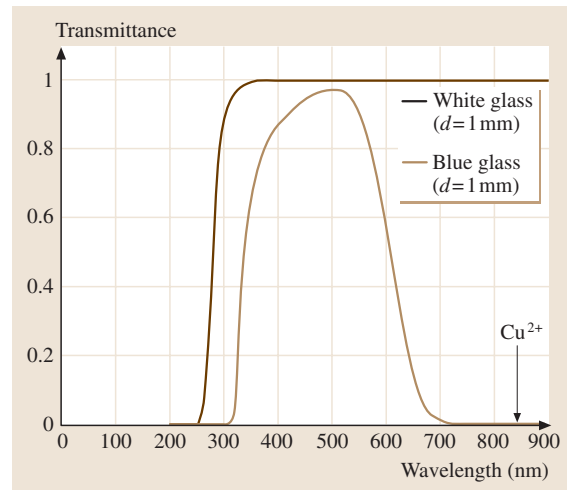


Fig. 5.40 Transmission spectra of white and blue glasses

transmission at a range of wavelengths is attenuated by any of the mechanisms discussed in Sect. 5.1.3, a color arises. The color depends on the part of the spectrum missing (Table 5.7).

Figure 5.40 shows the internal transmission plot of a white glass and a blue glass. The color of the blue glass is obtained by doping with Cu^{2+} . This kind of representation is chosen for scientific applications and construction of optical instruments because it advantageously shows the spectral distribution of transmitted light. Another way to describe the color of the glass is the use of color coordinates. When this method is used, the color is defined by two coordinates x and y , which are obtained by weighting the $\tau(\lambda)$ curve of the glass with the intensity distribution of the standardized light source for the colors red, green and blue. In the ideal case, the color portions of the light source add to clear white light. The procedure has been extensively described by Bamford [5.83].

Figure 5.41 shows the distribution coefficients for the red (P_x), green (P_y), and blue (P_z) part of a special standardized light source (illuminant D65) as has been set up by the International Commission on Light-

Table 5.7 Relation between color and absorbing wavelength range

$\lambda_{\text{absorbed}}$ (nm)	400	425	450	490	510	530	550	590	640	730
Color of light _{absorbed}	violet	indigo blue	blue	blue green	green	yellow green	yellow	orange	red	magenta
Color of light _{resulted}	green yellow	yellow	orange	red	magenta	violet	indigo blue	blue	blue green	green

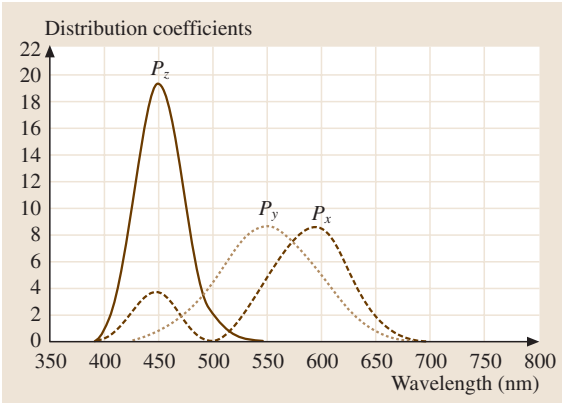


Fig. 5.41 Distribution coefficients for red (P_x), green (P_y), and blue (P_z) part of light of illuminant D65 (standardized light source)

ing (CIE). After multiplication of the transmission curve by the distribution coefficients, and normalization of the sum to unity, one obtains the color coordinates. The “z” coordinate has been omitted because it can easily be calculated from the condition that $x + y + z = 1$.

This method is used when the color impression is needed; e.g. for traffic signs and glasses for artists.

From the chemical point of view, the base-glass composition of colored glasses ranges within the compositional region of normal optical glasses, as discussed in Sect. 5.2. Historically, colored glasses have been developed from optical glasses by the empirical addition of so called “coloring elements”. There has been a lot of experimental work on this topic, of which a survey has been given by Weyl [5.84] and to some extent by Vogel [5.85]. The coloring elements additionally present in the base glasses, transition metals and lanthanides, contain partly filled d and f electronic shells. For some special applications, the elements of the sixth (S to Te) and seventh (Cl to I) main groups are used in combination with zinc and cadmium or copper and silver, respectively, in addition to the transition metals and lanthanides.

5.3.2 Color in Glass

The mechanisms of generating colors have been described in [5.86]. From these 15 mechanisms, only four are very important for practical use in inorganic oxidic bulk glasses: ligand field effects [5.82, 87–91], charge-transfer effects [5.82, 92], metal colloids [5.93–95] and semiconductor-doped glasses [5.85, 96–98]. These are in essence described in Sect. 5.1.3 and [5.82]. Therefore we will not explain this mechanism here.

The usual colors of some of the more important ions are listed in Table 5.8 [5.84], which can be described by ligand field effects.

The color of glass is determined by several parameters such as the kind of coloring species, its valency, the coordination of colored species, and the base glass.

The influence of the base glass on the spectra can be attributed to two items. The first is the intensity of the absorption band caused by the transition. This is influenced by the probability of electron transfer. Different base glasses show different coupling modes and therefore different extinction coefficients for a given coloring ion. Commonly, phosphate glasses show lower extinction and the color of a given ion is fainter than in silicate glasses [5.87]. The second reason is the difference of absorption peak maxima between silicate, phosphate and borate glasses. If it is only about 1–2% the arrangements of the coloring ion ligands stays unchanged. The differences between the peak maxima positions are much more pronounced. Some transitions show no correspondence at all. The explanation follows from changes in the coordination. The coordination, for instance, of Ni^{2+} in silicate glass is unexpectedly fourfold (tetrahedral), resulting in a purple color, whereas the dark yellow of Ni^{2+} in the phosphate glass results from the normal octahedral coordination [5.87].

Another coloring species of great importance is $Fe^{2+/3+}$. The interpretation of the spectra of iron is quite a complicated task. One not only has to consider two oxidation states with different peak positions but also that these two species each appear in tetrahedral as well as in octahedral positions. Adding further confusion, parts of these bands can occur at nearly the same wavelengths. Additionally, charge-transfer bands of Fe^{3+} with oxy-

Table 5.8 Usual colors of some ions in glasses, according to [5.84]

Element	Valency	Color
Fe	2+	Green, sometimes blue
Fe	3+	Brown
Cu	2+	Blue, turquoise
Cr	3+	Green
Ni	2+	Violet (tetrahedral)
Ni	2+	Yellow (octahedral)
Co	2+	Blue
Mn	2+	Pale yellow
Mn	3+	Violet
Pr	3+	Green
Nd	3+	Violet
Er	3+	Pale red

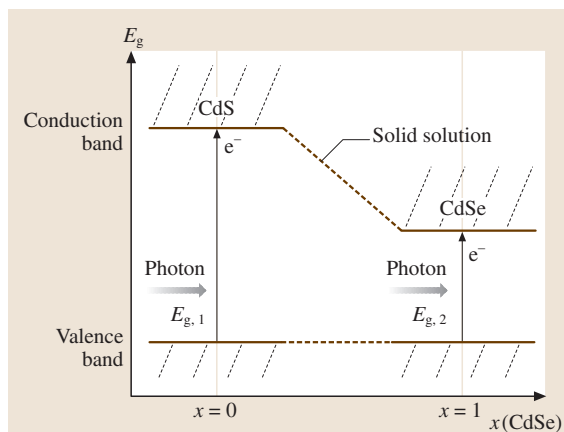


Fig. 5.42 Semiconductor-doped glasses: Band structure and absorption of light for II–VI semiconductors

gen occur. The color of the glass depends strongly on the redox conditions during melting. By the application of a low oxygen partial pressure the glass becomes very pale green to blue according to the transition at about 1100 nm. Under oxidizing conditions the glass turns brown because of the manifold bands of trivalent iron present all over the visible and UV range of the spectrum.

Beside the colored glasses described above, which are based on absorption bands, there are also long-pass filters, which are prepared by doping with semiconductors such as ZnS, ZnSe, ZnTe, CdS, CdSe, and CdTe followed by an additional annealing process to grow semiconductor crystals in the glass. These semiconductor-doped glasses show very differ-

Table 5.9 Band-gap energies for some semiconductor crystals (macrocrystals)

Compound	$E_g(\text{eV})/\lambda_c(\text{nm})$	$E_g(\text{eV})/\lambda_c(\text{nm})$
ZnS	3.7/335	3.65/340
ZnSe	2.7/459	–/–
ZnTe	2.2/563	–/–
CdS	2.4/517	2.42/512
CdSe	1.7/729	1.73/717
CdTe	1.5/827	–/–

ent behavior with interesting spectral properties. The absorption mechanism is explained in Sect. 5.1.3.

The essential properties of the transmission spectrum are the wavelengths λ_c , λ_s and λ_p . The wavelength λ_c is defined as the middle of the maxima between the blocking and the pass area ($\tau_i = 0.5$). In the blocking part ($\leq \lambda_s$) all light will be absorbed and at the pass part ($\geq \lambda_p$) the incident light will be transmitted.

Commercially available filter glasses of this series have absorption edges from about 400 nm to 850 nm. As indicated in Fig. 5.42 the band gap, and therefore the absorption edge, depend on the chemical composition and size of the crystals. The transition from Zn to Cd and from S over Se to Te leads to an increasing red shift of λ_c . Decreasing the size of the crystals also moves the effective band gap to higher energies and creates a corresponding blue shift of λ_c . The exact energies of the band gap can be taken from Table 5.9.

By annealing [5.99, 100], influencing the diameter of the crystals, and by mixing appropriate amounts of the pure compounds, all values for λ_c between 350 nm and 850 nm can be achieved.

5.4 Laser Glass

5.4.1 Common Laser Glasses and Properties

Laser glass is a solid-state material with the ability to amplify light by stimulated emission of radiation. In its most common form it is a multicomponent oxide glass doped with a lasing ion such as neodymium. In this section we deal predominantly with multicomponent glasses in contrast to the rare-earth-doped fused silica such as that commonly employed in Er-doped fiber amplifiers.

Although the dominant commercial market for laser glass is in large laser systems for inertial confinement fusion research with application to fusion energy and weapons physics science [5.101], these materials

have also found their way into a number of industrial and laboratory environments. For example, one leading application is in the field of laser shock peening [5.102].

Commercial laser glasses fall into three broad categories depending on the operating regime of the applied laser system. Laser glass composition development is a mature field and for each specific application glass compositions and/or processing details of these glasses have already been identified by suppliers that provide optimized properties for each situation.

There are, for example, laser glasses designed for high peak power. Here, laser properties have been op-



timized to offer the highest possible stored energy and extraction efficiency, resulting in high peak powers in single shots that are separated in time. Repetition rates of such systems are at most a few Hertz, and more commonly are one to several laser shots per day.

There are also laser glasses for high average power applications where repetition rates can be in the range of 1–20 Hz. Such laser systems are in general aggressively cooled to remove heat deposited into the glass during repeated optical pumping. In addition to good laser properties, these glasses also possess enhanced thermomechanical properties consistent with high thermal loading without part fracture.

Still another example are glasses developed for guided wave optics. Here, glass compositions are selected for high stability to devitrification during re-drawing into fibers or for compatibility with structuring technologies, for example ion exchange, used in preparing planar waveguide structures. Figure 5.43 shows representative laser glass parts, the largest being a slab more than 1 m along on its diagonal axis for use in high peak power laser fusion research.

Another way laser glasses are categorized is by the glass type. The first laser glass to be identified was a glass based on silica (commonly called a silicate glass) [5.103]. Today, the multicomponent glasses of greatest commercial value are phosphate types and the compositions are essentially free of any silica. In addition, there are glasses based on other glass forming systems, as well as glasses that contain, in addition or in place of oxygen, other anions such as fluorine.

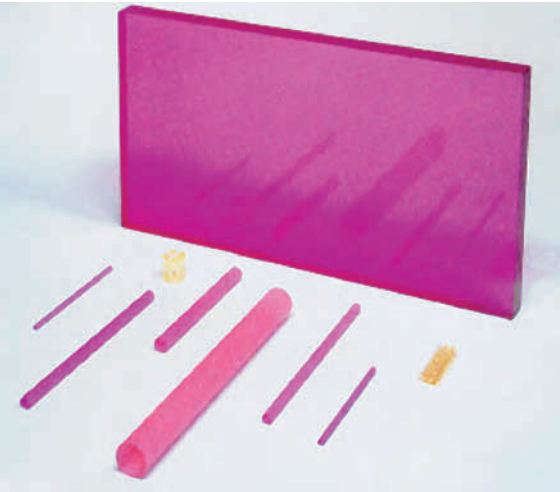


Fig. 5.43 Laser glasses

Today, the most commonly employed multicomponent laser glasses are based on phosphorous. These glasses are available in high optical quality from a number of commercial vendors with a wide range of dopants and doping levels, and in particular offer superior performance for resistance to laser damage when operated in high-fluence applications.

Although neodymium is the most common dopant and lasing ion, through selection of various dopant ions, either alone or in combination, a wide range of lasing wavelengths are possible. When more than one active ion is selected, usually one or more ions acts as a sensitizer for the primary lasing ion. The classical case is the codoping of laser glass containing erbium with ytterbium. Table 5.10 lists typical lasing wavelengths available in common glass hosts along with common sensitizing ions. The emission wavelength is, however, a function of glass composition so these values should only serve as a rough guide.

Specification of Laser Glass Doping Level

A common problem in identification of the optimal laser glass composition is the selection of the doping level of the lasing species or other sensitizing ions. The doping level is largely driven by two factors: the need to pump the entire laser glass volume uniformly and the avoidance of a phenomena called concentration quenching, in which two neighboring lasing ions in the glass can exchange energy by a radiationless process that steals available energy from the laser beam of interest. A rough guideline for dealing with the former case is provided in Table 5.11, which lists typical doping levels as a function of rod diameter for typical flashlamp pumped laser systems that provide nominally uniform excitation of the complete active volume.

Table 5.10 Laser wavelengths from selected active ions in glass

Active ion	Approximate emission wavelength (μm)	Sensitizing ion(s)
Nd ³⁺	0.93, 1.06, 1.35	Cr ³⁺ , Mn ²⁺ , Ce ³⁺ , Eu ³⁺ , Tb ³⁺ , U ³⁺ , Bi ³⁺
Er ³⁺	1.30, 1.54, 1.72, 2.75	Cr ³⁺ , Yb ³⁺ , Nd ³⁺
Yb ³⁺	1.03	Nd ³⁺ , Cr ³⁺
Dy ³⁺	1.32	
Sm ³⁺	0.65	
Ho ³⁺	0.55, 1.38, 2.05	Er ³⁺ , Yb ³⁺
Tm ³⁺	0.80, 1.47, 1.95, 2.25	Er ³⁺ , Yb ³⁺
Tb ³⁺	0.54	Ce ³⁺ , Cu ⁺
Pr ³⁺	0.89, 1.04, 1.34	

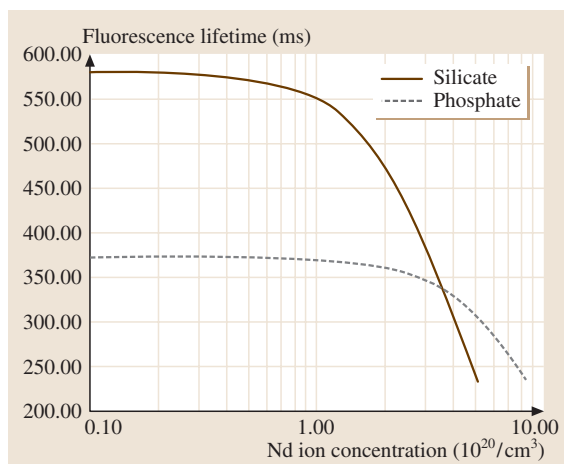
Table 5.11 Guidelines for neodymium dopant level for various rod diameters

Rod diameter (mm)	Nd concentration in phosphate laser glass (wt% Nd ₂ O ₃)	Nd concentration in silicate laser glass (wt% Nd ₂ O ₃)
≤ 5	8.0	3.5
5 to 7	6.0	3.0
6 to 10	4.0	3.0
9 to 13	3.0	3.0
12 to 16	2.5	3.0
15 to 20	2.0	2.0
20 to 26	1.5	2.0
≥ 27	1.0	2.0

Figure 5.44 shows the effect of concentration quenching for neodymium where the lifetime of the excited state used in the laser transition is shown to decrease with dopant level for some representative neodymium-doped laser glasses. The effect of concentration quenching is often estimated by the following equation

$$\tau = \tau_0 / [1 + (N/Q)^n], \quad (5.118)$$

where τ is the lifetime at neodymium concentration level N (in units of 10^{20} ions per cm^3), τ_0 is the effective lifetime for a negligible doping level in the glass and Q is a concentration quenching factor that corresponds to the doping level at which the lifetime value falls to $\frac{1}{2}$ of τ_0 . The value of n is generally equal to 1 or 2, and depends more on the amount of data available to determine the fit to (5.118) than on material considerations. The concentration quenching effect varies with glass type, and in

**Fig. 5.44** Typical concentration quenching curves

particular is less pronounced in phosphate compositions than in silicate glasses. Table 5.11 also lists different optimal doping levels for both the silicate and phosphate glass types.

Rare-earth ion concentration is expressed in a number of ways, most commonly as either a weight percent of oxide in the glass or as the number of active ions per cubic centimeter. It is straightforward to convert between these two values if one knows the density of the glass. For the case of various lanthanide cations the following conversion formulae between lanthanide (Ln) ion density, in units of $10^{20}/\text{cm}^3$, and weight percent of the lanthanide oxide can be employed:

$$\text{Ln ion density} = \rho F(\text{wt}\% \text{Ln}_2\text{O}_3), \quad (5.119)$$

$$\text{wt}\% \text{Ln}_2\text{O}_3 = \text{Ln ion density} / (\rho F), \quad (5.120)$$

where ρ is the glass density in g/cm^3 and the value of F for a number of common rare-earth elements is given in Table 5.12.

Glasses for High Peak Power

Glasses for high peak power are generally characterized by enhanced laser properties. This can perhaps be expressed as having a large value of the following laser figure of merit $\text{FOM}_{\text{laser}}$,

$$\text{FOM}_{\text{laser}} = \sigma Q / n_2, \quad (5.121)$$

where σ is the cross section for stimulated emission and Q is the concentration quenching factor (see p. 294), which together are a measure of the laser gain available in the glass, and n_2 is the nonlinear index, a measure of glass weakness to resist damage when transmitting a high-intensity pulse of laser light (see Sect. 5.4.2). These properties for some typical commercial high peak power laser glasses are given in Table 5.13.

Table 5.12 Rare earth active ion content conversion factors

Rare earth cation	F
Nd	0.358
Er	0.315
Yb	0.306
Tm	0.313
Ho	0.319
Pr	0.402

Table 5.13 Properties of some typical high peak power laser glasses

	Schott LG-770	Schott LG-750	Hoya LHG-80	Hoya LHG-8	Kigre Q-88
Cross section for stimulated emission σ (10^{-20} cm ²)	3.9	3.7	4.2	3.6	4.0
Concentration quenching factor Q (10^{20} cm ⁻³) [$n = 2$ in (5.118)]	8.8	7.4	10.1	8.4	6.6
Nonlinear refractive index n_2 (10^{-13} esu)	1.01	1.08	1.24	1.12	1.14
Fracture toughness K_{IC} (MPa m ^{1/2})	0.48	0.48	0.46	—	—
Thermal conductivity K (W/m K)	0.57	0.60	0.59	0.58	0.84
Poisson's ratio ν	0.25	0.26	0.27	0.26	0.24
Thermal expansion α (10^{-6} /K)	13.4	13.2	13.0	12.7	10.4
Young's modulus E (GPa)	47.3	50.1	50.0	50.1	69.8
Index at 1060 nm n_{1060}	1.499	1.518	1.534	1.521	1.536
$dn/dT_{20-40^\circ\text{C}}$ at 1060 nm	-4.7	-5.1	-3.8	-5.3	-0.5

Table 5.14 Properties of some typical high average power laser glasses

	Schott APG-1	Schott APG-2	Hoya HAP-4	Kigre QX-Nd
Cross section for stimulated emission σ (10^{-20} cm ²)	3.35	2.39	3.6	3.34
Concentration quenching factor Q (10^{20} cm ⁻³) [$n = 1$ in (5.118)]	16.7	10.6	—	—
Nonlinear refractive index n_2 (10^{-13} esu)	1.13	1.02	1.21	1.17
Fracture toughness K_{IC} (MPa m ^{1/2})	0.60	0.67	0.83	—
Thermal conductivity K (W/m K)	0.83	0.86	1.02	0.85
Poisson's ratio ν	0.24	0.24	0.24	0.24
Thermal expansion α (10^{-6} /K)	7.6	6.4	7.2	8.4
Young's modulus E (GPa)	71	64	70	71
Index at 1060 nm n_{1060}	1.529	1.503	1.534	1.530
$dn/dT_{20-40^\circ\text{C}}$ at 1060 nm	1.2	4.0	1.8	1.0

Glasses for High Average Power

Glasses for high average power are generally characterized by enhanced thermomechanical properties as well as good laser properties. This is often expressed as having a large value of the following thermomechanical figure of merit FOM_{tm} ,

$$FOM_{tm} = K_{IC}K(1 - \nu)/\alpha E, \quad (5.122)$$

where K_{IC} is the fracture toughness, K is the thermal conductivity, ν is the Poisson's ratio, α is the thermal expansion and E is the Young's modulus of the glass. These properties for some typical high average power

commercial laser glasses, along with the laser properties discussed earlier, are given in Table 5.14.

Planar and Fiber Waveguide Lasers and Amplifiers

The glasses described in the previous sections can also be drawn into fiber form for preparation of fiber lasers and amplifiers. In addition, a number of glasses are also available for preparing planar waveguide structures by techniques such as ion exchange. The ion-exchange structuring technique is normally performed at temperatures below the transformation point of the glass. Properties of some of these glasses specifically devel-

oped for planar waveguide applications are provided in Table 5.15. Since these glasses have been commonly employed in the preparation of active planar guided-wave devices based on the erbium emission in the 1540 nm telecommunications region, Table 5.15 contains some associated properties for this active ion.

Estimating Refractive Index

Glass manufacturers typically report refractive index and dispersion as measured at wavelengths near 587.6 nm. It is a common requirement to estimate the refractive index of laser glass at additional wavelengths, for example for purposes of coating design at a pump or lasing wavelength. In addition, refractive index is a function of doping identity and content, so that catalog index values and those listed here in Tables 5.13, 5.14, and 5.15 are only rough guides. For this, a useful relationship for the refractive index at a different wavelength $n(\lambda)$ is

$$n(\lambda) = n_D - [(n_D - 1)/V_D](1.5079 - 523\,640/\lambda^2) \quad (5.123)$$

where n_D is the refractive index at 589.3 nm, V_D is given by

$$V_D = (n_D - 1)/(n_F - n_C), \quad (\text{see Sect. 5.1.2}) \quad (5.124)$$

and n_F is the index at 486.1 nm, n_C is the index at 656.3 nm, and λ is in nm. The accuracy of (5.123) is given as ± 0.001 in the cases where index measurements are accurate to ± 0.0001 , but there is no applicable wavelength range provided [5.104].

Alternatively, a similar equation based on index data measured at the mercury d-line at 587.6 nm is the following:

$$n(\lambda) = n_d - [(n_d - 1)/V_d](1.5079 - 523\,640/\lambda^2), \quad (5.125)$$

where n_d is the refractive index at 587.6 nm, V_d is given by

$$V_d = (n_d - 1)/(n_F - n_C), \quad (5.126)$$

and n_F is the index at 486.1 nm, n_C is the index at 656.3 nm, and λ is again in nm. Equation (5.125) is expected to be accurate to ± 0.003 for IR wavelengths up to 2.3 μm in the cases where index measurements are accurate to ± 0.00002 .

5.4.2 Laser Damage

Advances in high-power and high-energy laser systems have made power densities in excess of 10^9 W/cm^2 available. At such power levels a number of different damage mechanisms come into play that can lead to temporary or even permanent changes in the optical quality of optical materials, including laser glass. Such changes may occur at a surface or within the bulk material, and can both render an optic unusable and/or alter the transmitted beam profile in such a way as to place other downstream components in an optical system at increased risk of damage.

Transient Thermal Effects

Since the refractive index is generally a function of temperature, thermal gradients within laser glass parts can lead to optical distortion of propagating light through inhomogeneous changes in the optical pathlength. For some applications, it is possible to have laser glasses that behave essentially as an athermal component in which the thermal expansion of the glass offsets the change in optical pathlength due to the variation of index with temperature.

For a laser glass component within a cavity of an otherwise fixed length (for example as defined by a frame constructed from near zero expansion materials such as Zerodur or Invar), the variation in optical pathlength

Table 5.15 Properties of typical waveguide laser glasses

Glass type	Schott IOG-10 Silicate	Schott IOG-1 Phosphate	Hoya LHG-5 Phosphate	Kigre MM-2 Phosphate
n_d	1.530	1.523	1.541	1.54
V_d	56.6	67.5	63.5	—
Thermal expansion $\alpha_{20-40^\circ\text{C}}$ ($10^{-6}/\text{K}$)	6.8	9.3	8.4	7.3
Er peak emission wavemlength (nm)	1536	1534	—	1535
Er FWHM fluorescence linewidth, $\Delta\lambda_{\text{em}}$ (nm)	18.5	26.5	—	55
Er fluorescence lifetime (ms)	17.8	10.7	—	7.9
Glass transformation point T_g ($^\circ\text{C}$)	569	474	455	506

with temperature, dS/dT , is given by

$$dS/dT = \alpha(n-1) + dn/dT, \quad (5.127)$$

where the glass is characterized by a thermal expansion value of α , refractive index of n , and temperature change of refractive index, relative to air, with temperature of dn/dT .

An alternative situation occurs if the laser glass part is coated in such a way as to define the length of the optical cavity, as in the case where the end mirrors of a laser oscillator cavity are directly applied to the ends of a laser rod. In this case the cavity length variation with temperature is given by

$$dS/dT = n\alpha + dn/dT. \quad (5.128)$$

Typical values for α , n , and dn/dT are included with other laser glass properties in Tables 5.13–5.15.

Surface Damage

Surface damage on active (and also passive) optical materials in high-intensity laser systems normally appears as circular elevations (pustules) or depressions (pits) on transmission surfaces. The number density and size of these features is somewhat determined by the laser energy density and pulse length, respectively. Laser glass surfaces are vulnerable to laser damage since they typically become contaminated through routine storage and handling as well as the gradual accumulation over time of airborne particles. Surface contamination serves as a nucleation point for damage. The initial preparation of the surface can potentially play a critical role in determining the likelihood of experiencing laser damage. After polishing of a surface is completed, there can still exist residual scratches, defects, and subsurface flaws that can retain contaminants of polishing compounds and cleaning materials. Such sites on optical material surfaces serve as additional nucleation points for laser damage.

Induced damage on beam exit surfaces is often correlated with absorption of laser energy at such surface defect sites, followed by formation of a plasma which in turn can enhance reflection of the laser light back on itself, further increasing the local electric field intensity associated with the laser beam. Finishing vendors for high-damage-threshold optics need to avoid leaving residue of polishing compounds that absorb strongly at the intended wavelength of exposure. Surface damage threshold is also clearly improved by proper cleaning of optical surfaces and subsequent protection from atmospheric contaminants, including water, which can alter surface properties through condensation and diffusion.

Self-Focusing Damage

Self-focusing occurs due to the dielectric breakdown of an optical material from localized focusing of a propagating laser beam by increase of the refractive index with the applied light intensity. This effect leads to one or more threadlike damage sites oriented along the main optical axis that are often referred to by a number of different terms including *tracks* or *angel hair*.

The damage event often begins as a localized hot spot within a laser beam that causes an increase in local refractive index within the glass. Since the optical pathlength is increased at such a hot spot, and drops towards the original value away from the high intensity region, this portion of the laser glass essentially acts as a positive lens. This artificial lens further converges the laser beam, compounding the effect. This convergence continues until the electric field associated with the laser increases to a point that atoms are ionized and a plasma is produced within the glass. An example of such self-focusing damage is shown in Fig. 5.45.

The phenomena of self-focusing is driven by the laser glass nonlinear refractive index n_2 , the accumulated optical pathlength, and the presence/absence of localized hot spots in the laser beam. Internal optical quality of laser glass has improved over time to the point that meter-class sections are available that are essentially free of all bubbles and inclusions that can cause localized fluctuations in beam intensity through diffraction effects, so the nonlinear index plays a prominent role in determining if laser damage of this type will occur. It is for this reason that n_2 appears in (5.121) for FOM_{laser} in Sect. 5.4.1. In addition to selection of a laser glass with low n_2 , laser cavity and beamline designs need to minimize the possibilities of high-fluence locations in

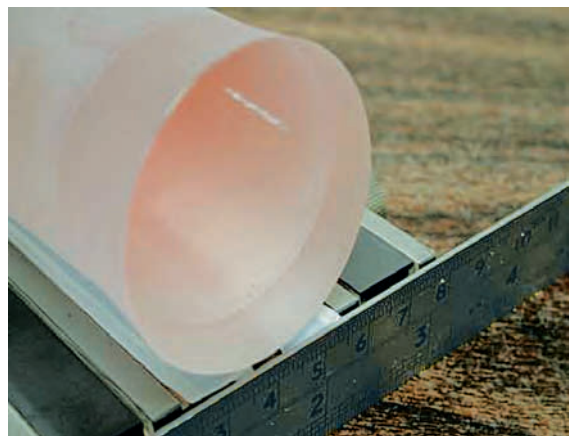


Fig. 5.45 Self-focusing damage in laser glass

beams passing through laser glass components, i. e. the laser energy should be spread out over as large an area as practically possible.

Multiphoton Induced Damage

Multiphoton transitions involve two or more photons whose energies are combined to a value sufficient to excite a real transition in a material. High-fluence laser sources provide high photon flux levels that make these low-probability multiphoton transitions in glass observable. Damage from multiphoton absorption can occur when electrons are excited into an optical material's effective conduction band, where they then migrate to stable traps in the material and induce structure and property changes. Damage takes two principle forms: as a localized discoloration and/or as a permanent change in the refractive index of the damaged region. This type of laser damage is almost always viewed as problematic and the observed absorption characteristics as a function of wavelength are generally similar to that observed with high-energy radiation damage, such as that found with UV, X-ray, or particle (proton, electron, etc.) exposure.

For a particular laser design, the extent of multiphoton damage is determined by the number of laser photons required to achieve a combined energy sufficient to lead to damage. As a rough guide, one can use an effective band gap of the laser glass, as calculated from the onset of absorption in the high-energy end of the electromagnetic spectrum, compared to the energy available from a single photon at the laser wavelength.

Point-Defect Laser Damage

Nearly all optical materials, including laser glass, contain some level of localized microscopic bulk defects including bubbles, dielectric inclusions, and precious metal particles. All such defects have some relationship to the original manufacturing process. Most inclusions and metal particles represent either unmelted compounds from the original glass melt or portions of the manufacturing equipment that were incorporated into the final laser glass component. Bubbles by themselves are not typically linked with the creation of localized damage sites, however they can cause diffraction of a propagating laser beam with associated small-spatial-scale high-intensity power spikes that can then damage other optical elements in a laser system through non-linear effects (see for example the discussion in the subsection on surface damage).

Metal particles, chiefly platinum, remain in laser glass since for optical quality reasons these glasses are produced within pots and crucibles made of precious

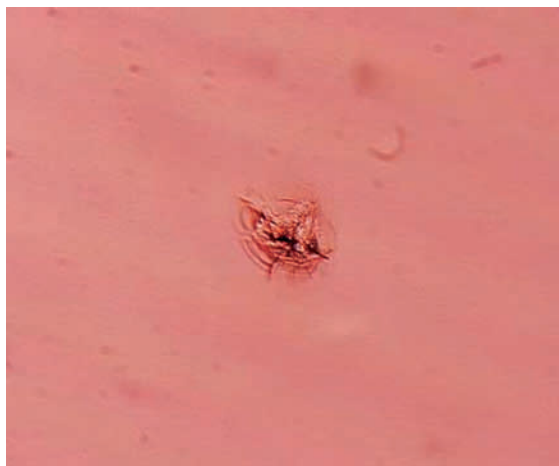


Fig. 5.46 Particulate damage in laser glass due to a platinum particle

metals and their alloys. Subsequent particulate-related damage from these inclusions appears as isolated damage sites within a bulk optic. Figure 5.46 shows a magnified image of such a damage site that measures 0.25 mm in diameter. The damage mechanism is believed to be direct absorption of laser radiation by the dielectric or conducting-defect site, followed by heating of the particle to above its boiling point, with subsequent crack nucleation and growth from both the vaporization-related shock wave and localized thermal-induced stress occurring [5.105].

Since the initial discovery of the laser in 1960, the largest single improvement in laser glass damage-threshold performance has been the development of manufacturing technologies capable of producing meter-class phosphate laser glass components completely free of all inclusions or internal defects capable of becoming particulate damage sites [5.106]. The specification of such platinum-particle-free glass should be made when laser fluence levels approach the damage threshold, a condition approximated for pulsed 1.064 μm laser exposure by

$$\text{Damage threshold in J/cm}^2 = 2.5 \times (\text{pulse length in ns})^{0.3}. \quad (5.129)$$

Although laser damage is an extensively studied field, there is no comprehensive database of the enormous amount of accumulated experimental data on this subject that would make it possible to predict the laser damage level for a particular situation. Interested readers are referred to several textbooks that deal exclusively with this field of investigation [5.107, 108]



5.4.3 Storage and Handling of Laser Glass

Laser glasses are typically brittle materials that are easily chipped or broken upon thermal or mechanical loading. As a result, heating and cooling of glass parts should be at rates not exceeding 20–30 °C/h. Laser glasses with good laser properties are generally of compositions that also exhibit low chemical durability. The low durability can sometimes be used to an advantage, since deliberate chemical etching [5.109] or chemical ion-exchange processes [5.110] can be used to enhance mechanical strength.

There exists no universal method for evaluating all possible chemical attack paths and laser glass compositions. However, as a general rule, the presence of water is nearly always a prerequisite for the chemical attack of glass. As a consequence, long-term storage of sensitive glasses is best accomplished within a closed, and preferably evacuated, environment containing a desiccant material.

Methods to Enhance Part Strength

Laser glasses can be strengthened by techniques such as acid etching and ion exchange. In the former, surface removal by etching is accompanied by blunting of crack tips left following fabrication or mechanical handling. Blunted crack tips require a higher energy level to propagate into cracks or fractures, effectively increasing the strength of the glass part to breakage. Ion exchange involves the substitution of a smaller cation within the glass structure by a larger cation, effectively placing the surface of the glass under compression. To experience

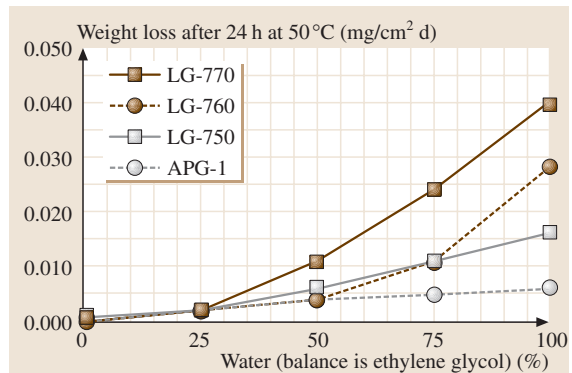


Fig. 5.47 Chemical durability in EtOH/water mixtures

fracture, the applied stress must now exceed not only the initial strength level of the glass but the compressive surface stress as well. Both processes involve some level of chemical attack to the glass surfaces and are not suitable for polished optical surfaces.

Liquid Cooling of Laser Glasses

Solid-state laser systems employing laser glass as the gain medium are typically cooled by a recirculating liquid coolant system. Chemical attack by the coolant solution is a variable to be considered, in particular when phosphate laser glasses are employed and the component is to be left in contact with coolant during periods of no operation. In this case, it has been shown that addition of ethylene glycol to water can provide improved protection to chemical attack, as seen in Fig. 5.47.

5.5 Glass–Ceramics for Optical Applications

5.5.1 Overview

Glass–ceramics – glasses that are crystallized via a deterministic and controlled process – combine the relative ease of manufacturing associated with glass with the added functionalities that exist only in crystalline states of matter [5.111, 112]. For example, the combination of a crystalline phase that exhibits negative thermal expansion, a relatively rare occurrence in nature, with the normal positive expansion for glass has led to the development of ultra-low thermal expansion glass–ceramics used for such things as telescope mirrors, IR-transparent cooker tops, and reflectors for digital projection [5.113]. Dental glass–ceramics combine high strength with the proper amount of controlled translucency to mimic

natural teeth precisely [5.114]. And a wide variety of nonlinear phenomena have been demonstrated in glass–ceramics, including ferroelectric and pyroelectric behavior [5.115], second-harmonic generation [5.116], upconversion [5.117], and scintillation [5.118].

Contrary to one's first impression, the presence of crystals in a glass framework need not significantly inhibit transmission. The ability to control the crystallization process often allows one to keep crystal sizes small relative to the wavelength of the incident radiation, thereby minimizing scattering losses, particularly at longer wavelengths. In addition, the precipitation of very small, insoluble crystals in a soluble matrix allows one to form crystals in compositions that are otherwise difficult if not impossible to control. For ex-

ample, *Beall* [5.119] developed a Cr-doped forsterite (Cr:Mg₂SiO₄) glass–ceramic; stoichiometric forsterite glass by itself is extremely unstable and only very special processing techniques are able to form it [5.120]. However, by relying on phase separation to create Cr-doped, forsterite-like sub-domains which subsequently crystallized, Beall was able to combine the ease of glass production with the formation of a desired crystal. This tactic is a familiar one in glass–ceramic development.

Glass–ceramic production begins with the formation of suitable glass castings using now-standard manufacturing techniques [5.124]. Detailed inspection techniques are then used to determine the internal quality of the part owing to the inherent high transparency of glass. After inspection, a controlled thermal cycling program, termed “ceramization”, is used, whereby the glass part is partially or wholly transformed into a glass–ceramic. Typical thermal histories take advantage of the disparate temperature regimes in which crystal nucleation and subsequent growth take place [5.125]. At a minimum, three heating/cooling ramp rates and two pairs of temperature/time conditions need to be specified in a ceramization process (Fig. 5.48). This, coupled with the fact that both nucleation and growth are thermally activated processes with inherently strong nonlinear dependencies on temperature [5.126], can make ceramization a challenging process.

5.5.2 Properties of Glass–Ceramics

Compared to their precursor glasses, glass–ceramics typically have increased strength and higher elastic constants. Thermal expansion, on the other hand, can be either higher or lower than the precursor glass. Glass–ceramics normally exhibit less dielectric loss than their glassy precursors, due in part to tying up the more mobile species in a more rigid, crystalline framework (though

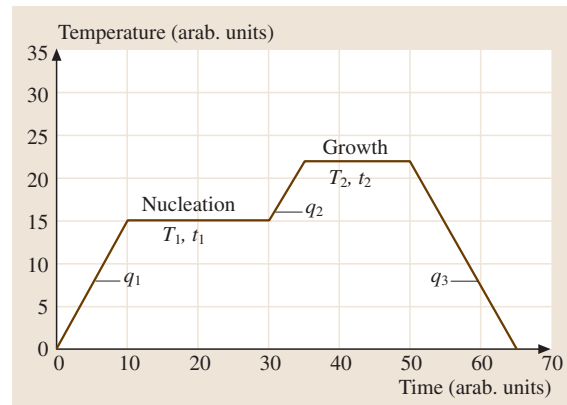


Fig. 5.48 Schematic temperature–time path used for the production of glass–ceramics, a process termed ceramization; q_i refers to heating and cooling rates, T_i and t_i refer to temperatures and time used during crystal nucleation and growth stages, respectively

exceptions are known [5.111, 127]) whereas relative permittivities are generally higher in glass–ceramics. Finally, chemical durability is typically enhanced over glass, in some cases quite dramatically [5.112].

Thermomechanical Properties

Thermal Expansion. Coefficient of thermal expansion values (CTE) for glass–ceramics range from nearly -10 to $+25 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$, a remarkably wide range few materials can approach (Table 5.16) [5.111]. Expansion matching to most metals (including Si) has been accomplished thereby allowing for the production of multi-material devices under stress-free or stress-controlled conditions (Fig. 5.49) [5.111, 128, 129]. A possible complexity with glass–ceramics arises from the often nontrivial expansion curves for the crystalline phases. For example, glass–ceramics that contain a silica polymorph typically show a pronounced change

Table 5.16 Comparison of optical and physical parameter values for a range of glass–ceramic materials

Material	n	T_i (%)	λ (nm)	Thickness (mm)	CTE (ppm/ $^\circ\text{C}$)	T range ($^\circ\text{C}$)	E (GPa)
Zerodur ^a	1.52	99.6	1550	1	0 ± 0.05	0 to 50	90
SA-O2 ^b	1.56	≥ 98	1550	1	1.8	30 to 500	98
WMS-15 ^b	1.52	99.9	1550	1	11.4	-30 to $+70$	96
NEX-C ^b		99.3	1550	10	-2	-40 to $+80$	94
Oxyfluoride ^c		99.99	1500	1			
Telluride ^d	2.1	≈ 80	800	1			
Pyroceram ^e					5.7	20 to 320	120

(Sources: ^a Zerodur product brochure, ^b Ohara product brochure, ^c [5.121], ^d [5.122], ^e [5.123]; note that Pyroceram, while opaque in the visible region, is transparent in its operative, radio-frequency region)



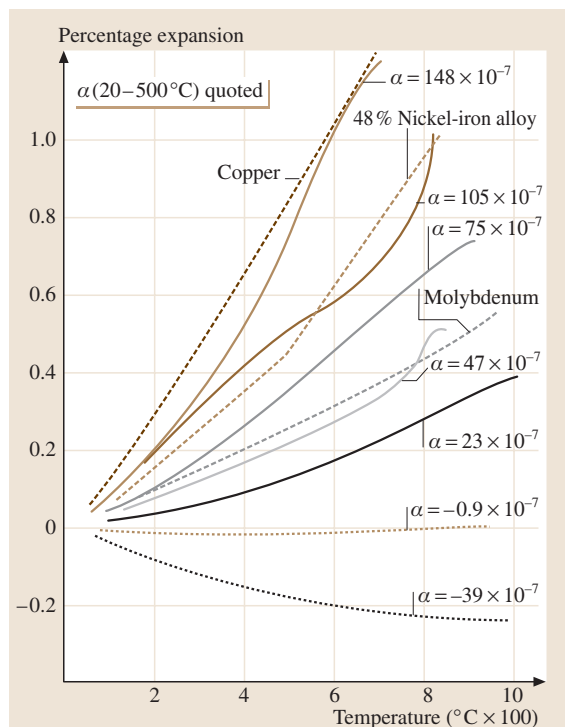


Fig. 5.49 Range of thermal expansion curves ($\Delta L/L$) attainable with glass-ceramics. Also plotted are expansion curves for typical metals (here, α is the CTE of the material). (After [5.111])

in slope in a plot of CTE versus temperature, associated with a polymorphic transition of the silica phase (Fig. 5.50). This can be dealt with, in principle, through a suitable choice of the particular crystalline phases that most modify the CTE relative to the glass precursor. Additionally, the value of CTE can be tuned over fairly substantial ranges for a single bulk composition through proper choice of heat treatment conditions (Fig. 5.50). To a fairly good degree, the CTE of a glass-ceramic depends linearly on the relative amounts of the various phases, though more complicated treatments exist [5.130]. Cryogenic thermal expansion data are known for only a few glass-ceramics [5.131].

Elastic Constants. Due to their lack of crystalline order, most glasses have elastic moduli and related parameters (e.g., Young's modulus E , sound velocities) considerably lower than those of their crystalline counterparts. Thus, glass-ceramics commonly have E values between those of crystals and glasses (e.g., 80 to 150 GPa),

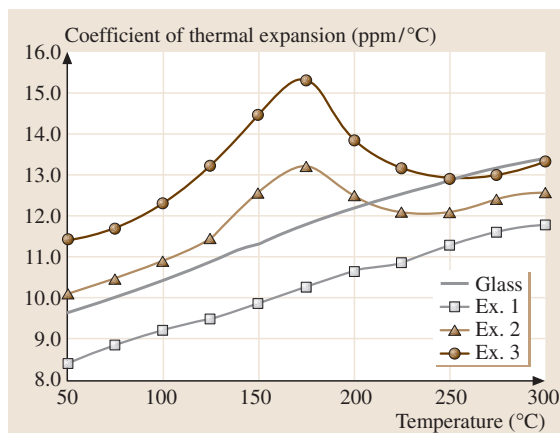


Fig. 5.50 Example of the range of CTEs that can be attained in glass-ceramics for a single composition. The curve labeled “glass” is for the unceramized precursor glass whereas those labeled “Ex. 1” and so on are for 3 different ceramization protocols. The large bump in CTE near 175 °C is due to the polymorphic transition of a cristobalite solid-solution phase

though in some cases nearly double that of their precursor glasses (Table 5.16) [5.111]. As for CTE, the Young's modulus also displays a nearly additive relationship with respect to the phase content [5.132].

Strength, Hardness, and Fracture Toughness. Enhanced strength is one of the hallmarks of glass-ceramics. Here, the moduli of rupture measured via four-point bend tests or associated techniques can be double or more those of the precursor glasses [5.133], though surface flaws still ultimately control the maximum attained strength values. Unlike thermal expansion and elastic properties, strength is not only controlled by the proportion of the various phases present, but also by the internal microstructure, particularly at or near the surface [5.112, 134]. The presence or absence of interlocking crystals, preferred crystalline orientation, or even the slip system associated with the crystalline phases present can all play a role in dictating glass-ceramic strength.

As a brittle material, the fracture behavior of glass-ceramics is commonly characterized by hardness and fracture toughness values. The former is normally quantified via diamond micro-indentations and the latter by the propagation of either a preexisting flaw of known dimensions or the development of indentation fractures [5.135, 136]. Here too crystal content and microstructure play dominant roles [5.137].

Electrodynamical Properties

Dielectric Properties. Dielectric property characterization – typically restricted to sub-optical frequencies and often grouped with physical properties – is analogous to refractive index and absorption characterization at optical frequencies. Dielectric properties are generally restricted to GHz and lower frequencies (microwaves and radio-frequency), whereas optical characterization is normally restricted to tens of THz and above (infrared, visible and UV primarily). Little is known about the response of glass–ceramics, or of many substances for that matter, in the so called terahertz regime (0.1 THz to 10 THz) [5.138].

The most important dielectric parameters in the GHz and below regime are the dielectric constant (relative permittivity) and dielectric loss (loss tangent). The power absorbed per unit volume P may be written:

$$P = \frac{1}{2} E_0^2 \omega \varepsilon_0 \varepsilon_r \tan \delta, \quad (5.130)$$

where E_0 is the magnitude of the electric field, ω is the frequency of the field, ε_0 is the permittivity of vacuum, ε_r is the *relative* permittivity of the material (not to be confused with the permittivity of the material, given by the product of ε_r and ε_0), and $\tan \delta$ is the loss tangent [5.139]. The relative permittivity of the dielectric material is related to its refractive index via

$$\varepsilon_r = n^2. \quad (5.131)$$

Note that this latter expression is only strictly true if the frequency with which both permittivity and refractive index are made are equivalent, or, somewhat more loosely, if the polarization loss mechanisms are unchanged between the two measurements. Thus, ε_r is related to the speed of propagation within the dielectric whereas $\tan \delta$ is a measure of the absorption. Typical values for permittivity start at values similar to glass (≈ 5) but can exceed 1000 for glass–ceramics containing ferroelectric crystals [5.140–142]. Loss tangents also

start close to glass values (e.g., 0.0002) and range up to nearly 1 (Table 5.17) [5.111]. The dielectric strength of glass–ceramics – the magnitude of the electric field just sufficient to initiate breakdown of the dielectric – is generally superior to that of ceramics and comparable to that of glasses (Table 5.17).

Refractive Index and Absorption. Optical properties of glass–ceramics are quite similar to those of their glass precursors and the crystalline phases contained therein. As introduced in Sect. 5.1.1, the two fundamental quantities, refractive index and absorption, can be described collectively via the complex refractive index

$$\tilde{n}(\omega) = n(\omega) + i\kappa(\omega), \quad (5.132)$$

where n , the refractive index, and κ , the bulk absorption, are both frequency dependent [5.143]; κ is in turn related to the more often used absorption per unit length α via

$$\alpha = \frac{2\omega\kappa}{c}, \quad (5.133)$$

(see Sect. 5.1.3) where c is the speed of light and α is used to characterize the total transmission T through a sample via

$$\frac{I_t}{I_0} = T = RT_i = R \exp(-\alpha z), \quad (5.134)$$

(Fig. 5.10) where I_t is the transmitted intensity (W/m^2), I_0 is the incident intensity, T_i is the internal transmission (not including surface reflection losses), R is a factor that accounts for reflection losses, and z is the sample thickness [5.144].

Refractive indices of most glass–ceramics are in the range 1.5 to 2.0 across much of the visible and infrared region, whereas transmission values are strongly dependent on the wavelength of interest (Table 5.16). In some applications, absorption across a wide range of

Table 5.17 Dielectric properties of various glass, glass–ceramic, and ceramic materials

Material	ε_r	$\tan \delta$	Test frequency	Dielectric strength (kV/mm)	Test frequency (Hz)
Lithium–zinc silicate glass-ceramic ^a	5.0	0.0023	1 MHz	47	50
Fused silica ^b	3.8	0.0004	1 MHz	36	100
Alumina (99.5%) ^c	9.7	0.0001	1 MHz	8.7	60
Y-stabilized zirconia ^c	9.0	0.001	1 MHz	9.0	60
Steatite ^d	6.1	0.0008	1 MHz	9.1	60
Pyroceram ^e	5.5	0.00033	8.5 GHz		

(Sources: ^a [5.111], ^b GE Quartz commercial literature, ^c Coors Ceramics Company commercial literature, ^d CeramTec commercial literature, ^e [5.123])



wavelengths is of interest. For example, in the characterization of dental glass–ceramics with controlled translucency, standardized tests exist that quantify a material's integrated absorption in the visible region when recorded against a white or black background [5.114] and may be quantified with respect to standardized color coordinates [5.145].

Scattering. Volume scattering (see Sect. 5.1.7) of radiation by crystals is one of the most interesting aspects of glass–ceramics. Glass–ceramics have in fact been used recently as test materials for wavelength-dependent scattering experiments and analysis in the UV–VIS portion of the spectrum [5.146]. Depending on the application, the wavelength of light may be greater or less than the average crystal size. This distinction, among other factors, leads to a variety of phenomena that dictate the ultimate utility of glass–ceramics for optical applications. The theoretical basis for scattering in condensed matter began over a hundred years ago with the work of Rayleigh and with important generalizations some twenty years later by Mie (Sect. 5.1.7). More recent efforts relevant to glass–ceramics have been devoted to tying the early, theoretical work to the particular set of conditions that glass–ceramics present.

Based on the theoretical arguments of Sect. 5.1.7, we consider here the most important type of scattering for optical glass–ceramics: pure Rayleigh scattering, in which the size of the particle is less than that of the wavelength of light and the scattering cross section is defined by [5.147–149]:

$$C_{\text{scat}} = \pi a^2 Q_{\text{scat}}, \quad (5.135)$$

where a is the particle radius and Q_{scat} is the scattering efficiency factor. The latter is in turn defined by

$$Q_{\text{scat}} = \frac{8}{3} \left| \frac{m^2 - 1}{m^2 + 2} \right|^2 x^4, \quad (5.136)$$

where m is often approximated by the ratio of the refractive index of the particle to that of the matrix (but see [5.149]), and x is a size parameter defined by

$$x = \frac{2\pi a}{\lambda}, \quad (5.137)$$

where λ is the light wavelength. Taken together, we can write for the scattering cross section (cm^2):

$$C_{\text{scat}} = \frac{128}{3} \pi^5 \frac{a^6}{\lambda^4} \left| \frac{m^2 - 1}{m^2 + 2} \right|^2. \quad (5.138)$$

The λ^{-4} dependence is a hallmark of Rayleigh scattering as is the strong dependence on particle size.

An alternative definition, based on a quasi-continuum model due to Hopper [5.150], is given by

$$C_{\text{scat}} = (6.3 \times 10^{-4}) k^4 \bar{w}^3 (\bar{n} \Delta n)^2 N_V^{-1}, \quad (5.139)$$

where k is the wavenumber ($2\pi/\lambda$), $\bar{w}$ is a mean width, here taken as the particle radius (but see explanation in [5.150]), $\bar{n}$ is the mean refractive index, and N_V is the number of particles per unit volume. Here we have used Hopper's equation (70) and the definition of the absorption due solely to scattering, sometimes called the turbidity τ (cm^{-1})

$$C_{\text{scat}} = \frac{\tau}{N_V}. \quad (5.140)$$

The often cited subcase of Rayleigh–Gans scattering has a further stipulation that there is negligible reflection at the particle–matrix interface. In this case, (5.136) is modified [5.149]

$$Q_{\text{scat}} = |m - 1|^2 x^4 \int_0^\pi G^2(u) (1 + \cos^2 \theta) \sin \theta d\theta, \quad (5.141)$$

where

$$G(u) = \frac{2}{u^3} (\sin u - u \cos u) \quad (5.142)$$

and

$$u = 2x \sin \frac{\theta}{2}. \quad (5.143)$$

Another definition of the scattering cross section has no dependence on particle size, but is dependent on the particle number density [5.151]

$$C_{\text{scat}} = \frac{8}{3} \left(\frac{\pi (n^2 - 1)}{N_V \lambda^2} \right)^2. \quad (5.144)$$

Finally, work by Hendy takes recent findings in structure factors in phase-separated media into account to arrive at a scattering cross section with quite different dependencies on both particle size and wavelength but which explicitly includes the volume fraction crystallized φ [5.152]

$$C_{\text{scat}} = \frac{14}{15\pi} \varphi (1 - \varphi) k^8 a^7 \left(\frac{\Delta n}{\bar{n}} \right)^2 N_V^{-1}. \quad (5.145)$$

To relate scattering cross sections to experimental measurables, we make use of the equation that relates the normalized transmitted intensity to the path length z [5.151]

$$\frac{I_{\text{scat}}}{I_0} = \exp(-C_{\text{scat}} N_V z), \quad (5.146)$$

where here (5.146) ignores losses due to absorption and reflection.

As an example of scattering in a glass–ceramic, we will analyze data from the nanocrystalline glass–ceramic Zerodur. This material, described more fully below, contains approximately 70 vol. % of 50 nm crystals. The number density for the microstructure is $\approx 10^{16}$ crystals/cm³ [5.153]. Figure 5.51 plots uncorrected transmission data for both glassy and ceramized Zerodur. Note that the composition of these two samples is identical and so the difference between the two curves should be dominated by scattering losses. Figure 5.52 plots this difference, here normalized to remove the reflection losses which are virtually identical for both glassy and ceramized material. Rearranging (5.146) we have

$$-\ln\left(\frac{I_{\text{scat}}}{I_0}\right) = C_{\text{scat}} N_V z. \quad (5.147)$$

The measured quantity on the left-hand side (LHS) of this expression is directly proportional to the scattering cross section. Dividing the LHS of (5.147) by the path length (≈ 5 mm) and by the crystal number density for Zerodur, we arrive at the scattering cross section itself (small symbols in Fig. 5.53). Note the nearly linear appearance when plotted on this semilog scale, suggesting a power-law dependence on wavelength (the best-fit slope to data in the range 450 to 800 nm is -4.3 , consistent with Rayleigh scattering). The rea-

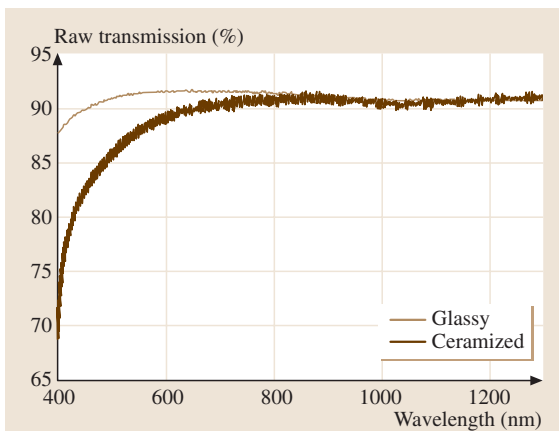


Fig. 5.51 Raw transmission data for glassy and ceramized Zerodur (thickness = 5.05 mm for both samples). Note the high transmission of both samples through the visible region (400 to 800 nm) and the nearly identical values for both ceramized and unceramized (glassy) samples at wavelengths greater than ≈ 1000 nm

son for the small deviation from the λ^{-4} dependence at short wavelengths (< 425 nm) is not clear. Also plotted in Fig. 5.53 is the prediction of the Hopper cross section (5.139). We used an estimated ratio of indices (m) of $1.55/1.5 \approx 1.03$ for this example. For comparison, note that the observed cross section at 500 nm was 1.2×10^{-17} cm². Although both the Rayleigh and Hopper equations predict the observed λ^{-4} dependence of the cross section, the pure Rayleigh model is some three orders of magnitude too high (3.3×10^{-14} cm²), whereas the Hopper model predicts 1.0×10^{-17} cm², very close to

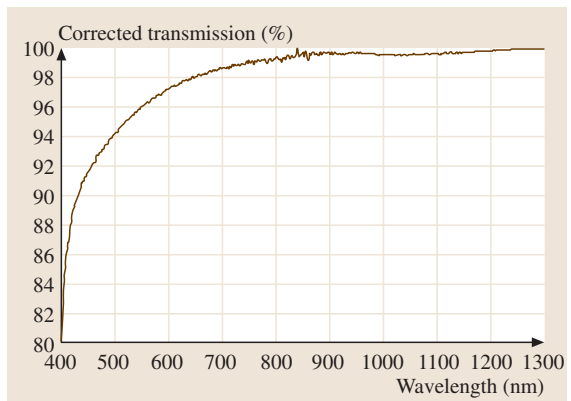


Fig. 5.52 Pure scattering effect on the transmitted intensity in Zerodur. The difference spectrum from the transmission curves in Fig. 5.51 were renormalized to remove reflection losses common to both samples. The subtraction process itself removes common absorption losses. What then remains, and is plotted here, is only the modification to the observed intensity due to scattering

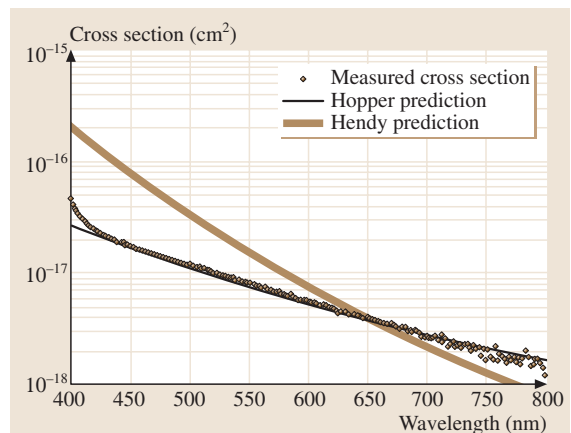


Fig. 5.53 Measured and calculated scattering cross sections in Zerodur

the observed value. The particle-size-independent model (5.144) gave comparable results to the pure Rayleigh model at 500 nm ($6.6 \times 10^{-14} \text{ cm}^2$). Numerically integrating the Rayleigh–Gans equations (5.141–5.143) yielded $1.4 \times 10^{-14} \text{ cm}^2$ at 500 nm, a fairly small correction from the pure Rayleigh value. The model of Hendy yielded $3.3 \times 10^{-17} \text{ cm}^2$, quite close to the measured one near 500 nm. However, the Hendy model predicts a too steep, λ^{-8} dependence, refuted by the very close agreement between the data in Fig. 5.53 and those models that predict a λ^{-4} dependence.

Nonlinear Properties (Electro-Optic, Pyroelectric, SHG, etc.). Although nonlinear crystals have been in use for over 100 years [5.143], the use of glass–ceramics as a host for such crystals has only been known for some 50 years. Herczog [5.140] reported the crystallization of microcrystalline BaTiO₃ from suitable borosilicate glasses and the production of glass–ceramics with relative permittivities exceeding 1000, whereas Borrelli et al. [5.159] and Borelli [5.160] reported quadratic electro-optic behavior in NaNbO₃-containing glass–ceramics. Nanometer-sized crystals in related transparent glass–ceramics that previously failed to exhibit ferroelectricity were shown to exhibit pyroelectricity, thereby confirming their status as active crystals but revealing a size dependence on the glass–ceramic’s measured functionality [5.115]. More recent work regarding transparency requirements in systems containing ferroelectric crystals has focused on crystal number densities, sizes and refractive index contrasts [5.161, 162].

A wide variety of compositional families that contain nonlinear crystals have been explored, including transparent titanate- and tantalate-bearing glass–ceramics [5.161, 163]. Telluride systems offer the advantage of a relatively small refractive index contrast between the high-index active crystals and the high-index residual glass matrix [5.122, 164–166]. Transparent bismuth–borate glass–ceramics have been developed to yield materials that exhibit second-harmonic

generation [5.155, 167, 168] as have potassium–niobate [5.154] and potassium–titanium phosphate compositions [5.169]. Results on glass–ceramics containing superconducting Bi₃Sr₂Ca₂Cu₃O_{12–x} crystals have been reported [5.170] as have glass–ceramics for X-ray scintillators [5.118, 171].

Glass–ceramics have been developed to serve as frequency upconverters, luminescent materials, laser hosts, Q-switches, and optical amplifiers (Table 5.18) [5.117, 172–177]. The typical design goal is to partition all elements that participate actively in the nonlinear process into the crystal phase, as in the Beall example noted earlier [5.119], thereby increasing the efficiency of the active process (e.g., narrowing a laser linewidth). Although early reports of Nd-doped Li₂O–Al₂O₃–SiO₂ glass–ceramics reported lasing [5.178, 179], subsequent work showed that Nd was not incorporated into the crystal [5.180], thereby limiting the advantage of the glass–ceramic route. Much recent work has focused on high-transparency rare-earth-doped fluoride- and oxyfluoride-based compositions for optical amplifiers [5.181–191]. Q-switching has been demonstrated in Co-doped MgO–Al₂O₃–SiO₂ glass–ceramics [5.177, 192, 193]. Long-lasting (> 12 h) phosphorescence has been reported for Mn²⁺-doped ZnO–B₂O₃–SiO₂ glass–ceramics [5.194]. Cr-doped mullite-containing glass–ceramics have been developed for solar energy concentrators and laser hosts [5.195]. These have met the criteria for good absorption in the visible with luminescence in the near-IR suitable for conversion to electrical energy by Si photovoltaic strips, although the conversion efficiency is too low at present to make a viable alternative to existing solutions [5.112].

5.5.3 Applications

This section will highlight optical glass–ceramic applications that have actually made it to the commercialization stage, a stringent feat for any materials development program. The list includes glass–ceramics

Table 5.18 Examples of active functionalities attained in glass–ceramics

Material	Active feature	Characteristic feature
K ₂ O–Nb ₂ O ₅ –4SiO ₂ ^a	Second-harmonic generation	Up to 600 times that of quartz
Li ₂ B ₄ O ₇ –Bi ₂ WO ₆ ^b	Relative permittivity	Up to 10 ⁶ at 673 K (100 Hz)
(Na, Sr)NbO ₃ ^c	Kerr constant (electrooptic)	$\approx 2 \times 10^{-17} \text{ m}^2/\text{V}^2$
BaO–GeO ₂ –TiO ₂ ^d	Piezoelectric	$d_{33} \approx 6 \text{ pC/N}$
Li ₂ O–SiO ₂ –B ₂ O ₃ ^d	Pyroelectric	$p_3 \approx -8 \text{ } \mu\text{C/m}^2/\text{K}$
BaO–B ₂ O ₃ ^e	Second-harmonic generation	0.12 times that of urea

(Sources: ^a [5.154], ^b [5.155], ^c [5.156], ^d [5.157], ^e [5.158])

used for both reflective (e.g., mirror) and transmissive (e.g., coating substrates) applications.

Reflective

One of the cornerstone applications for glass–ceramics is for mirror substrates. Typically this calls for ultra-low-expansion materials due to the exacting dimensional tolerances that must be attained to ensure satisfactory optical resolution in semi-controlled ambient environments such as an astronomical observatory. The use of negative-expansion high-quartz solid-solution crystals embedded in a positive expansion glass matrix to induce a nearly zero CTE has a long history [5.112, 196]. One of the most widely used low-expansion glass–ceramics for mirrors is Zerodur, developed by Schott Glas nearly 40 years ago [5.153]. In addition to control of the dependence of CTE on temperature [5.197], studies have reported precision polishing of Zerodur substrates using both conventional and ion-beam finishing [5.198, 199] although much knowledge on this topic remains proprietary to individual finishing houses. A common and nonproprietary technique to remove microcracks formed during grinding is the use of an HF–HCl acid etch to physically and chemically remove material [5.124]. Although HF is a well known and effective etchant of silicate glasses and glass–ceramics, the use of HCl is equally important to prohibit the precipitation of insoluble fluoride compounds on the surface of the material being etched. Recipes are also known for the removal of standard mirror coatings [5.124].

Although there are many examples of large Zerodur mirror blanks currently in service in astronomical observatories around the world, one is particularly noteworthy: the very-large telescope (VLT) array in Chile. There, four 8.2 m Zerodur-based telescopes use state-of-the-art adaptive optics and interferometric techniques to attain very high optical resolution with tremendous light-gathering capability [5.200, 201].

Transmissive

One of the earliest applications of glass–ceramics coincides with the emergence of glass–ceramics as a new material. *Stookey* [5.202] at Corning was able to fabricate a radio-frequency-transparent high-strength cordierite glass–ceramic called Pyroceram. This material is still used on the radar domes (radomes) of hypersonic missiles and has extremely low levels of rain erosion at these speeds; a chemical etch is used to increase the strength of the material [5.123].

Robax, a VIS-transparent, low-expansion glass–ceramic manufactured by Schott Glas, is widely used in oven and fireplace sight panels. The high transparency of the material (similar to Zerodur noted above), high thermal shock resistance and extremely high maximum service temperature (680 °C continuous, 750 °C for short times) has allowed Robax to replace borosilicate glass in many applications [5.124].

Telecommunication technologies have also driven glass–ceramic developments. Ohara has released IR-transparent high-Young’s-modulus (*E*) high-CTE glass–ceramics called WMS-13 and WMS-15 for dense-wavelength-division multiplexing (DWDM). These materials also feature excellent chemical durability and the ability to achieve low RMS roughness values through superpolishing. The somewhat atypical combination of high *E* and high CTE while maintaining good chemical durability – required for the use of these materials as substrates for thin-film interference filters in field-based applications – is considerably more difficult to achieve in glass.

Thin-film, transparent protective glass–ceramic coatings couple good transparency with extremely good wear resistance. *Rother* and *Mucha* [5.203] report using SiO₂ and Al₂O₃ powders and an electron beam evaporator to deposit a 3 μm coating on automobile wheel demonstrators. The mechanically stable coatings will undergo further testing before widespread commercialization.

5.6 Nonlinear Materials

5.6.1 Overview on Nonlinear Optical Materials

In Sect. 5.1.5, we have mentioned that, even among the second-order nonlinear (NL) materials [5.204–211], not many made their way into industrial application (because of disadvantages that NL materials are suffering

due to other intensity dependent effects besides the desired one).

The materials with the highest nonlinearities can be found among the (photonic) crystals, such as the ferroelectrics ADP and KDP used in electrooptical and acoustooptical devices where the first crystals were applied for frequency conversion. Due to their lack



of a center of symmetry, these crystals show rather high second-order susceptibilities, which are suitable for applications like frequency upconversion and others. Substitution of cations of an intrinsic nonlinear crystal (e.g.: Gd in Gd–Ca–Oxyborate) by optical active cations, like rare-earth cations (e.g. Nd^{3+}), results in self-frequency doubling crystallites, which combine the active ion with the nonlinear host. Photorefractive crystals can be used for many applications including phase conjugation, image amplification, information processing and optical computing. In the past, most of the experiments concerning optical nonlinearity have been performed on BaTiO_3 single crystals, which have excellent photorefractive properties, but are difficult to grow. One of the present leading crystals is K–Na–Sr–Ba–Niobate (KNSBN) because of its large electrooptic coefficients. Two further advantages of this material are that there is no room temperature phase transition and that there is a possibility of doping with Mn, which enables the formation of deep level centers for charged carrier recombination. On the other hand, single crystals, which are very expensive, are needed to avoid scattering effects [5.205, 212–216]. The high costs, and thus economic disadvantages, of crystalline materials arise due to the extreme difficulties to grow them. Nevertheless crystalline materials, especially the niobates will still be among the major materials for all-optical applications. But on the other hand, the limitations of Li-niobates will result in the research into materials like novel ceramic crystals, polymers and even fullerenes.

Other materials with high nonlinear refractive index and near resonance nonlinear effects are organic molecules, such as dyes, polymers (p-toluene-sulphonate), aromatic molecules, and molecules of the pyridine family, the stilbene family, and others which have extremely high $\chi^{(3)}$ values in the direction of the polymer chain, or chiral conjugated polymers and dendrimers (e.g. polydiacetylene). The disadvantages of these materials are their thermal instability, their softness, and the difficulty to get crystals of sufficient size. Furthermore, because of the mostly resonant nonlinearity, the response time for optical switching of these materials is rather low. A possible way to stabilize the organic molecules against degradation and to achieve larger sample sizes is to dope a glass host with these molecules. Glasses doped with organics will be further discussed Sect. 5.6.5 [5.205, 217, 218].

Currently it seems rather clear that the main improvements will come out of the field of organic crystals, as there exist the best possibilities to de-

velop new high potential systems, whereas no real developments are expected to come from the field of inorganic crystals. Composites may also offer new possibilities to overcome the problems of the current materials.

An overview over many common NL materials is given in Table 5.19. Table 5.20 then lists some of the most common NL materials and their proposed (and sometimes already commercially realized) applications. Table 5.21 presents nonlinear susceptibilities for a wide range of exemplary inorganic and organic crystals.

According to Hirao [5.204, 219, 220] and others [5.221, 222], for the development of high efficient optical devices the following points have to be taken into account:

- Increase of the upconversion-efficiency and the transition probability of the radiating transition of RE-ion dopands by selection of glass hosts with low multiphonon emission rates.
- Processing speed (“turn-off” times in a time range < ps)
- Phase matching
- Operating power and fraction of power, which can be switched: low switching energy and rapid switching times
- Nondispersive
- Heating stability
- Improvement of the crystallization stability
- Enhancement of IR-transparency and chemical resistance
- Avoiding release of toxic and aggressive substances
- Development of low cost fabrication methods
- Control and measurement of traces of contamination during the fabrication process

In contrast to the materials named above, materials which possess inversion symmetry (isotropic materials), like glass, have no second-order contribution of the optical susceptibility. Nonlinear glasses can be found among the oxide- halide- and chalcogenide glasses (intrinsic nonlinearity) and the semiconductor (SC), metal particle and organic dye-doped glasses (extrinsic nonlinearity). These materials have been intensively investigated in bulk structure as well as in waveguide and fiber structure and have shown many interesting properties: SC-micro-crystal doped films emit electroluminescence, dye-doped polymers exhibit SHG and Faraday effect glasses can be used for magneto-optical switches. Furthermore, methods for the fabrication of long period optical fiber gratings have been developed [5.204, 219, 220].

Table 5.19 The most important and other less important nonlinear optical materials [5.210]

Specialty ceramics: Li-, K-Niobate, K-Titanylphosphate (KTP, also very successful: used for doubling to Nd:YAG laser frequency to 532 nm), Li-Tantalate, Sr-Ba-Niobate, B-Ba-Borate, Li-Triborate (Other ceramic NLO materials: Ce-doped Sr-Ba-Niobate, Bi-silicon dioxide, Ba-Titanate)

Rare-earth elements doped in silica: Er and Pr in optical fiber amplifier (OFA)

Nanophase materials made from micrometer-sized powders: especially important, when used in composites

Polymers (including third-order NL-optical polymers, NL-dye functionalized thermoplastics)

Silicon-plastics: polysilicates and polysilynes

Liquid crystal polymers

Compound SC: transparent in the IR region

Ceramics

Semiconductors: excitation of carriers changes the optical properties, but difficult to find materials with low activation energies and large NLO properties, also difficult production.

Superlattices: excitation of carriers in the narrow bandgap part leads to migration of the carriers into the wide bandgap part, which creates a large charge separation. Possibility of tailoring the optical properties. But also difficult to produce.

Organic materials: disadvantages: thermal (operation temperatures about 80 °C) and chemical instabilities, but easy manipulation. Leading candidates are the Methacrylates (properties similar to NiBO_3 , low cost production, but poor strength and easy evaporation), Poly- (di-) acetylenes (good NLO properties, but unstable, hard to dissolve), Nitroanilins, ...

Table 5.20 Applications and proposed nonlinear materials [5.222]

Application	Material
Communications	
Modulation and switching	LiNbO_3 , KTP, II–V-SC, MQW's
Directional couplers	LiNbO_3
Lenses	LiNbO_3 with As_2S_3 -glass
EO Grating modulator	Ti: LiNbO_3
Sensors	
Imaging/processing	KTP, BaTiO_3 , KNBO_3
Single mode fiber optic sensors	LiNbO_3
Optical imiting	LCPs, (in-) organic materials
Instrumentation	
Fabry–Perot interferometers	LiNbO_3
Mach–Zahnder interferometers	LiNbO_3
Autocorrelators	KDP
Two photon spectroscopy	KTP, KDP
Holographic interferometry	LiNbO_3 , KTP, BaTiO_3 , ...
Computing	
Holographic gratings	BaTiO_3 , KTP, BBO, ...
Storage media	LiNbO_3
Laser frequency doubling	SrBaNbO_3 , Fe: LiNbO_3 , Bi: SiO_2 , III–V- and II–VI-SC



Table 5.21 Nonlinear susceptibilities of exemplary inorganic and organic crystals [5.211]

Material	Crystal symmetry neg.: $n_o > n_e$ pos: $n_e > n_o$	Point group	Transparency range (0.5 μm)		Breakdown threshold (10 ⁹ W/cm ²)		NL coefficient d_{36} (10 ⁻¹³ m/V)		Second harmonic generation: 1064 $\rightarrow$ 532 nm	
			from	to	$\lambda = 1064 \text{ nm}$	τ (ns)	1.06 μm	10.6 μm	Active length (mm)	Conversion eff. (%)
KDP: KH_2PO_4	neg. uniaxial	42m	0.1765	1.7	23	0.2	4.35/6.3		14	82
DKDP: KD_2PO_4	neg. uniaxial	42m	0.2	2.0	6	0.25	4.02		40	70
ADP: $\text{NH}_4\text{H}_2\text{PO}_4$	neg. uniaxial	42m	0.184	1.5	6.4	15	5.28			
DADP: $\text{ND}_4\text{D}_2\text{PO}_4$	neg. uniaxial	42m	0.22	1.7			5.2			
RDP: RbH_2PO_4	neg. uniaxial	42m	0.22	1.5	>0.3	12	4.02			
CDA: CsH_2AsO_4	neg. uniaxial	42m	0.26	1.43	0.5	10	4.02		17.5	57
KDA: KH_2AsO_4	neg. uniaxial	42m	0.216	1.7			5.2			
LiIO_3	neg. uniaxial	6	0.3	6	3–8	0.12	−55.3(d_{15})/ −72(d_{33})		18	44
LiNbO_3	neg. uniaxial	3m	0.33	5.5	0.05	10	−59.5(d_{31})/ 27.6(d_{22})		20	40
$\text{K}_3\text{Li}_2\text{Nb}_5\text{O}_{15}$	neg. uniaxial	4mm	0.4	5			62(d_{15})/ 127(d_{33})			
BBO: $\beta\text{-BaB}_2\text{O}_4$	neg. uniaxial	3m	0.198	2.6	10	0.1	22.2(d_{22})/ 1.6(d_{31})		6.8	70
Proustite: Ag_3AsS_3	neg. uniaxial	3m	0.6	13	0.02	20		113(d_{15})/ 180(d_{22})		
Pyrargite: Ag_3SbS_3	neg. uniaxial	3m	0.7	14	0.02	17.5		84(d_{15})/ 92(d_{22})		
AgGaS_2	neg. uniaxial	42m	0.5	13	0.025	35		134		
AgGaSe_2	neg. uniaxial	42m	0.71	18	0.03	35		330		
ZnGeP_2	pos. uniaxial	42m	0.74	12	0.003	30		754		
CdGeAs_2	pos. uniaxial	42m	2.4	18				2350		
GaSe	neg. uniaxial	62m	0.65	18	0.035	10		544(d_{22})		
CdSe	pos. uniaxial	6mm	0.75	20	>0.05	10		180(d_{15})		
HgS	pos. uniaxial	32	0.63	13.5	0.04	17		502(d_{11})		
Se	pos. uniaxial	32	0.7	21				970(d_{11})		
Te	pos. uniaxial	32	3.8	32				6500(d_{11})		
Quartz: $\alpha\text{-SiO}_2$	pos. uniaxial	32	0.15	4.5			3.64(d_{11})			
Li-formate: $\text{LiCOOH} \cdot \text{H}_2\text{O}$	neg. biaxial	mm2	0.23	1.2		cw	1.0(d_{31})/ −1.16(d_{32})/ 1.68(d_{33})			
Ba-Formate: $\text{Ba}(\text{COOH})_2$	pos. biaxial	222	0.245	2.2			1.17(d_{36})/ 1.13(d_{14})/ 1.17(d_{25})			
Na-formate: NaCOOH	neg. biaxial	mm2	0.26	1.28			0.22(d_{31})/ −2.2(d_{32})/ 3.3(d_{33})			
LiB_3O_5	neg. biaxial	mm2	0.16	2.6	25	0.1	10.9(d_{31})/ 11.7(d_{32})/ 0.65(d_{33})			

Table 5.21 Nonlinear susceptibilities of exemplary inorganic and organic crystals [5.211], cont.

Material	Crystal symmetry neg.: $n_o > n_e$ pos: $n_e > n_o$	Point group	Transparency range (0.5 μm)		Breakdown threshold (10^9 W/cm^2)		NL coefficient d_{36} (10^{-13} m/V)		Second harmonic generation: 1064 $\rightarrow$ 532 nm	
			from	to	$\lambda = 1064 \text{ nm}$	τ (ns)	1.06 μm	10.6 μm	Active length (mm)	Conversion eff. (%)
KTP: KTiOPO_4	pos. biaxial	$mm2$	0.35	4.5	15	1	$65(d_{31})/$ $50(d_{32})/$ $137(d_{33})$		4	60
$\text{Ba}_2\text{NaNb}_5\text{O}_{15}$	neg. biaxial	$mm2$	0.37	5	0.003	10	$-132(d_{31})/$ $-182(d_{33})$		3	20
KNbO_3	neg. biaxial	$mm2$	0.4	4.5	0.012	10	$158(d_{31})/$ $-132(d_{32})/$ $-201(d_{33})$			
MgBaF_4	neg. biaxial	$mm2$	0.185	10			$0.248(d_{31})/$ $0.37(d_{32})/$ $0.1(d_{33})$			
Urea: $\text{CO}(\text{NH}_2)_2$	pos. uniaxial	$42m$	0.2	1.8	5	10	13			
3-Methyl-4-nitro-pyridine-1-oxide	pos. biaxial	222	0.4	3	2	0.02	10			
MAP: $\text{C}_{10}\text{H}_{11}\text{N}_3\text{O}_6$	pos. biaxial	2	0.5	2	3	10	$167(d_{21})/$ $184(d_{22})/$ $36.8(d_{23})/$ -5.44			
COANP: $\text{C}_{13}\text{H}_{19}\text{N}_3\text{O}_2$	pos. biaxial	$mm2$	0.47	1.5			$150(d_{31})/$ $260(d_{32})/$ $100(d_{33})$			

The third-order susceptibility of glasses is generally much smaller than the second-order contributions of the crystalline material mentioned above. Additionally, the third-order nonlinearity makes a variety of processes possible, such as **THG**, **FWM**, **DFWM** (optical Kerr effect), as well as intensity dependent refractive index changes. But glass also shows some advantages compared to other second and third-order nonlinear materials [5.204, 206, 207, 222]:

- roducible with high optical quality and transparency/homogeneity even in large sizes
- easy fabrication of fiber and film waveguides with high quality
- high transparency over large spectral ranges
- high chemical, thermal and mechanical stability and durability, high optical damage threshold
- adjustable compositions
- quicker production process
- potential lower cost

A correlation between nonlinear properties and material structure would allow us to predict the nonlinear prop-

erties of new materials. A kind of quality coefficient can be used as a measure for the nonlinear properties of the materials. There are many suggestions for appropriate figures of merit for nonlinear materials, most of them reflect besides $\chi^{(3)}$, the absorption α and the response time τ of the material

$$F = \chi^3 / \tau \alpha. \quad (5.148)$$

For instance, the optical glass SF59 has a relative high figure of merit (**FoM**), not because of high nonlinearities (compared to typical second-order materials), but because of extremely low absorption and insensitivity to thermal effects [5.206, 207].

Due to the usually low third-order nonlinearities of glassy materials, nonlinear optical devices for signal processing or logic circuits for optical computing have to be produced in waveguide geometry to increase the power density. Optical waveguides can be produced via sputtering, **CVD** (Chemical vapor deposition), sol-gel coating, ion implantation and ion exchange. Besides the planar or channel waveguides, optical fibers are ideal for the study of nonlinear interactions as well as for



the desired applications, because they combine strong beam confinement and low loss for propagation over long distances [5.206].

We now describe in more detail two representative research areas for nonlinear materials, all optical switching and observation of second-harmonic generation in the isotropic material glass. Following this is a discussion of the many glass systems, which have been intensely investigated for nonlinear effects over the last 20 years.

5.6.2 Application: All Optical Switching

All-optical switching [5.204–206, 219, 220, 223] is a technique based on the optical Kerr effect (linear electrooptical effect) and means the switching of a light beam by using a pump beam which changes the optical communication signal to be switched. The first ultra fast all-optical switch was demonstrated on a silica glass fiber. Silica is known to have only a small nonlinear susceptibility, but also a very small absorption coefficient at the operating wavelength. This allows all-optical switching between two waveguides embedded in the same fiber simply by controlling the pulse intensity. Generally, for optical switching at THz frequencies, there are several important material requirements besides $\chi^{(3)}$, interaction length, and light intensity that need to be present [5.205, 212]

- high damage threshold or thermal stability,
- low power switching threshold and minimal (two photon) absorption,
- wavelength tunability,
- sub-ps- nonlinear response times,
- compatibility with waveguide fabrication process.

The efficiency of a switching device is induced by confinement of the beam into the waveguide over a macroscopic distance. The “turn-on”-time of all nonlinearities is instantaneous, even for the thermal related effects. Thus, the most important parameter for optical switching is the relaxation time of the nonlinear index change. To avoid crosstalk, this relaxation time has to be significantly shorter than the time between two pulses. Two of the fastest switching speeds among the nonlinear materials, besides the organic para-toluene sulphonate, are found in the systems Au:SiO₂ and Cu:SiO₂. Generally, chalcogenide glasses, as well as metal and semiconductor doped glasses also are prospective materials for all-optical switching devices.

A figure of merit (FOM) for ultrafast photonic switching expresses the ratio of fast index change to the resultant thermal index change and is an indication for the number of switching operations that can be performed within the thermal lifetime [5.206, 207]:

$$F = n_2 c_p \rho / \tau \alpha (dn/dT) . \quad (5.149)$$

Due to this FOM, the ideal material for all-optical switching has high nonlinearities, fast response and minimal absorption. For instance, the lead silicate glass SF59 has a relative high FOM because of extremely low absorption and insensitivity to the thermal effects. SC-doped glasses show large and relatively fast NL-absorption changes, but the switching value of these SC-doped glasses is too low for practical application. Dye doped glasses have higher nonlinearity, but their response times are too slow for application. Another limitation of the resonant nonlinear materials for all-optical switching is two-photon-absorption (TPA).

Among the other nonresonant, nonlinear glasses, Bi₂O₃-glass with response times of ≈ 100 fs and a repetition rate of switching of about 1.6 THz is a very prospective candidate for all-optical switching. Other candidates are chalcogenide glass fibers (response time ≈ 200 fs) One disadvantage of these glasses may be a structural change occurring when irradiated with light near the bandgap wavelength. Additionally, photonic bandgap materials, like Au-doped, poled glasses are considered for optical switches [5.204, 219, 220].

Like the diffractive and the holographic optical elements, the switchable optical elements can be used in a great variety of advanced devices; from everyday consumer products (eyeglasses, cameras, CD-players, animated holograms and real time holographic display systems) to commercial instrumentation (tunable grating elements, fiber optic crossbar switch, tunable Bragg grating for DWDM) Many patents exist for applications from biochip reader to flexible flat-panel displays. Three existing kinds of optical switches are:

- **Directional coupler:** switching into one or the other channel; characteristics are controlled by the beam itself.
- **Mach-Zehnder interferometer:** switching is controlled by one or two separate control beams.
- **Nonlinear mode sorter:** a multi-mode beam is split into two channels. By varying the input power, the phase difference between two modes can be tuned. Thus, depending on the power of the signal, the

modes were separated and the output switched from one channel to another.

5.6.3 Second Harmonic Generation in Glass

Besides the Ge:SiO₂ fibers discussed in Sect. 5.1.5, other glasses which show photo-induced second-harmonic generation are Tm-doped aluminosilicate fibers (resonant enhanced SHG) [5.224] and semiconductor microcrystallites-doped glasses [5.225], whereas Ce-doped lead germanate glasses show only low intensities (Table 5.22) [5.226].

SHG in HMO-Glass

Ionic bonds are believed to play a major role for encoded photoinduced SHG. Thus, it is not surprising that the SHG intensity in binary lead germanate glasses is found to be zero in contrast to binary lead silicates. Addition of Ce to the glass resulted in observation of saturable SHG. Low-Pb glass has a very low photoinduced SHG efficiency saturation level. On the other hand, for higher Pb-concentration the damage threshold of the glass decreases significantly. Further, the induced grating can be erased by an IR reading beam after some seconds. Comparison of lead silicate, Ce-Pb-germanate and Ge:silicate shows that high SH efficiencies can be achieved with Pb-containing glass at much lower preparation powers.

SHG in Rare Earth-Doped Glasses [5.227, 228]

Due to the absorption peaks near the SH wavelength, only a limited number of REs are suited for SHG with 1064 nm as fundamental wavelength. Furthermore, two or three photon absorption processes have been observed e.g. for Tm-doped glass, which lead to luminescence e.g. at 470 nm. Temporal and thermal stability is also a crucial factor in these glasses. As the formation of the $\chi^{(2)}$ grating is based on photooxidation, thermal and temporal stability of the gratings can be rather low, as the electron traps are usually rather shallow.

SHG in CdSSe-Doped Glasses [5.226]

SHG with pumping at 1064 nm was found to be more intensive for OG550 than for GG495. But GG495 is expected to be less sensitive to pumping and readout index changes. For instance, the SHG-intensity for OG550 ($7.7 \times 10^{-2} \text{ s}^{-1}$ for 2 W) is found to rapidly decrease with time (optical erasure), whereas there is only a small decay for GG495 ($1.5 \times 10^{-2} \text{ s}^{-1}$ for 2 W). Smaller decay effects due to heating are also found. Damage threshold of the glass is about $500 \text{ W}/\mu\text{m}^2$.

5.6.4 Glass Systems Investigated for Nonlinear Effects

Oxide Glasses Without Heavy Cations

The glasses of this group usually show only small nonlinearities, which are directly correlated with the nonbridging oxygen content [5.205, 206, 212, 229]. Measurements of various groups showed that the nonlinearity of nonbridging oxygen is significantly higher than that of bridging oxygens.

Despite the low nonlinearities of the oxide glass system, there have been recent experiments with silicon nanocrystals embedded in SiO₂ matrix. These samples show a $\chi^{(3)}$ of about 10^{-9} esu, dependent on the nanocrystal size. A much more interesting system, which is a kind of link to the heavy metal oxide glasses, is silica doped with various amounts of germania. The increase of the linear refractive index with increasing germania content consequently increases the nonlinear part of the refractive index. For about 50 wt % of GeO₂, a nonlinear refractive index of 1.1×10^{-13} esu has been found. Hydrogen-treatment even increases this value. Germania-doped silica fibers show photo-induced refractive index changes of up to 10^{-3} through exposure to UV light [5.230].

Heavy Metal Oxide Glasses (HMO)

The HMO-cations [5.231–233], like Pb, Bi and Ga possess large polarizabilities and therefore strong optical nonlinearities. Elements heavier than Bi might have larger NL-properties, but have disadvantages, such as their high toxicity, radioactivity and the sometimes poor optical quality of the resulting glasses, which make them useless for any practical application. Ba and La seem not to be responsible for nonlinearities. Tl⁺ contributes strongly to the nonlinearity, for instance in the Tl-germanate, Tl-tellurite and Tl-antimonite glasses. It is the only metal which is a post-transition element and therefore forms bonds with large polarizability. In spite of this, Tl-containing glasses are not suitable for practical application, mainly due to the high toxicity.

Among the HMO-glasses, the Bi-oxide glasses and the Bi-Ga-oxide glasses are the most prospective for nonlinear devices. For example a Pb-Bi-Gallate glass was found to have the largest nonlinearity among the oxide glasses. In this glass system, Pb and Bi have an equivalent influence on the nonlinearity. Optical Kerr shutter experiments on Bi-glasses have shown that the response times of these glasses are in the range of 150 fs and THz optical switching can be performed. Thus, the conclusion was drawn, that these glasses are prospec-



tive materials for the realization of ultra fast response and high frequency switching (ultrafast switches, optical isolators and in advanced computing).

As a rule of thumb, one can say that the larger the linear refractive index, the higher the third-order susceptibility. Glasses with refractive indices higher than 2 have a $\chi^{(3)}$ larger than 10^{-13} esu. Typical ranges of $\chi^{(3)}$ for HMO-glasses are 10^{-12} to 10^{-11} esu, which are almost the same for all HMO-glasses. Due to the electronic nature of the nonlinearity, the response times of HMO-glasses lie in the sub-ps range. This makes the HMO-glasses good candidates for all-optical switching applications. For incident light with optical frequencies, possible present nuclear contributions would be mitigated, because of slower response times of nuclear motion.

Transition Metal-Doped Glasses

Oxide glasses show an increase in the NL-indices when they are doped with transition metal cations or index raising conditional network formers, like Ti, Ta and Nb, due to the high hyperpolarizability of these ions [5.206, 234–236]. Ions with an empty or unfilled d-shell (d^0 electron configuration), like Nb or Ti (i.e. Ti–O entities), contribute most strongly to the hyperpolarizabilities [5.231]. On the other side doping, for instance with rare earths, can lead to a strong increase of the absorption coefficient in some of the commercially interesting spectral regions (e.g. telecommunication windows) Thus, rare-earth- (RE) doped glasses were are prominent for their application as laser glasses (upconversion lasers) or optical amplifiers.

Compared to the heavy metal cations, the nonlinearity of this group seems to be caused by another mechanism. This is due to the fact that (e.g.) TiO_2 -dopands are found to be responsible for nonresonant NL-effects in glass. However, the combined effect of Ti and HM-cations, like Bi and Tl, show NL-effects not higher than $10^{-18} \text{ m}^2/\text{W}$. An explanation might be the fact, that glasses with high concentrations of Ta, Ti and Nb are found to have negative electrooptic coefficients, in contrast to Pb^{2+} . Therefore Ti^{4+} can be used to balance the positive electrooptic contribution of Pb^{2+} .

Studies with Nb-containing oxide glasses have shown that the d^0 -transition metal is located in a (Met–O)⁶-configuration, with the oxygen electrons strongly delocalized towards the transition metal. This delocalization results in a large hyperpolarizability and thus large nonlinearity, which is found to be directly proportional to the concentration of (Met–O)⁶-entities [5.235].

Non-Oxide Glasses

Among the halide glasses, the fluoride glasses have the smallest NL, and the bromide and iodide glasses have the largest NL [5.206, 231, 237]. However, bromide and iodide glasses show very low glass-transition temperatures and poor chemical durability. Thus, they are not attractive for nonlinear applications, although the significant laser oscillation of the fluoride glasses at 1340 nm cannot be covered by oxide glasses.

Among the chalcogenide glasses, S glasses as well as Se and Te glasses show much higher nonlinearities than oxide glasses. As_2S_3 glass has shown the highest third-order susceptibility of all intrinsic nonresonant nonlinear glasses. Further examples of chalcogenide glasses with very high nonlinear susceptibilities are: $\text{As}_{40}\text{S}_{57}\text{Se}_3$, $\text{As}_{40}\text{S}_{60}$ and $\text{As}_{24}\text{S}_{58}\text{Se}_{38}$. Closer inspection of the origin of the nonlinearity of this glass exhibits a ratio $\chi_{2121}^{(3)}/\chi_{1111}^{(3)}$ of nearly zero, which is much smaller than the expected value for nonresonant NL of 1/3, the expected value for purely electronic contribution. The result of further investigations showed, that a kind of resonant contribution due to two-photon absorption is responsible for the high nonlinearity of this glass. Due to their very short response times, chalcogenide glasses are prospective materials for use as optical memory, switching devices, and (Bragg-) gratings in planar waveguides ($\Delta n \approx 10^{-4}$). Experiments with these glasses have shown that high quality, low loss waveguides for optical applications can be easily formed.

Tellurite and antimonate are both known as network formers, which contribute to $\chi^{(3)}$ via their stereochemical lonstanding electron pair in 5s (e.g. structural unit TeO_4) Both tellurite and antimonate glasses show a very good IR-transmission as well as very good chemical durability. Tellurite glasses have a nonlinear refractive index of 30–70 times that of silica. Pure tellurite containing glasses are difficult to prepare, thus, most of the investigated glasses have tellurite concentrations of about 70% [5.231].

5.6.5 NL-Effects in Doped Glasses

With the exception of the transition metal-doped glasses, all the glasses discussed above have intrinsic nonresonant nonlinearities [5.206]. Due to the fast response time of the usually electronic nonlinearity and the low absorption coefficient, the figures of merit of these materials can be rather high, despite the low $\chi^{(3)}$. In contrast to these glasses, the glasses discussed in the next part of the chapter have extrinsic resonant nonlinear optical prop-

erties, which are usually higher than the nonlinear ones at least in the spectral regions of fundamental absorption (Table 5.23).

Semiconductor-Doped Glasses

Among the studied nonlinear glasses, semiconductor-doped glasses [5.205, 208, 212, 223, 238, 239], especially the Schott filter glasses based on $\text{Zn}_y\text{Cd}_{1-y}\text{S}_x\text{Se}_{1-x}$, have been studied extensively. Other semiconductor dopants, besides CdSSe, are CuCl, CuBr, Cu_xSe , Ag(Cl, Br, I), CdTe, III–V compounds, Bi_2S_3 , PbS, HgSe, In_2O_3 , In_2Se_3 , SnO_2 and AgI. Among all these glasses, the best materials for ultrafast switching, based on a FOM, are ZnTe, CdTe and GaAs. In some experiments, the optical switching is done with different pump and probe wavelengths. As a result, it has been found that one gets a larger $\chi^{(3)}$ for the small loss in the near resonance region.

Semiconductor-doped glasses exhibit large nonlinearities based on optical Kerr effect at photon energies near the bandgap. The NL-behavior of these materials is characterized by intensity dependent changes in absorption and related changes in the refractive index ($\Delta n \cong 10^{-4}$) with relaxation times in the range of ns or ps. A strong dispersion in the electronic nonlinearity is found near the two-photon absorption (TPA) edge. The nonlinear refractive index increases as the TPA-edge is approached and eventually becomes negative at higher energies. The nonlinear properties of CuCl-doped glasses are based on an enhancement of the nonlinearity in the resonance region of the CuCl Z_3 -exciton [5.207]. The advantages of semiconductor-doped glasses (SDG) compared to the bulk semiconductor (SC) are [5.238, 239]:

- fast (ns–ps) response time,
- high threshold,
- weak absorption losses (for working wavelengths still near high absorption wavelengths!)
- possible usage at room temperature (but binding energies of these glasses are usually in the range of a few meV, thus, devices based on these materials better have to be operated at low temperatures).

On the other hand, the nonlinear optical effects in SDGs are smaller than those in bulk SC: $\chi^{(3)}$ of the doped glasses is typically in the range of 3×10^{-9} to 10^{-4} esu. For example for CuCl-doped glasses, $\chi^{(3)}$ is in the range of 10^{-3} to 10^{-8} esu, depending on the crystallite size. Changing the crystallite size, doping with other semiconductors or metal colloids like Au or Ag, or changing the glass host composition may improve

the nonlinear properties. Decreasing the size of the nanocrystallites, and thus enhancing the confinement, enhances the nonlinearity.

Metal-doped Glasses

Glasses containing very small metal particles exhibit an efficient, fast optical Kerr-effect [5.204, 206, 212, 219, 220]. The enhancement of the intrinsic $\chi^{(3)}$ arises from the surface mediated plasmon resonance, analogous to the SER-effect. The observed Kerr susceptibility or electronic nonlinearity of these materials includes contributions from intraband (transition into the conduction band), interband (saturation of interband transitions between d levels and the conduction band) and hot electron transitions (creation of photoexcited hot electrons due to the surface plasmon absorption). The maximum of the nonlinear susceptibility is found to be near the surface plasmon resonance.

Metal-doped glasses are expected to be good candidates for optical bistability and optical switching. Silver and gold particles dispersed in glass show NL effects, such as optical phase conjugation.

Gold-doped glasses: For gold spheres, $\chi^{(3)}$ is positive and found to be independent of particle size, with response times in the ps-regime. The nonlinear response of gold nanoparticle-doped glasses has been found to be as fast as 240 fs [5.240].

Silver-doped glasses: Glasses doped with silver nanoparticles exhibit an ultrafast nonlinear response of 360 fs and a $\chi^{(3)}$ of about 7.6 to 15×10^{-8} esu [5.204, 208, 241].

A nonlinearity of thermal origin has been reported for Mn doped glass. Due to a theoretical calculation, Al is suggested to have a larger nonlinearity than Ag and Au. By preparing core-shell systems, an enhancement of the nonlinearity by a factor of 10^8 might be possible [5.206]. Metal-doped glasses can be produced via sol–gel-technique, ion exchange, ion implantation or electrolytic coloration. Various metal-doped silica glasses, such as copper-, silver-, gold-, platinum-, lead-, tin-, iron-, germanium-, and phosphorous-doped glasses have been investigated. Of these glasses, the tin-implanted glasses have the highest nonlinear susceptibility. The size and the shape of the metal clusters can be modified by thermal treatment or by ion implantation with another element [5.206, 241].

Organic (Dye-) doped Glasses

Organic dyes, polymers and solvents, like 4-(N,N-diethylamino)- β -nitrostyrene (DEANST), acridine, fluorescein and rhodamine 590, have intense singlet–singlet–



Table 5.22 Poling of glass for second-harmonic generation [5.225, 227, 228, 242]

Glass	Poling	Parameters		SHG coefficient d (pm/V)
Fused silica, containing Na, OH	Thermal	275 °C	5 kV	1 (in a 10 μm layer)
GeO ₂ –SiO ₂ , Herasil	Thermal + electric	280 °C	4.3 kV	1
Ge-silicate “half”-fiber	Thermal + electric	275 °C	–5 kV	0.3
Ge-silicate optical fibers	Electric, thermal electric and UV			0.002 to 6
GeO ₂ –SiO ₂	Optical + electric field	ArF (193 nm)	Electric field	3.4–5.2
GeO ₂ –SiO ₂ sputtered film	Optical + electric field	ArF 100 mJ/cm ² /pulse	Electric field	up to 12.5
15.7 GeO ₂ –84.2 SiO ₂ (mol%)	Optical + electric field	ArF: 10–100 mJ/cm ² /pulse	3 $\times 10^5$ V/cm	7 $\times 10^{-3}$ –2
Ge-silicate	fs-encoding	810 (960 mW)	405 nm (200 fs, 200 kHz)	
Er-, Tb- and Sm-doped Al-silicate fibers, Er-doped Ge:SiO ₂ fiber	Optical	1065 nm (30–50 GW/cm ²)	Seed: 532 nm	Conversion effic.: 10 ^{–6} to 10 ^{–3} , 8 $\times 10^{-3}$ in Ge-silicate
Ce- and Eu-doped Al-silicate fibers	Optical	Nd:YAG: 1064 nm	Seed: 532 nm	Conversion effic.: 2 $\times 10^{-6}$ to 1.5% for Ce-doped fiber for 200 W pump
Tm-doped aluminosilica fiber	Optical	Nd:YAG: 1064 nm	Seed: 532 nm	SH output e.g. 4.6 $\times 10^{-2}$ W for 240 W pump at 1064 nm and 4.6 $\times 10^{-2}$ W for 55 W at 844 nm
Ce-doped lead germanate glass	Optical	1065 nm Nd:YAG	Seed: 532 nm KTP	
CdSSe-doped: GG495–RG630	Optical	1065 nm (3 W)	532 nm (1 W) pulsed laser	10 ^{–4}
Chalcogenide glasses: GeAsS	Optical	Nd:YAG 1064 nm	Seed: 532 nm	$\alpha(532 \text{ nm}) = 1.99 \text{ cm}^{-1}$ conversion effic.: 10 ⁴ x that of TeO ₂ –Nb ₂ O ₅ -glass
Fused silica fibers containing impurities: OH, Na, Fe	Gamma ray	⁶⁰ Co source (10 ⁶ rad at 400 rad/s)	1064 nm (Nd:YAG, 50 ns, 10 Hz, 1 kW) and 532 nm seed (300 W), 1 h	2 $\times 10^{-4}$ (natural SiO ₂) 9 $\times 10^{-3}$ (Ge:SiO ₂ + Ce) 2 $\times 10^{-2}$ (Ge:SiO ₂ + Ce)

Table 5.23 Nonlinear susceptibilities of commercial glasses

Glass system	$\chi^{(3)}$ (10 ^{–14} esu)	Wavelength range
Heavy metal (Tl–Pb–Cd–Ga)	up to 56	High frequency
Pb-silicate	up to 16	
Ge–Ga–S	3000–27	600–1250 nm
Semiconductor doped (CdSSe)	up to 5 000 000	Near exciton absorption peak
Nb–Te-oxide	9.4	
Bi-oxide	930	
Chalcogenide	174	
Noble metal doped glass	up to 10 000 000	Near plasmon absorption
Dye doped glass	up to 1.4 $\times 10^{14}$	Near absorption

transitions, which contribute to a high second-order susceptibility [5.204, 206]. The doping of glasses with dyes, polymers, and other organic materials also results in high nonlinearities, several orders of magnitude larger than that of inorganic glasses. Thus, these glasses

are attractive candidates for various low power optical devices, such as optical power limiters, phase conjugated mirrors or optical logic gates. The large nonlinearities are caused by saturation of the near resonant intensity dependent ground state absorption. The

excited dyes may return to their ground state either radiatively or nonradiatively or be transferred to the lowest triplet state by intersystem crossing. Triplet-singlet-transitions are forbidden, thus the triplet state has a lifetime many orders of magnitude longer than the singlet states. For sufficient intense excitation, the electrons accumulate in the triplet state and the ground state absorption is saturated. This produces

a change in the refractive index via the Kramers–Kronig relation.

Dye-doped glasses are made by impregnation of porous silica or sol–gel-methods or require low melting glass hosts, such as lead–tin–fluorophosphate- (good optical quality and chemical stability) or boric acid-glass, because of the thermal degradation or decomposition of the dyes.

5.7 Plastic Optics

Injection moulded precision plastic optics in high volumes were first produced during the 1960's. After the development of sophisticated measuring and manufacturing methods in the late sixties also precise aspheric surfaces were as easy to make as spheric contours. Today plastic optics are a widely used low-cost option compared to glass with even more degrees of freedom for optical and component design. Polymer optical systems at present are used in sensor applications, visual systems, cameras (mobile phones, video-conferencing cameras), scanners, security systems and so on.

Of course the properties of glass materials are very different from that of plastic materials. But glass and plastic optics both offer unique advantages and help to solve various engineering problems.

Generally speaking, glass materials are harder and more durable than plastic materials. Glass materials are also more stable (temperature and humidity) than plastic. The variety of optical glass available from well-known suppliers comprises hundreds of different materials. Compared with this the choice for plastic materials is limited only to about 10 different materials (and even less optical parameter variations). The large selection of glass materials allows the designer to choose materials with desirable optical properties to gain better optical performance. However, plastic optics offers other design freedoms that are not achievable with glass optics.

The manufacturing technologies for glass and plastic optics are totally different. Glass lenses are made by a grinding and polishing process whereas precision plastic lenses are in the whole made by injection-moulding or compression moulding. Because of the materials characteristics and the manufacturing process plastic optics have some unique advantages.

- **High production numbers at low costs**

Injection moulding is ideal for high volume production with low unit costs. Moderate raw material costs

and multi cavity moulds (up to 32 cavities) allow large production volumes at a reasonable unit price. In spite of normally considerable tooling costs the cost break-even compared to glass design versions can be relatively low.

- **Lightweight and hardness**

For a given volume glass is much heavier than plastic (by a factor of 2.3 to 4.9). Plastic materials are relatively shatter and impact resistant. These features are important, e.g. for head-mounted systems.

- **Design potentials**

Injection moulding makes it economical to produce sophisticated optical shapes such as aspheres, diffractive optical elements or even freeform surface structures. From the design point of view, the more sophisticated surface shapes reduce costs or obtain better performance or even performance un-realizable by optics in glass (Fresnel structures, lens arrays, diffractive optical elements).

- **Optical systems and component assembly**

For typical optical systems, the optical components (mirrors, lenses, prisms etc.) must be fixed in a mounting. With plastic optics, it is possible to mould mounting elements, posts or alignment notches integrally with the optical component. This can reduce part and assembly costs considerably.

Technologies adapted to plastic materials like ultrasonic and laser welding, gluing and integrated snap-in structures allow fast and cost efficient automated and manually operated assembling solutions. Two component injection moulding is the choice for integrating optical elements and mounts in one injection cycle.

5.7.1 Moulding Materials

As already mentioned, the choice for plastic optical material is limited only to about 10 different types



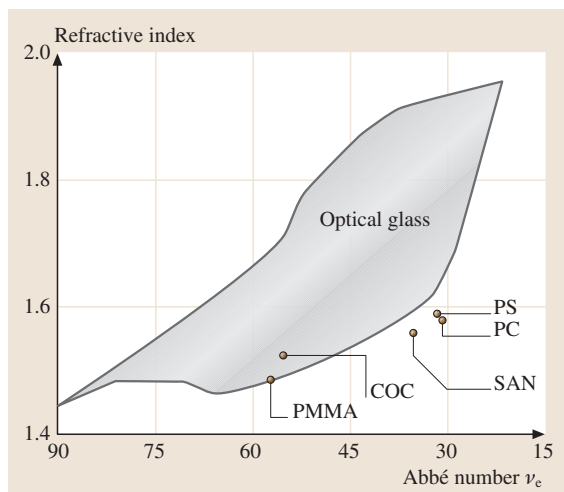


Fig. 5.54 Abbe diagram

of material. Optical properties (Abbe value, refractive index, Fig. 5.54) as well as mechanical, thermal and humidity boundary conditions are decisive for the material choice. Different to glass during plastic injection moulding the process affects not only the geometry but also the inner properties like refractive index and birefringence.

Optical Properties

Fundamental optical properties are defined by optical transmission, refractive index and dispersion. For plastic optics birefringence is an important parameter too. Although the total number of plastic materials has increased in recent years, the range of refractive index and dispersion characteristic is limited to almost two major groups – crown-like materials as acrylic (PMMA), polyolefin (COC, COP) and Flint-like materials as polystyrene, polycarbonate and SAN. This strongly limited variety of plastic optical materials restricts the optical design that can be made in plastic considerably.

Physical Properties

Important physical properties are weight, impact and abrasion resistance and thermal properties like temperature resistance and thermal expansion (Table 5.24).

Already during system design both mechanical and thermal properties have to be taken into consideration. The thermal expansion of optical plastics is approximately ten times higher than that of glass materials. In an optical system this effect has to be compensated by optical design and mounting.

Typically most of optical plastics can withstand temperatures up to 90 °C. The maximum service temperature of polycarbonate and polyolefine materials reaches 120 °C. The specific gravity of plastic optical materials ranges from 1 to 1.3. Polycarbonate has the highest impact resistance of all optical plastics and is used for windshields and crash helmets. Acrylic has the best abrasion resistance.

Most Common Materials

Some of the most common optical polymers are:

PMMA (Polymethyl methacrylate). Acrylic is the most commonly used optical plastic. It is moderately priced, easy to mould, scratch resistant and not very water absorptive. Its transparency is greater than that of most optical glasses. Additives to acrylic (as well as to several other plastics) considerably improve its ultraviolet transmittance and stability. So acrylic is used in almost 80% of all plastic optical applications.

(Poly-)Styrene. Polystyrene is a low-cost material with excellent moulding properties. Styrene has a higher index and a lower numerical dispersion value than other plastics. It is often used as the flint element in color-corrected plastic optical systems.

Compared with acrylic, styrene has lower transmission in the UV portion of the spectrum: It does not have the UV radiation resistance nor the scratch resistance of acrylic.

Because its surface is less durable, styrene is more typically used in nonexposed areas of a lens system.

NAS Copolymer. NAS is a copolymer of certain fractions of polystyrene and acrylic (typically 70/30). This allows to adjust refractive index is a low-cost material with excellent moulding properties.

PC (Polycarbonate). Polycarbonate is very similar to styrene in terms of such optical properties as transmission, refractive index and dispersion. Polycarbonate, however, has a much broader operating temperature band of up to 120 °C. For this reason, it is used as the flint material for systems that have to withstand severe thermal conditions.

Another advantage is the high impact resistance of polycarbonate. Safety glasses and systems requiring durability often consist of polycarbonate.

Cyclic Olefin Polymer and Copolymer (COP/COC). Cyclic olefin (co-)polymer provides a high temperature alterna-

Table 5.24 Plastic material properties

Material	Characteristics	Acrylic (PMMA)	Poly-styrene (PS)	Poly-carbon-ate (PC)	Styrol–Acryl-nitril (SAN)	Cyclo-olefine (ZEONEX)	Polyether-sulfone (PES)	Acrylnitril–Butadien–Styrolcopoly-mere (ABS)	Optical glass (BK7)
Optical	Spectral passing band (nm)	390–1600	400–1600	360–1600	395–1600	300–1600			
	Refractive index at 587 nm and 20 °C	1.4918	1.5905	1.5855	1.5674	1.5261	1.6600	1.538	1.517
	Abbé value $(n_D - 1)/(n_F - n_C)$	57.2	30.7	30	34.8	56	19.4		64.4
	Transmittance (%) thickness 3.2 mm	92	88	90	88	92	80	85	
	Haze (%) thickness 3.2 mm	1.3	1.5	1.7	1.5	1.5			
Physical	Specific gravity (g/cm ³)	1.18	1.06	1.25	1.07	1.02	1.37	1.05	2.53
	Max. service temperature (°C)	80	90	120	95	125	200	90	400
	Linear expansion coefficient (1/K)	6.8×10^{-5}	8.0×10^{-5}	6.6×10^{-5}	7.0×10^{-5}	7.0×10^{-5}	5.5×10^{-5}	8.5×10^{-5}	1.1×10^{-5}
	Abrasion resistance (1–10)	10	4	2		6			
	Izod impact strength (kJ/m ²)	2.0	2.0			2.4	7.0	25.0	
Environ-mental	dn/dT (10 ^{−6})	−105							
	Sensitivity to humidity	high	low	low	medium	low	high	medium	
	Water absorbtion (weight %)	0.60	0.10	0.15	0.30	0.01	0.70	0.45	0
	23 °C, ISO 61								
Manufac-turability	Processability	Excellent	Good	Poor	Excellent	Good			
	Birefringence	Low	High	High		Low			
Chemical	Resitance to alcohol	Limited	Good	Limited			Good		
Costs	Approx. material costs (EUR/kg)	3.3	2.5	4.4	4.4	27.1	21.0	3.5	25

tive to acrylic. Its refractive index and transmittance are similar but the heat distortion temperature is about 30 °C higher than for acrylic. Its water absorbtion capability is significant smaller.

5.7.2 Manufacturing Methods

The manufacturing processes for glass and plastic optics are totally different [5.243–247]. Glass lenses are made by grinding and polishing process. By contrast typical manufacturing methods for precise technical plastic optical parts are diamond turning, injection and compression moulding. Other methods like casting have their application in special fields like ophthalmics (eye-glasses). In any case, processing transparent plastic

materials for optical purposes should take place in a clean environment.

Diamond Point Turning

Diamond point turning is an ultraprecision machining process carried out on special high performance, modular set up, multi-axis, linear motor driven, ultraprecision diamond CNC-machining systems.

In combination with vibration isolation systems, digital signal processor based machine control and integrated measuring systems an one nanometer programming resolution is possible.

This method can be expanded to full 3-D-milling systems. With such an arrangement the generation of free-form surface profiles is possible. Because of



the long production time and the high machine costs, this technology is used for prototyping in plastic and nonferrous metals, mould inserts and series production plastic parts – which can not to be manufactured by injection moulding (because of size or precision demands).

Injection Moulding

Injection moulding produces several parts per shot (moulding cycle) in single or multicavity tools. This production method is used for most of the plastic optic parts. Other fabrication techniques are used only when moulding is inappropriate.

A plastic injection-moulding machine consists of a fixed platen, a moving platen, a clamping unit and an injection unit. Moulding of optical parts requires special machine configurations and auxiliary equipment.

Compression Moulding

Compression moulding is used for making of Fresnel lenses or other microoptical structures. The material is pressed between heated platens with accurately defined temperature cycling during pressing. Mould inserts are formed by electroplated copies, replicated from master structures. These masters can be, for instance, diamond turned structures or diffraction gratings made by holographic methods. The compression moulding process allows to realize small structures with high aspect ratios and tight angular and positioning tolerances.

5.7.3 Manufacturing Process

Optical and System Design

On the one hand optic design [5.243–249] for plastic optics in principle uses the same mathematical algorithms than optic design for glass. On the other hand, designing plastic optics requires a profound understanding of the material properties and the manufacturing processes [5.245, 248, 249]. Knowledge of production technologies, material characteristics and assembling methods together with design expertise are needed to fully exhaust what precision plastic optics can be.

Simply substituting the indices of refraction and re-optimising the design will not succeed. Expert design assistance is essential at this stage.

Designing plastic optics with modern design tools allows great design freedom. The advantage of combining integral mounting structures with the optical surfaces to create mounting flanges, alignment and snap features provides the ability to automated assembly. Aspheric, cylindric, or toroidal surfaces are as easy to realize

as spherical ones. Microstructures such as diffractive optical elements can be integrated too.

Prototyping

After designing a plastic optical system, lens prototypes can be made by diamond point turning in various plastic materials. The best surface finishes can be obtained with PMMA. Materials such as polycarbonate, pleximid or Zeonex do not yield very smooth surface finishes.

A major problem is the availability of semifinished plastic blocks in various materials. For PMMA a wide variety of bars or plates is available. For other materials or colored options semifinished items have to be made by injection moulding.

Because of the high manufacturing costs diamond point turning is only recommended for making a limited number of prototypes to verify functionality of the optical design and to perform first tests.

In this stage optical systems are normally assembled from single elements. Housing parts often are made from aluminium. The resulting surface quality and system performance can not be a validation of the manufacturability by injection moulding. To get reliable knowledge about this, making a moulded prototype from a single-cavity prototype mould is recommended.

The Injection Mould

A high quality injection mould is obviously essential for precise plastic optic parts. The parts never can be better than the tool - but good tooling, however, does not guarantee good parts. A strong understanding of the whole manufacturing process is the key to producing precision plastic components.

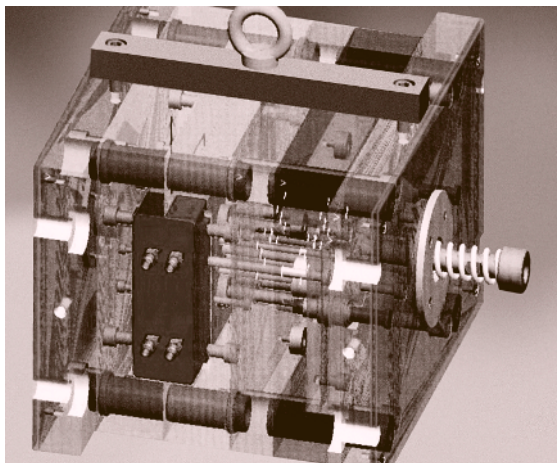


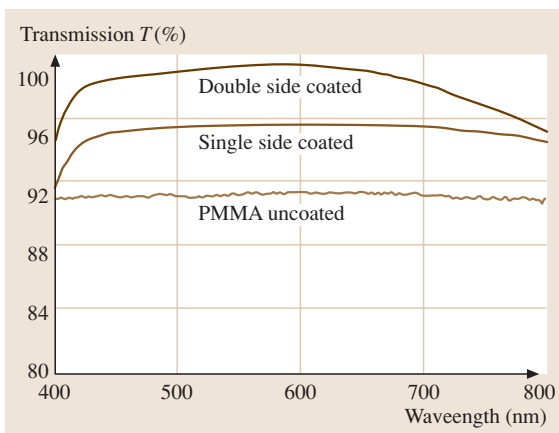
Fig. 5.55 Injection mould

Table 5.25 Tolerances of optical components

	Unit	Injection molding	High-end injection molding	Single-point diamond turning
Radius	% (+/-)	5	2	0.5
Refractive index	(+/-) (PMMA)	< 0.0005	< 0.0005	< 0.0005
EFL	% (+/-)	5	2	1
Thickness	mm (+/-)	0.13	0.05	0.02
Diameter	mm (+/-)	0.13	0.05	0.02
Surface figure	Fringes per inch of diameter	< 10 <i>f</i> (5 λ)	< 8 <i>f</i> (4 λ)	< 2 <i>f</i> (1 λ)
Surface irregularity	Fringes per inch of diameter	< 5 <i>f</i> (2.5 λ)	< 3 <i>f</i> (1.5 λ)	< 1 <i>f</i> (0.5 λ)
Surface rms error	nm	< 10	< 5	< 2
Surface S/D quality		80/50	60/40	40/20
Wedge (TIR)	mm	< 0.025	< 0.015	< 0.010
Radial displacement	mm	< 0.100	< 0.050	< 0.020
Diameter/thickness ratio		> 2 : 1	> 3 : 1	> 5 : 1
Repeatability	Focal length (part to part in one cavity)	< 2%	< 1%	< 0.5%
DOE depth	μm (+/-)	***	0.25	0.1
DOE min groove	mm	***	0.05	0.01

Any injection mould consists of three main parts. The upper half, which is affixed to the injection-side platen, the lower mould half, which is affixed to the ejector-side platen and the mould-ejection mechanism. Guide pins and taper locks ensure proper alignment of the mould halves.

Production volume and precision required from a tool influences the selection of mould materials and built-in maintenance features.

**Fig. 5.56** Abbe values

Single-cavity and multi-cavity moulds from 2 to 8, 16, or even 32 cavities are used for plastic optic parts.

Thermoplastic material shrink during cooling down in the mould. This geometric effect has to be compensated for in the injection mould. Exact shrink rates can be calculated and the tool can be modified. For manufacturing optical plastic parts by injection moulding the optical surfaces (plano or aspherical shapes, diffractive, conical, lenticular and cylindrical surfaces) are generated as separate inserts in the tool.

Aspheric inserts are manufactured by two steps. First a best-fit curve is generated on a stainless steel substrate. The substrate is then subjected to a nickel plating process (electroless nickel) that deposits a thin layer of nickel (up to 500 μm). In the second step single-point diamond turning produces the final aspheric or diffractive curve in the nickel. Because the hardness of a nickel-plated insert is less than that of a steel spherical insert, it will be more susceptible to scratch defects.

Pre-Production

A pre-production stage is recommended to check the manufacturing process. Typically this is done by using a single-cavity prototype mould. This mould can be used to find optimal moulding conditions. Optical and mechanical design can be verified with real moulded



components and design revision is possible to affordable conditions. Often the prototype moulds is used to start limited production since production tooling may take much more time.

Series Production

For series production of high volumes multi-cavity production moulds are required. Depending on quality, volume, throughput and cost, the production tool may have 2, 4, 8 up to 32 cavities. The production moulds function for at least several hundred of thousands of injection cycles.

5.7.4 Coating and Component Assembly

Coating

Because of their limited temperature and UV resistance plastic lenses must not be coated in an elevated temperature and radiation environment. During the deposition of thin films onto plastic, the coating chamber temperature is significantly lower than that for glass optics. This requires deposition techniques such as ion-assisted deposition to apply antireflective, conductive, mirror and beamsplitter coatings.

Today multi-layer dielectric coatings are routinely deposited on plastic components. Typical broad band anti-reflection coatings reduce reflection to about 0.5% per surface across the entire visible spectrum. Narrowband, multilayer antireflection coatings can achieve surface reflectances less than 0.2%.

Multi-layer dielectric coatings can be modified easily to scratch resistant designs for front lenses and windows.

Several front and back surface reflector coatings are available for plastic substrates. Standard coating metals include aluminium, silver and gold. Aluminium coatings provide surface reflectances greater than 88% across the visible spectrum, and gold coatings greater than 95% for the near infrared region.

Component Assembly

Many advantages of using plastic optics will come out in particular in manufacturing components. The integration of optical, mechanical and electronic elements allows to build low cost polymer optical components like small camera lenses or scanner heads f.i.

Plastic optical assemblies are usually done in a clean room environment by manual, semi or fully automated processes. The components have to be designed to ensure ease of assembly. Snap-on features, UV-cementing, heat-staking and ultrasonic and laser welding can also be employed with plastic materials.

Since most optical tolerances are additive, it is essential to establish quality check points in the manufacturing process to sort out non-conforming sub-assemblies. Automatic in-line optical performance monitoring such as MTF testing can be implemented. SPC techniques should be used here to ensure the process is not drifting out of the controllable range.

Component Tolerances

The high repeatability of the injection moulding process yields components with little dimensional variability. But the specific tolerances for an optical component strongly depend upon the part geometry and size, the plastic material, the mould design and the production process stability.

5.7.5 New Developments

In the last years several new developments in the field of plastic optics took place. The permanent improvement process in tooling technologies in combination with diamond turning, milling and grinding led to new design and tolerance levels. New materials like the cycloolefins led to better material performance and coating stability.

The fabrication process for diffractive elements (done primarily by single-point diamond turning) has improved considerably. Micro- and nanostructures like moth-eyed surfaces can be achieved by injection moulding and compression moulding and allow functionalised surfaces.

Even calibration structures for scanning probe microscopy applications can be replicated by injection moulding up to 25 nm structures.

Plastic optic technology permanently expands its traditional limits and fields of application. On the one hand plastic optic is still limited by material properties but on the other hand design and assembling freedoms allow new approaches.

5.8 Crystalline Optical Materials

In the following section we will discuss a few crystalline materials [5.250–252] that are used for optical applications. Among the oxides we want to focus on Sapphire, we mention the class of halides where CaF_2 is the most prominent example and we discuss a few semiconductors. The following subsections summarize material properties and fields of application. In each subsection, tables summarize the optical and mechanical materials properties.

5.8.1 Halides, CaF_2

Halides are crystals whose anions are halogens, the elements of group seven in the periodic table, namely fluorine, chlorine, bromide and iodine. Halides tend to have rather simple ordered structures with a high degree of symmetry. The most famous halide is rock salt, NaCl . Halides can generally be synthesized as single crystals from molten salt by Czochralski process.

Table 5.26 Basic properties of CaF_2 crystals

Formula	CaF_2					
Max. sizes (diameter (mm))	180					
Transmission range, μm	0.15–9.0					
Density (g/cm^3)	3.18					
Melting point ($^\circ\text{C}$)	1418					
Hardness (Mohs)	4					
Thermal expansion coefficient ($10^{-6}/\text{K}$)	16.2–19.4					
Thermal conductivity ($\text{W}/\text{m K}$)	9.17					
Specific heat capacity ($\text{J}/\text{kg K}$)	888					
Solubility in water ($\text{g}/100 \text{ cm}^3$)	0.0016					
Wavelength (μm)	0.2	0.5	1.0	5.0	10.0	12.0
Refractive index	1.4951	1.4365	1.4289	1.3990	1.3002	1.2299
Absorption coefficient (cm^{-1})	0.10 at 0.2 μm ; 0.01 at 0.4 μm ; 0.03 at 2.7 μm					

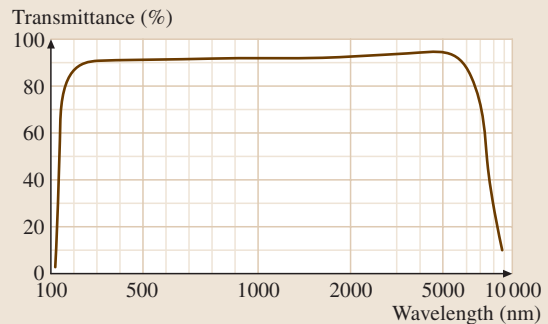
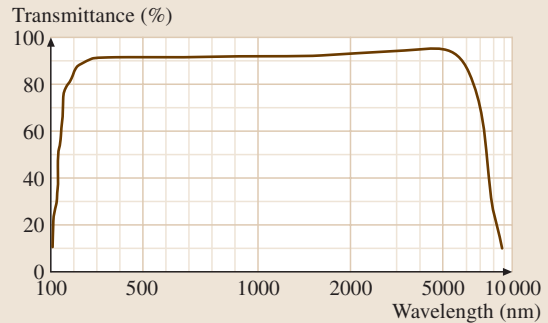


Table 5.27 Basic properties of MgF_2 crystals

Formula	MgF ₂						
Maximum sizes (diameter)	80 mm						
Transmission range, μm	0.13–7.0						
Density (g/cm ³)	3.18						
Melting point (°C)	1255						
Hardness (Mohs)	6						
Specific heat capacity (J/kg K)	920						
Solubility in water (g/100 cm ³)	0.0076						
Thermal expansion coefficient (10 ^{−6} /K)	parallel <i>c</i> -axis: 10.86–14.54, vertical <i>c</i> -axis: 6.23–9.25						
Wavelength (μm)	0.2	0.5	1.0	3.0	5.0	7.0	
Refractive index	<i>n</i> _o	1.4231	1.3797	1.3736	1.3618	1.3400	1.3044
	<i>n</i> _e	1.4367	1.3916	1.3852	1.3724	1.3487	1.3101
Absorption coefficient (cm ^{−1})	0.07 at 0.2 μm; 0.02 at 5.0 μm						

Transmittance (%)

Wavelength (nm)



In optical applications two candidates are of particular importance: MgF_2 and CaF_2 (Table 5.26 and Table 5.27). These materials transmit well into the UV region and are used for most UV-optics. MgF_2 is an excellent material for excimer lasers. MgF_2 is

slightly birefringent as can be seen from the data in Table 5.27.

CaF_2 is also an excellent material in laser optic field. It has a wide transmission range from 0.13–9.5 μm and an especially high transmission in the IR range. CaF_2 is widely used in laser, IR, and UV optics. It is slightly soluble in water and is susceptible to thermal shock.

Table 5.28 Main properties of Si

Chemical formula	Si
Molecular weight	28.09
Crystal class	Cubic
Lattice constant (Å)	5.43
Density at 293 K (g/cm ³)	2.329
Dielectric constant for 9.37×10^9 Hz	13
Melting point (K)	1690
Thermal conductivity (W/m K)	
at 125 K	598.6
at 313 K	163
at 400 K	105.1
Thermal expansion (1/K)	
at 75 K	-0.5×10^{-6}
at 293 K	2.6×10^{-6}
at 1400 K	4.6×10^{-6}
Specific heat (cal/g K)	
at 298 K	0.18
at 1800 K	0.253
Debye temperature (K)	640
Bandgap (eV)	1.1
Solubility in water	None
Knoop hardness (kg/mm ²)	1100
Mohs hardness	7
Young's modulus (GPa)	130.91
Shear modulus (GPa)	79.92
Bulk modulus (GPa)	101.97
Poisson's ratio	0.28
Wavelength (μm)	Refractive index
1.40	3.49
1.50	3.48
1.66	3.47
1.82	3.46
2.05	3.45
2.50	3.44
3.50–5.00	3.43
6.00–25.00	3.42

Table 5.29 Main properties of GaAs

Chemical formula	GaAs
Molecular weight	144.63
Density at 300 K (g/cm ³)	5.32
Absorption coefficient	
at 10.6 μm (1/cm)	< 0.02
Useful transmission range (μm)	1–11
Reflection losses	
for 2 surfaces at 12 μm (%)	45
Dielectric constant at 300 K	
high frequency	10.88
static	12.85
Melting point (K)	1511
Thermal conductivity	
at 300 K (W/m K)	55
Thermal expansion at 300 K (1/K)	5.7×10^{-6}
Specific heat at 273 K (cal/g K)	0.076
Debye temperature (K)	360
Knoop hardness (kg/mm ²)	731
Young's modulus (GPa)	82.68
Bulk modulus (GPa)	75.5
Poisson ratio	0.31
Bandgap (eV)	1.4
Solubility in water	None
Wavelength (μm)	Refractive index
8.0	3.34
10.0	3.13
11.0	3.04
13.0	2.97
13.7	2.89
14.5	2.82
15.0	2.73
17.0	2.59
19.0	2.41
21.9	2.12

CaF_2 and all halides are rather soft materials which are generally not very dense (3.18 g/cm^3). CaF_2 belongs to the cubic class and its cleavage plane is the 111-plane. In particular CaF_2 has been focused on as the material for future micro-lithography applications in the semiconductor industry because of its high transmission in the UV region. The band gap is larger than 11 eV. Further applications are laser windows and optical elements in the visible region. In addition CaF_2 has a low refractive index and its mechanical properties allow for rather easy polishing.

5.8.2 Semiconductors

Semiconductors have a wide range of applications in optical devices. In the following we will focus on Si (Table 5.28), Ge and CdTe as materials of interest in IR-transmitting elements. The crystal structure is cubic, and in the case of Si the diamond structure.

Silicon is commonly used as substrate material for infrared reflectors and windows in the $1.5\text{--}8 \mu\text{m}$ region. The strong absorption band at $9 \mu\text{m}$ makes it unsuitable for CO_2 laser transmission applications, but it is frequently used for laser mirrors because of its high thermal conductivity and low density. Silicon is also useful as a transmitter in the $20 \mu\text{m}$ range.

Germanium is used primarily in the $2\text{--}12 \mu\text{m}$ spectral region. Ge has a good thermal conductivity and excellent hardness and strength. It is the material of choice as a substrate for a variety of filters. One disadvantage of Ge over Si is its greater tendency to become opaque at higher temperatures due to its relatively low band-gap of 0.75 eV. This lower band-gap results in easier stimulation of electrons from the valence band to the conduction band. Cadmium telluride (CdTe) has an IR-transparent region from $1\text{--}25 \mu\text{m}$. It is a very soft material and a low thermal conductivity. It is an excellent substrate material for filters in the $12\text{--}25 \mu\text{m}$ region in which many other materials have decreased transmittance due to the presence of absorption bands.

Gallium arsenide (GaAs) used for lenses and beam splitters provides an alternative to ZnSe in medium and high power CW CO_2 laser systems (Table 5.29). It is most useful in applications where toughness and durability are important. Its hardness and strength make it a good choice where dust or abrasive particles tend to build up on or bombard the optical surfaces. When frequent cleaning by wiping is required, GaAs is excellent. The material is nonhygroscopic, safe to use in laboratory and field conditions and is chemically stable except when in contact with strong acids.

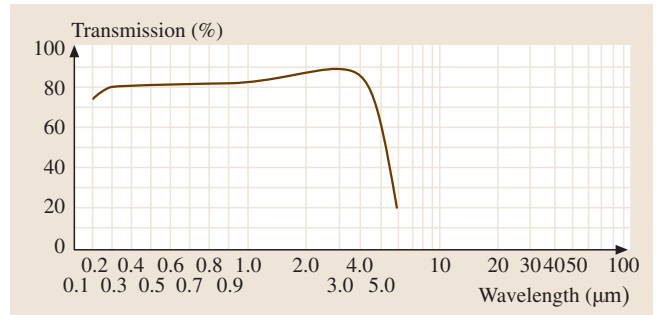


Fig. 5.57 Transmission of sapphire

Further areas of interest where semiconductor materials are used as optical materials are the LEDs and laser diodes. Depending on the spectral properties one is interested in the semiconductors of III–V type. Famous candidates are InAs or GaN as material for blue LEDs.

5.8.3 Sapphire

Sapphire is the leading material for highly sophisticated optical applications that require reliability, strength, and a wide range of light transmittance. Sapphire transmits light over a broad wavelength range spanning from 0.15 to $5 \mu\text{m}$ (Figs. 5.57, 5.58 and Table 5.30).

This unique ability to transmit over a broad range combined with its mechanical strength makes sapphire the material of choice for many applications. For imaging optics it is desirable for the refractive index of an optical material to have a low dependence on temperature. Because of its low dn/dt , a temperature gradient across a window will not cause image blur and foresight error. The United States National Bureau of Standards has extensively researched the index of refraction of sapphire. Recently a model was developed by Thomas et al. at the Advanced Physics Laboratory to predict dn/dt from 0.7 to $5 \mu\text{m}$.

Sapphire is a single crystal material from aluminium oxide, Al_2O_3 . It has a hexagonal/rhombohedral structure. Optically it is a negative uniaxial crystal in the visible region with a small birefringence.

Sapphire has a band-gap of approximately 10 eV, which is one of the largest for an oxide crystal, permitting optical transmission in the above mentioned range. Like other crystals sapphire can be synthesized by a Czochralski process. Sapphire has a high melting point and is a very hard material which allows it to be used in applications of extrem temperature conditions. Sapphire is used in laser windows – in particular for ap-



plications where high transmission in the IR-region is required. Its hardness and stiffness make it a suitable substrate material. In the semiconductor industry sapphire is used as a substrate to grow GaN on it for the production of blue LEDs.

5.8.4 Optic Anisotropy in Cubic Crystals

According to the Sect. 5.1.4 optical anisotropy would not be expected in a cubic crystal. Birefringence in cubic solids should only be caused by deviations from the cubic symmetry. There is, however, an effect that makes cubic solids optically anisotropic. In these crystals

Table 5.30 Main Properties of sapphire

Chemical formula	Al ₂ O ₃
Crystal class	Trigonal
Molecular weight	101.94
Density (20 °C) (g/cm ³)	3.98
Reflection loss for two surfaces at 4 μm (%)	12
Dielectric constant for 10 ² –10 ⁸ Hz at 298 K	
parallel	10.55
perpendicular	8.6
Dielectric strength (kV/mm)	17
Resistivity at 20 °C (Ω cm)	> 10 ¹⁶
Melting temperature (K)	2300
Thermal conductivity at 300 K (W/m K)	
parallel	35.1
perpendicular	33.0
Thermal expansion at 293 K (1/K)	
parallel 5.6 × 10 ^{−6}	
perpendicular	5.0 × 10 ^{−6}
Specific heat at 298 K (cal/g K)	0.18
Bandgap (eV)	9.9
Solubility in water	None
Mohs hardness	9
Knoop hardness (kg/mm ²)	1370
Young’s modulus (GPa)	335
Shear modulus (GPa)	148
Bulk modulus at 273 K (GPa)	240
Apparent elastic limit (MPa)	275
Poisson’s Ratio	0.25

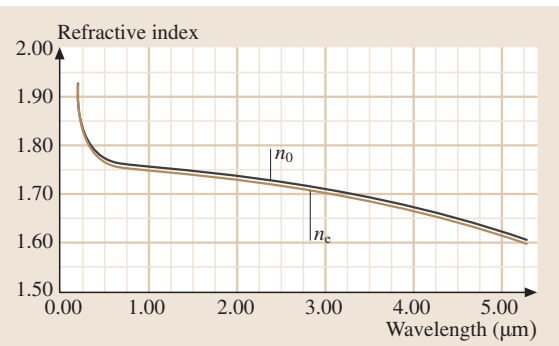


Fig. 5.58 Refractive index of the ordinary (n_o) and extraordinary (n_e) ray in sapphire

a symmetry breaking is provided by the electromagnetic radiation of the light wave. This is called induced spatial dispersion. Here the dependency of the dielectric function on the wavevector of the incident radiation k becomes important.

Let us try to estimate the order of magnitude of optical anisotropy which we can get due to the symmetry breaking of the light. Because of the extremely large velocity of the light, $c = 2.9979 \times 10^8$ m/s the corresponding wavevectors are very small

$$E_{\text{light}} = \hbar\omega = ck . \tag{5.150}$$

For visible light, e.g. at 600 nm, the photon energy is about 2.1 eV and this translates, using the dispersion relation (5.150), to $k = 2\pi/\lambda = 0.01 \text{ nm}^{-1}$. This has to be compared with the size and usual wavevectors occurring for electronic degrees of freedom in the solid. For a lattice constant of $a = 5.2 \text{ Å} = 0.52 \text{ nm}$ (CaF₂) the maximum wavevector at the edge of the Brillouin zone results to $k_{\text{BZ}} = 2\pi/a = 12.1 \text{ nm}^{-1}$. There are more than three orders of magnitude between the length scales on which the electromagnetic field is varying (λ) and the length at which the lattice is varying (a).

The spatial dispersion is the influence of the inhomogeneity of the electromagnetic field on the optical response of a system. This causes a non-locality. For perfectly homogeneous fields ($\lambda \rightarrow \infty, k \rightarrow 0$) the optical response of the cubic system is isotropic.

Let us assume now that our system allows for an expansion of the dielectric function around the zero wavevector [5.253]. Taylor expanding the (inverse) dielectric function with respect to the wavevector of the light gives an idea how the anisotropy enters into the optical response

$$\epsilon_{ij}^{-1}(\mathbf{k}, \omega) = \epsilon_{0ij}^{-1}(\omega) + h_{ijkl}k_kk_l + O(k^4) . \tag{5.151}$$

The term linear in k vanishes due to the inversion symmetry of the cubic crystal. In order to obtain an upper boundary for the expected anisotropy one can do the following estimate [5.254]. Responsible for the anisotropy is the deviation of the electromagnetic field from uniformity along the elementary cell of the crystal. This is the quantity

$$\frac{a}{\lambda} = \frac{k}{k_{BZ}}. \quad (5.152)$$

This means we get, using the k^2 dependency from (5.151) and the magnitude from (5.152) as an upper boundary for the magnitude of the expected anisotropy

$$\left| \epsilon_{ij}^{-1}(\mathbf{k}, \omega) - \epsilon_{0ij}^{-1}(\omega) \right| \ll \left(\frac{k}{k_F} \right)^2 = \left(\frac{a}{\lambda} \right)^2. \quad (5.153)$$

Since $k/k_{BZ} = 8 \times 10^{-4}$, the anisotropy in ϵ is smaller than 10^{-6} . This effect can be only observed with extremely accurate high resolution interference measurements.

In the vicinity of a strong absorption in the material the physics changes. This was first described in 1958 by the 2003 nobel laureate *Ginzburg* [5.253]. The reason is that not the wavelength in vacuum as in (5.152)

has to be compared with the length of the elementary cell but the wavelength in the medium $\lambda_m = \lambda/n$. Since the refractive index n becomes large in the vicinity of a strong absorption (Sect. 5.1.1) the spatial dispersion induced optical anisotropy becomes better observable near an absorption in the material.

In semiconductor materials, the enhancing resonance can be provided by the vicinity of a band edge and has been shown in [5.255] for silicon and [5.256] for GaAs. It has also been measured in alkali halide crystals [5.257]. In strongly ionic crystals deep excitonic electron-hole bound states are present. For the strongly ionic crystal CuCl the exciton resonance was identified as the enhancing absorption [5.258]. Recently the effect of spatial dispersion induced birefringence became of special importance for the crystal CaF₂ [5.259]. This material is used for UV lenses employed for microlithographic structuring of semiconductors. Here a direct connection between the exciton resonance and the spatial dispersion was made [5.260].

Finally, one has to note that – opposite to usual birefringence with according to Sect. 5.1.4 a maximum of two – the spatial dispersion induced birefringence in a cubic crystal has seven optical axes.

5.9 Special Optical Materials

5.9.1 Tunable Liquid Crystal Electronic Lens

The lens is a key element for use in optical systems. A conventional lens made by shaping a medium with a fixed refractive index brings light to focus. Glass, polymers and other transparent solid materials are commonly used for fabricating lenses. One major characteristic of the solid lens is that it has only one focus. To change the focal length, a zoom lens has been developed. A zoom lens system usually consists of groups of lenses. The separation distances between the lenses are mechanically adjusted. However, the adjusting processes are complicated, bulky and even inefficient. Therefore, developing a compact, lightweight, low-cost, and efficient tunable lens is highly desirable and urgently needed.

Due to high birefringence of nematic liquid crystals (LCs), which can be controlled by the applied voltage, an electrooptical effect based on the reorientation of LC molecules can be easily obtained. In addition to displays, adaptive optical components using LC materials have been demonstrated and application increased during the past decades. Among these photonic devices,

tunable-focus LC lens is particularly attractive for optoelectronics, machine vision, imaging, and eyeglasses applications.

The use of nematic LCs for making tunable lenses was proposed as early as 1977 by *Bricot* [5.261]. Since then, various approaches have been developed, such as surface relief profile [5.262–265], hole-, line- or spherical patterned electrode [5.266–269], modal addressing [5.270–272], Fresnel zone lens [5.273–275], as well as LC/polymer UV pattern irradiation technique [5.276–278]. Each method has its own merits and demerits. In this chapter, we classify those LC lenses into three groups according to the underlying lens structure. In each group, we will choose a typical cell structure as an example for discussion. Especially, we put emphasis on the newly developed LC lens using the simple UV exposure technique. We describe the fabrication method and performance of this kind of lens in details.

Types of LC Lens

Similar to a conventional LC cell for displays, the LC lens is composed of LC material sandwiched between



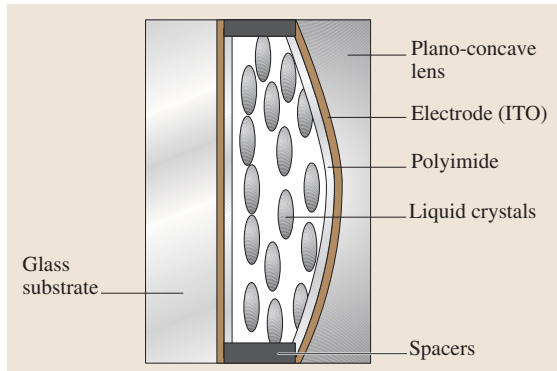


Fig. 5.59 A tunable-focus lens with an inhomogeneous LC layer

two glass substrates. To obtain a switchable gradient profile within LC layer, several methods have been proposed. According to the device structure, we can classify LC lens into three different categories.

Curved Cell Gap. In some LC lenses, LC layers can have concave, convex or Fresnel lens structure [5.261–265]. Figure 5.59 shows such a structure. One glass substrate has plano-concave surface and the other is flat. The inner surfaces of the indium-tin-oxide (ITO) glass substrates are coated with thin polyimide layers and then rubbed in order to get a homogeneous alignment throughout the LC bulk. LC lens with such a structure can be operated in two modes.

In the voltage-off state, the incident unpolarized light can be separated into two polarization components: parallel and perpendicular to the LC director. The beam with polarization parallel to the LC directors is called extraordinary ray and the perpendicular one is called ordinary ray. The extraordinary and ordinary rays will be focused onto two different but fixed focal planes. Their focal lengths are given by $f_e = R/(n_e - n_g)$, and $f_o = R/(n_o - n_g)$, respectively, where R is the radius of the concave lens substrate, n_g the refractive index of the glass substrate, and n_e and n_o are the extraordinary and ordinary refractive indices of the LC material.

When a certain voltage is applied to the electrodes, the LC directors are reoriented along the electric field direction. Under such a circumstance, the extraordinary ray sees an effective refractive index (n_{eff}). In this case, the focal length is given by

$$f_{\text{eff}} = R / (n_{\text{eff}} - n_g). \quad (5.154)$$

In (5.154), n_{eff} is equal to n_e at $V = 0$ and is reduced to n_o as $V \rightarrow \infty$. Therefore, the focal length can be

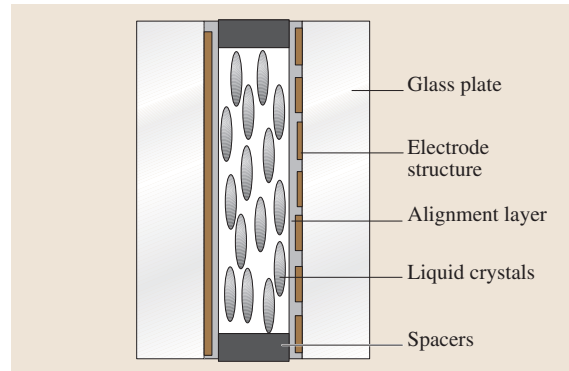


Fig. 5.60 A tunable-focus lens with striped electrodes and a homogeneous LC layer

tuned between f_e and f_o by the applied voltage. For the ordinary ray, it always sees the ordinary refractive index n_o regardless of the applied voltage. Thus, its focal length is not tunable by the applied voltage.

Because of the curved cell gap, liquid crystal in different positions will exhibit a different switching speed. The LC in the thicker area usually has a slower response time and presents light scattering due to the worse alignment. Another drawback of this kind of lens is the focal length can only be tuned between f_o and f_e . To widen the focal length, one has to use a high birefringence LC.

Planar Cell Gap. Several methods have been reported using a planar LC cell to realize the tunable-focus lens. To create a gradient refractive index profile within the LC layer, three different approaches have been explored. **Patterned Electrodes.** Several patterned electrodes, such as line-, hole-, spherical pattern electrode [5.266–269], and modal addressing have been considered [5.270–272]. A classical electrode structure is shown in Fig. 5.60. A homogeneous nematic LC layer is sandwiched between an ITO plate and a ground ITO plate. The ITO plate is a glass plate with a group of addressable discrete transparent ITO stripes. The ground plate is a glass plate with uniform ITO electrode covering the entire plate. Each electrode stripe on the glass substrate is individually addressed with voltages chosen so that the refractive index across the cell forms parabolic shape such that the incident light is brought to focus at a selected plane. From Fig. 5.60, one can see that only the light with its polarization axis parallel to the LC directors is affected. This kind of lens with linear electrode belongs to the cylindrical lens. The focal length has the following

expression:

$$f = \frac{x_0^2}{2d(n_e - n_{\text{eff}})}, \quad (5.155)$$

where x_0 is the maximum aperture distance and n_{eff} is the effective refractive index.

Since $n_o \leq n_{\text{eff}} \leq n_e$, f can be tuned in the following range

$$\frac{x_0^2}{2d(n_e - n_o)} \leq f \leq \infty. \quad (5.156)$$

To get a true spherical lens, four cylindrical lenses are required to combine together. By designing the electrode with hole-, hybrid- or spherical pattern, or using modal addressing, a spherical LC lens is realized. The focal length of the spherical lens is expressed by

$$f = \frac{\pi r^2}{(\delta_c - \delta_e)\lambda}, \quad (5.157)$$

where r is the radius of the lens, λ is the wavelength, and $\delta_c - \delta_e$ is the phase difference at the lens center and edge.

The lens with a planar cell gap has several advantages. For instance, the focal length can be tuned within a wide range, be fabricated with large size, and the switching speed is uniform across the whole lens. The drawback of this type of LC lens is that the cell fabrication process is complicated and the addressing technique is difficult.

Spherical Electrode within a Planar Substrate. To fabricate an adaptive lens with a wide range tunable focal length, a new lens structure as shown in Fig. 5.61 is considered. The upper planar substrate actually consists of a convex and a matched concave spherical surface. One of the spherical surfaces is coated with indium-tin-oxide (ITO) as electrode. The lower substrate has a transparent electrode coated on its inner surface. Homogeneous LC layers are aligned between the substrates. In the null voltage state, no focusing effect occurs provided

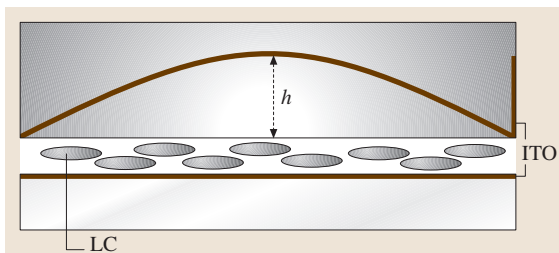


Fig. 5.61 A tunable-focus lens with a spherical electrode and a homogeneous LC layer

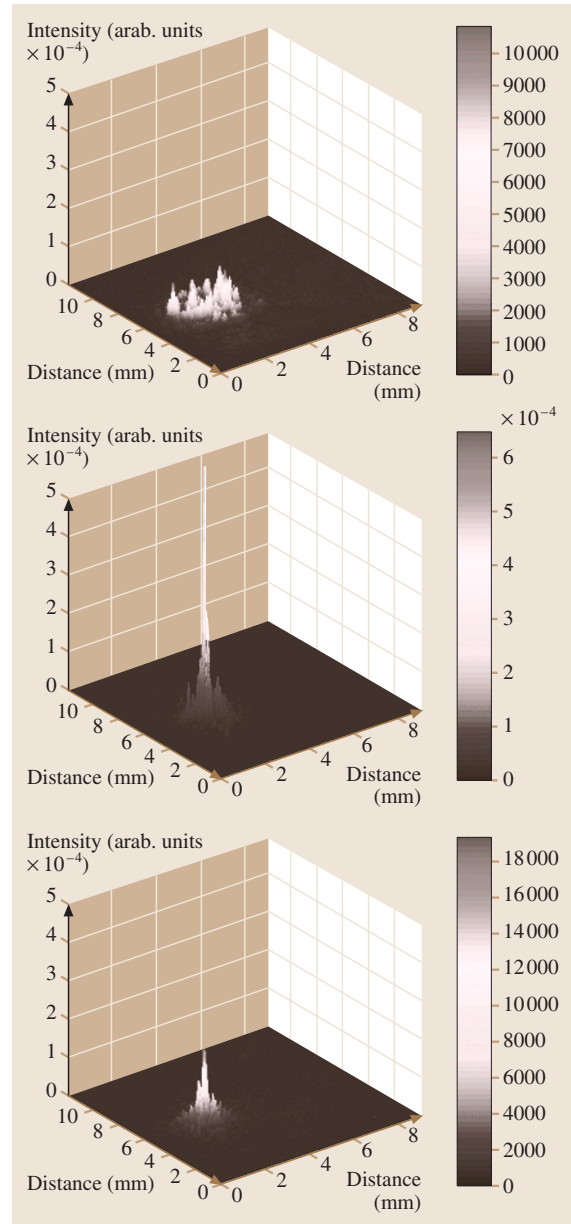


Fig. 5.62 Measured spot intensity profiles for the LC lens shown in Fig. 5.61. *Top:* $V = 0$; *middle:* $V = 18 \text{ V}_{\text{rms}}$; and *bottom:* $V = 35 \text{ V}_{\text{rms}}$. Lens diameter: 6 mm

that the refractive index of the upper convex and concave substrates is the same. When a voltage is applied across the LC layer, a centro-symmetrical gradient distribution of refractive index is generated. As a result, the LC layer causes light to focus. By controlling the ap-



plied voltage, the focal length of the lens is continuously tunable.

A sample LC lens was fabricated according to the design sketched in Fig. 5.61. The diameter of the lens is 6 mm and the LC cell gap is 40 μm which is filled with UCF-2 ($\Delta n = 0.4$) nematic LC mixture. The height from the top spherical electrode to the liquid crystal layer is $d = 1.03$ mm. The 3-D focal length intensity was measured using a CCD camera at different voltages. The distance between the LC lens and CCD camera is 78 cm. Results are shown in Fig. 5.62. At $V = 0$, the light intensity of a collimated He-Ne laser passing through the lens cell is very weak. As the voltage is increased to $V = 18 V_{\text{rms}}$, a tightly focused spot appears. As the voltage is increased further, say to $V = 35 V_{\text{rms}}$ the measured light intensity decreases. This means that the focus is somewhere between the LC cell and CCD camera. The focal length of this spherical lens can be tuned continuously from ∞ to 0.5 m by controlling the applied voltage. The response time is ≈ 200 ms for the 40 μm cell gap employed.

UV Exposure through Photomask. Tunable lenses using polymer dispersed liquid crystal (PDLC) and polymer network liquid crystal (PNLC) have been demonstrated recently [5.276, 277]. This kind of lenses is prepared using photo-polymerization induced phase separation

technique. A unique feature of these polymer dispersed LCs is that the larger LC domain exhibits a lower threshold voltage. If LC presents an inhomogeneous centrosymmetric LC domain dispersed in a polymer matrix, then the applied homogeneous electric field would induce a gradient refractive index profile, which reveals a lens-like character.

A PNLC lens using UV exposure fabrication technique has been proposed. Fig. 5.63a illustrates the fabrication method of a positive PNLC lens. The key element is the patterned photomask. To obtain a PNLC cell with an inhomogeneous centrosymmetric polymer network profile, a circular continuously variable optical density is used as the photomask, as shown in Fig. 5.63b. Its optical density increases outward radially from a minimum at the center to the maximum at the edges. When a uniform UV light passes through the photomask, the output intensity varies in parabolic profile across the LC sample. The area with a stronger UV light would accelerate the polymerization process and produce a higher polymer concentration. Conversely, the area with a weaker UV exposure would have a lower polymer network concentration. As a result, an inhomogeneous centro-symmetric polymer distribution is formed.

The PNLC cell with convex refractive index profile functions as a positive lens. When a uniform electric field is applied, the LC directors are reoriented to different degrees. The area with looser network packing (edges) has a lower threshold voltage than that of a denser zone (center). Thus, the convex refractive index is generated. As the applied voltage increases, the curvature of the gradient lens decreases. In the high voltage regime, nearly all the LC directors are aligned along the electric field direction. The gradient direction no longer exists and the lens effect vanishes.

To fabricate a positive PNLC lens, a mixture of 5 wt% UV curable monomer BAB-6 (containing 2% photoinitiator IRG-184) in a LC host (W-1331, $\Delta n = 0.229$) was injected to a homogeneous ITO coated LC cell. The monomer has a common rod-like structure with a reactive double bond at both sides. The inner surfaces of the LC cell were coated with thin polyimide layers. The cell gap of the homogeneous cell is 16.5 μm . The diameter of the photomask is 1.7 cm. The measured UV intensity after the photomask is 14 mW/cm², and the exposure time is 60 min.

The prepared sample is highly transparent in the voltage-off state. By observing the birefringence colors on a light table, one can check the gradient refractive index. Three photographs of the PNLC sample

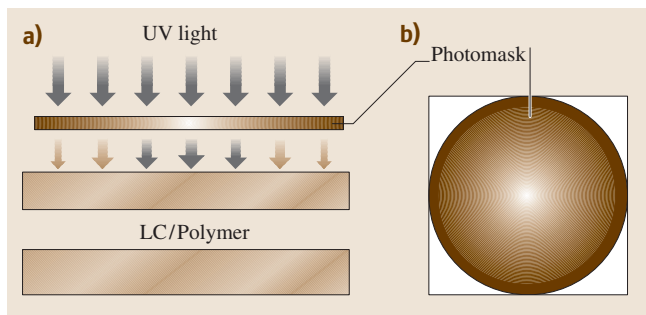


Fig. 5.63a,b Device fabrication of an inhomogeneous PNLC using a patterned photomask

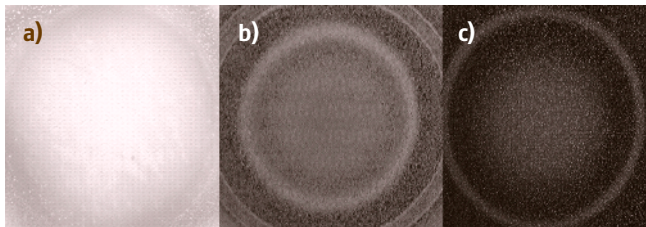


Fig. 5.64a-c Photographs of the PNLC sample at different operating voltages. (a) $V = 0$, (b) $V = 2 V_{\text{rms}}$, and (c) $V = 35 V_{\text{rms}}$

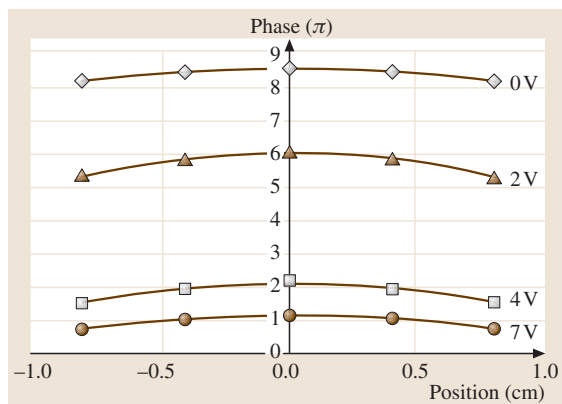


Fig. 5.65 Spatial phase profile of the positive lens at various voltages

cell at $V = 0, 2$ and $35 V_{\text{rms}}$ were taken, as shown in Fig. 5.64a–c, respectively. The rubbing direction of the cell was orientated at 45°C with respect to the fast axis of the linear polarizer. The analyzer is crossed to the polarizer. At $V = 0$, the color is relatively uniform. As the applied voltage exceeds the threshold, color starts to change from the borders and gradually extends to the center, like a swallowing spot. A circular color ring was shown in Fig. 5.64b at $V = 2 V_{\text{rms}}$. This implies that the central area has a higher threshold voltage than that of the edges. On the contrary, decreasing voltage changes the birefringence color from center to borders, like a vomiting spot. This cell acts as a positive lens.

The measured phase retardation of the cell is plotted in Fig. 5.65. At $2 V_{\text{rms}}$, $\Delta\delta = 0.8\pi$ which corresponds to 120 meter focal length. As the voltage increases, $\Delta\delta$ decreases so that the focal length increases. Due to the long focus, this lens is suitable for satellite imaging and astronomy applications. To shorten the focal length, several methods can be chosen, such as improving the photomask pattern, using high birefringence LC material, increasing the cell gap, or reducing the diameter of the curved spot.

To fabricate a negative lens, one can choose a photomask with reversed optical density. By using patterned UV exposure method, both positive and negative lenses can be fabricated. Since the lens is a single pixel device, the fabrication process is simple and the lens size depends upon the photomask design.

Like the aforementioned LC lens, PNLC lens is polarization dependent too. To overcome this polarization dependence, two cells with orthogonal alignment directions should be considered. The PDLC with sub-

wavelength droplet size is another technique for solving the polarization dependent problem for LC lens [5.276]. The drawbacks of the nano-PDLC lens are the high operating voltage and small refractive index change which results in a long focal length.

Besides the patterned photomask, exposure using a UV laser beam with Gaussian intensity distribution to the LC/monomer cell has been studied [5.278]. The cured spot presents a lens-like character under the action of a homogeneous electric field. The concern for this technology is that a high power laser is needed for scaling up the lens aperture.

Fresnel Zone Plate. In comparison with the above refractive lens, Fresnel zone plate lens belongs to the diffraction lens. It can be fabricated with large aperture size. Fresnel lenses are suitable for long distance optical communication, optical distance measurement and space navigation. In a binary-phase planar Fresnel lens, the phase difference between the adjacent zones can be tuned by the applied voltage. Maximum diffraction occurs when the phase difference is equal to an odd multiple of π while no diffraction occurs for even multiple of π . Through diffraction, the Fresnel zone plate works as a focusing element.

Due to higher-order Fourier components, Fresnel lens has multiple foci at $f, f/3, f/5$, etc. However, the majority of the incident light diffracts into the primary focus. The primary focal length can be expressed by $f = r_1^2/\lambda$, where r_1 is the innermost zone radius. The radius of the n th zone (r_n) is given by $r_n^2 = nr_1^2$, n is the zone number. Theoretically, the diffraction efficiency of the primary focus for the binary phase Fresnel lens is 41%.

A method of fabricating LC Fresnel lens was proposed by Patel and Rastani [5.273]. The idea to fabricate the lens is to make the neighboring zones with orthogonal LC directors on one substrate, and homeotropic alignment on the other. Another method of fabricating LC Fresnel lens is to etch the ITO electrode by photolithographic to form a zone plate [5.274]. However, the two fabrication methods are rather complicated.

Using UV pattern irradiation technique, one can simplify the device fabrication process [5.275]. A method of fabricating Fresnel lens using patterned photomask is similar to that as shown in Fig. 5.63, except that the photomask was produced with transparent odd zones and opaque even zones.

To fabricate a nanoscale PDLC Fresnel lens, 26 wt % LCE48 and 74 wt % UV curable prepolymer NOA81 were mixed and injected into a homogeneous empty

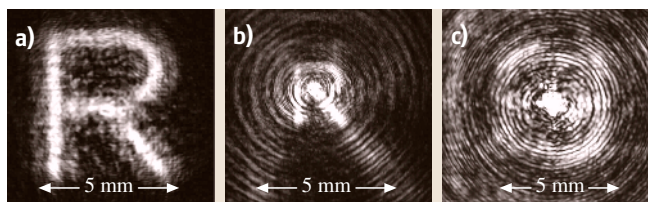


Fig. 5.66a–c Imaging properties of the PDLF Fresnel lens recorded by a CCD camera. **(a)** without sample, **(b)** with sample near the focal spot, and **(c)** with sample at focal point

cell. The cell gap is 15 μm . A photomask was produced by etching a chromium oxide layer using electron beam lithography. The radius of the innermost zone is 0.5 mm, and the zone plate consists of 80 zones in 1 cm aperture. The photomask is in proximity contact with the LC cell substrate during exposure to the UV light.

To evaluate the image quality of the PDLF Fresnel lens, a black cardboard with transparent alphabet **R** was placed before the sample. A collimated He–Ne laser has 1 cm diameter which fills the entire zone plate. A CCD camera was set at ≈ 25 cm behind the sample. Figure 5.66a and Fig. 5.66b show the photos without and with sample, respectively. When the sample is absent, no focusing effect occurs. Once the LC Fresnel lens is in position, a clear but smaller image is observed, although some circular noises exist due to diffraction. When the CCD camera was moved to the focal point, a tight spot appeared in the center, as shown in Fig. 5.66c. These results indicate that the sample indeed behaves like a lens.

The light efficiency of the PDLF Fresnel lens is about 32% at $V = 0$. As the applied voltage is increased, the diffraction efficiency is increased. At 180 V_{rms} (or 12 $V/\mu\text{m}$), the diffraction efficiency reaches nearly $\approx 39\%$. Further increasing the voltage, the diffraction efficiency will decrease. Because of the sub-wavelength PDLF droplets, the Fresnel lens is transparent in the voltage-off state, polarization independent, and its switching speed is in the sub-millisecond range. However, the operating voltage of this kind of lens is relatively high. To decrease the operating voltage, PNLC

can be considered. The PNLC contains a lower polymer concentration than PDLF. As a result, its operating voltage is much lower except the PNLC lens is polarization dependent.

MicroLens Arrays

Two-dimensional microlens array has a wide range of applications, including information processing, optoelectronics, integrated optics components and

optical communication systems. The hole-patterned electrode, hybrid-patterned electrode as well as surface relief profile can be used to demonstrate microlens arrays. To fabricate microlens arrays, we continue to use the UV exposure through a patterned photomask. According to diffraction theory, when UV light passes through a tiny hole, it will be diffracted and the diffracted intensity presents a parabolic-like profile. Therefore, when the UV curable monomer doped in LC host is exposed using the diffracted UV light, an inhomogeneous centro-symmetrical PNLC can be obtained. Similar to the results as shown in Fig. 5.64, the cured spot acts as a lens.

An example of fabricating LC lens array is demonstrated using PNLC technique. A mixture of 3 wt % monomer BAB6 (with a small amount of photoinitiator IRG-184) and 97 wt % LC E48 was injected to a homogeneous cell composing of ITO glass substrates. The inner surfaces of the cell were coated with polyimide layers and then buffed in the antiparallel direction. The LC cell gap and the substrate thickness are 15 μm and 1.1 mm, respectively. A chromium layer with circular aperture array deposited on a glass substrate is then placed on the top of the cell. The diameter of each aperture is 25 μm and the distance between the neighboring pixels is 85 μm from center to center. The cured UV intensity from the photomask side was 40 mW/cm^2 and the curing time was 45 min.

The cured sample was observed using a polarizing microscope. The rubbing direction of the cell was oriented at 45° with respect to the axis of a linear polarizer and the analyzer was crossed to the polarizer. Three photographs were taken at $V = 0, 5$ and 20 V_{rms} and results are shown in Fig. 5.67a–c, respectively.

At $V = 0$, the UV exposed array spots is very weak and cannot be observed clearly. The cured spot size is about 50 μm due to the diffraction effect. When the sample cell was rotated gradually, the array spots will disappear, this means that the PNLC cell is homogeneous and polarization dependent. When the applied electric field is increased slowly, a clear spot array appears as shown in Fig. 5.67b. In the mean time, a color

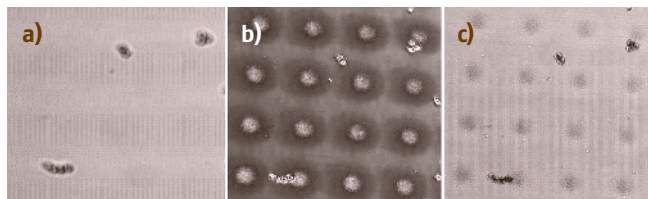


Fig. 5.67a–c Photographs of the PNLC sample cell at different operating voltages: **(a)** $V = 0$, **(b)** $V = 4 V_{\text{rms}}$ and **(c)** $V = 30 V_{\text{rms}}$

change occurring starts from the spot borders, and gradually shrinks to its center. This implies that the spot center has a higher threshold voltage than that of its borders and the cured spot acts as a positive lens. As the applied voltage is sufficiently higher than the threshold, most of the bulk LC directors are orientated perpendicular to the substrates. Under such a circumstance, the spot profile becomes very weak, as shown in Fig. 5.67c.

To characterize the light focusing properties of the sample cell, A collimated He-Ne laser beam ($\lambda = 633 \text{ nm}$) was normally incident on the sample with its polarization direction parallel to the cell rubbing direction. The laser beam was expanded to 5 mm diameter before illuminating the LC sample. A 10 times beam expander was positioned between the CCD and the LC sample in order to clearly resolve the output intensity. A CCD camera at 2 cm behind the sample cell was used to record the light focusing ability. Figure 5.68a–c show the photos for $V = 0, 5$ and $20 \text{ V}_{\text{rms}}$, respectively. One can see in the voltage-off state, no focusing effect occurs at the exposed spots. At 5 V_{rms} , the light passing through the exposed spots was focused. The spots become bright, as shown in Fig. 5.68b. However, the focusing point is relatively large. When the voltage was increased to $20 \text{ V}_{\text{rms}}$, a tightly focused spot appeared in the center of each cured spot, as shown in Fig. 5.68c. These results indicate that each spot indeed behaves like a lens.

The voltage-dependent focal length of the microlens arrays was measured and results were shown in Fig. 5.69. As the applied voltage increases, the focal length increases too. This is because the lens profile becomes shallower with increasing voltage.

The focal length of the lens is expressed as $f = \pi r^2 / \lambda \Delta\delta$, where r is the radius of the lens, λ the wavelength and $\Delta\delta$ is the phase difference between the lens center and border. Since $\Delta\delta$ can be adjusted by the applied voltage, the focal length is tunable. Theoretically, sufficiently high electric field would reorient the bulk LC directors perpendicular to the substrates so that $\Delta\delta \approx 0$ and $f \approx \infty$. However, in our microlens arrays the polymer networks exert the weakest stabilization on LC molecules at the border of the microlens, thus the radius of the microlens has the tendency to shrink at high voltage. In this condition, the focal length cannot be lengthened, it will gradually saturate as the voltage exceeds $10 \text{ V}_{\text{rms}}$, as shown in Fig. 5.69.

Comparing with other microlens technologies, the PNLC advantages are in uniform cell gap and using single electrode on both substrates. The tunable gradient profile is induced with the help of electric field. Like conventional PNLC cells, the PNLC lens has reasonably

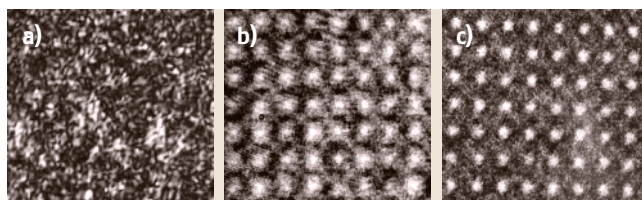


Fig. 5.68a–c Focusing properties of the PNLC lens arrays recorded by the CCD camera. **(a)** $V = 0$, **(b)** $V = 4 \text{ V}_{\text{rms}}$, and **(c)** $V = 6.7 \text{ V}_{\text{rms}}$

fast switching speed, wide range tunable focal length by optimizing the cell structure and UV curing conditions, and relatively low operating voltage. This kind of microlens is easily realized with a long focal length and, therefore, is suitable for optical fiber switches and incoherent correlator applications.

In summary, three basic mechanisms have been proposed to fabricate LC lens. They are: inhomogeneous electric field on inhomogeneous LC media, inhomogeneous electric field on homogeneous LC media, and homogeneous electric field in inhomogeneous LC media. Each technique has its own merits and demerits. In terms of fabrication process, the patterned UV exposure technique seems to be the simplest. Depending upon the specific photomask patterns, positive and negative spherical lens, elliptical lens, cylindrical lens, microlens array, prism, and prism grating can be fabricated. The nano-PDLC approach offers advantages in polarization independency and fast response time. However, its operating voltage exceeds $10 \text{ V}/\mu\text{m}$. On the other hand, polymer network has a lower polymer concentration. As a result, its operating voltage is reduced to $2 \text{ V}/\mu\text{m}$, except that the focusing effect is dependent on the light polarization.

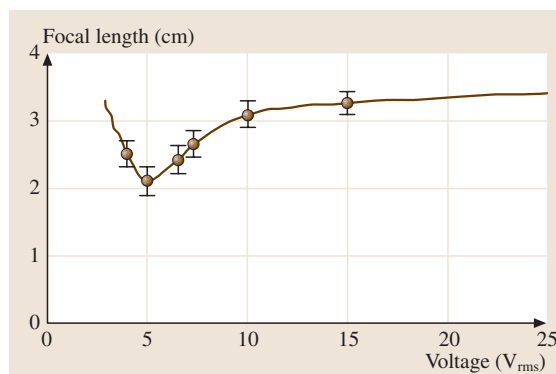


Fig. 5.69 Voltage dependent focal length of the PNLC microlens arrays. LC: E48, cell gap $d = 15 \mu\text{m}$, and $\lambda = 633 \text{ nm}$

5.9.2 OLEDs

For over 100 years, people have been obtaining light from light bulbs. Unfortunately, the majority of the energy involved is converted into heat and only a tiny fraction actually produces light. In an attempt to overcome this inefficiency, very competitive technologies such as sodium pressure and fluorescent lamps and inorganic light-emitting diodes (LED) were developed. LEDs are highly efficient and available in all colors. Recently, a promising technology to convert electricity into light was developed by depositing thin layers of appropriate materials between two electrodes and applying an electric field (Fig. 5.70) [5.279–281].

The emissive material between the electrodes is an organic semiconducting material, and hence the entire device is now called an organic light-emitting diode (OLED). OLEDs combine the advantages of LEDs with those of organic materials, which include mechanical flexibility, full color availability and large lighting areas. To couple light out from the device one electrode must be transparent. Indium-tin-oxide (ITO) on glass or plastic substrates with a layer thickness of 100–200 nm is the most common transparent and conductive electrode.

This brief review cannot replace a physical textbook and not all elementary processes in an operating OLED can be examined herer intensively. However, the physics of a single layer OLED can be simplified. ITO works well, with its large work-function, as an anode and injects positive charge carriers (holes) into the organic semiconductor. On the other side of the organic layer, a metal electrode simultaneously injects electrons. After migration in the emissive layer, a hole and an electron combine to form an exciton. This neutral excited state is

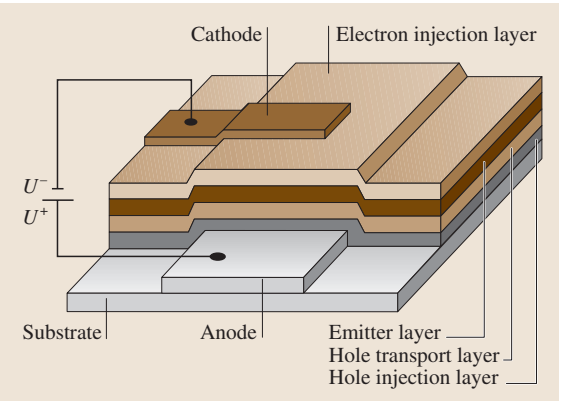


Fig. 5.70 Multi layer setup of an organic light-emitting diode

Table 5.31 Relation of energy gap of HOMO and LUMO, emission color, and wavelength of emitted light

Energy gap in emitters (eV)	Color	Wavelength (nm)
1.9–1.6	Red	650 780
2.3–1.9	Yellow	530–650
2.5–2.3	Green	490–530
3.1–2.5	Blue	420–490

a high energy state far from thermal equilibrium. Unfortunately, this surplus energy cannot be released solely as radiation (light), but is also dissipated as heat. The emission color (wavelength, Table 5.31) is a material property and depends on the difference between the energy levels of the first electronic excited state and the ground state, i. e. the energy difference between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) of the molecule.

An excited state can be generated by the absorption of light of suitable wavelength as well as by applying a voltage to an OLED device. Independent of the kind of excitation, the radiative decay of the excited state is physically identical and thus both emission spectra have the same shape. The processes are called photoluminescence (PL) and electroluminescence (EL), respectively.

Properties such as flatness and brightness, combined with the possibility of high resolution (pixelation) of the device, make OLEDs a very promising and competitive technology for liquid crystal (LC) displays. Extremely lightweight, full color and flexible displays are likely to be produced within a few years to penetrate the huge market dominated by LC, LEDs, and plasma technology. In recent years, enormous improvements upon OLED emitters allow the production of large bright areas, which will be interesting and useful for lighting applications. Developing a suitable coating technology is the key for making self-luminating large areas with low manufacturing costs. Although a few products equipped with OLED-technology are already available, several technical challenges still remain to be solved.

Emitter Materials

In the pioneering work by Tang and van Slyke, EL was first discovered in 1987 by using an aluminum quinolate (Alq₃) as an emitter material [5.282]. A second milestone in the field of organic EL occurred when polymers were brought into the action in 1990, after workers at Cambridge University evidenced the generation of light from poly-para-phenylenevinylene (PPV) [5.283].

The discovery of light emission from organic materials has led to tremendous research activity. As a consequence, the number of materials employed as the emitter material in **OLEDs** is vast, and a comprehensive list can not be published within the scope of this book. All organic materials used as emitters in **OLEDs** must combine several properties. First of all, they need to be of high purity and chemically and thermally stable. Secondly, they must exhibit a high quantum yield. Finally, the materials need to be processible and have the ability to form thin films.

The most common classification for emitter materials used in **OLEDs** is “Small Molecules” (SM) versus “Polymers” (Fig. 5.71 and Fig. 5.72). The main difference between the two is the processing of the materials for fabrication of devices. SMs (compounds with a low molecular weight) are evaporated in a high vacuum process to deposit a film – ranging in thickness from a few layers to a few nanometers – on a surface. Due to the large molecular weight of polymer chains, a vacuum process cannot be used. Advantage can be taken of good solubility in various solvents to obtain homogeneous films by spin-coating or printing techniques. These solution coating processes offer enormous advantages concerning thin films on large areas. It is difficult to compare both classes so only advantages and drawbacks with regard to processability and **OLED** performance are considered.

Purification of the materials is an important issue since it is well known that tiny amounts of impurities can dramatically impair device performance. Deposition of pure low molecular weight compounds in high vacuum serves as an extra purification step during the **OLED** process itself. In contrast, polymers must be purified as well as possible before the coating process. For efficient devices, the number of positive and negative charge carriers in the emitter layer must be equal. This goal can be achieved by pursuing the following two strategies described. Using an evaporation process for SM, several materials can easily be deposited on top of each other. This process allows the manufacture of multilayer devices, in which each layer has its own function, such as hole injection or transport, light emission, electron transport or blocking, etc. Efficiency is optimized by material choice and order, layer thickness, co-evaporation, etc.

Polymeric multilayer devices are more difficult to produce due to polymer solubility in multiple solvents. Most two layer devices are made utilizing water soluble polyethylenedioxythiophene/polystyrylsulfonat (**PEDT/PSS**), which is rendered insoluble

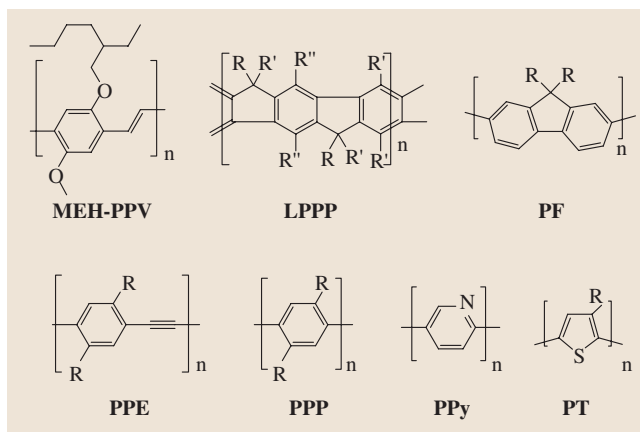


Fig. 5.71 Chemical structures of representative luminescent conjugated polymers

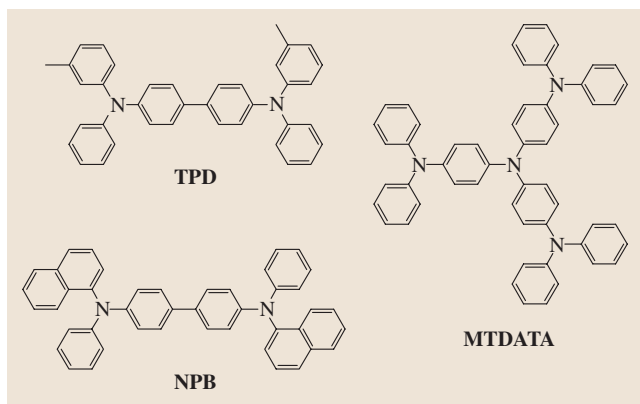


Fig. 5.72 Chemical structures of representative hole transport materials

(even in water) after baking the film. In this case the emissive polymer can easily be deposited on top of **PEDT/PSS** by a solution coating process. Due to limitations on the number of polymeric layers that can be stacked, the ideal emitter should have a balanced number of holes and electrons during operation. Furthermore using appropriate electrodes gives nearly balanced electron and hole injection leading to efficient **OLEDs**.

Fluorescence and Phosphorescence

An important point to keep in mind is the fact that singlet and triplet excitons exist. Whereas during photoexcitation generally singlet excitons are formed, the spin statistics predict the ratio of singlet to triplet excitons created in an operating **OLED** device should be 1 : 3.



After the formation of singlet excitons by the combination of oppositely charged carriers, it is impossible to distinguish singlet excitons created by charge carriers from single excitons formed by photoexcitations. Without going into further details, the radiative decay of a singlet state is physically permitted and fast (lifetime $\approx 10^{-9}$ s). The transition from a triplet excited state to the ground state is strictly forbidden and only occurs under certain circumstances like spin–orbit coupling in the presence of heavy atoms, with lifetimes ranging from 10^{-6} s up to hours. To improve the efficiency of devices it is desirable to include radiative decay of both kinds of excitons by adding phosphorescence emitters like organic-heavy metal complexes.

Color

Yellow, red, and green emitters are already commercially available in large quantities. Recently, material suppliers are focusing on synthesizing long-lifetime blue and white emitters. In fact, it is the lack of stability in both blue and white emitters that prevents full color products with long lifetimes even under moderate operating conditions. Due to the high energy band gap of blue emitting materials, they are particularly unstable and side-reactions often damage the molecular structure.

Problems for full color displays are caused by the different lifetimes of the emissive colors, leading to a color shift over a short period of time. For example, white emitting devices – which are always a combination of several emission colors – show a slow change of hue during device operation.

Conjugated Polymers

Film formation, quantum yield, device efficiency, lifetime, and emission color can be tuned by modifying the chemical structure of the emitting materials. Since the discovery of EL in polymers many derivatives of poly-para-phenylenevinylene (PPV) as well as other classes of luminescent polymers have been synthesised and used successfully in OLEDs. PPV itself is an insoluble polymer. To make devices with PPV, a soluble non-conjugated precursor form must be coated on the device. The precursor is subsequently thermally transformed to PPV. PPV emits yellow-green light with a peak wavelength of 510 nm. For the sake of purification and processing a soluble polymer is highly desirable. Introducing side groups onto the polymer chain cannot only make the polymer soluble, but can also modify the electrical and optical emission characteristics. Poly[2-methoxy-5(2-ethylhexyloxy)-1,4-phenylenevinylene] (MEH-PPV), a material

investigated and used in OLEDs in 1991 is one of the first examples of this approach. This molecule shows an emission in the orange-red region of the spectrum [5.284].

A disadvantage of PPV is the easy oxidation of the vinylene bridge between two phenylene-rings. This undesired side-reaction, which can occur during synthesis or device operation, breaks the conjugation and leads to a reduction in device efficiency. Again, two ways were identified to improve the oxidative stability: substitution at the vinylene carbon position or replacing the double bond by a triple bond ending with poly[2,5-bis(hexyloxy)-1,4-phenylene-(1-cyanovinylene)] (CN-PPV) [5.285] and poly(para-phenyleneethylene) (PPE), respectively [5.286].

Poly-para-phenylene (PPP) derivatives were synthesized to obtain stable blue emitters [5.287], however, conjugation in the polymer chain is reduced due to non-planar alignment of adjacent phenylene rings. Improved planarity is realised by bridging two phenyl rings with a methylene-group to form polyfluorene (PF). PF is a blue emitter which can be color tuned over the visible spectral range by adding different monomers during the polymerization of PF-precursors [5.288–290].

Researchers in Germany synthesised a PPP derivative where all adjacent rings are bridged, namely the ladder-PPP (LPPP) [5.291, 292]. The phenylene-rings are completely planarised and the stiff chains show high persistence lengths that are responsible for the relatively high mobility of charge carriers in this semiconducting polymer.

Another approach in polymer OLEDs is the application of polymer blends [5.293–295]. At least two polymers are mixed: a polymer for charge carrier transport and an emissive polymer. Recent research has been done on avoiding the problem of phase separation. For the polymer blend approach, either a copolymer possessing all properties of the blend itself is synthesized or the chains of the two polymers are crosslinked after processing as a thin film in the device.

The synthesis of polymer chains with defined chain ends has led to more stable derivatives in comparison to the polymer without defined chain ends. Such “end-capped” emitter polymers (for instance attached to a hole transport molecule at the chain end) show better device performance.

Along with conjugated polymers, other classes of organic materials are noteworthy. Heterocyclic polymers like polythiophene (PT) [5.296], polypyridine [5.297], and many others were synthesized and countless derivatives were investigated in OLEDs.

Small Molecules

Thermal stability is required in an evaporation process to deposit a molecule on a surface without decomposition. Furthermore, formation of stable and amorphous films of one or more functions, such as charge injection and transport, and emission must be combined within the material (Fig. 5.73) [5.298].

Promising compounds with reasonably efficient emission are organo-metallic complexes. In this case emission comes from either the ligands or the metal-center. The prototype for ligand emission is the aluminum triquinolate (Alq_3) [5.282] shown in Fig. 5.72. Many materials have been found to generate light of all colors. Rare-earth-complexes emit through electron transition in the metal atom [5.299]. These spectra are very narrow compared with polymer emission spectra, however short wavelength emission is hard to realise and quantum efficiency seems to be lower than in ligand-emission complexes, too.

In efficient SM devices multilayer systems are used. For a balanced number of oppositely charged carriers in the emissive layer and for decreasing the on-set voltage, injection and transport materials are necessary for holes and electrons in addition to device setup optimization.

Material improvements for SM-OLEDs can be shown exemplary at hole transport materials, where the most widely used compounds are arylamines. To ensure the final materials are high temperature stable glassy films (instead of crystalline layers) many derivatives have been synthesised, including N,N'-diphenyl-N,N'-bis-(3-methylphenyl)-(1,1')-biphenyl-4,4'-diamine (TPD) [5.300] and N,N'-diphenyl-N,N'-denaphthyl-(1,1')-biphenyl-4,4'-diamine (NPB). An other large arylamine, the so-called starburst amine (MTDATA), shows a glass transition temperature (T_g) around 160 °C [5.301].

Two other approaches to increasing the T_g are raising the molecular weight or modifying the molecular structure of the material. The former approach is limited by the need to increase the evaporation temperature for higher molecular weights. In recent years *spiro*-compounds have been synthesized [5.302] to obtain materials with a high T_g . An advantage of this molecular element is the inhibition of crystallization to produce materials in a stable glassy state.

For generating light in an OLED, small fluorescent molecules are also used for doping both electron and hole transport materials. Due to the lower excited state energy of the dopants (as compared to the host), excitons formed in the binder are immediately transferred to a dopant which acts like an

exciton trap. By optimizing the dopant/host ratio, emission only comes from the dopant. However, if the dopant concentrations are too high (> 2–3%), performance actually suffers due to concentration quenching of the luminescence. Examples of fluorescent dyes are perylene [5.303], 4-(dicyanomethylene)-2-methyl-6-(p-dimethylaminostyryl)-4H-pyran (DCM) [5.304, 305], and chinacridone [5.306].

Mechanism of Degradation and Device Failure

Depending on the desired properties of an OLED a long lifetime (ten thousands of hours) and/or a high brightness (1000 to 5000 Cd/m^2) have to be achieved (Fig. 5.74).

Two kinds of failure can be observed: a sudden death, where a short circuit prevents light emission, or a slow decrease, where the lighting area becomes inhomogeneous (black spots) and/or darker.

Immediate failure of the device occurs by connecting the ITO anode directly to the metal cathode during fabrication. In this case a large amount of current flows through this defect and the area becomes hot, damaging the metal and polymer alike. Interestingly, it was found that shorts can also arise during operation by a currently unknown mechanism. Non-emissive regions on the lighting area have to be avoided. These so-called black spots are very tiny in the beginning and grow larger with time. It was shown by different spectroscopic techniques that a large particle (in comparison to the film thickness) in the middle of such a spot can cause a pinhole in the cathode. These damages are supposed to be preferred channels for the penetration of reactive species like oxygen and water into the sensitive materials. This idea is also supported by the constant circular growth of these spots [5.307]. Clean room conditions during OLED preparation and manufacturing of a thin film of suitable polymer like PEDT/PSS to smooth the ITO-surface can prevent failure by shorts.

Lifetime is typically defined as the reduction of brightness to 50% of the initial brightness at constant current and depends strongly on the operation conditions. The brighter the device and the more charge carriers go through the layers the shorter the lifetime due to generating more light and heat in the device. Lifetime depends also on the emission color. Red, yellow and green emitters are stable for more than 20 000 h. Blue and white emitters are being improved intensively, but are still not appropriate for long life products yet.

A typical lifetime curve for a PPV-derivative such as that shown in Fig. 5.73 reveals two areas. During the first phase the brightness shows an “initial drop”. After this short period the reduction of brightness be-



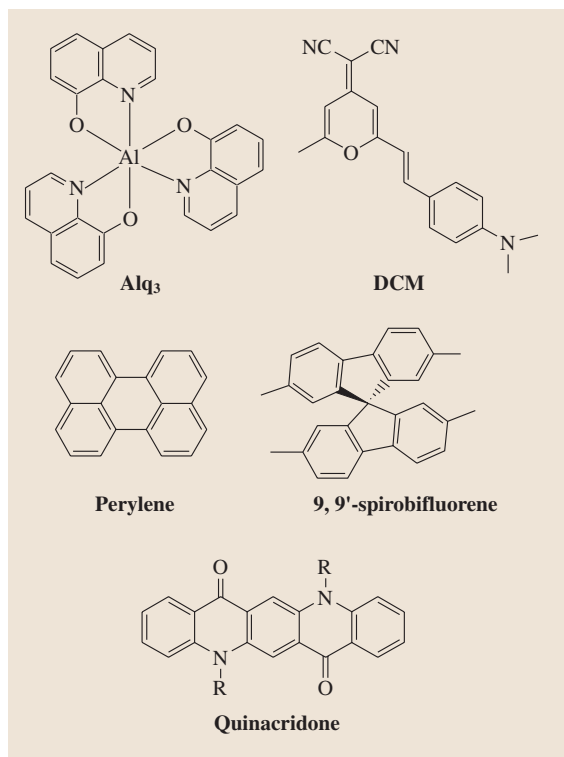


Fig. 5.73 Chemical structures of representative Small-Molecules luminescent dyes

comes slower. Although a stable level is never reached and the brightness decreases continuously, lifetimes of PPV derivatives can still reach several 10 thousand hours under moderate conditions (100 Cd/m² at room temperature). The initial drop is often called burn-in and is not fully understood, whereas the slow reduction of brightness is attributed to slow degradation mechanisms.

OLEDs are generally very sensitive to water and oxygen; therefore the whole device must be hermetically encapsulated to minimize material damages. Thin metal cathodes are supposed to react with traces of water to decrease the ability of electron injection, particularly under operating conditions. Oxygen reacts with emitter polymers due to its potential for photooxidation of the conjugated polymer. Cleavages of polymer chains have been observed, consequently breaking the conjugated system [5.308–310]. But air is not the only oxygen source – ITO is another [5.311]. Again a dramatic improvement in lifetime was found by introducing a PEDT/PSS layer on top of ITO. A further advantage of this polymer coating is the prevention of indium migration in the device. Ion migration is also known to be

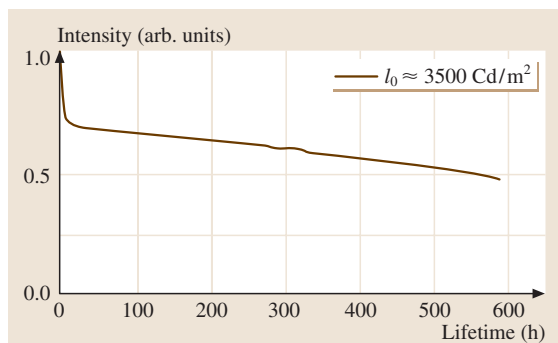


Fig. 5.74 Typical accelerated lifetime measurement of a PPV derivative at a high and constant current and high brightness (started at 3500 Cd/m² and cancelled at 1750 Cd/m² after several 100 h segments)

a general problem in operating devices [5.312]. Several scenarios are described concerning this issue. First, the ion remains in his charge state and counteracts the applied electric field. Second, ions damage the materials by oxidizing and reducing. Third, the ions can react at the electrode and generate impurities which prevent the charge carrier injection. These descriptions of ion migration fit the observation that devices driven in a pulsed mode (forward/reverse bias) show slower aging than direct current driven devices.

A further drawback of organic materials is their sensitivity to UV-light. Formation of highly reactive radicals, cleavage of chemical bonds and polymer chain decomposition have negative effects on device performance. UV protection of OLEDs is necessary particularly for outdoor applications.

Summary

LEDs can be made of organic materials. These OLEDs possess a layer setup and are made of either Small Molecules or polymers, or a mixture of both. Thin amorphous films of Small Molecules are deposited chiefly by vacuum processes, whereas polymer films are obtained by wet coating techniques. All emission colors are available, including white. However, lifetimes of blue and white emitters (compared to red, yellow and green) are still too short for long life products and full color applications, respectively. A further issue is the color shift during device operation, which depends strongly on the device operation conditions. An encapsulation increases the device lifetime tremendously – however, hermetic protection against environmental impact is not available yet. Emitter materials, particularly polymers, have been improved astonishingly in the last decade. Detailed

understanding of the device physics and the degradation mechanisms is necessary to further develop material properties and device design.

For high-end products, some technical problems must still be solved. But visible progress has been made over the last decade in all the important fields like effi-

ciency, color portfolio brightness, and lifetime to make **OLED** devices already competitive in some niches. Time will tell whether the replacement of spot lights (like light bulbs) by lighting areas made of diffuse **OLEDs** in applications will take place.

5.9.3 Photorefractive Crystals

Light-induced refractive index changes, so-called photorefractive effects, in inorganic electrooptic crystals were discovered by *Ashkin* et al. in 1966 [5.313]. Though in the beginning these effects seemed to be very undesirable (“optical damage”), *Chen* et al. recognized only two years later the significance for holographic storage [5.314]. In 1975, *Staebler* et al. reported the recording of 500 thermally fixed volume phase holograms in $\text{LiNbO}_3 : \text{Fe}$, each hologram with more than 2.5% readout efficiency [5.315]. The method is based on the Bragg condition allowing the superposition of many volume (“thick”) holograms at the same site under different angles.

During hologram recording a light pattern has to be transposed into a refractive index pattern. Interfering light beams generate bright and dark regions in an electrooptic crystal. When light of suitable wavelength is chosen, charge carriers – usually electrons [5.316] – are excited in the bright regions and become mobile. The charge carriers migrate in the crystal and are subsequently trapped at new sites. By this means electronic space-charge fields are set up that give rise to a modulation of refractive index via the electrooptic effect. Figure 5.75 illustrates the photorefractive process. Index changes up to 10^{-3} are obtained. The trapped charge can be released and the field pattern erased by uniform illumination or by heating. On the one hand, this reversibility is highly desired for, e.g., erasable holographic memories or adaptive optical components; on the other hand, the problem of destructive readout arises.

Photorefractive effects have been observed in many electrooptic crystals, among them LiNbO_3 , LiTaO_3 , the ferroelectric perovskites BaTiO_3 , $\text{Ba}_{1-x}\text{Ca}_x\text{TiO}_3$, KNbO_3 , and $\text{KTa}_{1-x}\text{Nb}_x\text{O}_3$ (KTN), the tungsten-bronze-type crystals $\text{Ba}_2\text{NaNb}_5\text{O}_{15}$ and $\text{Sr}_{1-x}\text{Ba}_x\text{Nb}_2\text{O}_6$ (SBN), the nonferroelectric sillenites $\text{Bi}_{12}\text{TiO}_{20}$ (BTO), $\text{Bi}_{12}\text{SiO}_{20}$ (BSO), and $\text{Bi}_{12}\text{GeO}_{20}$ (BGO), the semiconductors GaAs and InP, and others. Very early crucial influences of dopants and thermal treatments were discovered [5.317, 318]. Photorefractive crystals are of particular interest for many unique devices, such as self-pumped phase-conjugating mirrors, parametric amplifiers and oscillators as well as static and dynamic

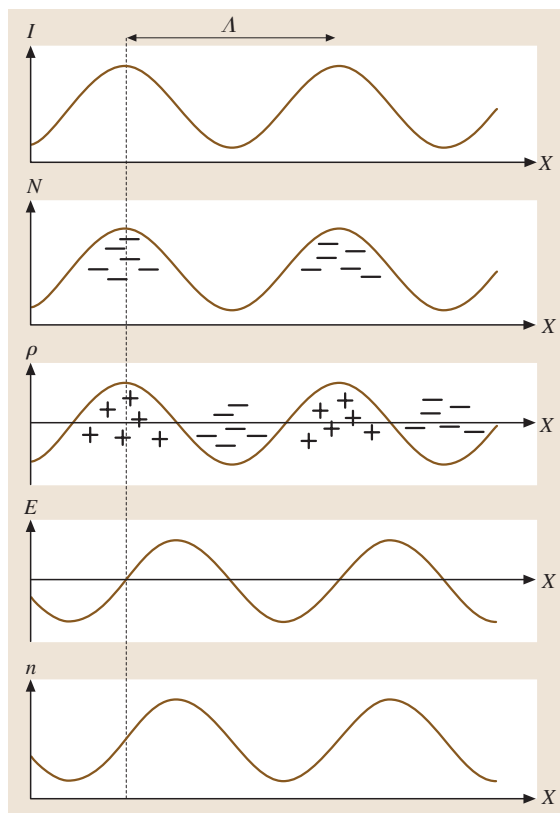


Fig. 5.75 Photorefractive process in electrooptic materials. A light intensity pattern $I(x)$ illuminates a photorefractive material. Here as an example we use a sinusoidal light pattern with the period length Λ as it is generated by interference of two plane waves. The light excites free charge carriers; e.g., electrons from traps are excited into the conduction band. The free-carrier distribution $N(x)$ yields currents due to, e.g., diffusion. For pure diffusion charge transport, the charge carriers move into the darker areas where they are captured by empty traps. An effective charge density distribution $\rho(x)$ builds up with missing electrons and effective positive charge in the bright and excess electrons and effective negative charge in the dark areas. This charge distribution yields a space-charge field $E(x)$ and, because of the linear electrooptic effect, also a refractive-index modulation $n(x)$



optical filters [5.319, 320]. They are also considered for holographic data storage [5.321].

In the following we describe the charge transport processes that critically determine the performance of the material. Figures of merit for the hologram storage properties will be introduced, the limits will be evaluated, and the crystals mentioned above will be discussed. Finally, fixing techniques will be briefly reviewed.

Charge Transport

The charge driving forces are well known: diffusion, bulk photovoltaic currents [5.322], and drift of charge carriers in external or pyroelectric fields [5.323] generate the space-charge field. Compensating drift currents arise, and saturation is achieved if they are as large as the driving currents. Application of large external electric fields is of interest mostly for crystals with a relatively high conductivity, and large pyroelectric fields occur only in the case of high light intensities. Thus diffusion and the bulk photovoltaic effect are of major importance.

Extrinsic or intrinsic defects occurring in more than one valence state are sources and traps of the charge carriers. Figure 5.76 shows different charge transport situations in a band diagram. The energetic positions of the levels shown correspond to the thermal energy required to excite a charge carrier from the filled level. Photon energies necessary to release electrons are typically larger because of the Franck–Condon principle. The simplest and for many applications the desired mechanism is provided by a “one-center system” with monopolar conductivity (shown in Fig. 5.76 for electron conductivity). One impurity type occurs in two different valence states, and charge is redistributed via

either conduction or valence band [5.324]. An example is iron-doped LiNbO_3 at usual continuous-wave laser intensities. Iron ions occur only in the valence states 2+ and 3+ [5.325]. Electrons are excited from Fe^{2+} to the conduction band and trapped by Fe^{3+} elsewhere.

Electron–hole competition may complicate the situation [5.316, 326, 327]. Electrons and holes can be created at the same center, as shown in Fig. 5.76, but it is also possible that an additional independent center is present that generates holes. In any case, diffusion currents of electrons and holes compensate each other which is disadvantageous for hologram recording.

Many materials show more complicated transport mechanisms, because two or more photorefractive levels participate in the charge transport simultaneously. Additional independent centers [5.328, 329] or impurities occurring in three or more valence states [5.330] may be the origin of these additional levels. We have to distinguish between so-called deep and shallow levels. “Shallow” indicates here that charge can be released thermally from this level, which is practically impossible for a deep site. A system with the simultaneous presence of a deep and a shallow energy level (shown in Fig. 5.76 for electron conduction) has detrimental consequences for hologram recording. The shallow levels are empty in the dark because all charge carriers have been thermally excited and trapped by deep sites. Upon illumination, the shallow levels trap charge carriers from either the conduction or the valence band. As a consequence, the response time grows (smaller photoconductivity [5.331]) and, moreover, often the absorption increases, too, because the filled shallow traps absorb light effectively [5.329].

Deep and shallow levels exchange charge via conduction or valence band. Direct exchange of charge is possible, if the concentrations are large enough that different centers are located closely together. This may occur in the case of intrinsic defects [5.332]. For very high defect concentrations additional bands are created, because centers of the same type couple and exchange charge directly [5.333, 334]. More details about the charge transport may be found in a review [5.335].

The charge transport situation can be influenced by doping, thermal annealing, and selection of experimental parameters (light intensity and wavelength, crystal temperature). Thus it is possible to establish in many materials the desired one-center charge transport situation. We will concentrate in the following on a one-center system, because we are interested in the performance limits of different materials. Iron-doped lithium niobate ($\text{LiNbO}_3 : \text{Fe}$) will be used as an example to describe

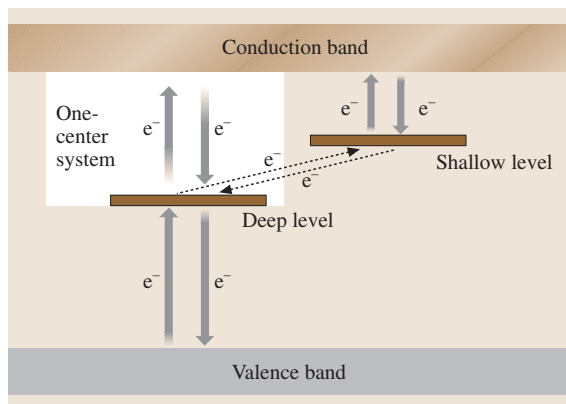


Fig. 5.76 Charge transport processes in a photorefractive crystal. Arrows indicate excitation and recombination of electrons (for details see text)



in more detail the ongoing processes and the resulting dependences. However, most of the theoretical consideration is valid for any photorefractive crystal.

Quantitative description of the formation of the space-charge pattern under nonuniform illumination requires the solution of the Maxwell equations, together with the current and the rate equations. Generation and recombination of conduction band electrons are described by $(dN_e)/(dt) = +SIc_{Fe^{2+}} - \gamma c_{Fe^{3+}}N_e$, where N_e is the concentration of electrons in the conduction band, S is the photon absorption cross-section, I is the light intensity, $c_{Fe^{2+}}$ and $c_{Fe^{3+}}$ are the concentrations of Fe^{2+} and Fe^{3+} ions, and γ is the recombination coefficient. The general problem leads to a system of coupled nonlinear differential equations that cannot be solved analytically [5.324]. Reasonable approximations reveal that simple exponential evolutions govern build-up and decay of the space-charge field. Analysis yields the steady-state amplitude E_{sc} of the space-charge field [5.324]

$$E_{sc} = m(E_D^2 + E_{phv}^2)^{1/2}, \quad E_D = \frac{k_B T}{e} K, \\ E_{phv} = \frac{\beta \gamma}{e \mu S} c_{Fe^{3+}}, \quad (5.158)$$

where m is the modulation degree (“visibility”, “contrast”) of the interference pattern, E_D is the diffusion field, E_{phv} is the bulk photovoltaic field, k_B is the Boltzmann constant, T is the temperature, e is the elementary charge, K is a typical spatial frequency of the hologram (2π /period length of the interference pattern), β is the bulk photovoltaic coefficient [bulk photovoltaic current density/ $(c_{Fe^{2+}}I)$], and μ is the charge carrier mobility. In the following we will assume a fully modulated light pattern, i.e., $m = 1$. The field E_{sc} might be reduced by space-charge limiting effects ($E_q < E_D$ or $E_q < E_{phv}$ with the space-charge limiting field $E_q = [e/(\epsilon\epsilon_0)](1/K)N_{eff}$, dielectric con-

stant ϵ , permittivity of free space ϵ_0 , and effective trap density N_{eff}).

The bulk photovoltaic effect dominates in $LiNbO_3$ and $LiTaO_3$ crystals containing 0.01 wt % or more Fe, but diffusion is the major transport mechanism in many other materials. We have to distinguish between different geometries, e.g., transmission and 90° geometry (Fig. 5.77). Typical spatial frequencies in these geometries are $2\pi/0.4 \mu m^{-1}$ and $2\pi/0.15 \mu m^{-1}$, respectively. These values will be used in the following if we refer to transmission or 90° geometry. The corresponding diffusion fields at room temperature are 4 kV/cm and 10 kV/cm. Bulk photovoltaic fields in $LiNbO_3$ can reach values up to 100 kV/cm [5.322]. Larger fields cannot be created because electric breakdown takes place [5.336]. We will treat in the following the situation where the light is polarized perpendicularly to the plane of incidence (ordinary polarization), if nothing else is mentioned. In-plane polarization yields recording beams of crossed polarization in the case of the 90° geometry, and no intensity pattern is present. Extraordinary polarization can be used in the transmission geometry, and electrooptic coefficients, dynamic range and sensitivity might be higher, but extraordinarily polarized light tends to create strong holographic scattering [5.337], which is unacceptable for many applications.

The refractive index is modulated via the linear electrooptic effect [5.338], and the refractive index changes are

$$\Delta n = -\frac{1}{2}n^3 r E_{sc}, \quad (5.159)$$

where n is the refractive index and r is the effective electrooptic coefficient. Proper n and r values must be selected considering the material symmetry, the crystal cut, the polarization of light, and the direction of the space-charge field E_{sc} . For example, $LiNbO_3$ with ordinarily (o) or extraordinarily (e) polarized light and a field along the optical axis corresponds to $n = n_o$ and $r = r_{113}$, or $n = n_e$ and $r = r_{333}$, respectively.

Useful figures of merit are required in order to allow comparison and evaluation of the materials in terms of storage performance.

The *dark decay* of the refractive index modulation typically follows a monoexponential function $\Delta n(t) = \Delta n(t=0) \exp(-t/\tau_{dark})$. Large dark storage times τ_{dark} are usually desired.

Monoexponential growth of the space-charge fields and refractive index changes, $\Delta n(t) = \Delta n_s[1 - \exp(-t/\tau)]$, is characterized by the *response time* τ , which depends on light intensity according to $\tau = \epsilon\epsilon_0/\sigma_{ph}(I)$, where σ_{ph} is the photoconductivity.

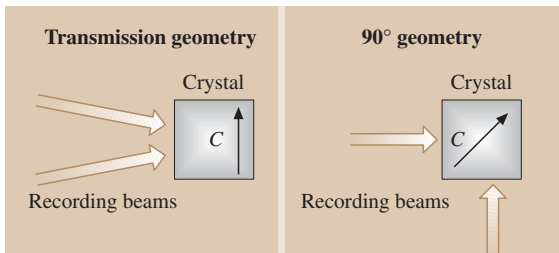


Fig. 5.77 Transmission and 90° recording geometries. The black arrows show the direction of the crystallographic c -axis



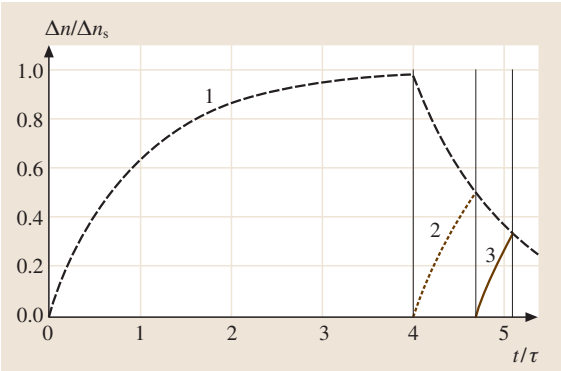


Fig. 5.78 Normalized refractive index modulation $\Delta n/\Delta n_s$ vs. normalized time t/τ for an exposure schedule that yields holograms of equal strength. Hologram Nr. 1 (dashed curve) is recorded into saturation. Recording of hologram Nr. 2 (dotted curve) erases part of hologram Nr. 1. Recording is stopped as soon as both holograms approach equal strength. At this point both holograms have both a Δn of $\Delta n_s/2$. Recording of hologram Nr. 3 (full curve) then erases the holograms Nr. 1 and 2 and so forth

Here Δn_s is the saturation value of the amplitude of a single hologram.

Subsequent superposition of several holograms requires a special exposure schedule, if holograms of equal efficiency are the target [5.352]. The diffraction efficiency (ratio between the intensities of diffracted and incident light) of an individual of M multiplexed holograms will be $\eta(M) = [(M\#)/M]^2$, if M is large enough to ensure that η is much smaller than 1. Figure 5.78 illustrates the required exposure schedule and derives graphically that for M holograms of equal

strength the Δn per hologram is just $\Delta n_s/M$. Considering $\eta \propto \Delta n^2$ [5.353], we end up with the formula $\eta(M)$ given above.

The $M\#$ is a useful measure for the *storage density* and *capacity* of photorefractive crystals [5.354]. It is given by

$$M\# = \frac{\pi \Delta n_s d}{\lambda \cos \Theta} \exp(-1) \quad \text{for } \alpha d = 2, \quad (5.160)$$

if the time constants for recording and erasure are equal. Here d is the thickness of the crystal, λ is the vacuum wavelength of the light, Θ is half the angle between the recording beams in the medium, and α is the intensity absorption coefficient. Build-up of Δn requires the presence of excitable electrons, which cause absorption. Considering this trade-off, the largest $M\#$ is achieved for $\alpha d = 2$. The photon budget determines the minimum efficiency η necessary to read a hologram. Assuming, e.g., $\eta = 10^{-6}$, an $M\#$ of 10 indicates that 10 000 holograms can be multiplexed. However, the number of useful holograms can be reduced significantly owing to hologram cross-talk, light scattering, or homogeneous bulk photovoltaic fields.

There are various ways of defining the *sensitivity*. A useful measure is $W_{\eta=1\%}$, the exposure density (laser-light intensity multiplied by the recording time) required to record a hologram of 1% diffraction efficiency. Recording and erasure of the refractive index changes Δn are governed by monoexponential functions [5.324]. Using *Kogelnik's* equation [5.353] for small diffraction efficiencies ($\eta \approx [(\pi \Delta n d)/(\lambda \cos \Theta)]^2 \exp(-\alpha d)$) we end up with

$$W_{\eta=1\%} = \frac{1}{10} \frac{\tau(I)I}{M\#} \quad \text{for } \alpha d = 2. \quad (5.161)$$

Table 5.32 Refractive index n_o for light polarized perpendicularly to the optical axis, electrooptic coefficient r , and dielectric coefficient ϵ for fields along the optical axis for different photorefractive crystals (light wavelength 514 nm; room temperature values)

Material	n_o		r (pm/V)			ϵ_{33}	
LiNbO ₃	2.33	[5.339]	11	*1	[5.339]	28	[5.339]
LiTaO ₃	2.21	[5.339]	8	*1	[5.339]	43	[5.339]
BaTiO ₃	2.42	[5.340]	22	*	[5.341]	83	[5.341]
BCT ^a	2.47	[5.342]	36	*	[5.343]	230	[5.344]
KTN ^b	2.31	[5.345]	21	*	[5.346]	485	[5.345]
KNbO ₃	2.33	[5.339]	40	*	[5.341]	35	[5.341]
SBN ^c	2.37	[5.347]	47	*	[5.348]	880	[5.339]
BTO ^d	2.65	[5.349]	5.75	+	[5.350]	52	[5.349]
GaAs	3.45	[5.351]	1.4	+2	[5.339]	12	[5.339]

^a Ba_{0.77}Ca_{0.23}TiO₃, ^b KTa_{0.52}Nb_{0.48}O₃, ^c Sr_{0.61}Ba_{0.39}Nb₂O₆, ^d Bi₁₂TiO₂₀, * r_{113} , + r_{231} , ¹ 633 nm, ² 1150 nm

Small response times τ and large values of $M\#$ are required to achieve good sensitivities (small $W_{\eta=1\%}$ values). In addition to $W_{\eta=1\%}$ another measure for the sensitivity is also commonly used

$$S = \frac{\sqrt{\eta}}{Idt} \bigg|_{t \ll \tau}, \quad (5.162)$$

where η is the diffraction efficiency that the hologram reaches after the recording time t . For small recording times ($t \ll \tau$) we can write $\sqrt{\eta} = M/\# \times (t/\tau)$. From this the simple relation $S = (10dW_{\eta=1\%})^{-1}$ can be derived. For a sample of, e.g., $d = 1$ cm thickness and a $W_{\eta=1\%}$ of 10 mJ/cm^2 we end up with $S = 10 \text{ cm/J}$.

Performance Limits

There is no obvious theoretical limit to the *dark storage time*; very large values up to 10 y have been extrapolated [5.355]. For strongly doped crystals tunnelling of electrons between traps may reduce the dark storage time [5.333].

Crystals with thicknesses d up to 1 cm are available, and are supposed to have the highest *storage density*. Absorption is limited to $\alpha = 2/d = 2 \text{ cm}^{-1}$, which determines the concentration $\text{Fe}^{2+} = 4.3 \times 10^{17} \text{ cm}^{-3}$ [5.325]. Thus the largest possible E_{sc} (all electrons are redistributed) is $E_{sc} = E_q = [e/(\epsilon\epsilon_0)](1/K)N_{\text{eff}} = 180 \text{ kV/cm}$ and 66 kV/cm for transmission and 90° geometry, respectively (ϵ , Table 5.32). A field of 180 kV/cm is above the break-down threshold, but the $M\#$ describes the multiplexing properties, and the amplitude of each single

hologram is well below the break-down field if many holograms are superimposed. We get $\Delta n_o = 13 \times 10^{-4}$ and $\Delta n_o = 5 \times 10^{-4}$ ((5.159), n_o and r_{113} from Table 5.32), which yields according to (5.160) the $M\#$ values 30 and 16 for transmission and 90° geometry respectively ($\lambda = 514 \text{ nm}$, $\Theta = 45^\circ$, $\alpha d = 2$). A similar estimate holds for LiTaO_3 . However, in diffusion-controlled media the space-charge fields are lower, and limited by $E_{sc} = E_D$. The $M\#$ can be estimated using (5.158–5.160). $M\#$ s range from 0.3 to 10 for different materials (Table 5.33).

Regarding *response time and sensitivity*, we have to consider the quantum limit, i. e., each incident photon moves in the optimum situation one electron to the desired position. Considering green light (photon energy 2.4 eV) and an energy difference of about 1 eV between the impurity level and the band, about 1.4 eV remains to move the electron against the space-charge field. Supposing a typical period length of $0.4 \mu\text{m}$ (transmission geometry), we have to move the electron about $0.2 \mu\text{m}$ against the field. Thus the energy of one photon is sufficient to move the electron against fields of up to 70 kV/cm . The fields are larger in the 90° geometry, but the transport lengths are smaller. From this perspective it is reasonable to treat the situation where each photon redistributes one electron. In this case we can replace $\tau(I)$ by $(\epsilon\epsilon_0/e)K(hc/\lambda)dE_{sc}(1/I)$, where h is the Planck constant and c is the vacuum speed of light, and get for $I = 1 \text{ W/cm}^2$ and parameters of LiNbO_3 the result $\tau \approx 100 \text{ ms}$ (ϵ from Table 5.32, $K = 2\pi/0.4 \mu\text{m}^{-1}$, $\lambda = 514 \text{ nm}$). This yields $W_{\eta=1\%} \approx 0.3 \text{ mJ/cm}^2$.

Table 5.33 Various storage parameters for different photorefractive crystals (light wavelength 514 nm ; room temperature values; Trans., transmission geometry; 90° , 90° geometry). The values for $M\#$ and $W_{\eta=1\%}$ are calculated from (5.158–5.161)

Material	Dark storage time		Response time (ms) @ 1 W/cm^2		M# Storage capacity		W _{η=1%} Sensitivity (mJ/cm ²)	
					Trans.	90°	Trans.	90°
LiNbO ₃	1 a	[5.355]	3000	[5.356]	30	16	10	19
LiTaO ₃	10 a	[5.355]	250 ¹	[5.355]	30	16	0.8	1.6
BaTiO ₃	1 h – 1 a	[5.357]	500	[5.358]	1.5	5	33	10
BCT ^a	10 s	[5.359]	400	[5.359]	2.5	9	16	4
KTN ^b	0.6 a	[5.360]	200	[5.361]	1.2	4	16	5
KNbO ₃	1 h – 30 d	[5.362, 363]	100	[5.364]	2.4	8	4	1.3
SBN ^c	1 h – 30 d	[5.365, 366]	100	[5.365]	3.0	10	3	1
BTO ^d	10 s	[5.367]	2 ²	[5.368]	0.5	1.7	1.2	0.35
GaAs	1 ms	[5.369]	0.2 ³	[5.370]	0.12	0.41	0.02	0.006

^a Ba_{0.77}Ca_{0.23}TiO₃, ^b KTa_{0.52}Nb_{0.48}O₃, ^c Sr_{0.61}Ba_{0.39}Nb₂O₆, ^d Bi₁₂TiO₂₀, ¹ 351 nm, ² 633 nm, 3 W/cm² and Bi₁₂SiO₂₀.

³ 1160 nm, 0.13 W/cm²



Various Crystals

Table 5.32 summarizes important parameters, and Table 5.33 presents typical hologram recording characteristics of various inorganic photorefractive crystals.

Hologram recording performances of LiNbO_3 and LiTaO_3 , of the perovskites BaTiO_3 , $\text{Ba}_{0.77}\text{Ca}_{0.23}\text{TiO}_3$ (BCT), $\text{KTa}_{0.52}\text{Nb}_{0.48}\text{O}_3$ (KTN), and KNbO_3 , of the tungsten-bronze structure $\text{Sr}_{0.61}\text{Ba}_{0.39}\text{Nb}_2\text{O}_6$ (SBN), of the sillenite $\text{Bi}_{12}\text{TiO}_{20}$ (BTO), and of the semiconductor GaAs are summarized in Table 5.33. *Dark storage* and *response time* may vary by orders of magnitude for the same material depending on doping (elements, concentrations), annealing (oxidation, reduction), and experimental conditions (light intensity, crystal temperature, etc.). Thus the values shown are only typical numbers. It is possible to improve one value at the expense of another. For example, the response time of KNbO_3 can be decreased to a few ms by electrochemical reduction [5.371], but simultaneously the dark storage time is decreased by orders of magnitude.

The largest possible *storage densities* are calculated [(5.158–5.160), Table 5.32] considering bulk photovoltaic charge transport for LiNbO_3 and LiTaO_3 , and diffusion for the other materials. LiNbO_3 and LiTaO_3 have larger storage densities in the transmission geometry because more charge per interference fringe is available to build-up the space-charge field. However, the 90° geometry is advantageous for diffusion recording, because higher spatial frequencies yield larger diffusion fields. Using the response time τ and the storage capacity already determined, the *sensitivity* is estimated based on (5.161).

The number of holograms that can be multiplexed may be reduced significantly in many samples by several undesired effects. For example, light scattering reduces the visibility of the interference pattern, and therefore the fields and the dynamic range are also lowered. The required exposures $W_{\eta=1\%}$ [5.355, 372] for recording in LiNbO_3 and LiTaO_3 are one order of magnitude larger than the values presented in Table 5.33. It is also known that the measured refractive index changes in, e.g., BaTiO_3 and KTN crystals are often about 4 to 5 times below the calculated values [5.346, 373]. The numbers for storage capacity and sensitivity in Table 5.33 are optimum values which may be achieved in carefully selected and optimized samples.

Table 5.33 clearly shows which material is the best for which application. Crystals of the lithium-niobate family (LiNbO_3 and LiTaO_3) are the favorites for long-term high-capacity storage of holograms. The

performance of the rarely used LiTaO_3 exceeds that of LiNbO_3 , but the drawback in this case is that ultraviolet light is required. Good availability, excellent homogeneity, and high robustness are further advantages of LiNbO_3 and LiTaO_3 . Furthermore, the processes occurring in LiNbO_3 are well understood, e.g., the proportionality between the space-charge field and the concentration of Fe^{3+} ions (5.158) is verified [5.356]. The crystals can be tailored for applications by optimization of the Fe concentration, by annealing and by additional doping. For example, doping with two deep independent levels which can be addressed with light of different wavelength allows optical fixing of the information [5.374]. Doping with large amounts of Mg increases the conductivity and improves the sensitivity [5.356]. More details about the specific properties of LiNbO_3 and other photorefractives can be found in a review [5.375].

The perovskites have shorter hologram lifetimes, but they are well suited for, e.g., holographic double-exposure interferometry [5.376]. The properties of all sillenite-type crystals $\text{Bi}_{12}\text{TiO}_{20}$ (BTO), $\text{Bi}_{12}\text{SiO}_{20}$ (BSO), and $\text{Bi}_{12}\text{GeO}_{20}$ (BGO) are very similar with one exception: The optical activity of BTO is much smaller than that of BSO and BGO [5.349], which is of importance especially for thick samples. Parameters of the semiconductor GaAs are shown for completeness. Small dark storage times and small storage capacities make GaAs unattractive for many photorefractive applications. However, perovskites, SBN crystals, crystals of the sillenite type, and photorefractive semiconductors can be of great interest for dynamic components because of the relatively large sensitivity that allows fast optical processing, fast adaption of optical components, and fast refreshment of storage systems.

The sensitivities of the materials are orders of magnitude away from the quantum performance limits, and the low sensitivity is indeed the major drawback of inorganic photorefractives. A detailed study of the properties of cerium-doped SBN revealed the reasons [5.377] which apply for many photorefractive crystals. Only a few percent of the photons release electrons; most of the light energy is immediately transferred to heat. Trapping of electrons is very efficient because of Coulomb attraction which limits the life-time and the transport lengths. It is unlikely that these problems can be completely overcome by a new inorganic photorefractive crystal.

Laser technology may solve the remaining problems of speed. The available output power of continuous-wave lasers increases rapidly. Pulsed systems become smaller and cheaper. Recording and reconstruction of

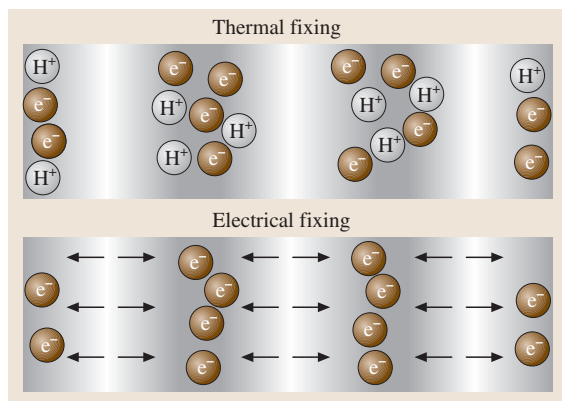


Fig. 5.79 *Thermal fixing.* During or after recording the crystal is heated. Ions become mobile and screen the electronic space-charge field. In the example shown, electrons are moved into the darker areas. This charge is compensated by H^+ ions. After cooling back to room temperature the ions are almost immobile. Now illumination moves part of the electrons back because of spatially inhomogeneous excitation, and a space-charge field appears that modulates the refractive-index through the electrooptic effect. Further erasure of the electronic pattern is impossible because drift of electrons in the space-charge field cancels all erasure currents. *Electrical fixing.* Similar to thermal fixing, but instead of heating and ions, external electrical fields and switched ferroelectric domains are involved. The arrows indicate the directions of the switched ferroelectric domains

holograms in photorefractives by, e.g., femtosecond light pulses has been demonstrated [5.378]. Another approach to solve the problem of insufficient sensitivity is the usage of multiple-quantum wells [5.379, 380]. Electric space-charge fields yield large absorption changes which modulate the refractive index due to the Kramers–Kronig relations. Multiple-quantum wells are fast, and there seems to be a lot of room for further improvements.

Nondestructive Readout

The retrieval of information stored in electrooptic crystals requires homogeneous illumination and thus leads to erasure effects. In many cases asymmetries in the recording and read-out processes that are simply achieved by a reduction of the read-out light intensity are not sufficient. For this reason several methods have been developed to stabilize volume phase holograms versus the readout light.

Thermal fixing has already been discovered in $LiNbO_3$ in 1971 [5.381]. By heating, a hologram re-

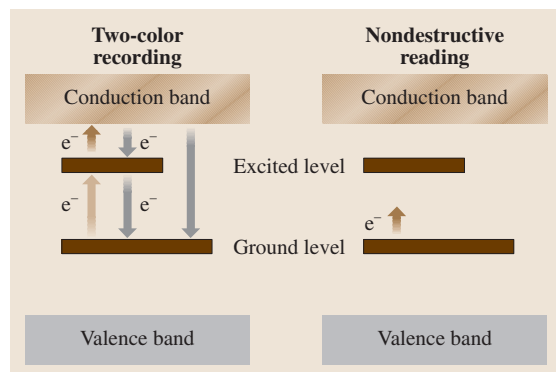


Fig. 5.80 Two-photon persistent holographic recording: The first photon excites an electron from a ground state into an excited state. This illumination can be, e.g., spatially homogeneous. Exposure with a holographic interference pattern of the recording wavelength excites in the illuminated areas electrons from the excited state into the conduction band where they move and recombine elsewhere with traps either through the excited state into the ground state or directly into the ground state. The wavelengths for the first and second transition must be carefully selected to make excitations through real levels possible. For reading, homogeneous light of the recording wavelength must be present. However, this light fails to excite the electrons from the ground state into the excited state because of insufficient photon energy. Thus no electrons are released, and the hologram is stable against further illumination

sulting from electronic and ionic charges is formed which is stable against homogeneous illumination. Protons (H^+) play the major role as mobile ions in the thermal fixing process [5.336, 382]. Electrical fixing has also been demonstrated [5.383, 384]. An external electric field slightly smaller than the coercive field has been applied to convert the space-charge pattern into a domain pattern. Figure 5.79 illustrates these fixing processes.

To preserve the possibility of desired optical erasure, the use of two-photon excitation has been proposed for hologram recording [5.385]. Then readout without erasure is possible using reduced light intensity [5.386]. Similar methods that avoid large recording intensities are based on the population of shallow levels [5.387] or of photochromic centers [5.374]. A general picture about two-photon persistent holographic recording is shown in Fig. 5.80.

There seems to exist a simple possibility for reading volume holograms nondestructively: the use of readout

light of low photon energy, insufficient to excite charge carriers. However, information losses then arise because of the Bragg condition. A method has been proposed to improve the situation with the help of anisotropic diffraction [5.388]. Another solution to increase the bandwidth of the readout light has been developed [5.389, 390], based on a spectrum of spatially adapted wave vectors. Finally, the use of frequency-difference holograms has been suggested and demonstrated for nondestructive readout [5.391, 392]. The spatial frequency shift is achieved by nonlinear mixing of a hologram with a carrier frequency grating, but recording with two different wavelengths is required.

Conclusions

Good availability, excellent homogeneity, robustness, large dark storage times, large storage capacities, reversibility and the availability of efficient and convenient fixing techniques make LiNbO_3 and LiTaO_3 the favorites among the photorefractive crystals for applications that involve long-term storage of many strong holograms. This is the reason why many demonstrations are based on this material. Tungsten-bronze- and sillenite-type crystals are more sensitive, but small dark storage times and low storage capacities limit their applicability to short-term, high-resolution storage of images, e.g. for the purpose of holographic interferometry. Knowledge about the charge transport nowadays enables tailoring of the crystals considering system issues such as the desired Bragg selectivity, the light wavelength and the light intensity. Concentrations and valence states of impurities may be adjusted by doping or thermal annealing.

Holographic storage is only one of many possible applications of photorefractive crystals. Lithium niobate and lithium tantalate are, e.g., excellent to realize fixed Bragg wavelength filters for telecommunication. Materials like the perovskites barium titanate and potassium niobate, tungsten-bronze type and sillenite type photorefractive crystals as well as semiconductor photorefractive crystals can be of high interest for dynamic holography, optical signal processing, realization of adaptive optical components, for phase conjugation, and for many other advanced applications.

5.9.4 Metal Mirrors

Mirrors can be fabricated from any number of materials with an emphasis for high performance applications on near-zero expansion materials such as ULE^{TM} [5.394] and Zerodur[®] [5.395]. However, metals and ceram-

ics, primarily silicon carbide, are now being used in many applications. When designing a mirror, there are many metals from which to choose. After considering the properties with respect to the requirements, fabricability, compatibility with other components and dimensional stability must be considered. Metal and silicon carbide mirrors and support structures need to be as stable as their dielectric counterparts. After a brief discussion of dimensional stability – its causes and prevention, properties, selection criteria and recommended fabrication methods are given. More complete treatments have been given by Paquin [5.396, 397] and others [5.398].

Dimensional Stability

Dimensional stability of a component is determined by the amount of instability, recognized as dimensional change or distortion leading to reduction of system performance. Stability can only be achieved and controlled when the causes and sources are understood.

Instability Types. Basically there are two types of dimensional instability. First there are permanent changes such as temporal and hysteresis instabilities, and second, changes that may be permanent that happen when the environment is changed. Temporal instability is a dimensional change or distortion that occurs over time in a constant environment, while hysteresis is a change measured at constant conditions before and after changes in the environment such as temperature cycling or vibration (mechanical cycling). Thermal instability can be a change that is seen at a different temperature but returns to the original at the original temperature, or a change that depends on the rate of change of temperature and is not the same when returned to the original conditions.

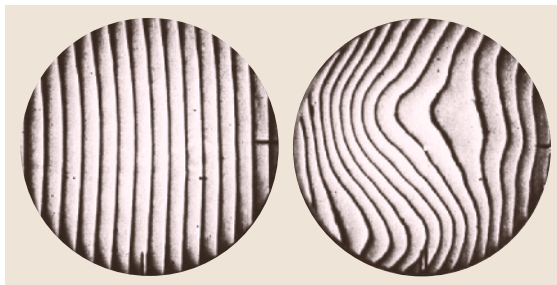


Fig. 5.81 Interferograms of an optical flat at 26 °C (left) and 85 °C showing thermal instability due to severe inhomogeneity of thermal expansion. The figure repeats no matter how many cycles are performed [5.393]

Instability Sources. The sources of these dimensional instabilities are the material, the fabrication process, the environment or a combination of them.

The Material

The material from which a component is fabricated is often the source of instability due to anisotropy and/or inhomogeneity. Anisotropy is direction dependent property variation and inhomogeneity is random spatial property variation looking like the material is “poorly mixed” as shown in Fig. 5.81 [5.393]. The result is thermal instability that is permanent in that the inhomogeneity and/or anisotropy is permanently in the material and cannot be changed, and repeatable in that the distortion at any temperature is repeatable time after time. To avoid this form of thermal distortion, the material selected must be homogeneous and isotropic. Single crystals are intrinsically homogeneous, but are uniformly anisotropic in some properties depending on crystal structure.

Metals can have microstructural changes from exposure to heat from any number of sources, such as processing operations or exposure in use. Amorphous metals such as as-deposited electroless nickel can crys-

tallize and shrink, heat treatable metals such as 6061-T6 aluminum can overage and lose strength and metals with a low annealing temperature such as OFC copper can soften. Fabrication methods to eliminate or minimize these effects are discussed for each material in Sect. 5.9.4.

Another form of thermal instability is thermal mapping where the distortion is proportional to the intensity of impinging radiation and thermal properties of the material. This instability can be minimized by choosing materials that have low coefficient of linear thermal expansion (CTE or α), high thermal conductivity (k or λ) and high specific heat (c_p).

Fabrication

The most common instabilities are temporal and hysteresis. They are both caused by decrease of residual stress and associated strain. Machining and/or grinding operations induce residual stress, sub-surface damage and strain in the cut surface. This layer must be removed or the part will be unstable and subject to further surface damage from high energy laser impingement. Chemical etching can remove the damaged layer and heat treatment can reduce or eliminate the stress.

Temporal instability is the dimensional change associated with stress relaxation. The rate of decrease in stress (s) is proportional to the amount of stress present, that is,

$$(-ds/dt)\tau = s \quad (5.163)$$

and

$$s = s_0 e^{(-t/\tau)}. \quad (5.164)$$

The relaxation time (τ) is the time (t) at which the stress has dropped to 0.37 of the original stress (s_0) and the decay is exponential. Similar equations can also be formulated for change in strain or surface figure. Stress relaxation is also a thermally sensitive process so that the rate at which the stress decreases, increases exponentially with increasing temperature,

$$1/\tau \propto e^{(-E/kT)}. \quad (5.165)$$

These relationships are used in fabricating components to relieve stress, thereby increasing dimensional stability. Lokshin [5.399] has shown the effects of both thermal cycling and isothermal exposure on stress relaxation as shown in Fig. 5.82. Note that thermal cycling is more effective in reducing stress than isothermal treatment for the same elevated temperature.

Since thermal cycling effectively reduces residual stress it is obvious that hysteresis instabilities are caused by reduction of residual stress.

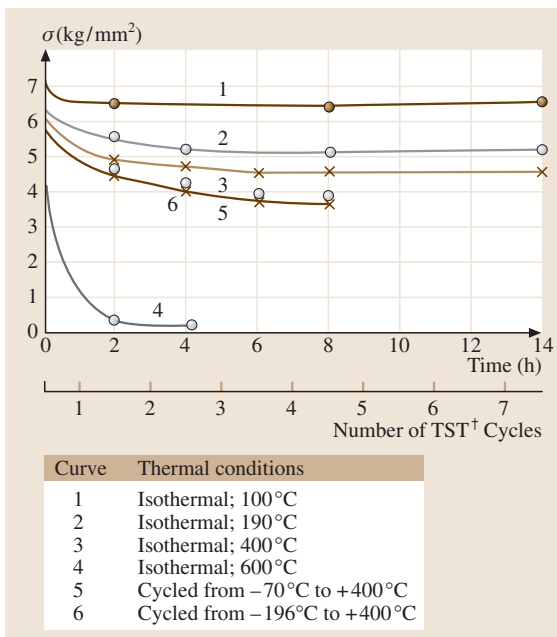


Fig. 5.82 Comparison of the effects of thermal cycling and isothermal exposure on stress relaxation in stressed pure beryllium. For Be, thermal cycling reduces stress more than isothermal treatment at the same upper temperature. (Thermocyclic stabilizing treatments) (After [5.399])

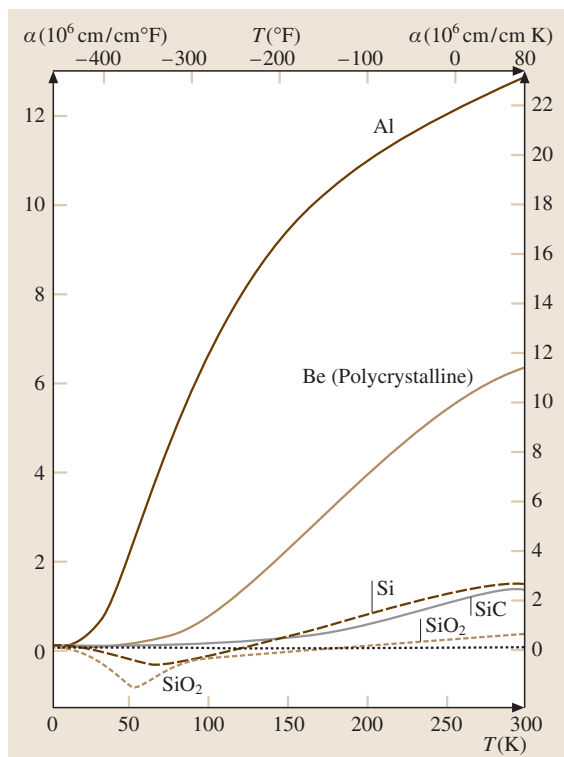


Fig. 5.83 Dependence of coefficient of linear thermal expansion on temperature for several mirror materials [5.400]

Minimizing residual stress in fabrication of metal optics is best achieved in the fabrication process by using low stress machining/grinding techniques, chemical etching to remove damaged layers, annealing to eliminate stress, and thermal cycling to stabilize. Thermal cycling should be carried out between temperature extremes that are higher and lower than those the part will see in subsequent fabrication and assembly operations, shipping and storage, and in service.

Environment

A mirror's environment includes externally applied stresses incurred during fabrication and assembly, the attachment stresses, thermal stresses in storage and finally the use environment. To resist deformation due to these stresses, the material must be strong enough, both on a macro and micro scale and not be subject to temperatures that will cause any of the instabilities discussed above. Athermalization of systems to minimize thermal distortions is described elsewhere in this volume.

Mirror Materials. The room temperature properties of metal mirror materials are compared with those of

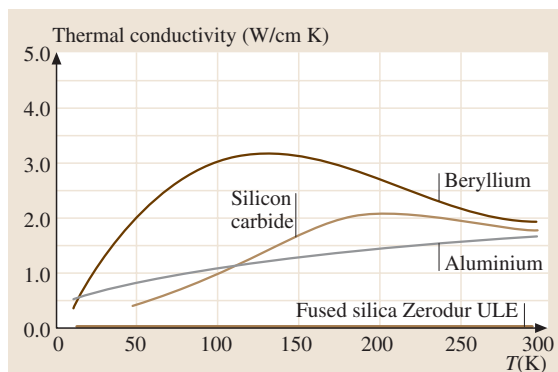


Fig. 5.84 Dependence of thermal conductivity on temperature for several mirror materials. (After [5.400])

other common mirror and structural materials in Table 5.34. The properties were obtained from many sources and are believed to be the most accurate at the time of preparation. Care should be taken in using these properties as they are typical and should not be used for design purposes. Properties vary with temperature, as shown in Figs. 5.83 and 5.84 [5.400], and the table values should not be assumed for other temperatures.

Some of the properties that may be unfamiliar are:

- Yield strength: stress at which plastic strain is 0.2% or 2 parts in 10^3
- Microyield strength: stress at which plastic strain is 1 ppm or 1 microstrain
- Steady state thermal distortion coefficient: distortion per unit of power in micrometers/watt
- Transient thermal distortion coefficient: related to the time in seconds for a distortion to reach steady state, per unit volume and unit temperature

Aluminum (Al)

The most readily available of the metals for use in mirrors and structures, Al is available as wrought or cast and heat treatable or not. Properties and heat treat temperatures are given in Table 5.35.

The most versatile alloy is the wrought, heat treatable 6061, the commonly used alloy for optical systems. The similar, but purer 6081 yields higher quality diamond turned (SPDT) surfaces. For either alloy the preferred fabrication sequence, as described in a study by NASA Goddard Space Flight Center [5.401], is: rough shape (forging preferred), solution treat, quench (in polyalkylene glycol/water solution) and age, followed by further machining, re-age, thermal cycle and then complete optical fabrication. The aircraft alloys 2024 and 7075 have

Table 5.34 Properties of mirror and structural materials

Preferred	Density ρ (g/cm ³) Small	Young's modulus E (GPa) Large	Poisson's ratio ν Small	Yield strength σ_{YS} (MPa) Large	Microyield strength MYS (MPa) Large	Thermal expansion α (10 ⁻⁶ /K) Small	Thermal conduct- ivity k (W/m K) Large	Specific heat C_p (W s/kg K) Large	Thermal diffusivity D (10 ⁻⁶ m ² /s) Large	Thermal distortion coefficients	
										Steady state α/k (μ m/W) Small	Transient α/D (s/m ² K) Small
Pyrex TM 7740	2.23	63	0.2	—	—	3.3	1.13	1050	0.65	2.92	5.08
Fused silica	2.19	72	0.17	—	—	0.5	1.4	750	0.85	0.36	0.59
ULE TM fused silica	2.21	67	0.17	—	—	0.003	1.31	766	0.78	0.02	0.04
Zerodur [®]	2.53	91	0.24	—	—	0.05	1.64	821	0.77	0.03	0.07
Aluminum: 6061-T6	2.70	68	0.33	276	160.0	22.5	167.0	896	69.0	0.13	0.33
MMC: 30% SiC/Al	2.91	117	0.29	190	>150.0	12.4	123.0	870	57.0	0.10	0.22
AlBeMet [®] 162	2.10	193	0.17	320	—	13.9	210	1506	62.3	0.06	0.22
Beryllium: O-30	1.85	303	0.043	296	35.0	11.38	216.0	1925	57.2	0.05	0.20
Beryllium: I-220-H	1.85	303	0.043	531	90.0	11.3	216.0	1925	57.2	0.05	0.20
Copper: OF	8.94	117	0.343	195	12.0	17.5	391.0	385	113.6	0.04	0.15
GlidCop [®] AL-15	8.75	131	0.35	352	>100.0	16.6	365.0	384	108.6	0.05	0.15
Invar 36	8.05	141	0.259	276	70.0	1.0	10.4	515	2.6	0.10	0.38
Super invar	8.13	148	0.26	280	75.0	0.3	10.5	515	2.5	0.03	0.12
Molybdenum	10.21	324	0.293	600	280.0	5.0	140.0	247	55.5	0.04	0.09
Silicon	2.33	131	0.42	—	—	2.6	156.0	710	94.3	0.02	0.03
SiC: sintered	3.16	420	0.14	—	—	2.0	190.0	700	85.9	0.02	0.04
SiC: CVD beta	3.21	465	0.21	—	—	2.4	198.0	733	84.2	0.01	0.03
SiC: RB-30% Si	2.89	330	0.24	—	—	2.6	155.0	670	80.0	0.02	0.03
Cesic [®]	2.65	197	0.25	—	—	2.0	125	700	71.5	0.02	0.03
SuperSiC [®]	2.55	214	—	—	—	—	143	—	—	—	—
Si infiltrated SuperSiC [®]	2.93	232	—	—	—	—	158	—	—	—	—
Stainless steel: 304	7.90	193	0.27	241	>300.0	14.7	16.2	500	4.2	0.91	3.63
Stainless steel: 416	7.70	215	0.283	950	>300	8.5	24.9	460	7.0	0.34	1.21
Stainless steel: 17-4PH	7.80	200	0.28	870	>300	10.4	22.2	460	6.2	0.47	1.68
Titanium: 6Al4V	4.43	114	0.31	830	>150.0	8.8	7.3	560	2.9	1.21	3.03
Electroless Ni: 12%	7.9	140	0.41	615	—	14.0	5.0	—	—	—	—
Electroplated Al	2.64	69	0.33	—	—	22.7	218	900	96.3	—	—

PyrexTM and ULETM are registered trademarks of Corning Glass Works, Corning, N.Y., USA; Zerodur[®] is a registered trademark of Schott Glaswerke, Mainz, Germany; GlidCop[®] is a registered trademark of OMG Americas Corp., Research Triangle Park, N.C., USA; Cesic[®] is a registered trademark of ECM Ingenieur-Unternehmen, Munich, Germany; SuperSiC[®] is a registered trademark of POCO Graphite, Inc., Decatur, TX, USA



Table 5.35 Properties of representative aluminum alloys

	Density ρ (g/cm ³)	Young's modulus E (GPa)	Yield strength σ_{YS} (MPa)	Thermal expansion ^a α (10 ⁻⁶ /K)	Thermal conductivity K (W/m K)	Anneal temperature (°C)	Solution treatment temperature (°C)	Aging temperature (4–5 h) (°C)
Wrought alloys								
1100-O	2.71	69	34	23.6	222	343	—	—
5056-O	2.64	71	152	24.1	117	415	—	—
2024-T6	2.77	72	393	23.2	151	385	500	190
6061-T6	2.70	68	276	23.6	167	415	530	170
7075-T6	2.81	72	462	23.6	130	415	470	120
Cast Alloys								
A201-T7	2.80	71	414	19.3	121	315	528	188
A356-T6	2.685	72	186	21.5	159	315	538	152
A357-T6	2.68	72	276	21.6	152	315	540	170
713	2.81	67	152	23.9	140	450 ^b	N/A	120 ^c
771-T7	2.823	71	372	24.7	138	—	588	140 ^d
^a 20–100 °C								
^b Stress relief temperature of 6 h and air cool yields stress free full strength component								
^c for 16 h; alternate treatment: room temperature for 21 d								
^d for 15 h								

much higher strength, but are harder to stabilize and undergo dimensional changes during aging [5.402]. For large components with section thickness greater than 7.0 cm, the non-heat treatable alloys of the 5000 series are recommended, e.g., 5056 [5.403]. These alloys can be repeatedly annealed and still retain yield strengths of 150 MPa or more. For applications requiring minimal strength, 1100 is the preferred alloy for SPDT since it is 99% Al with no alloying that forms hard particles.

Cast alloys are generally more difficult to stabilize. The most commonly cast Al is A356, A357 being a cleaner version with tighter specifications. A201 has lower alloying and when carefully cast can be SPDT finished directly. Alloys 713 (Tenzaloy) and 771 (Precedent 71A) are Al/Zn alloys that machine well and can be precision cast. Tenzaloy is usually not heat treated whereas Precedent 71A is. Precedent is the most stable of these alloys but requires special considerations in casting i.e., silicon free crucibles, chlorine purging and ceramic filters. Castings can be hot isostatically pressed (HIP'ed) to close porosity and increase properties, but is only cost effective in production.

Mirrors can be fabricated with bare surfaces finished by SPDT and/or polishing, but are usually plated with either electroless nickel (EN) that can be turned and/or polished, or electroplated Al for elimination of

bimetallic thermal distortions [5.404]. Systems with Al mirrors and structures are a low cost approach for fully athermalized performance.

Beryllium (Be)

Beryllium is the lightest structural metal, with high elastic modulus, thermal conductivity and infrared reflectance, and is transparent to X-rays. The primary optical applications are space optics and structures, applications requiring low moment of inertia, and X-ray windows. All Be materials are made from powder, with HIP the preferred consolidation method [5.405]. The optical grade, O-30, is HIP'ed from spherical powder produced by gas atomization [5.406], is the most homogeneous and isotropic Be material, and is recommended for bare polished mirrors. Instrument grade I-220H, as used for the 1.1 m chopping secondary mirrors of the VLT telescopes [5.407], is a fine grained, high strength Be material made from HIP'ed impact ground powder that is often used for electroless nickel coated mirrors requiring higher strength than can be obtained with O-30 or other grades, e.g., S-200FH, S-65 or I-70H. Prior to the availability of O-30, I-70H, HIP'ed from impact ground powder, was the preferred mirror material as used for SIRTf [5.408], an all Be telescope launched in 2003.

The typical fabrication method for Be components requires machining the shape from a HIP'ed billet,

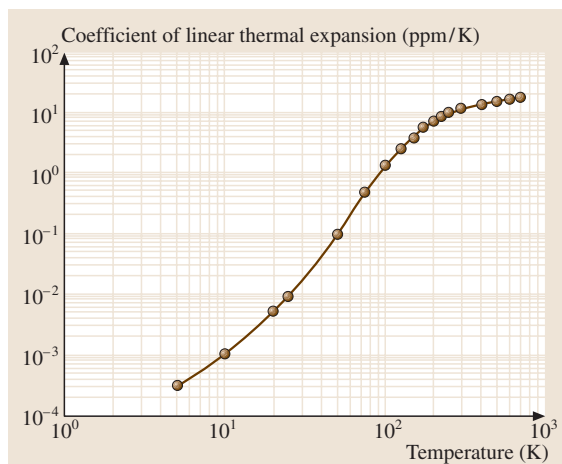


Fig. 5.85 Coefficient of linear thermal expansion of I-70H HIP'ed beryllium. Low-temperature data from Swenson [5.412]. High-temperature data from Touloukian et al. [5.413]

followed by chemical etch and anneal before finish machining. There are other near-net-shape fabrication techniques that can produce open or closed back mirrors or even integral cooling channels [5.405, 406, 409–411], but as yet, none of these methods has been extensively employed. Beryllium mirrors can be polished bare or with an EN coating that can be SPDT finished and/or polished.

Although Be has high CTE of 11.4 ppm/K at room temperature, it drops rapidly to 0.1 ppm/K at 50 K as shown in Fig. 5.85 [5.412, 413]. Combined with its exceptionally high thermal conductivity, peaking at 150 K with 300 W/mK, as shown in Fig. 5.83, Be has exceptional cryogenic performance when bare polished and can be made very stable when careful fabrication methodology is strictly followed [5.414].

AlBeMet[®] 162 is an alloy of 62% Be, 38% Al, that has been used for space structural applications [5.415] but is now also being used as a mirror substrate material [5.411]. It is typically produced from atomized powder and extruded. It can be machined and welded much like Al and in mirror form must be plated to achieve an optical surface. Modifications of this alloy have been cast, but with mixed success due to porosity and segregation of the Be and Al phases.

Handling Be materials in solid form poses no health risk but inhalation of airborne Be may cause a serious lung disorder in susceptible individuals. The Brush Wellman website [5.416] contains extensive information on

Be health and safety, specifications for all Be grades and a complete guide to designing with Be.

Copper

Copper has traditionally been used for high energy applications because of its high thermal conductivity. The two principal Cu alloys in use are commercially pure oxygen free copper (OFC) and dispersion strengthened GlidCop[®]. While OFC has the highest thermal conductivity, it loses most of its strength when brazed. It is this loss of strength that drove the selection of GlidCop[®], a high strength Cu alloy with near OFC thermal conductivity, for brazed heat exchanger components [5.417, 418]. The preferred grade is GlidCop[®] Al-15 LOX, dispersion strengthened with 0.3% Al₂O₃ in the form of 5–15 nm particles that retains > 70% of strength at temperatures up to 800 °C. Other grades have higher Al₂O₃ content but are more difficult to braze. Brazing is readily accomplished, but takes some care. Several brazing methods have been demonstrated [5.417, 418] both without and with a pure Cu plated layer, and with either Ti/Cu/Ni or Au/Cu braze alloys. The former method seems simpler. Mirrors made from GlidCop[®] are usually EN plated and polished.

Ferrous Alloys

There are many uses for ferrous alloys in optical systems, but few mirrors are fabricated from these alloys. Among these are 17-4 PH precipitation hardening stainless and AISI 1010 mild steel [5.417]. The processing of 17-4 PH is simple when material is obtained in the solution treated and cold finished form. After machining to shape, a precipitation treatment of one hour at 480 °C with air cool to condition H900 provides a dimensionally stable substrate that can then be ground, thermal cycled, lapped and polished bare to very low roughness levels. Mild steel in the fully annealed condition has good strength and reasonably good thermal conductivity, but requires an EN coating for polishability.

Invar 36 and super invar have the lowest thermal expansion and therefore best thermal distortion performance of any metal in the temperature range of 0–80 °C. To achieve this low CTE and dimensional stability at the same time, the material must have < 0.02% carbon, low silicon and manganese and be given the three-step heat treatment developed by Lement et al. [5.419]. The recommended treatment is:

1. heat to 830 °C for 30 min and quench in polyalkylene glycol/water solution,
2. heat to 315 °C for 1 h and air cool,
3. heat to 95 °C for at least 48 h and air cool.

The fabrication/heat treat schedule then follows [5.417]. Shaping is be done before step 1) and finish machining after step 2), after which step 2) is repeated. Thermal cycling, not exceeding 300 °C, should also be performed in the final stages of fabrication. Step 3) should only be carried out after all fabrication steps are complete, and then that temperature, 95 °C, should never be exceeded.

Super invar has a somewhat lower expansion coefficient by virtue of a substitution of about 5% cobalt for some of the normal 36% Ni. It is fabricated and heat treated the same as regular invar 36, but undergoes a phase change at low temperatures that results in a sudden expansion and substantial increase in CTE. The phase change temperature is normally $\leq 80\text{ }^{\circ}\text{C}$, but if the material is to be used below about $-40\text{ }^{\circ}\text{C}$, it should be dilatometrically tested.

Other Metals

There are many other metals that have been utilized in optical systems, chief among them being molybdenum (Mo). Many water cooled and uncooled Mo mirrors have been fabricated for use in high power laser optical systems. The preferred grade is low carbon vacuum arc cast, where fine grain material can be brazed and polished bare. Other grades (Mo 0.5% Ti

and TZM) containing titanium (Ti) and/or zirconium (Zr) have fine particulates that make brazing more difficult and a fine polish impossible. Sputtered Mo coatings on Mo mirrors have produced the best finishes. For further information on Mo and its fabrication see the many papers in the proceedings of the Laser-Damage Conferences [5.420–424].

Titanium has also been used for some mirrors, primarily in high stress, low mass applications. In the 6Al4V alloy it has a very high strength-to-weight ratio. Very little has been published, but it is known to be difficult to polish well and also difficult to plate with Ni. Various schemes to improve plating adhesion include grit blasting the substrate, putting on a thin Ni or other electroplated metal strike followed by a diffusion anneal, and other proprietary methods used by various plating companies.

Polishable Coatings

Electroless nickel (EN) is the primary coating used to make metal mirror surfaces more polishable (including SPDT). It has been used on a variety of substrate materials including Al, Be, Cu, invar, steels, Ti and others. It is an alloy of Ni and phosphorous (P) and its properties depend on a number of factors including % P and post-plate bake time and temperature [5.425]. It deposits

Table 5.36 Major types of silicon carbide

SiC type	Structure/composition ^a	Density	Fabrication process	Remarks
Hot pressed	$\geq 98\% \alpha + \text{others}^b$	$> 98\%$	Powder pressed in heated dies	Simple shapes only
Hot isostatic pressed	$> 98\% \text{ SiC} + \text{others}$	$> 99\%$	HIP encapsulated preform	Complex shapes can be formed; high equipment cost for large sizes
Chemically vapor deposited	100% β	$\geq 99.9\%$	Deposition on hot mandrel	Thin shell or plate forms and coatings
Sintered	$\geq 98\% \alpha + \text{others}$	$> 98\%$	Preform with sintering aids vacuum fired	Preform machined from cold pressed billet; $\approx 15\%$ shrinkage in firing; CVD SiC clad for mirrors
Reaction bonded/sintered	$50\text{--}90\% \alpha + \approx 2\% \beta + \text{Si}$	$> 99\%$	Prefired porous preform fired to infiltrate with Si	Complex shapes can be formed; properties depend on % Si, and SiC grain size; CVD Si or SiC clad for mirrors
Cesic [®]	$50\text{--}60\% \alpha + 20\text{--}30\% \text{ Si} + 10\text{--}20\% \text{ C}$	$> 99\%$	Porous carbon/carbon preform fired to infiltrate with Si partly reacted to SiC	Complex shapes can be formed; properties depend on fibers and SiC:Si:C proportions; slurry clad for mirrors
SuperSiC [®]	100% β	$\approx 80\%$	Graphite preform converted to SiC	Complex shapes can be formed; CVD SiC clad to seal
Si-infiltrated SuperSiC [®]	$82\% \beta + \text{Si}$	$> 96\%$	Graphite preform converted to SiC	Complex shapes can be formed; CVD SiC clad for mirrors

Cesic[®] is a registered trademark of ECM Ingenieur-Unternehmen, Munich, Germany, for its short carbon fiber reinforced SiC
SuperSiC[®] is a registered trademark of POCO Graphite, Inc., Decatur, TX, USA for its converted graphite grades of SiC

^a α is hexagonal SiC, β is face centered cubic SiC
^b others are sintering aids such as Al₂O₃, Al, B, Be and C

as an amorphous coating at P contents of 6–15%, but with baking at moderate temperatures of 150–190 °C, the lower concentration alloys devitrify. Alloys above 10.5% P remain amorphous to greater than 300 °C and with a 2 h, 190 °C bake, they are ideal for SPDT. Lower concentration coatings are harder and are better for polishing. The CTE of EN varies inversely with % P from approximately 18 ppm/K at 6% P to 9 ppm/K at 13% P, all after baking.

All EN coatings have residual stresses [5.426] that come from three sources: intrinsic from deposition process, bimetallic from cooling from plating temperature (≈ 90 °C) with CTE mismatch to substrate, and from shrinkage during post plate baking. The stresses can be measured [5.427] and controlled. For optical applications, great care must be taken to control the process parameters tightly to prevent variation in % P as plating progresses and to provide a cover on the bath to prevent contamination. Further details on processing are given by *Howells* and *Paquin* [5.417].

A more recent finishing method for virtual elimination of residual stress and bimetallic stress is electrolytic Al plating, a 99.9% pure Al coating called *Alumiplat*TM [5.428] that can be diamond turned. It was used on cryogenic Al mirrors for the Gemini Near Infrared Spectrometer with great success [5.429]. It has also been applied to Be mirrors.

Silicon Carbide

There are many types of silicon carbide (SiC) as can be seen from those shown in Table 5.34. The differences are substantial, both in the properties and in the fabrication methods as shown in Table 5.36.

Silicon carbide exists in two forms: alpha (α), the more common form with hexagonal crystal structure and anisotropic CTE and elastic properties, and beta (β), the face centered cubic form, isotropic in CTE but anisotropic in elastic properties. The α form is stable at all temperatures while β converts to α above about 1600 °C. Most properties of the two forms are within a few percent of each other. The fabrication methods described in Table 5.36 for the many SiC types yield combinations of α , β , Si and C that provide substantial differences in the usefulness and/or applicability for optical components.

Hot pressed SiC has been used for substrates in heat exchanger mirrors but while it is not normally used for optics, some applications have been reported [5.430]. HIP methods can provide small complex components, but have not been used because the extremely high temperatures and pressures needed require a large capital investment for an autoclave of practical size. Chem-

ically vapor deposited (CVD) SiC [5.431] is extremely pure β , so it can be polished to very low roughness levels [5.432, 433], but can only be deposited as thin shells or plates. Many small mirrors have been fabricated, but attempts to produce large mirrors with lightweight back structure have been only partially successful. CVD SiC is used extensively to coat multiphase SiC types and has many other uses.

Sintered SiC has been used to fabricate a number of different optical forms and systems [5.434] including the 3.5 m diameter Herschel primary mirror of twelve brazed segments and the RocSat2 all SiC telescope with 600 mm primary. Fabrication consists of cold isostatic pressing a billet of α SiC with sintering aids, machining the part to near finish quality allowing for shrinkage ($\approx 15\%$, but known precisely) vacuum sintering and final grinding of critical surfaces. The finished surface contains approximately 2% voids of 2 μ m average size and is typically coated with CVD SiC for final optical surface finishing.

Reaction bonded SiC, also known as reaction sintered or silconized SiC, is a two phase material consisting of interpenetrating networks of Si and α -SiC. Properties depend on the amount of Si that can be 10 to 50%. The process is flexible as practiced by numerous companies. The simplest form, siliconized SiC [5.435], uses a cast preform that is prefired, machined and final fired in the presence of Si to infiltrate to near full density. Higher SiC content provides improved properties, but requires more steps in the process. Carbonaceous material can be added at several points in the process, but these can lead to inhomogeneity. Preforms can be fabricated from reusable tooling to reduce cost of production parts. Meter-class mirrors have been fabricated [5.436]. Finishing can be performed on the bare two phase surface, but is usually performed on an amorphous PVD Si or CVD SiC surface [5.435, 437].

The multiphase short carbon fiber reinforced ceramic Cescic[®] [5.438] is an innovative approach to relatively rapid, low cost SiC optics and structures. Short fiber carbon felt is processed to graphitize it to a lightweight porous body that is readily machined. It is then infiltrated with Si that reacts with the carbon matrix and surfaces of the carbon fibers to form SiC. There is no noticeable shrinkage in the conversion process. Surfaces are not suitable for polishing, and components are slurry clad with material of the same composition but with a much finer microstructure allowing polishing to surface roughness of 2 nm rms. Mechanical properties, as shown in Table 5.34, are somewhat lower than other SiC types.



The SuperSiC[®] materials [5.439] are perhaps the simplest and quickest to get from raw material to finished optical component. A proprietary graphite body is machined to near finish dimensions and then converted

to pure β SiC that has about 18% porosity. The porosity can be filled with Si. In either case, a coating of CVD SiC is applied to all surfaces. A complete list of properties is not yet available for this material.

5.10 Selected Data

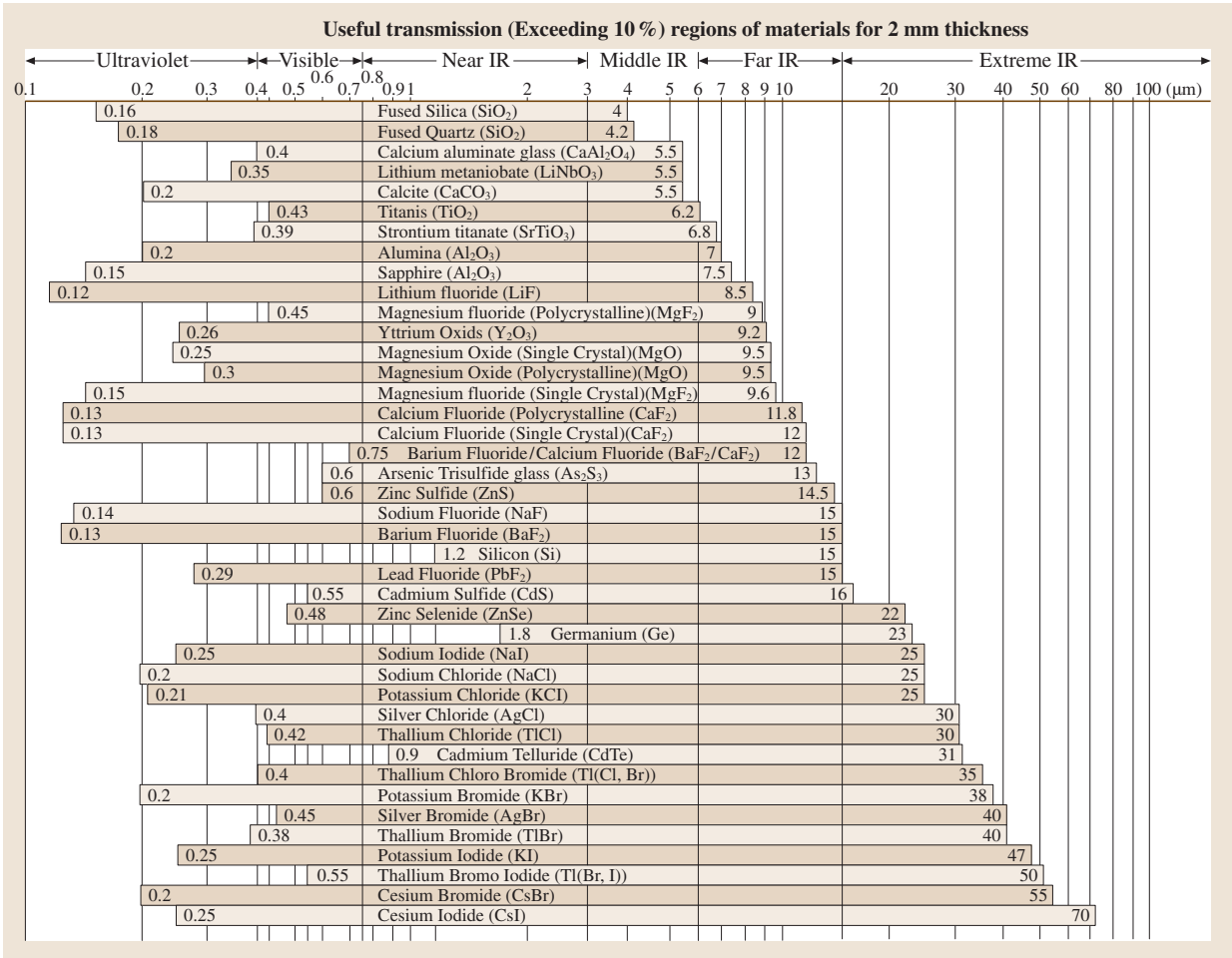


Fig. 5.86 Overview on optical material classes for different wavelength applications (After [5.440])



Table 5.37 Commercially available optical glasses by code, glass type and manufacture (H = Hoya, O = Ohara, and S = SCHOTT) [5.441–443]

Code	Glass type	n_d	ν_d	Producer	Code	Glass type	n_d	ν_d	Producer
434950	N-FK 56	1.43430	95.0	S	571530	S-BAL 3	1.57135	53.0	O
439950	S-FPL 53	1.43875	95.0	O	573575	N-BAK 1	1.57250	57.5	S
456903	S-FPL 52	1.45600	90.3	O	573578	S-BAL 11	1.57250	57.8	O
457903	FCD 10	1.45650	90.3	H	575415	S-TIL 27	1.57501	41.5	O
487702	S-SL 5	1.48749	70.2	O	580539	N-BALF 4	1.57956	53.9	S
487704	FC 5	1.48749	70.4	H	581407	S-TIL 25	1.58144	40.7	O
487704	N-FK 5	1.48749	70.4	S	581408	N-LF 5	1.58144	40.8	S
487845	N-FK 51	1.48656	84.5	S	581409	E-FL 5	1.58144	40.9	H
497816	FCD 1	1.49700	81.6	H	581409	LF 5	1.58144	40.9	S
497816	S-FPL 51	1.49700	81.6	O	583464	BAM 3	1.58267	46.4	O
497816	N-PK 52	1.49700	81.6	S	583466	N-BAF 3	1.58272	46.6	S
498670	N-BK 10	1.49782	67.0	S	583594	S-BAL 42	1.58313	59.4	O
501564	K 10	1.50137	56.4	S	583595	BACD 12	1.58313	59.5	H
508612	N-ZK 7	1.50847	61.2	S	583595	M-BACD 12	1.58313	59.5	H
511604	K 7	1.51112	60.4	S	589612	S-BAL 35	1.58913	61.2	O
516641	S-BSL 7	1.51633	64.1	O	589613	BACD 5	1.58913	61.3	H
517522	E-CF6	1.51742	52.2	H	589613	M-BACD 5N	1.58913	61.3	H
517524	S-NSL 36	1.51742	52.4	O	589613	N-SK 5	1.58913	61.3	S
517642	BSC 7	1.51680	64.2	H	592684	N-PSK 57	1.59240	68.4	S
517642	N-BK 7	1.51680	61.2	S	593353	S-FTM 16	1.59270	35.3	O
518590	E-C 3	1.51823	59.0	H	593355	FF 5	1.59270	35.5	H
518590	S-NSL 3	1.51823	59.0	O	596392	E-F 8	1.59551	39.2	H
522595	N-K 5	1.52249	59.5	S	596392	S-TIM 8	1.59551	39.2	O
522598	S-NSL 5	1.52249	59.8	O	603380	E-F 5	1.60342	38.0	H
523515	N-KF 9	1.52346	51.5	S	603380	S-TIM 5	1.60342	38.0	O
529770	N-PK 51	1.52855	77.0	S	603380	F 5	1.60342	38.0	S
532488	E-FEL 6	1.53172	48.8	H	603606	N-SK 14	1.60311	60.6	S
532489	S-TIL 6	1.53172	48.9	O	603607	BACD 14	1.60311	60.7	H
532489	N-LLF 6	1.53169	48.9	S	603607	S-BSM 14	1.60311	60.7	O
540595	S-BAL 12	1.53996	59.5	O	603655	S-PHM 53	1.60300	65.5	O
540597	N-BAK 2	1.53996	59.7	S	606437	S-BAM 4	1.60562	43.7	O
541472	E-FEL 2	1.54072	47.2	H	606437	N-BAF 4	1.60568	43.7	S
541472	S-TIL 2	1.54072	47.2	O	606637	LBC 3N	1.60625	63.7	H
547536	N-BALF 5	1.54739	53.6	S	607567	BACD 2	1.60738	56.7	H
548458	E-FEL 1	1.54814	45.8	H	607567	N-SK 2	1.60738	56.7	S
548458	S-TIL 1	1.54814	45.8	O	607568	S-BSM 2	1.60738	56.8	O
548458	LLF 1	1.54814	45.8	S	609466	N-BAF 52	1.60863	46.6	S
548459	N-LLF 1	1.54814	45.9	S	613370	E-F 3	1.61293	37.0	H
551495	SBF 1	1.55115	49.5	H	613370	PBM 3	1.61293	37.0	O
552635	N-PSK 3	1.55232	63.5	S	613443	BPM 51	1.61340	44.3	O
558540	N-KZFS 2	1.55836	54.0	S	613443	KZFSN 4	1.61340	44.3	S
564607	S-BAL 41	1.56384	60.7	O	613444	E-ADF 10	1.61310	44.4	H
564608	BACD 11	1.56384	60.8	H	613445	N-KZFS 4	1.61336	44.5	S
564608	N-SK 11	1.56384	60.8	S	613586	BACD 4	1.61272	58.6	H
567428	E-FL 6	1.56732	42.8	H	613586	N-SK 4	1.61272	58.6	S

Table 5.37 Commercially available optical glasses by code, glass type and manufacture, cont.

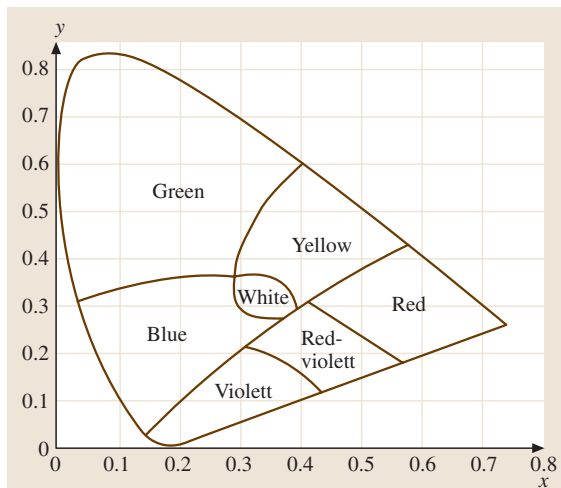
Code	Glass type	n_d	v_d	Producer	Code	Glass type	n_d	v_d	Producer
567428	PBL 26	1.56732	42.8	O	613587	S-BSM 4	1.61272	58.7	O
569560	BAC 4	1.56883	59.0	H	614550	BSM 9	1.61405	55.0	O
569560	N-BAK 4	1.56883	56.0	S	617366	F 4	1.61659	36.6	S
569563	S-BAL 14	1.56883	56.3	O	618498	S-BSM 28	1.61772	49.8	O
569712	N-PSK 58	1.56907	71.2	S	618498	N-SSK 8	1.61773	49.8	S
571508	S-BAL 2	1.57099	50.8	O	618634	PCD 4	1.61800	63.4	H
618634	S-PHM 52	1.61800	63.4	O	667330	S-TIM 39	1.66680	33.0	O
620363	E-F 2	1.62004	36.3	H	667483	BAF 11	1.66672	48.3	H
620363	S-TIM 2	1.62004	36.3	O	667483	S-BAH 11	1.66672	48.3	O
620364	F 2	1.62004	36.4	S	670393	BAH 32	1.66998	39.3	O
620364	N-F 2	1.62005	36.4	S	670471	N-BAF 10	1.67003	47.1	S
620603	BACD 16	1.62041	60.3	H	670472	BAF 10	1.67003	47.2	H
620603	S-BSM 16	1.62041	60.3	O	670473	S-BAH 10	1.67003	47.3	O
620603	N-SK 16	1.62041	60.3	S	673321	S-TIM 25	1.67270	32.1	O
620622	ADC 1	1.62000	62.2	H	673322	E-FD 5	1.67270	32.2	H
620635	N-PSK 53	1.62014	63.5	S	673322	SF 5	1.67270	32.2	S
621603	SK 51	1.62090	60.3	S	673323	N-SF 5	1.67271	32.3	S
622532	BSM 22	1.62230	53.2	O	678506	LACL 9	1.67790	50.6	H
622533	N-SSK 2	1.62229	53.3	S	678507	S-LAL 56	1.67790	50.7	O
623569	E-BACD 10	1.62280	56.9	H	678549	LAKL 12	1.67790	54.9	S
623570	S-BSM 10	1.62280	57.0	O	678552	N-LAK 12	1.67790	55.2	S
623570	N-SK 10	1.62278	57.0	S	678553	S-LAL 12	1.67790	55.3	O
623580	N-SK 15	1.62296	58.0	S	678555	LAC 12	1.67790	55.5	H
623581	BACD 15	1.62299	58.1	H	689311	S-TIM 28	1.68893	31.1	O
623582	S-BSM 15	1.62299	58.2	O	689312	E-FD 8	1.68893	31.2	H
624471	E-BAF 8	1.62374	47.1	H	689313	N-SF 8	1.68894	31.3	S
626357	E-F 1	1.62588	35.7	H	691547	LAC 9	1.69100	54.7	H
626357	S-TIM 1	1.62588	35.7	O	691547	N-LAK 9	1.69100	54.7	S
638424	N-KZFSN 11	1.63775	42.4	S	691548	S-LAL 9	1.69100	54.8	O
639449	S-BAM 12	1.63930	44.9	O	694508	LACL 5	1.69350	50.8	H
639509	S-BSM 18	1.63854	55.4	O	694508	LAL 58	1.69350	50.8	O
639554	N-SK 18	1.63854	55.4	S	694532	M-LAC 130	1.69350	53.2	H
639555	BACD 18	1.63854	55.5	H	694532	S-LAL 13	1.69350	53.2	O
640345	S-TIM 27	1.63980	34.5	O	694533	LAC 13	1.69350	53.3	H
640346	E-FD 7	1.63980	34.6	H	694533	LAKN 13	1.69350	53.3	S
640601	S-BSM 81	1.64000	60.1	O	697485	LAFL 2	1.69700	48.5	H
640601	N-LAK 21	1.64049	60.1	S	697485	LAM 59	1.69700	48.5	O
640602	LACL 60	1.64000	60.2	H	697554	N-LAK 14	1.69680	55.4	S
648338	E-FD 2	1.64769	33.8	H	697555	LAC 14	1.69680	55.5	H
648338	S-TIM 22	1.64769	33.8	O	697555	S-LAL 14	1.69680	55.5	O
648338	SF 2	1.64769	33.8	S	699301	E-FD 15	1.69895	30.1	H
649530	E-BACED20	1.64850	53.0	H	699301	S-TIM 35	1.69895	30.1	O
649530	S-BSM 71	1.64850	53.0	O	699301	SF 15	1.69895	30.1	S
650557	LACL 2	1.65020	55.7	H	699302	N-SF 15	1.69892	30.2	S
651562	S-LAL 54	1.65100	56.2	O	700481	S-LAM 51	1.70000	48.1	O
651669	N-LAK 22	1.65113	55.9	S	702402	BAFD 15	1.70200	40.2	H

Table 5.37 Commercially available optical glasses by code, glass type and manufacture, cont.

Code	Glass type	n_d	v_d	Producer	Code	Glass type	n_d	v_d	Producer
652450	N-BAF 51	1.65224	45.0	S	702412	BAFD 7	1.70154	41.2	H
652584	LAC 7	1.65160	58.4	H	702412	S-BAH 27	1.70154	41.2	O
652585	S-LAL 7	1.65160	58.5	O	704394	N-BASF 64	1.70400	39.4	S
652585	N-LAK 7	1.65160	58.5	S	706302	N-SF 64	1.70591	30.2	S
654396	E-ADF 50	1.65412	39.6	H	713538	N-LAK 8	1.71300	53.8	S
654396	KZFSN 5	1.65412	39.6	S	713539	LAC 8	1.71300	53.9	H
654397	BPH 5	1.65412	39.7	O	713539	S-LAL 8	1.71300	53.9	O
658509	BACED 5	1.65844	50.9	H	717295	E-FD 1	1.71736	29.5	H
658509	S-BSM 25	1.65844	50.9	O	717295	PBH 1	1.71736	29.5	O
658509	N-SSK 5	1.65844	50.9	S	717295	SF 1	1.71736	29.5	S
664360	N-BASF 2	1.66446	36.0	S	717296	N-SF 1	1.71736	29.6	S
717479	S-LAM 3	1.71700	47.9	O	762401	S-LAM 55	1.76200	40.1	O
717480	LAF 3	1.71700	48.0	H	773496	TAF 1	1.77250	49.6	H
717480	N-LAF 3	1.71700	48.0	S	773496	S-LAH 66	1.77250	49.6	O
720347	BPH 8	1.72047	34.7	O	773496	N-LAF 34	1.77250	49.6	S
720420	LAM 58	1.72000	42.0	O	785257	FD 110	1.78472	25.7	H
720437	S-LAM 52	1.72000	43.7	O	785257	S-TIH 11	1.78472	25.7	O
720460	LAM 61	1.72000	46.0	O	785258	SF 11	1.78472	25.8	S
720502	S-LAL 10	1.72000	50.2	O	785261	FDS 30	1.78470	26.1	H
720503	LAC 10	1.72000	50.3	H	785261	N-SF 56	1.78470	26.1	S
720506	N-LAK 10	1.72003	50.6	S	785261	SF 56A	1.78470	26.1	S
722292	S-TIH 18	1.72151	29.2	O	785263	S-TIH 23	1.78470	26.3	O
723380	BAFD 8	1.72342	38.0	H	786439	NBFD 11	1.78590	43.9	H
723380	S-BAH 18	1.72342	38.0	O	786441	N-LAF 33	1.78582	44.1	S
724381	BASF 51	1.72373	38.1	S	786442	S-LAH 51	1.78590	44.2	O
728283	E-FD 10	1.72825	28.3	H	788474	S-LAH 64	1.78800	47.4	O
728284	SF 10	1.72825	28.4	S	788475	TAF 4	1.78800	47.5	H
728285	S-TIH 10	1.72825	28.5	O	788475	N-LAF 21	1.78800	47.5	S
728285	N-SF 10	1.72828	28.5	S	795454	TAF 2	1.79450	45.4	H
729547	TAC 8	1.72916	54.7	H	795455	N-LAF 32	1.79457	45.5	S
729547	S-LAL 18	1.72906	54.7	O	800422	S-LAH 52	1.79952	42.2	O
731405	M-LAF 81	1.73077	40.5	H	800423	NBFD 12	1.79950	42.3	H
734511	TAC 4	1.73400	51.1	H	801350	S-LAM 66	1.80100	35.0	O
734515	S-LAL 59	1.73400	51.5	O	801351	N-LASF 45	1.80100	35.1	S
740283	PBH 3	1.74000	28.3	O	804396	S-LAH 63	1.80440	39.6	O
741278	E-FD 13	1.74077	27.8	H	804465	TAF 3	1.80420	46.5	H
741278	S-TIH 13	1.74077	27.8	O	804465	N-LASF 44	1.80420	46.5	S
741526	TAC 2	1.74100	52.6	H	804466	S-LAH 65	1.80400	46.6	O
741527	S-LAL 61	1.74100	52.7	O	805254	S-TIH 6	1.80518	25.4	O
743492	NBF 1	1.74330	49.2	H	805254	N-SF 6	1.80518	25.4	S
743493	M-NBF 1	1.74330	49.3	H	805254	SF 6	1.80518	25.4	S
743493	S-LAM 60	1.74320	49.3	O	805255	FD 60	1.80518	25.5	H
744448	S-LAM 2	1.74400	44.8	O	805396	NBFD 3	1.80450	39.6	H
744449	LAF 2	1.74400	4.9	H	806333	NBFD 15	1.80610	33.3	H
744449	N-LAF 2	1.74397	44.9	S	806406	N-LASF 43	1.80610	40.6	S
749348	N-LAF 7	1.74950	34.8	S	806407	M-NBFD 130	1.80610	40.7	H

Table 5.37 Commercially available optical glasses by code, glass type and manufacture, cont.

Code	Glass type	n_d	v_d	Producer	Code	Glass type	n_d	v_d	Producer
750350	E-LAF 7	1.74950	35.0	H	806407	NBFD 13	1.80610	40.7	H
750350	LAFN 7	1.74950	35.0	S	806409	S-LAH 53	1.80610	40.9	O
750353	LAM 7	1.74950	35.3	O	808228	S-NPH 1	1.80809	22.8	O
752251	FF 8	1.75211	25.1	H	815370	M-NBFD 82	1.81474	37.0	H
754524	N-LAK 33	1.75398	52.4	S	816445	TAFD 10	1.81550	44.5	H
755274	N-SF 4	1.75513	27.4	S	816466	TAF 5	1.81600	46.6	H
755275	E-FD 4	1.75520	27.5	H	816466	S-LAH 59	1.81600	46.6	O
755275	S-TIH 4	1.75520	27.5	O	834372	S-LAH 60	1.83400	37.2	O
755276	SF 4	1.75520	27.6	S	834373	NBFD 10	1.83400	37.3	H
755523	TAC 6	1.75500	52.3	H	834373	N-LASF 40	1.83404	37.3	S
755523	S-YGH 51	1.75500	52.3	O	835427	S-LAH 55	1.83481	42.7	O
757477	NBF 2	1.75700	47.7	H	835430	TAFD 5	1.83500	43.0	H
757478	S-LAM 54	1.75700	47.8	O	835431	N-LASF 41	1.83501	43.1	S
762265	S-TIH 14	1.76182	26.5	O	847236	SFL 57	1.84666	23.6	S
762265	SF 14	1.76182	26.5	S	847238	FDS 90	1.84666	23.8	H
762266	FD 140	1.76182	26.6	H	847238	S-TIH 53	1.84666	23.8	O
847238	SF 57	1.84666	23.8	S	850322	LASFN 9	1.85025	32.2	S
881410	N-LASF 31	1.88067	41.0	S	883408	TAFD 30	1.88300	40.8	H
883408	S-LAH 58	1.88300	40.8	O	901315	LAH 78	1.90135	31.5	O
923209	E-FDS 1	1.92286	20.9	H	923209	SF 66	1.92286	20.9	S
923213	PBH 71	1.92286	21.3	O	1022291	N-LASF 35	2.02204	29.1	S

**Fig. 5.88** Color chart (color coordinates) with some glass types and illuminant D65 (standardized light source)

References

- 5.1 J. Jackson: *Classical Electrodynamics* (Wiley, New York 1975)
- 5.2 L. D. Landau, E. M. Lifshitz: *The Classical Theory of Fields* (Addison Wesley, New York 1971)
- 5.3 Ch. Kittel: *Introduction to Solid State Physics* (Oldenbourg, Munich 1988)
- 5.4 H. Haken: *Quantum Field Theory of Solids* (Teubner, Stuttgart 1973)
- 5.5 A. Sommerfeld: *Optics: Lectures on Theoretical Physics*, Vol. IV (Academic, New York 1954)
- 5.6 A. C. Hardy, F. H. Perrin: *The Principles of Optics* (McGraw-Hill, New York 1932)
- 5.7 C. S. Williams, O. A. Becklund: *Optics: A Short Course for Engineers and Scientists* (Wiley, New York 1972)
- 5.8 W. G. Driscoll, W. Vaughan (Eds.): *Handbook of Optics* (McGraw-Hill, New York 1978)
- 5.9 D. Halliday, R. Resnick, J. Walker: *Fundamentals of Physics*, 4th edn. (Wiley, New York 1993)
- 5.10 J. H. Simmons, K. S. Potter: *Optical Materials* (Academic, New York 2000)
- 5.11 E. Hecht: *Optics*, 4th edn. (Addison Wesley, New York 2002)
- 5.12 S. Singh: Refractive index measurement and its applications, *Phys. Scr.* **65**(2), 167–180 (2002)
- 5.13 S. Tominaga, N. Tanaka: Refractive index estimation and color image rendering, *Pattern Recognition Lett.* **24**(11), 1703–1713 (2003)
- 5.14 J. E. Shelby: *Introduction to Glass Science and Technology* (The Royal Society of Chemistry, Cambridge 1997)
- 5.15 H. Bach, N. Neurorth (Eds.): *The Properties of Optical Glass* (Springer, Berlin, Heidelberg 1998)
- 5.16 J. V. Hughes: A new precision refractometer, *J. Rev. Sci. Instrum.* **18**, 234 (1941)
- 5.17 A. B. P. Lever: *Inorganic Electronic Spectroscopy* (Elsevier, New York 1968)
- 5.18 C. R. Bamford: *Colour Generation and Control in Glass* (Elsevier, New York 1977)
- 5.19 A. Paul: *Chemistry of Glass* (Chapman Hall, New York 1990)
- 5.20 D. C. Harris, M. D. Bertolucci: *Symmetry, Spectroscopy: An Introduction to Vibrational, Electronic Spectroscopy* (Dover, New York 1978)
- 5.21 M. Born, E. Wolf: *Principles of Optics* (Cambridge Univ. Press, Cambridge 1999) p. 218
- 5.22 I. B. Bersuker: *Electronic Structure and Properties of Transition Metal Compounds: Introduction to the Theory* (Wiley, New York 1996)
- 5.23 J. E. Shelby: *Introduction to Glass Science and Technology* (The Royal Society of Chemistry, Cambridge 1997)
- 5.24 B. Douglas, D. McDaniel, J. Alexander: *Concepts and Models of Inorganic Chemistry*, 3rd edn. (Wiley, New York 1994)
- 5.25 C. K. Jorgensen: Spectroscopy of transition-group complexes, *Adv. Chem. Phys.* **5**, 33–145 (1963)
- 5.26 N. W. Ashcroft, N. D. Mermin: *Solid State Physics* (Thomson, Stamford 2000)
- 5.27 J. D. Jackson: *Classical Electrodynamics* (Wiley, New York 1975)
- 5.28 J. F. Nye: *Physical Properties of Crystals* (Oxford Univ. Press, Oxford 1957)
- 5.29 M. Born, E. Wolf: *Principles of Optics* (Pergamon, Oxford 1986)
- 5.30 H. Vogel: *Gerthsen Physik* (Springer, Berlin, Heidelberg 1997)
- 5.31 B. E. A. Saleh, M. C. Teich: *Fundamentals of Photonics* (Wiley, New York 1991)
- 5.32 Y. R. Shen: *The Principles of Nonlinear Optics* (Wiley, New York 1984)
- 5.33 W. Nie: Optical nonlinearity: Phenomena, applications and materials, *Adv. Mater.* **5**, 520–545 (1993) and cited papers
- 5.34 C. F. Klingshirn: *Semiconductor Optics* (Springer, Berlin, Heidelberg 1997)
- 5.35 N. F. Borrelli, D. W. Hall: *Optical Properties of Glass*, ed. by Uhlmann, Kreidl (American Ceramic Society, Westerville 1991) pp. 87–124
- 5.36 P. Chakraborty: Metal nanoclusters in glasses as nonlinear photonic materials, *J. Mater. Sci.* **33**, 2235–2249 (1998)
- 5.37 E. M. Vogel, M. J. Weber, D. M. Krol: Nonlinear optical phenomena in glass, *Phys. Chem. Glasses* **32**, 231–250 (1991) and cited papers
- 5.38 E. M. Vogel: Glasses as nonlinear photonic materials, *J. Am. Ceram. Soc.* **72**, 719–724 (1989)
- 5.39 I. Kang, T. D. Krauss, F. W. Wise, B. G. Aitken, N. F. Borrelli: Femtosecond measurement of enhanced optical nonlinearities of sulfide glasses and heavy-metal-doped oxide glasses, *J. Opt. Soc. Am. B* **12**, 2053–2059 (1995)
- 5.40 H. Tanaka, K. Kashima, K. Hirao, N. Soga, A. Mito, H. Nasu: Second harmonic generation in poled tellurite glasses, *Jpn. J. Appl. Phys* **32**(2), 843 (1993)
- 5.41 K. Hirao, T. Mitsuyu, J. Si, J. Qiu: *Active Glasses for Photonic Devices* (Springer, Berlin, Heidelberg 2001)
- 5.42 K. Hirao: *Active Glass Project NEWS*, 99.8 Final Rep. No. 3 (Japan Science and Technology Agency, Kawaguchi City 1999)
- 5.43 H. Nasu, J. D. MacKenzie: Nonlinear optical properties of glass and glass or gel based compositions, *Opt. Eng.* **26**, 102–106 (1987)
- 5.44 J. S. Aitchinson, J. D. Prohaska, E. M. Vogel: The nonlinear optical properties of glass, *Met. Mater. Proc.* **8**, 277–290 (1996)
- 5.45 B. G. Potter, M. B. Sinclair: Photosensitive, rare earth doped ceramics for optical sensing, *J. Electroceram.* **2**, 295–308 (1998)



- 5.46 F. Ishih, T. Sawatari, A. Odajima: NMR study of chain orientation in drawn poly(vinylidene fluoride) films I. The effects of poling on the double orientation distribution function, *Jpn. J. Appl. Phys* **27**, 1047–1053 (1988)
- 5.47 M. G. Kuzyk, K. D. Singer, H. E. Zahn, L. A. King: Second-order nonlinear-optical tensor properties of poled films under stress, *J. Opt. Soc. Am. B* **6**, 742 (1989)
- 5.48 S. Miyata, H. Sasabe: *Poled Polymers, their Applications to SHG and EO Devices* (Taylor Francis, New York 1997)
- 5.49 W. Koechner: *Solid-State Laser Engineering* (Springer, Berlin, Heidelberg 1976)
- 5.50 H. C. Van de Hulst: *Light Scattering by Small Particles* (Dover, New York 1981)
- 5.51 G. Mie: *Beiträge zur Optik trüber Medien, speziell kolloidaler Metallösungen*, *Ann. Physik*, Vol. 25, ed. by E. Grüneisen (Barth, Leipzig 1908) pp. 377–445
- 5.52 A. Ishimaru: *Wave Propagation and Scattering in Random Media* (IEEE, Piscataway 1997)
- 5.53 C. Bohren, D. Huffman: *Absorption, Scattering of Light by Small Particles* (Wiley, New York 1983)
- 5.54 M. Born, E. Wolf: *Principles of Optics*, 7th edn. (Cambridge Univ. Press, Cambridge 1999)
- 5.55 P. Debye, A. M. Bueche: *J. Appl. Phys.* **20**, 518 (1949)
- 5.56 C. S. Johnson, D. A. Gabriel: *Laser Light Scattering* (Dover, New York 1981)
- 5.57 A. Dogariu: Volume Scattering in Random Media. In: *Handbook of Optics, Volume III, 2nd edition, Part 1 Classical optics*, ed. by M. Bass (McGraw-Hill, New York 2001) Chap. 3
- 5.58 P. D. Kaplan, A. D. Dinsmore, A. G. Yodh: Diffuse-transmission spectroscopy: A structural probe of opaque colloidal mixtures, *Phys. Rev. E* **50**, 4827 (1994)
- 5.59 J. M. Elson, H. E. Bennett, J. M. Bennett: Scattering from optical surfaces. In: *Applied Optics and Optical Engineering*, Vol. 7, ed. by R. Shannon, J. Wyant (Academic, New York 1979) Chap. 7
- 5.60 A. Ishimaru: *Wave Propagation and Scattering in Random Media* (IEEE, Piscataway 1997)
- 5.61 A. Duparré: Surface characterization techniques for determining the root-mean-square roughness and power spectral densities of optical components, *Appl. Opt.* **41**, 154–171 (2002)
- 5.62 M. Born, E. Wolf: *Principles of Optics*, 7th edn. (Cambridge Univ. Press, Cambridge 1999)
- 5.63 S. Musikant: *Optical Materials* (Dekker, New York 1985)
- 5.64 H. Bach, N. Neuroth: *The Properties of Optical Glass*, Schott Series on Glass and Glass Ceramics (Springer, Berlin, Heidelberg 1995)
- 5.65 W. H. Armistead, S. D. Stookey: Photochromic silicate glasses sensitized by silver halides, *Science* **144**, 150–158 (1964)
- 5.66 F.-T. Lentes: Refractive index and dispersion. In: *The Properties of Optical Glass*, ed. by H. Bach, N. Neuroth (Springer, Berlin, Heidelberg 1998) pp. 19–57
- 5.67 H. G. Pfänder: Optische Gläser und Brillengläser. In: *Schott Glaslexikon* (mvg, Landsberg 1997) pp. 129–142
- 5.68 B. Jaschke: Einfluß der Wissenschaften. In: *Glasherstellung* (Deutsches Museum, Munich 1997) pp. 79–86
- 5.69 H. G. Pfänder: Geschichte des Glases. In: *Schott Glaslexikon* (mvg, Landsberg 1997) pp. 13–23
- 5.70 N. J. Kreidl: Optical properties. In: *Handbook of Glass Manufacture*, ed. by F. V. Tooley (Books for Industry, New York 1974) pp. 957–997
- 5.71 B. Jaschke: Neuerungen in der Glasherstellung. In: *Glasherstellung* (Deutsches Museum, Munich 1997) pp. 65–78
- 5.72 M. K. Th. Clement: The chemical composition of optical glasses and its influence on the optical properties. In: *The Properties of Optical Glass*, ed. by H. Bach, N. Neuroth (Springer, Berlin, Heidelberg 1998) pp. 58–81
- 5.73 G. F. Brewster, N. J. Kreidl, T. G. Pett: Lanthanum and barium in glass-forming system, *J. Soc. Glass Technol.* **31**, 153–169 (1947)
- 5.74 W. Jahn: Mehrstoffsyste zu Aufbau Optischer Gläser, *Glastechn. Ber.* **43**, 107–120 (1961)
- 5.75 W. H. Zachariasen: Die Struktur der Gläser, *Glastechn. Ber.* **11**, 120–123 (1933)
- 5.76 W. H. Zachariasen: The atomic arrangement in glass, *J. Am. Chem. Soc.* **54**, 3841–3851 (1932)
- 5.77 B. E. Warren, A. D. Loring: X-ray diffraction study of the structure of soda-silica glasses, *J. Am. Ceram. Soc.* **18**, 269–276 (1935)
- 5.78 K. Fajans, N. J. Kreidl: Stability of lead glasses, polarization of ions, *J. Am. Ceram. Soc.* **31**, 105–114 (1948)
- 5.79 J. E. Stanworth: On the structure of glass, *J. Soc. Glass Technol.* **32**, 154–172 (1948)
- 5.80 A. M. Bishay, P. Askalani: Properties of antimony glasses in relation to structure, C. R. Vile Congrès International du Verre, Bruxelles 1965 (Maison d'Édition, Marcinelle 1965) Part 1, Nr. 24
- 5.81 S. Wolff, U. Kolberg: Environmental friendly optical glasses. In: *The Properties of Optical Glass*, ed. by H. Bach, N. Neuroth (Springer, Berlin, Heidelberg 1998) pp. 144–148
- 5.82 H. Bach, N. Neuroth: *The Properties of Optical Glass*, Schott Ser. Glass (Springer, Berlin, Heidelberg 1998)
- 5.83 C. R. Bamford: *Colour Generation and Control in Glass* (Elsevier, Amsterdam 1977)
- 5.84 W. A. Weyl: *Coloured Glasses* (Society of Glass Technology, Sheffield 1951)
- 5.85 W. Vogel: *Glaschemie* (Springer, Berlin, Heidelberg 1992) pp. 51–313
- 5.86 K. Nassau: The varied causes of colour in glass, *Mat. Res. Soc. Symp. Proc.* **61**, 427–439 (1986)
- 5.87 T. Bates: *Ligand field theory, absorption spectra of transition-metal ions in glasses*, Vol. 2, ed. by



- J. D. Mackenzie (Butterworth, London 1962) pp. 195–254
- 5.88 L. E. Orgel: *An Introduction to Transition Metal Chemistry and Ligand Field Theory* (Methuen, London 1960)
- 5.89 H. L. Schläfer, G. Gliemann: *Einführung in die Ligandenfeldtheorie* (Akademische Verlagsgesellschaft, Frankfurt/Main 1967)
- 5.90 C. J. Ballhausen: *Introduction to Ligand Field Theory* (McGraw-Hill, New York 1962)
- 5.91 C. K. Jorgensen: *Absorption Spectra and Chemical Bonding in Complexes* (Pergamon, Oxford 1962)
- 5.92 A. Bishay, A. Kinawi: Absorption spectra of iron in phosphate glasses and ligand field theory, *Phys. Non-Cryst. Solids* **2**, 589–605 (1965)
- 5.93 C. F. Bohren: *Absorption and Scattering of Light by Small Particles* (Wiley, New York 1983)
- 5.94 M. Kerker: *The Scattering of Light and Other Electromagnetic Radiation* (Academic, New York 1969)
- 5.95 M. Born, E. Wolf: *Principles of Optics* (Pergamon, Oxford 1986)
- 5.96 H. P. Rooksby: The colour of selenium ruby glasses, *J. Soc. Glass Technol.* **16**, 171–181 (1932)
- 5.97 G. Schmidt: Optische Untersuchungen an Selenrubingläsern, *Silikattechnik* **14**, 12–18 (1963) (in German)
- 5.98 A. Rehfeld, R. Katzschmann: Farbbildung und Kinetik von Steilkanten-Anlaufgläsern, *Silikattechnik* **29**, 298–302 (1978) (in German)
- 5.99 G. Walter, R. Kranold, U. Lemke: Small angle X-ray scattering characterization of inorganic glasses, *Makromol. Chem. Makromol. Symp.* **15**, 361–372 (1988)
- 5.100 T. Yanagawa, Y. Sasaki, H. Nakano: Quantum size effects, observation of microcrystallites in coloured filter glasses, *Appl. Phys. Lett.* **54**, 1495–1497 (1989)
- 5.101 J. L. Emmett, W. F. Krupke, J. B. Trenholme: *The Future Development of High-Power Solid State Laser Systems* (Lawrence Livermore National Laboratory, Livermore 1982)
- 5.102 A. H. Clauer: New life for laser shock processing, *Ind. Laser Rev.*, 7–9 (March 1996)
- 5.103 E. Snitzer: Optical laser action of Nd^{3+} in a barium crown glass, *Phys. Rev. Lett.* **7**, 444–446 (1961)
- 5.104 S. E. Stokowski, R. A. Saroyan, M. J. Weber: *Laser Glass Nd-Doped Glass Spectroscopic and Physical Properties* (Lawrence Livermore National Laboratory, Livermore 1981) pp. 1–9 M–95, Rev. 2
- 5.105 J. H. Pitts: Modeling laser damage caused by platinum inclusions in laser glass, laser induced damage in optical materials, *Techn. Dig. Boulder Damage Symp.*, Boulder 1985 (NBS, Boulder 1986) 537–542
- 5.106 J. H. Campbell, E. P. Wallerstein, J. S. Hayden, D. L. Sapak, D. E. Warrington, A. J. Marker III, H. Toratani, H. Meissner, S. Nakajima, T. Izumitani: *Elimination of Platinum Inclusions in Phosphate Laser Glasses* (Lawrence Livermore National Laboratory, Livermore 1989)
- 5.107 R. Wood: *Laser Damage in Optical Materials* (IOP Publishing Limited, Bristol, Great Britain 1986)
- 5.108 D. C. Brown: *High-Peak-Power Nd:Glass Laser Systems* (Springer, Berlin, Heidelberg 1981) Chap. 6
- 5.109 D. H. Roach, A. R. Cooper: The effect of etch depth on strength of indented soda lime glass rods. In: *Strength of Inorganic Glass*, ed. by C. R. Kurkjian (Plenum, New York 1985) pp. 185–195
- 5.110 W. C. LaCourse: The strength of glass. In: *Introduction to Glass Science*, ed. by L. D. Pye, H. J. Stevens and W. C. LaCourse (Plenum, New York 1972) pp. 451–512
- 5.111 P. W. McMillan: *Glass-Ceramics* (Academic, London 1979) p. 285
- 5.112 W. Holand, G. H. Beall: *Glass-Ceramic Technology* (The American Ceramic Society, Westerville 2002) p. 372
- 5.113 W. Pannhorst: Low expansion glass ceramics – current developments, *Proc. 7th Int. Otto Schott Coll.*, Jena, Germany 2002, ed. by C. Ruessel, G. Volksch (Verlag der Deutschen Glastechnischen Gesellschaft, Frankfurt 2002) 78
- 5.114 S. Cramer von Clausbruch, M. Schweiger, W. Hoeland, V. Rheinberger: Effect of ZnO on the crystallization, microstructure, properties of glass-ceramics in the $\text{SiO}_2\text{--Li}_2\text{O--ZnO--K}_2\text{O--P}_2\text{O}_5$ system, *Glotech. Ber.-Glass Sci. Technol.* **74**, 223 (2001)
- 5.115 M. M. Layton, J. W. Smith: Pyroelectric response in transparent ferroelectric glass, *J. Am. Ceram. Soc.* **58**, 435 (1975)
- 5.116 T. Komatsu, J. Onuma, H. G. Kim, J. R. Kim: Formation of Rb-doped crystalline phase with second harmonic generation in transparent $\text{K}_2\text{O--Nb}_2\text{O}_5\text{--TeO}_2$ glass ceramics, *J. Mater. Sci. Lett.* **15**, 2130 (1996)
- 5.117 F. C. Guinhos, P. C. Nobrega, P. A. Santa-Cruz: Compositional dependence of up-conversion process in $\text{Tm}^{3+}\text{--Yb}^{3+}$ codoped oxyfluoride glasses and glass-ceramics, *J. Alloy. Compd.* **323**, 358 (2001)
- 5.118 M. Secu, S. Schweizer, J. M. Spaeth, A. Edgar, G. V. M. Williams, U. Rieser: Photostimulated luminescence from a fluorobromozirconate glass-ceramic and the effect of crystallite size and phase, *J. Phys. Cond. Matter* **15**, 1097 (2003)
- 5.119 G. H. Beall: Glass-ceramics for photonic applications, *Glass Sci. Technol.-Glotech. Ber.* **73**, 3 (2000)
- 5.120 J. A. Tangeman, B. L. Phillips, A. Navrotsky, J. K. R. Weber, A. D. Hixson, T. S. Key: Vitreous forsterite (Mg_2SiO_4): Synthesis, structure, thermochemistry, *Geophys. Res. Lett.* **28**, 2517 (2001)
- 5.121 P. A. Tick, N. F. Borrelli, I. M. Reaney: The relationship between structure and transparency in glass-ceramic materials, *Opt. Mater.* **15**, 81 (2000)
- 5.122 K. Shioya, T. Komatsu, H. G. Kim, R. Sato, K. Matusita: Optical properties of transparent glass-ceramics in $\text{K}_2\text{O--Nb}_2\text{O}_5\text{--TeO}_2$ glasses, *J. Non-Cryst. Solids* **189**, 16 (1995)

- 5.123 H. A. Miska: Aerospace and military applications. In: *Ceramics and Glasses*, ed. by S. R. Lampman, M. S. Woods, T. B. Zorc (ASM Int., Materials Park 1991) p. 1016
- 5.124 H. Bach (Ed.): *Low Thermal Expansion Glass Ceramics* (Springer, Berlin, Heidelberg 1995) p. 223
- 5.125 P. F. James: Kinetics of crystal nucleation in lithium silicate glasses, *Phys. Chem. Glasses* **15**, 95 (1974)
- 5.126 A. C. Lasaga: *Kinetic Theory in the Earth Sciences* (Princeton Univ. Press, Princeton 1998) p. 811
- 5.127 P. W. McMillan, G. Partridge: Dielectric properties of certain ZnO–Al₂O₃–SiO₂ glass-ceramics, *J. Mater. Sci.* **7**, 847 (1972)
- 5.128 A. P. Tomsia, J. A. Pask, R. E. Loehman: Glass/metal and glass-ceramic/metal seals. In: *Ceramics and Glasses*, ed. by S. R. Lampman, M. S. Woods, T. B. Zorc (ASM Int., Materials Park 1991) p. 493
- 5.129 Ohara Corporation: *Glass-Ceramic Substrates for Planar Light Circuits (SA-02)*, Commercial Literature (Ohara Corporation, Japan 2003)
- 5.130 W. D. Kingery, H. K. Bowen, D. R. Uhlmann: *Introduction to Ceramics*, 2nd edn. (Wiley, New York 1976) p. 1032
- 5.131 R. B. Roberts, R. J. Tainsh, G. K. White: Thermal properties of Zerodur at low temperatures, *Cryogenics* **22**, 566 (1982)
- 5.132 S. W. Freiman, L. L. Hench: Effect of crystallization on mechanical properties of Li₂O–SiO₂ glass-ceramics, *J. Am. Ceram. Soc.* **55**, 86 (1972)
- 5.133 S. S. Bayya, J. S. Sanghera, I. D. Aggarwal, J. A. Wojcik: Infrared transparent germanate glass-ceramics, *J. Am. Ceram. Soc.* **85**, 3114 (2002)
- 5.134 Y. Tada, F. Kawano, M. Kon, N. Matsumoto, K. Asaoka: Influence of crystallization on strength and color of castable glass-ceramics containing two crystals, *Biomed. Mater. Eng.* **5**, 233 (1995)
- 5.135 B. R. Lawn, T. R. Wilshaw, T. I. Barry, R. Morrell: Hertzian fracture of glass ceramics, *J. Mater. Sci.* **10**, 179 (1975)
- 5.136 R. Morena, K. Niihara, D. P. H. Hasselman: Effect of crystallites on surface damage and fracture-behavior of a glass-ceramic, *J. Am. Ceram. Soc.* **66**, 673 (1983)
- 5.137 T. J. Hill, J. J. Melchiosky, K. J. Anusavice: Fractal analysis of toughening behavior in 3BaO.5SiO₂ glass-ceramics, *J. Am. Ceram. Soc.* **83**, 545 (2000)
- 5.138 D. Mittleman (Ed.): *Sensing with Terahertz Radiation* (Springer, Berlin, Heidelberg 2003) p. 337
- 5.139 A. J. Moulson, J. M. Herbert: *Electroceramics* (Chapman Hall, London 1990) p. 464
- 5.140 A. Herczog: Microcrystalline BaTiO₃ by crystallization from glass, *J. Am. Ceram. Soc.* **47**, 107 (1964)
- 5.141 N. F. Borrelli: Electrooptic effect in transparent niobate glass-ceramic systems, *J. Appl. Phys.* **38**, 4243 (1967)
- 5.142 O. P. Thakur, D. Kumar, O. M. Parkash, L. Pandey: Crystallization, microstructure development and dielectric behaviour of glass ceramics in the system SrO center dot TiO₂–2SiO₂ center dot B₂O₃–La₂O₃, *J. Mater. Sci.* **37**, 2597 (2002)
- 5.143 F. Agullo-Lopez, J. M. Cabera, F. Agullo-Rueda: *Electrooptics: Phenomena, Materials and Applications* (Academic, London 1994) p. 345
- 5.144 H. Bach, N. Neuroth (Eds.): *The Properties of Optical Glass* (Springer, Berlin, Heidelberg 1995) p. 410
- 5.145 K. J. Anusavice, N.–Z. Zhang, J. E. Moorhead: Influence of P₂O₅, AgNO₃, FeCl₃ on color, translucency of lithia-based glass-ceramics, *Dent. Mater.* **10**, 230 (1994)
- 5.146 V. M. Khomenko, K. Langer, R. Wirth: On the influence of wavelength-dependent light scattering on the UV-VIS absorption spectra of oxygen-based minerals: A study on silicate glass ceramics as model substances, *Phys. Chem. Min.* **30**, 98 (2003)
- 5.147 G. H. Beall, D. A. Duke: Transparent glass ceramics, *J. Mater. Sci.* **4**, 340 (1969)
- 5.148 C. F. Bohren, D. R. Huffman: *Absorption and Scattering of Light by Small Particles* (Wiley, New York 1983) p. 530
- 5.149 F. M. Modest: *Radiative Heat Transfer* (McGraw-Hill, New York 1993) p. 832
- 5.150 R. W. Hopper: Stochastic-theory of scattering from idealized spinodal structures 2 scattering in general and for the basic late stage model, *J. Non-Cryst. Solids* **70**, 111 (1985)
- 5.151 R. Menzel: *Photonics: Linear and Nonlinear Interactions of Laser Light and Matter* (Springer, Berlin, Heidelberg 2001) p. 873
- 5.152 S. Hendy: Light scattering in transparent glass ceramics, *Appl. Phys. Lett.* **81**, 1171 (2002)
- 5.153 W. Pannhorst: Zerodur – a low thermal expansion glass ceramic for optical precision applications. In: *Low Thermal Expansion Glass Ceramics*, ed. by H. Bach (Springer, Berlin, Heidelberg 1995) p. 107
- 5.154 P. Pernice, A. Aronne, V. N. Sigaev, P. D. Sarkisov, V. I. Molev, S. Y. Stefanovich: Crystallization behavior of potassium niobium silicate glasses, *J. Am. Ceram. Soc.* **82**, 3447 (1999)
- 5.155 G. S. Murugan, K. B. R. Varma: Dielectric, linear and non-linear optical properties of lithium borate-bismuth tungstate glasses and glass-ceramics, *J. Non-Cryst. Solids* **279**, 1 (2001)
- 5.156 N. F. Borrelli, M. M. Layton: Dielectric and optical properties of transparent ferroelectric glass-ceramic systems, *J. Non-Cryst. Solids* **6**, 197 (1971)
- 5.157 A. Halliyal, A. S. Bhalla, R. E. Newnham, L. E. Cross: Glass-ceramics for piezoelectric and pyroelectric devices. In: *Glass and Glass Ceramics*, ed. by M. H. Lewis (Chapman Hall, London 1989) p. 273
- 5.158 Y.–H. Kao, Y. Hu, H. Zheng, J. D. Mackenzie, K. Perry, G. Bourhill, J. W. Perry: Second harmonic generation in transparent barium borate glass-ceramics, *J. Non-Cryst. Solids* **167**, 247 (1994)



- 5.159 N. F. Borrelli, A. Herczog, R. D. Maurer: Electro-optic effect of ferroelectric microcrystals in a glass matrix, *Appl. Phys. Lett.* **7**, 117 (1965)
- 5.160 N. F. Borrelli: Electrooptic effect in transparent niobate glass, *J. Appl. Phys.* **38**, 4243 (1967)
- 5.161 S. Ito, T. Kokubo, M. Tashiro: Transparency of $\text{LiTaO}_3\text{--SiO}_2\text{--Al}_2\text{O}_3$ glass-ceramics in relation to their microstructure, *J. Mater. Sci.* **13**, 930 (1978)
- 5.162 A. Herczog: Phase distribution and transparency in glass-ceramics based on a study of the sodium niobate-silica system, *J. Am. Ceram. Soc.* **73**, 2743 (1990)
- 5.163 T. Kokubo, M. Tashiro: Fabrication of transparent lead titanate(IV) glass, *Bull. Inst. Chem. Res. Kyoto Univ.* **54**, 301 (1976)
- 5.164 Y. Fujimoto, Y. Benino, T. Fujiwara, R. Sato, T. Komatsu: Transparent surface and bulk crystallized glasses with lanthanide tellurite nanocrystals, *J. Ceram. Soc. Jpn.* **109**, 466 (2001)
- 5.165 H. G. Kim, T. Komatsu, K. Shioya, K. Matusita, K. Tanaka, K. Hirao: Transparent tellurite-based glass-ceramics with second harmonic generation, *J. Non-Cryst. Solids* **208**, 303 (1996)
- 5.166 R. T. Hart, M. A. Anspach, B. J. Kraft, J. M. Zaleski, J. W. Zwanziger, P. J. DeSanto, B. Stein, J. Jacob, P. Thiyagarajan: Optical implications of crystal-lite symmetry and structure in potassium niobate tellurite glass ceramics, *Chem. Mater.* **14**, 4422 (2002)
- 5.167 N. S. Prasad, K. B. R. Varma: Nanocrystallization of $\text{SrBi}_2\text{Nb}_2\text{O}_9$ from glasses in the system $\text{Li}_2\text{B}_4\text{O}_7\text{--SrO--Bi}_2\text{O}_3\text{--Nb}_2\text{O}_5$, *Mater. Sci. Eng. B* **90**, 246 (2002)
- 5.168 G. S. Murugan, K. B. R. Varma: Characterization of lithium borate-bismuth tungstate glasses, glass-ceramics by impedance spectroscopy, *Solid State Ionics* **139**, 105 (2001)
- 5.169 V. N. Sigaev, P. Pernice, A. Aronne, O. V. Akimova, S. Y. Stefanovich, A. Scaglione: KTiOPO_4 precipitation from potassium titanium phosphate glasses, producing second harmonic generation, *J. Non-Cryst. Solids* **292**, 59 (2001)
- 5.170 Y. Balci, M. Ceylan, M. E. Yakinci: An investigation on the activation energy and the enthalpy of the primary crystallization of glass-ceramic Bi-rich BSC-COHTc superconductors, *Mater. Sci. Eng. B* **86**, 83 (2001)
- 5.171 A. Edgar, S. Schweizer, S. Assmann, J. M. Spaeth, P. J. Newman, D. R. MacFarlane: Photoluminescence and crystallization in europium-doped fluorobromozirconate glass-ceramics, *J. Non-Cryst. Solids* **284**, 237 (2001)
- 5.172 G. Muller, N. Neuroth: Glass ceramic as an active laser material, US Patent 3843551 (Jenaer Glaswerk Schott and Gen., USA 1974)
- 5.173 A. Lempicki, M. Edwards, G. H. Beall, D. Hall, L. J. Andrews: *Transparent Glass Ceramics* (Optical Society of America, Arlington, VA 1985) Laser Prospects, in Topical Meeting on Tunable Solid State Lasers
- 5.174 R. Reisfeld, C. K. Jorgensen: Excited-states of chromium(III) in translucent glass-ceramics as prospective laser materials, *Struct. Bond.* **69**, 63 (1988)
- 5.175 R. Reisfeld: Potential uses of chromium(III)-doped transparent glass ceramics in tunable lasers and luminescent solar concentrators, *Mater. Sci. Engin.* **71**, 375 (1985)
- 5.176 P. A. Tick, N. F. Borrelli, L. K. Cornelius, M. A. Newhouse: Transparent glass ceramics for 1300 nm, *J. Appl. Phys.* **78**, 6367 (1995)
- 5.177 A. M. Malyarevich, I. A. Denisov, K. V. Yumashev, O. S. Dymshits, A. A. Zhilin, U. Kang: Cobalt-doped transparent glass ceramic as a saturable absorber Q switch for Er: Glass lasers, *Appl. Opt.* **40**, 4322 (2001)
- 5.178 C. F. Rapp, J. Chrysochoos: Neodymium-doped glass-ceramic laser material, *J. Mater. Sci. Lett.* **7**, 1090 (1972)
- 5.179 G. Muller, N. Neuroth: Glass ceramic – A new laser host material, *J. Appl. Phys.* **44**, 2315 (1973)
- 5.180 C. F. Rapp, J. Chrysochoos: Fluorescence lifetimes of Neodymium-doped glasses and glass ceramics, *J. Phys. Chem.* **77**, 1016 (1973)
- 5.181 Y. H. Wang, J. Ohwaki: New transparent vitroceramics codoped with Er^{3+} , Yb^{3+} for efficient frequency up-conversion, *Appl. Phys. Lett.* **63**, 3268 (1993)
- 5.182 P. A. Tick, N. F. Borrelli, L. K. Cornelius, M. A. Newhouse: Transparent glass ceramics for 1300 nm amplifier applications, *J. Appl. Phys.* **78**, 6367 (1995)
- 5.183 M. J. Dejneka: The luminescence and structure of novel transparent oxyfluoride glass-ceramics, *J. Non-Cryst. Solids* **239**, 149 (1998)
- 5.184 Y. Kawamoto, R. Kanno, J. Qiu: Upconversion luminescence of Er^{3+} in transparent $\text{SiO}_2\text{--PbF}_2\text{--ErF}_3$ glass ceramics, *J. Mater. Sci.* **33**, 63 (1998)
- 5.185 M. Takahashi, M. Izuki, R. Kanno, Y. Kawamoto: Up-conversion characteristics of Er^{3+} in transparent oxyfluoride glass-ceramics, *J. Appl. Phys.* **83**, 3920 (1998)
- 5.186 M. Mortier, G. Patriarche: Structural characterisation of transparent oxyfluoride glass ceramics, *J. Mater. Sci.* **35**, 4849 (2000)
- 5.187 L. L. Kukkonen, I. M. Reaney, D. Furniss, M. G. Pellatt, A. B. Seddon: Nucleation and crystallisation of transparent, erbium III-doped, oxyfluoride glass-ceramics, *J. Non-Cryst. Solids* **290**, 25 (2001)
- 5.188 M. Mortier, A. Monteville, G. Patriarche, G. Maze, F. Auzel: New progresses in transparent rare-earth doped glass-ceramics, *Opt. Mater.* **16**, 255 (2001)
- 5.189 M. Mortier: Between glass and crystal: Glass-ceramics, a new way for optical materials, *Philos. Mag. B* **82**, 745 (2002)
- 5.190 V. K. Tikhomirov, D. Furniss, A. B. Seddon, I. M. Reaney, M. Beggiora, M. Ferrari, M. Montagna, R. Rolli: Fabrication and characterization of nanoscale,



- Er³⁺-doped, ultratransparent oxy-fluoride glass ceramics, *Appl. Phys. Lett.* **81**, 1937 (2002)
- 5.191 J. Mendez-Ramos, V. Lavin, I. R. Martin, U. R. Rodriguez-Mendoza, V. D. Rodriguez, A. D. Lozano-Gorrin, P. Nunez: Optical properties of rare earth doped transparent oxyfluoride glass ceramics, *Radiat. Eff. Defects Solids* **158**, 457 (2003)
- 5.192 A. M. Malyarevich, I. A. Denisov, K. V. Yumashev, O. S. Dymshits, A. A. Zhilin: Optical absorption, luminescence study of cobalt-doped magnesium aluminosilicate glass ceramics, *J. Opt. Soc. Am. B* **19**, 1815 (2002)
- 5.193 A. M. Malyarevich, I. A. Denisov, Y. V. Volk, K. V. Yumashev, O. S. Dymshits, A. A. Zhilin: Nanosized glass ceramics doped with transition metal ions: Non-linear spectroscopy and possible laser applications, *J. Alloy. Compd.* **341**, 247 (2002)
- 5.194 C. Y. Li, Q. Su, S. B. Wang: Multi-color long-lasting phosphorescence in Mn²⁺-doped ZnO–B₂O₃–SiO₂ glass ceramics, *Mater. Res. Bull.* **37**, 1443 (2002)
- 5.195 G. H. Beall: Glass-ceramics: Recent developments and applications. In: *Nucleation and Crystallization in Liquids and Glasses*, ed. by M. C. Weinberg (The American Ceramic Society, Westerville 1993)
- 5.196 G. H. Beall, L. R. Pinckney: Nanophase glass ceramics, *J. Am. Ceram. Soc.* **82**, 5 (1999)
- 5.197 I. Mitra, M. J. Davis, J. Alkemper, R. Mueller, H. Kohlmann, L. Aschke, E. Moersen, S. Ritter, H. Hack, W. Pannhorst: Thermal expansion behavior of proposed EUVL substrate materials, *SPIE Proc.* 4688—Emerging Lithographic Technologies VI, Santa Clara, CA 2002, ed. by R. L. Englestad (SPIE, Belham 2002) 462–468
- 5.198 N. Reisert: Application and machining of Zerodur for optical purposes, *Proc. SPIE* **1400**, 171 (1991)
- 5.199 L. N. Allen: Progress in ion figuring large optics, *SPIE Proc.* 2428—Laser-induced Damage in Optical Materials, Soileau 1995, ed. by H. E. Bennett, A. H. Guenther, M. R. Kozlowski, B. E. Newnam, M. J. Soileau (SPIE, Belham 1995) 237–247
- 5.200 L. Noethe: Active optics in modern large optical telescopes, *Prog. Opt.* **43**, 1 (2002)
- 5.201 C. A. Haniff: High angular resolution studies of stellar atmospheres, *IAU Symp.: Galaxies and their Constituents at the Highest Angular Resolutions*, (2001) 288–295
- 5.202 S. D. Stookey: Catalyzed crystallization of glass in theory and practice, *Ind. Eng. Chem.* **51**, 805 (1959)
- 5.203 B. Rother, A. Mucha: Transparent glass-ceramic coatings: Property distribution on 3D parts, *Surf. Coat. Technol.* **124**, 128 (2000)
- 5.204 J. Hirao, T. Mitsuyu, J. Si, J. Qiu: *Active Glasses for Photonic Devices* (Springer, Berlin, Heidelberg 2001)
- 5.205 W. Nie: Optical nonlinearity phenomena, applications, materials, *Adv. Mater.* **5**, 520–545 (1993) and cited papers
- 5.206 E. M. Vogel, M. J. Weber, D. M. Krol: Nonlinear optical phenomena in glass, *Phys. Chem. Glasses* **32**, 231–250 (1991) and cited papers
- 5.207 A. J. Hayden, A. J. Marker: Glass as a nonlinear optical material, *SPIE Proc.* **1327**, 132–144 (1990)
- 5.208 R. Shechter, E. Millul, Y. Amitai, A. A. Friesem, V. Weiss: Hybrid polymer-on-glass integrated optical diffractive structures for wavelength discrimination, *Opt. Mater.* **17**, 165–167 (2001)
- 5.209 Transparencies and oral communication J. Hayden: SCHOTT North America
- 5.210 Frost and Sullivan (study): Photonic Materials (2000)
- 5.211 V. G. Dmitriev, G. G. Gurzadya, D. N. Nikogosyan: *Handbook of Nonlinear Optical Crystals* (Springer, Berlin, Heidelberg 1991)
- 5.212 P. Chakraborty: Metal nanoclusters in glasses as nonlinear photonic materials, *J. Mater. Sci.* **33**, 2235–2249 (1998)
- 5.213 H. R. Xia, J. H. Zou, H. C. Chen, D. L. Sun: Photorefractive properties of Co-doped potassium sodium strontium barium niobate crystals, *Cryst. Res. Technol.* **34**, 403–407 (1999)
- 5.214 H. R. Xia, C. J. Wang, H. C. Chen, X. L. Lu: Photorefractive properties of manganese-modified potassium sodium strontium barium niobate crystals, *Phys. Rev. B* **55**, 1292–1294 (1997)
- 5.215 S. Zhang, Z. Cheng, H. Chen: A new oxyborate crystal GdCa₄O(BO₃)₃: Defects and optical properties, *Defect Diff. Forum* **186–187**, 79–106 (2000)
- 5.216 V. Berger: Photonic crystals for nonlinear optical frequency conversion, *Confined Photon Systems* **531**, 366–392 (1999) *Lect. Notes Phys.*
- 5.217 H. Nasu, J. D. MacKenzie: Nonlinear optical properties of glass and glass or gel based compositions, *Opt. Eng.* **26**, 102–106 (1987)
- 5.218 F. Kajzar, J. Swalen: *Organic Thin Films for Waveguiding Nonlinear Optics*, Springer Ser. Adv. Nonlin. Opt., Vol. 3 (Springer, Berlin, Heidelberg 1996)
- 5.219 Hirao Active Glass Project NEWS, '99.8 Final Rep. No. 3 (1999)
- 5.220 Hirao Active Glass Project NEWS, '97.7 Final Rep. No. 2 (1997)
- 5.221 G. I. Stegeman, E. M. Wright, N. Finlayson, R. Zanoni, C. T. Seaton: Third order nonlinear integrated optics, *J. Lightwave Technol.* **6**, 953–967 (1988) and cited papers
- 5.222 R. W. Bryant: *Nonlinear Optical Materials: New Technologies, Applications, Markets* (Business and Corporation Inc., Norwalk 1989)
- 5.223 Y. Kondo, Y. Kuroiwa, N. Sugimoto, T. Manabe, S. Ito, T. Tokizaki, A. Nakamura: Third-order optical nonlinearities of CuCl-doped glasses in a near resonance region, *J. Non-Cryst. Solids* **6**, 90–94 (1996)
- 5.224 D. M. Krol, D. J. DiGiovanni, W. Pleibel, R. H. Stolen: Observation of resonant enhancement of photoinduced second-harmonic generation in Tm-doped aluminosilicate glass fibers, *Opt. Lett.* **18**, 1220 (1993)



- 5.225 N.M. Lawandy, R.L. MacDonald: Optically encoded phase-matched second-harmonic generation in semiconductor-microcrystallite-doped glasses, *J. Opt. Soc. Am. B* **8**, 1307 (1991)
- 5.226 E.M. Dianov, D.S. Starodubov, A.A. Izyneev: Photoinduced second-harmonic generation in fibers doped with rare-earth ions, *Opt. Lett.* **19**, 936 (1994)
- 5.227 E.M. Dianov, L.S. Kornienko, V.I. Stupina, P.V. Chernov: Correlation of defect centers with photoinduced second-harmonic generation in Er- and Sm-doped aluminosilicate fibers, *Opt. Lett.* **20**, 1253–1255 (1995)
- 5.228 D.M. Krol, D.J. DiGiovanni, K.T. Nelson, W. Pleibel, R.H. Stolen: Observation of resonant enhancement of photoinduced second-harmonic generation in Tm-doped aluminosilicate glass fibers, *Opt. Lett.* **18**, 1220–1222 (1993)
- 5.229 I.S. Fogel, J.M. Bendickson, M.D. Tocci, M.J. Bloemer, M. Scalora, C.M. Bowden, J.P. Dowling: Spontaneous emission and nonlinear effects in photonic bandgap materials, *Pure Appl. Opt.* **7**, 393–407 (1998)
- 5.230 F. Ouellette, K.O. Hill, D.C. Johnson: Enhancement of second-harmonic generation in optical fibres by hydrogen heat treatment, *Appl. Phys. Lett.* **54**, 1086 (1989)
- 5.231 J.S. Aitchinson, J.D. Prohaska, E.M. Vogel: The nonlinear optical properties of glass, *Met. Mater. Proc.* **8**, 277–290 (1996)
- 5.232 I. Kang, T.D. Krauss, F.W. Wise, B.G. Aitken, N.F. Borrelli: Femtosecond measurement of enhanced optical nonlinearities of sulfide glasses and heavy-metal-doped oxide glasses, *J. Opt. Soc. Am. B* **12**, 2053–2059 (1995)
- 5.233 J. Fu, H. Yatsuda: New families of glasses based on Bi_2O_3 , *Phys. Chem. Glasses* **36**, 211–215 (1995)
- 5.234 N.F. Borrelli, B.G. Aitken, M.A. Newhouse, D.W. Hall: Electric field induced birefringence properties of high-refractive-index glasses exhibiting large Kerr nonlinearities, *J. Appl. Phys.* **70**(5), 2774–2779 (1991)
- 5.235 S. Santran, L. Canioni, T. Cardinal, E. Fargin, G. Le Flem, C. Rouyer, L. Sarger: Precise and absolute measurements of the complex third-order optical susceptibility, *Proc. SPIE* **4106**, 349–359 (2000)
- 5.236 E. Fargin, A. Berthereau, T. Cardinal, G. Le Flem, L. Ducasse, L. Canioni, P. Segonds, L. Sarger, A. Ducasse: Optical nonlinearity in oxide glasses, *J. Non-Cryst. Solids* **203**, 96–101 (1996)
- 5.237 T. Cardinal, K.A. Richardson, H. Shim, A. Schulte, R. Beatty, K. Le Foulgoc, C. Meneghini, J.F. Viens, A. Villeneuve: Nonlinear optical properties of chalcogenide doped glasses in the system As–S–Se, *J. Non-Cryst. Solids* **256,257**, 353–360 (1999)
- 5.238 B. Speit, K.E. Remitz, N. Neuroth: Semiconductor doped glass as a nonlinear material, *SPIE Proc.* **1361**, 1128–1131 (1990)
- 5.239 B. Danielzik, K. Nattermann, D. von der Linde: Nanosecond optical pulse shaping in cadmium-sulfide-selenide glasses, *Appl. Phys. B* **38**, 31–36 (1985)
- 5.240 H. Inouye, K. Tanaka, I. Tanahashi, K. Hirao: Ultrafast dynamics of nonequilibrium electrons in a gold nanoparticle system, *Phys. Rev. B* **57**, 11334 (1998)
- 5.241 H. Inouye, K. Tanaka, I. Tanahashi, T. Hattori, H. Nakatsuka: Ultrafast optical switching in a silver nanoparticle system, *Jpn. J. Appl. Phys.* **39**, 5132–5133 (2000)
- 5.242 J. Khaled, T. Fujiwara, A.J. Ikushima: Optimization of second-order nonlinearity in UV-poled silica glass, *Opt. Mater.* **17**, 275–278 (2001)
- 5.243 B.G. Broome: *The Design of Plastic Optical Systems*, SPIE Short Course **384** (SPIE, San Diego 2001)
- 5.244 X. Ning, R.T. Hebert (Eds.): *Design, Fabrication and Application of Precision Plastic Optics*, SPIE Proc. **2600** (SPIE, Bellingham 1995)
- 5.245 Corning Precision Lens: *The Handbook of Plastic Optics*, 2nd edn. (Corning Precision Lens, OwensCorning 2000)
- 5.246 D.J. Butler: Plastic optics challenge glass, *Photon. Spectra* (May 2000) p. 168–171
- 5.247 Optical Coating Laboratory Inc. "An Introduction to the Design, Manufacture and Application of Plastic Optics" Technical Information, 2001
- 5.248 A. Ning: Plastic versus glass optics: Factors to consider, SPIE SC **384** (SPIE, San Diego 2001) short note
- 5.249 E. Bürkle, B. Klotz, P. Lichtinger: Durchblick im Spritzguss, *KU Kunststoffe* **11**, 54–60 (2001)
- 5.250 S. Musikant: *Optical Materials* (Dekker, New York 1990)
- 5.251 M.J. Weber: *Handbook of Optical Materials* (CRC, Boca Raton 2002)
- 5.252 K.S. Potter, J. Simmons: *Optical Materials* (Academic, New York 2000)
- 5.253 V.L. Ginzburg: Electromagnetic waves in isotropic and crystalline media characterized by dielectric permittivity with spatial dispersion, *JETP* **34**, 1096 (1958)
- 5.254 V.M. Agranovich, V.L. Ginzburg: *Crystal Optics with Spatial Dispersion and Excitons* (Springer, Berlin, Heidelberg 1984)
- 5.255 J. Pastrnak, K. Vedam: Optical anisotropy of silicon single crystals, *Phys. Rev. B* **3**, 2567 (1971)
- 5.256 P.Y. Yu, M. Cardona: Spatial dispersion in the dielectric constant of GaAs, *Solid State Commun.* **9**, 1421 (1971)
- 5.257 C. Zaldo, C. Lopez, F. Meseguer: Natural birefringence in alkali halide single crystals, *Phys. Rev. B* **33**, 4283 (1986)
- 5.258 E.G. Tsitsishvili: Optical anisotropy of cubic crystals induced by spatial dispersion, *Sov. Phys. Semicond.* **15**, 1152 (1981)
- 5.259 J.H. Burnett, Z.H. Levine, E.L. Shirley: Intrinsic birefringence in calcium fluoride and barium fluoride, *Phys. Rev. B* **64**, 241102R (2001)

- 5.260 M. Letz, L. Parthier, A. Gottwald, M. Richter: Spatial anisotropy of the exciton level in CaF_2 at 11.1 eV and its relation to the weak optical anisotropy at 157 nm, *Phys. Rev. B* **67**, 233101 (2003)
- 5.261 C. Bricot, M. Hareng, E. Spitz: Optical projection device and an optical reader incorporating this device, US Patent 4037929 (1977)
- 5.262 S. Sato: Liquid-crystal lens-cells with variable focal length, *Jpn. J. Appl. Phys.* **18**, 1679–1684 (1979)
- 5.263 S. Suyama, M. Date, H. Takada: Three-dimensional display system with dual-frequency liquid-crystal varifocal lens, *Jpn. J. Appl. Phys.* **39**, 480–484 (2000)
- 5.264 L.G. Commander, S.E. Day, D.R. Selviah: Variable focal microlenses, *Opt. Commun.* **177**, 157–170 (2000)
- 5.265 Y. Choi, J.H. Park, J.H. Kim, S.D. Lee: Fabrication of a focal length variable microlens array based on a nematic liquid crystal, *Opt. Mater.* **21**, 643–646 (2002)
- 5.266 S.T. Kowel, D.S. Cleverly, P.G. Kornreich: Focusing by electrical modulation of refraction in aliquid crystal cell, *Appl. Opt.* **23**, 278–289 (1984)
- 5.267 T. Nose, S. Sato: A liquid crystal microlens with a nonuniform electric field, *Liq. Cryst.* **5**, 1425–1433 (1989)
- 5.268 W. Klaus, M. Ide, Y. Hayano, S. Morokawa, Y. Arimoto: Adaptive LC lens array and its application, *SPIE Proc.* **3635**, 66–73 (1999)
- 5.269 B. Wang, M. Ye, M. Honma, T. Nose, S. Sato: Liquid crystal lens with spherical electrode, *Jpn. J. Appl. Phys.* **41**, L1232–1233 (2002)
- 5.270 A.F. Naumov, M.Yu. Loktev, I.R. Guealnik, G. Vdovin: Liquid crystal adaptive lenses with modal control, *Opt. Lett.* **23**, 992–994 (1998)
- 5.271 A.F. Naumov, G.D. Love, M.Yu. Loktev, F.L. Vladimirov: Control optimization of spherical modal liquid crystal lenses, *Opt. Express* **4**, 344–352 (1999)
- 5.272 G.D. Love, A.F. Naumov: Modal liquid crystal lenses, *Liq. Cryst. Today* **10**(1), 1–4 (2001)
- 5.273 J.S. Patel, K. Rastani: Electrically controlled polarization-independent liquid crystal Fresnel lens arrays, *Opt. Lett.* **16**, 532–534 (1991)
- 5.274 G. Williams, N.J. Powell, A. Purvis, M.G. Clark: Electrically controllable liquid crystal Fresnel lens, *SPIE Proc.* **1168**, 352–357 (1989)
- 5.275 H. Ren, Y.H. Fan, S.T. Wu: Tunable Fresnel lens using nanoscale polymer-dispersed liquid crystals, *Appl. Phys. Lett.* **83**, 1515–1517 (2003)
- 5.276 H. Ren, S.T. Wu: Inhomogeneous nanoscale polymer-dispersed liquid crystals with gradient refractive index, *Appl. Phys. Lett.* **81**, 3537–3539 (2002)
- 5.277 H. Ren, S.T. Wu: Tunable electronic lens using a gradient polymer network liquid crystal, *Appl. Phys. Lett.* **82**, 22–24 (2003)
- 5.278 V.V. Presnyakov, K.E. Asatryan, T.V. Galstian, A. Tork: Polymer-stabilized liquid crystal for tunable microlens applications, *Opt. Express* **10**, 865–870 (2002)
- 5.279 W. Helfrich, W.G. Schneider: Recombination radiation in anthracene crystals, *Phys. Rev. Lett.* **14**, 229 (1965)
- 5.280 M. Kawabe, K. Masuda, S. Nambu: Electroluminescence of green light region in doped anthracene, *Jpn. J. Appl. Phys.* **10**, 527 (1971)
- 5.281 C. Adachi, S. Tokito, S. Saito: Electroluminescence in organic films with three-layer structure, *Jpn. J. Appl. Phys.* **27**, L269 (1988)
- 5.282 C.W. Tang, S.A. van Slyke: Organic electroluminescent diodes, *Appl. Phys. Lett.* **51**, 913 (1987)
- 5.283 J.H. Burroughes, D.D.C. Bradley, A.R. Brown, R.N. Marks, K. Mackey, R.H. Friend, P.L. Burns, A.B. Holmes: Light-emitting-diodes based on conjugated polymers, *Nature* **347**, 539 (1990)
- 5.284 D. Braun, A.J. Heeger: Visible-light emission from semiconducting polymer diodes, *Appl. Phys. Lett.* **58**, 1982 (1991)
- 5.285 N.C. Greenham, S.C. Moratti, D.C.C. Bradley, R.H. Friend, A.B. Holmes: Efficient light-emitting-diodes based on polymers with high electron-affinities, *Nature* **365**, 628 (1993)
- 5.286 L.S. Swanson, J. Shinar, Y.W. Ding, T.J. Barton: Photoluminescence, electroluminescence and optically detected magnetic resonance study of 2,5-dialkoxy derivatives of poly(p-phenylene-acetylene) (PPA), PPA-based light emitting diodes, *Synth. Met.* **55–57**, 1–6 (1993)
- 5.287 M. Remmers, D. Neher, J. Grüner, R.H. Friend, G.H. Gelinck, J.M. Warman, C. Quattrocchi, D.A. dos Santos, J.-L. Brédas: The optical, electronic, and electroluminescent properties of novel poly(p-phenylene)-related polymers, *Macromol.* **29**, 7432 (1996)
- 5.288 M. Kreyenschmidt, G. Klaerner, T. Fuhrer, J. Ashenurst, S. Karg, W.D. Chen, V.Y. Lee, J.C. Schott, R.D. Miller: Thermally stable blue-light-emitting copolymers of poly(alkylfluorene), *Macromol.* **31**, 1099 (1998)
- 5.289 M.M. Grell, D.D.C. Bradley, M. Inbasekaran, E.P. Woo: A glass-forming conjugated main-chain liquid crystal polymer for polarized electroluminescence, *Adv. Mater.* **9**, 798 (1997)
- 5.290 Y. Ohmori, M. Uchida, K. Muro, K. Yoshino: Blue electroluminescent diodes utilizing poly(alkylfluorene), *Jpn. J. Appl. Phys.* **30**, 1941 (1991)
- 5.291 G. Grem, G. Leising: Electroluminescence of wide-bandgap chemically tunable cyclic conjugated polymers, *Synth. Met.* **55–57**, 4105 (1993)
- 5.292 U. Scherf, K. Müllen: Polyarylenes and poly(arylene-vinyls) a soluble ladder polymer via bridging of functionalized poly(para-phenylene)-precursors, *Makromol. Chem. Rapid Commun.* **12**, 489 (1991)
- 5.293 V. Cimrovà, D. Neher, M. Remmers: Blue light-emitting devices based on novel polymer blends, *Adv. Mater.* **10**, 676 (1998)
- 5.294 G. Yu, H. Nishino, A.J. Heeger, T.-A. Chen, R.D. Rieke: Enhanced electroluminescence from



- semiconducting polymer blends, *Synth. Met.* **72**, 249 (1995)
- 5.295 I.-N. Kang, D.-H. Hwang, H.-K. Shim, T. Zyung, J.-J. Kim: Highly improved quantum efficiency in blend polymer LEDs, *Macromol.* **29**, 165 (1996)
- 5.296 Y. Ohmori, M. Uchida, K. Muro, K. Yoshino: Effects of alkyl chain length and carrier confinement layer on characteristics of poly(3-alkylthiophene) electro-luminescent diodes, *Solid State Commun.* **80**, 605 (1991)
- 5.297 D. D. Gebler, Y. Z. Wang, J. W. Blatchford, S. W. Jessen, L. B. Lin, T. L. Gustafson, H. L. Wang, T. M. Swager, A. G. MacDiarmid, A. J. Epstein: Blue electroluminescent devices based on soluble poly(p-pyridine), *J. Appl. Phys.* **78**, 4264 (1995)
- 5.298 Y. Shirota: *Proc. SPIE-Int. Soc. Opt. Eng.* **186**, 3148 (1997)
- 5.299 J. Kido, H. Hayase, K. Hongawa, K. Nagai, K. Okuyama: Bright red-emitting organic electroluminescent devices having an europium complex as an emitter, *Appl. Phys. Lett.* **65**, 2124 (1994)
- 5.300 M. A. Abkowitz, D. M. Pai: Comparison of the drift mobility measured under transient and steady-state conditions in a prototypical hopping system, *Philos. Mag. B* **53**, 193 (1986)
- 5.301 Y. Shirota, Y. Kuwabara, H. Inada, T. Wakimoto, H. Nakada, Y. Yonemoto, S. Kawami, K. Imai: Multilayered organic electroluminescent devices using a novel starburst molecule 4,4',4''-tris(3-methylphenylphenylamine)triphenylamine, as a hole transport layer, *Appl. Phys. Lett.* **65**, 807 (1994)
- 5.302 N. Johansson, D. A. dos Santos, S. Guo, J. Cornil, M. Fahlman, J. Salbeck, H. Schenck, H. Arwin, J. L. Brédas, W. R. Salaneck: Electronic structure and optical properties of electroluminescent spiro-type molecules, *J. Chem. Phys.* **107**, 2542 (1997)
- 5.303 S. A. van Slyke, P. S. Bryan, C. W. Tang: *Inorganic and Organic Electroluminescence* (W&T Verlag, Berlin 1996) p. 195
- 5.304 V. Bulovic, A. Shoustikow, M. A. Baldo, E. Bose, V. G. Kozlov, M. E. Thompson, S. R. Forrest: Bright, saturated, red-to-yellow organic light-emitting devices based on polarization-induced spectral shifts, *Chem. Phys. Lett.* **287**, 455 (1998)
- 5.305 J. Kido: Recent advances in organic electroluminescent devices, *Bull. Electrochem.* **10**, 1 (1994)
- 5.306 J. Shi, C. W. Tang: Doped organic electroluminescent devices with improved stability, *Appl. Phys. Lett.* **70**, 1665 (1997)
- 5.307 S. F. Lim, L. Ke, W. Wang, S. J. Chua: Correlation between dark spot growth and pinhole size in organic light-emitting diodes, *Appl. Phys. Lett.* **78**, 2116 (2001)
- 5.308 L. J. Rothberg, M. Yan, F. Papadimitrakopoulos, M. E. Galvin, E. W. Kwock, T. M. Miller: Photophysics of phenylenevinylene polymers, *Synth. Met.* **80**, 41 (1996)
- 5.309 B. H. Cumpston, K. F. Jensen: Photo-oxidation of polymers used in electroluminescent devices, *Synth. Met.* **73**, 195-199 (1995)
- 5.310 M. Yan, L. J. Rothberg, F. Papadimitrakopoulos, M. E. Galvin, T. M. Miller: Defect quenching of conjugated polymer luminescence, *Phys. Rev. Lett.* **73**, 744 (1994)
- 5.311 J. C. Scott, J. H. Kaufman, P. J. Brock, R. DiPietro, J. Salem, J. A. Goitia: Degradation and failure of MEH-PPV light-emitting diodes, *J. Appl. Phys.* **79**, 2745 (1996)
- 5.312 S. A. van Slyke, C. H. Chen, C. W. Tang: Organic electroluminescent devices with improved stability, *Appl. Phys. Lett.* **69**, 2160 (1996)
- 5.313 A. Ashkin, G. D. Boyd, J. M. Dziedzic, R. G. Smith, A. A. Ballman, J. J. Levinstein, K. Nassau: Optically induced refractive index inhomogeneities in LiNbO₃ and LiTaO₃, *Appl. Phys. Lett.* **9**, 72 (1966)
- 5.314 F. S. Chen, J. T. LaMacchia, D. B. Fraser: Holographic storage in lithium niobate, *Appl. Phys. Lett.* **13**, 223 (1968)
- 5.315 D. L. Staebler, W. J. Burke, W. Phillips, J. J. Amodi: Multiple storage and erasure of fixed holograms in Fe-doped LiNbO₃, *Appl. Phys. Lett.* **26**, 182 (1975)
- 5.316 R. Orłowski, E. Krätzig: Holographic method for the determination of photo-induced electron and hole transport in electro-optic crystals, *Solid State Commun.* **27**, 1351 (1978)
- 5.317 G. E. Peterson, A. M. Glass, T. J. Negran: Control of the susceptibility of lithium niobate to laser-induced refractive index change, *Appl. Phys. Lett.* **19**, 130 (1971)
- 5.318 J. J. Amodi, W. Phillips, D. L. Staebler: Improved electrooptic materials and fixing techniques for holographic recording, *Appl. Opt.* **11**, 390 (1972)
- 5.319 P. Günter, J.-P. Huignard (Eds.): *Photorefractive Materials, Their Applications I, II*, *Top. Appl. Phys.*, Vol. 61, 62 (Springer, Berlin, Heidelberg 1989)
- 5.320 P. Boffi, D. Piccinin, M. C. Ubaldi: *Infrared Holography for Optical Communications*, *Top. Appl. Phys.*, Vol. 86 (Springer, Berlin, Heidelberg 2003)
- 5.321 H. J. Coufal, D. Psaltis, G. Sincerbox (Eds.): *Holographic Data Storage* (Springer, Berlin, Heidelberg 2000)
- 5.322 A. M. Glass, D. von der Linde, T. J. Negran: High-voltage bulk photovoltaic effect and the photorefractive process in LiNbO₃, *Appl. Phys. Lett.* **25**, 233 (1974)
- 5.323 K. Buse: Thermal gratings and pyroelectrically produced charge redistribution in BaTiO₃ and KNbO₃, *J. Opt. Soc. Am. B* **10**, 1266 (1993)
- 5.324 N. V. Kukhtarev, V. B. Markov, S. G. Odoulov, M. S. Soskin, V. L. Vinetskii: Holographic storage in electrooptic crystals, *Ferroelectrics* **22**, 949-961 (1979)
- 5.325 H. Kurz, E. Krätzig, W. Keune, H. Engelmann, U. Gonser, B. Dischler, A. Räuber: Photorefractive

- centers in LiNbO_3 , studied by optical-, Mössbauer- and EPR-methods, *Appl. Phys.* **12**, 355 (1977)
- 5.326 G. C. Valley: Simultaneous electron/hole transport in photorefractive materials, *J. Appl. Phys.* **59**, 3363 (1986)
- 5.327 F. P. Strohkendl, J. M. C. Jonathan, R. W. Hellwarth: Hole-electron competition in photorefractive gratings, *Opt. Lett.* **11**, 312 (1986)
- 5.328 D. L. Staebler, W. Phillips: Hologram storage in photochromic LiNbO_3 , *Appl. Phys. Lett.* **24**, 268 (1974)
- 5.329 G. A. Brost, R. A. Motes, J. R. Rotgé: Intensity-dependent absorption and photorefractive effects in barium titanate, *J. Opt. Soc. Am. B* **5**, 1879 (1988)
- 5.330 K. Buse, E. Krätzig: Three-valence charge-transport model for explanation of the photorefractive effect, *Appl. Phys. B* **61**, 27 (1995)
- 5.331 L. Holtmann: A model for the nonlinear photoconductivity of BaTiO_3 , *Phys. Stat. Solidi (a)* **113**, K89 (1989)
- 5.332 F. Jermann, J. Otten: The light-induced charge transport in $\text{LiNbO}_3\text{:Fe}$ at high light intensities, *J. Opt. Soc. Am. B* **10**, 2085 (1993)
- 5.333 I. Nee, M. Müller, K. Buse, E. Krätzig: Role of iron in lithium-niobate crystals for the dark storage time of holograms, *J. Appl. Phys.* **88**, 4282 (2000)
- 5.334 Y. P. Yang, I. Nee, K. Buse, D. Psaltis: Ionic and electronic dark decay of holograms in LiNbO_3 crystals, *Appl. Phys. Lett.* **78**, 4076 (2001)
- 5.335 K. Buse: Light-induced charge transport processes in photorefractive crystals I: Models and experimental methods, *Appl. Phys. B* **64**, 273 (1997)
- 5.336 K. Buse, S. Breer, K. Peithmann, S. Kapphan, M. Gao, E. Krätzig: Origin of thermal fixing in photorefractive lithium niobate crystals, *Phys. Rev. B* **56**, 1225 (1997)
- 5.337 R. Magnusson, T. K. Gaylord: Laser scattering induced holograms in lithium niobate, *Appl. Opt.* **13**, 1545 (1974)
- 5.338 J. F. Nye: *Physical Properties of Crystals* (Oxford Univ. Press, London 1979)
- 5.339 O. Madelung (Ed.): *Landolt-Börnstein – Numerical Data, Condensed Matter*, Vol. III/11, III/16, III/18, III/28 (Springer, Berlin, Heidelberg 1979, 1981, 1984)
- 5.340 K. Buse, S. Riehemann, S. Loheide, H. Hesse, F. Mersch, E. Krätzig: Refractive indices of single domain BaTiO_3 for different wavelengths and temperatures, *Phys. Stat. Solidi (a)* **135**, K87 (1993)
- 5.341 M. Zgonik, K. Nakagawa, P. Günter: Electro-optic and dielectric properties of photorefractive BaTiO_3 and KNbO_3 , *J. Opt. Soc. Am. B* **12**, 1416 (1995)
- 5.342 M. Simon, F. Mersch, C. Kuper, S. Mendricks, S. Wevering, J. Imbrock, E. Krätzig: Refractive indices of photorefractive bismuth titanate, barium-calcium titanate, bismuth germanium oxide, and lead germanate, *Phys. Stat. Solidi (a)* **159**, 559 (1997)
- 5.343 K. Buse, U. van Stevendaal, M. Weber, T. Leidlo, H. Hesse, E. Krätzig: Electrooptic and photorefractive properties of ferroelectric barium-calcium titanate crystals, *Ferroelectrics* **208**, 213 (1998)
- 5.344 R. Pankrath, H. Hesse: Growth and dielectric properties of congruently melting $\text{Ba}_{1-x}\text{Ca}_x\text{TiO}_3$ crystals, *Appl. Phys. A* **65**, 301 (1997)
- 5.345 S. Loheide, S. Riehemann, F. Mersch, R. Pankrath, E. Krätzig: Refractive indices, permittivities, and linear electrooptic coefficients of tetragonal potassium tantalate-niobate crystals, *Phys. Stat. Solidi (a)* **137**, 257 (1993)
- 5.346 S. Loheide, S. Riehemann, R. Pankrath, E. Krätzig: Influence of Fe doping on the photorefractive properties of $\text{KTa}_{1-x}\text{Nb}_x\text{O}_3$, *Ferroelectrics* **160**, 213 (1994)
- 5.347 D. Kip, S. Aulkemeyer, K. Buse, F. Mersch, R. Pankrath, E. Krätzig: Refractive indices of $\text{Sr}_{0.61}\text{Ba}_{0.39}\text{Nb}_2\text{O}_6$ single crystals, *Phys. Stat. Solidi (a)* **154**, K5 (1996)
- 5.348 S. Ducharme, J. Feinberg, R. R. Neurgaonkar: Electrooptic and piezoelectric measurements in photorefractive barium titanate and strontium barium niobate, *IEEE J. Quantum Electron.* **23**, 2116 (1987)
- 5.349 F. Mersch, K. Buse, W. Sauf, H. Hesse, E. Krätzig: Growth and characterization of undoped and doped $\text{Bi}_{12}\text{TiO}_{20}$ crystals, *Phys. Stat. Solidi (a)* **140**, 273 (1993)
- 5.350 J. P. Wilde, L. Hesselink: Measurement of the electrooptic and electrogyratory effects in $\text{Bi}_{12}\text{TiO}_{20}$, *J. Appl. Phys.* **67**, 2245 (1990)
- 5.351 D. T. F. Marple: Refractive index of GaAs, *J. Appl. Phys.* **35**, 1241 (1964)
- 5.352 D. Psaltis, D. Brady, K. Wagner: Adaptive optical networks using photorefractive crystals, *Appl. Opt.* **27**, 1752 (1988)
- 5.353 H. Kogelnik: Coupled wave theory for thick hologram gratings, *Bell Syst. Tech. J.* **48**, 2909 (1969)
- 5.354 F. H. Mok, G. W. Burr, D. Psaltis: System metric for holographic memory systems, *Opt. Lett.* **21**, 896 (1996)
- 5.355 E. Krätzig, R. Orłowski: LiTaO_3 as holographic storage material, *Appl. Phys.* **15**, 133 (1978)
- 5.356 R. Sommerfeldt, L. Holtmann, E. Krätzig, B. C. Grabmaier: Influence of Mg doping and composition on the light-induced charge transport in LiNbO_3 , *Phys. Stat. Solidi (a)* **106**, 89 (1988)
- 5.357 G. D. Bacher, M. P. Chiao, G. J. Dunning, M. B. Klein, C. C. Nelson, B. A. Wechsler: Ultralong dark decay measurements in BaTiO_3 , *Opt. Lett.* **21**, 18 (1996)
- 5.358 D. Rytz, M. B. Klein, R. A. Mullen, R. N. Schwartz, G. C. Valley, B. A. Wechsler: High-efficiency fast response in photorefractive BaTiO_3 at 120°C , *Appl. Phys. Lett.* **52**, 1759 (1988)
- 5.359 N. Korneev, D. Mayorga, S. Stepanov, H. Veenhuis, K. Buse, C. Kuper, H. Hesse, E. Krätzig: Holographic and non-steady-state photocurrent characterization of photorefractive barium-calcium titanate, *Opt. Commun.* **160**, 98–102 (1999)
- 5.360 L. A. Boatner, E. Krätzig, R. Orłowski: KTN as a holographic storage material, *Ferroelectrics* **27**, 247 (1980)



- 5.361 S. Loheide, D. Sabbert, F. Mersch, H. Hesse, E. Krätzig: Influence of annealing treatments on photorefractive properties of $\text{KTa}_{1-x}\text{Nb}_x\text{O}_3\text{:Fe}$ crystals, *Ferroelectrics* **166**, 99 (1995)
- 5.362 R. J. Reeves, M. G. Jani, B. Jassemnejad, R. C. Powell, G. J. Mizell, W. Fay: Photorefractive properties of KNbO_3 , *Phys. Rev. B* **43**, 71 (1991)
- 5.363 M. Ewart, M. Ryf, C. Medrano, H. Wüest, M. Zgonik, P. Günter: High photorefractive sensitivity at 860 nm in reduced rhodium-doped KNbO_3 , *Opt. Lett.* **22**, 781 (1997)
- 5.364 C. Medrano, M. Zgonik, N. Sonderer, S. Krucker, J. Seglins, H. Wüest, P. Günter: Photorefractive effect in Cu-, Ni-doped KNbO_3 in the visible and near infrared, *J. Appl. Phys.* **76**, 5640 (1994)
- 5.365 K. Buse, U. van Stevendaal, R. Pankrath, E. Krätzig: Light-induced charge transport properties of $\text{Sr}_{0.61}\text{Ba}_{0.39}\text{Nb}_2\text{O}_6$ crystals, *J. Opt. Soc. Am. B* **13**, 1461 (1996)
- 5.366 K. Megumi, H. Kozuka, M. Kobayashi, Y. Furuhashi: High-sensitive holographic storage in Ce-doped SBN, *Appl. Phys. Lett.* **30**, 631 (1977)
- 5.367 P. Tayebati, D. Mahgerefteh: Theory of the photorefractive effect for $\text{Bi}_{12}\text{SiO}_{20}$ and BaTiO_3 with shallow traps, *J. Opt. Soc. Am. B* **8**, 1053 (1991)
- 5.368 M. P. Petrov, I. A. Sokolov, S. I. Stepanov, G. S. Trofimov: Non-steady-state photo-electromotive-force induced by dynamic gratings in partially compensated photoconductors, *J. Appl. Phys.* **68**, 2216 (1990)
- 5.369 Y. Fainman, J. Ma, S. H. Lee: Non-linear optical materials and applications, *Mater. Sci. Rep.* **9**, 53 (1993)
- 5.370 S. Bian, J. Frejlich: Photorefractive response time measurement in GaAs crystals by phase modulation in two-wave mixing, *Opt. Lett.* **19**, 1702 (1994)
- 5.371 D. Fluck, P. Amrhein, P. Günter: Photorefractive effect in crystals with a nonlinear recombination of charge carriers: Theory and observation in KNbO_3 , *J. Opt. Soc. Am. B* **8**, 2196 (1991)
- 5.372 E. Krätzig, R. A. Rupp: Holographic storage properties of electrooptic crystals, *SPIE Proc.* **673**, 483 (1986)
- 5.373 U. van Stevendaal, K. Buse, H. Malz, H. Veenhuis, E. Krätzig: Reduction of light-induced refractive-index changes by decreased modulation of light patterns in photorefractive crystals, *J. Opt. Soc. Am. B* **15**, 2868 (1998)
- 5.374 K. Buse, A. Adibi, D. Psaltis: Non-volatile holographic storage in doubly doped lithium niobate crystals, *Nature* **393**, 665 (1998)
- 5.375 K. Buse: Light-induced charge transport processes in photorefractive crystals II: Materials, *Appl. Phys. B* **64**, 391 (1997)
- 5.376 D. Dirksen, F. Matthes, S. Riehemann, G. von Bally: Phase shifting holographic double exposure interferometry with fast photorefractive crystals, *Opt. Commun.* **134**, 310 (1997)
- 5.377 K. Buse, A. Gerwens, S. Wevering, E. Krätzig: Charge transport parameters of photorefractive strontium-barium niobate crystals doped with cerium, *J. Opt. Soc. Am. B* **15**, 1674 (1998)
- 5.378 L. H. Acioli, M. Ulman, E. P. Ippen, J. G. Fujimoto, H. Kong, B. S. Chen, M. Cronin-Golomb: Femtosecond temporal encoding in barium titanate, *Opt. Lett.* **16**, 1984 (1991)
- 5.379 Q. N. Wang, D. D. Nolte, M. R. Melloch: Two-wave mixing in photorefractive AlGaAs/GaAs quantum wells, *Appl. Phys. Lett.* **59**, 256 (1991)
- 5.380 Q. N. Wang, R. M. Brubaker, D. D. Nolte: Photorefractive phase-shift induced by hot-electron transport – multiple-quantum-well structures, *J. Opt. Soc. Am. B* **11**, 1773 (1994)
- 5.381 J. J. Amodei, D. L. Staebler: Holographic pattern fixing in electrooptic crystals, *Appl. Phys. Lett.* **18**, 540 (1971)
- 5.382 H. Vormann, G. Weber, S. Kapphan, E. Krätzig: Hydrogen as origin of thermal fixing in $\text{LiNbO}_3\text{:Fe}$, *Solid State Commun.* **40**, 543 (1981)
- 5.383 F. Micheron, G. Bismuth: Electrical control of fixation and erasure of holographic patterns in ferroelectric materials, *Appl. Phys. Lett.* **20**, 79 (1972)
- 5.384 J. Ma, T. Chang, J. Hong, R. Neurgaonkar: Electrical fixing of 1000 angle-multiplexed holograms in SBN:75, *Opt. Lett.* **22**, 1116 (1997)
- 5.385 D. von der Linde, A. M. Glass, K. F. Rodgers: Multiphoton photorefractive processes for optical storage in LiNbO_3 , *Appl. Phys. Lett.* **25**, 155 (1974)
- 5.386 H. Vormann, E. Krätzig: Two step excitation in $\text{LiTaO}_3\text{:Fe}$ for optical data storage, *Solid State Commun.* **49**, 843 (1984)
- 5.387 K. Buse, L. Holtmann, E. Krätzig: Activation of BaTiO_3 for infrared holographic recording, *Opt. Commun.* **85**, 183 (1991)
- 5.388 M. P. Petrov, S. I. Stepanov, A. A. Kamshilin: Holographic storage of information and peculiarities of light-diffraction in birefringent electrooptic crystals, *Opt. Laser Technol.* **11**, 149 (1979)
- 5.389 H. C. Kiliç: Reconstructing volume holograms without image field losses, *Appl. Opt.* **30**, 2850 (1991)
- 5.390 E. Chuang, D. Psaltis: Storage of 1000 holograms with use of a dual-wavelength method, *Appl. Opt.* **36**, 8445 (1997)
- 5.391 S. Fries, S. Bauschulte, E. Krätzig, K. Ringhofer, Y. Yacoby: Spatial frequency mixing in lithium niobate, *Opt. Commun.* **84**, 251 (1991)
- 5.392 S. Fries: Spatial frequency mixing in electrooptic crystals – application to nondestructive read-out of optically erasable volume holograms, *Appl. Phys. A* **55**, 104 (1992)
- 5.393 R. A. Paquin: Metal mirrors. In: *Handbook of Optomechanical Engineering*, ed. by A. Ahmad (CRC, Boca Raton 1996) p. 92
- 5.394 ULE™ is a registered trademark of the Corning Glass Works, Corning, New York, USA for their ultra low expansion fused silica



- 5.395 Zerodur® is a registered trademark of Schott Glaswerke, Mainz, Germany, for their zero expansion glass ceramic
- 5.396 R. A. Paquin: Properties of metals. In: *Handbook of Optics, Devices, Measurements, and Properties*, Vol. II, 2nd edn., ed. by M. Bass (McGraw-Hill, New York 1994) pp. 35.1–35.78
- 5.397 R. A. Paquin: Materials for Optical Systems and Metal Mirrors. In: *Handbook of Optomechanical Engineering*, ed. by A. Ahmad (CRC, Boca Raton 1997) pp. 69–110
- 5.398 M. A. Ealey, R. A. Paquin, T. B. Parsonage (Eds.): *Advanced Materials for Optics and Precision Structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67 (SPIE, Bellingham 1997)
- 5.399 I. Kh. Lokshin: Heat treatment to reduce internal stresses in beryllium, Metal. Sci. Heat Treat. **426**, 426 (1970)
- 5.400 R. A. Paquin: Advanced materials: An overview. In: *Advanced materials for optics and precision structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67, ed. by M. A. Ealey, R. A. Paquin, T. B. Parsonage (SPIE, Bellingham 1997) p. 10
- 5.401 R. G. Ohl, M. P. Barthelmy, S. W. Zewari, R. W. Toland, J. C. McMann, D. F. Puckett, J. G. Hagopian, J. E. Hyland, J. E. Mentzell, R. G. Mink, L. M. Sparr, M. A. Greenhouse, J. W. Mac Kenty: Cryogenic optical systems and instruments IX, comparison of stress relief procedures for cryogenic aluminum mirrors, Proc. SPIE. **4822**, 51 (2002)
- 5.402 H. Y. Hunsicker: The metallurgy of heat treatment. In: *Aluminum 1: Properties, Physical Metallurgy and Phase Diagrams*, ed. by J. E. Hatch (American Society for Metals, Metals Park 1967)
- 5.403 J. B. C. Fuller Jr., P. Forney, C. M. Klug: Design and fabrication of aluminum mirrors for a large aperture precision collimator operating at cryogenic temperatures, Los Alamos Conference on Optics '81, Proc. SPIE **288**, 104 (1981)
- 5.404 D. Vukobratovich, K. Don, R. Sumner: Improved cryogenic aluminum mirrors, Cryogenic Optical Systems and Instruments VIII, Proc SPIE **3435**, 9–18 (1998)
- 5.405 R. A. Paquin: Hot isostatic pressed beryllium for large optics, Opt. Eng **25**, 1003 (1986)
- 5.406 D. Saxton, T. Parsonage: Advances in near net shape beryllium manufacturing technologies, Optical design, materials, fabrication, and maintenance, Proc. SPIE **4003**, 80 (2000)
- 5.407 M. Cayrel, R. A. Paquin, T. B. Parsonage, S. Stanghellini, K. H. Dost: Use of beryllium for the VLT secondary mirror, Advanced Materials for Optics and Precision Structures, Proc. SPIE **2857**, 86 (1996)
- 5.408 D. R. Coulter, S. A. Macenka, M. T. Stier, R. A. Paquin: ITTT: A state-of-the-art ultra-lightweight all-beryllium telescope. In: *Advanced Materials for Optics and Precision Structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67, ed. by M. A. Ealey, R. A. Paquin, T. B. Parsonage (SPIE, Bellingham 1997) p. 277
- 5.409 G. Gould: Method and means for making a beryllium mirror, US Patent 4,492,669 (1985)
- 5.410 R. A. Paquin: New technology for beryllium mirror production, Current Developments in Optical Engineering and Commercial Optics, Proc. SPIE **1168**, 83 (1989)
- 5.411 T. Parsonage, J. Benoit: Advances in beryllium and AlBeMet® optical materials, Optomechanical Design And Engineering 2002, Proc. SPIE **4771**, 222 (2002)
- 5.412 C. A. Swenson: HIP beryllium: Thermal expansivity from 4 to 300 K and heat capacity from 1 to 108 K, J. Appl. Phys. **70**, 3046 (1991)
- 5.413 Y. S. Touloukian, R. K. Kirby, R. E. Taylor, P. D. Desai: *Thermal expansion, metallic elements and alloys*, Thermophys. Prop. Matter, Vol. 12 (IFI/Plenum, New York 1977) p. 23
- 5.414 R. A. Paquin, D. R. Coulter, D. D. Norris, A. C. Augason, M. T. Stier, M. Cayrel, T. Parsonage: New fabrication processes for dimensionally stable beryllium mirrors, Specification, Production, and Testing of Optical Components and Systems, Proc. SPIE **2775**, 480 (1996)
- 5.415 T. B. Parsonage: Development of aluminum beryllium for structural applications. In: *Advanced Materials for Optics and Precision Structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67, ed. by T. B. Parsonage M. A. Ealey, R. A. Paquin (SPIE, Bellingham 1997) p. 236
- 5.416 www.brushwellman.com
- 5.417 M. R. Howells, R. A. Paquin: Optical substrate materials for synchrotron radiation beam lines. In: *Advanced Materials for Optics and Precision Structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67, ed. by T. B. Parsonage M. A. Ealey, R. A. Paquin (SPIE, Bellingham 1997) p. 339
- 5.418 R. Valdiviez, D. Schrage, F. Martinez, W. Clark: The use of dispersion strengthened copper in accelerator designs, Proc. XX Int'l Linac Conf., Monterey 2000) 956
- 5.419 B. S. Lement, B. L. Averbach, M. Cohen: The dimensional behavior of Invar, Trans. Am Soc. Met. **43**, 1072 (1951)
- 5.420 W. J. Spawr: Standard industrial polishing of high energy laser optics, NBS Spec. Pub. **435**, 10–12 (1975)
- 5.421 P. A. Temple, D. K. Burge, J. M. Bennett: Optical properties of mirrors prepared by ultraclean dc sputter deposition, NBS Spec. Pub. **462**, 195–202 (1976)
- 5.422 G. E. Carver, B. O. Seraphin: CVD molybdenum films for high power laser mirrors. In: *Laser Induced Damage in Optical Materials* (NBS, Bolder 1979) pp. 287–292
- 5.423 H. Okamoto, M. Matsusue, K. Kitazima, K. Yoshida, Y. Ichikawa, M. Yamanaka, T. Yamanaka, Y. Tsunawaki: Laser-induced Mo mirror damage for high



- power CO₂ laser. In: *Laser Induced Damage in Optical Materials* (NBS, Bolder 1985) pp. 248–260
- 5.424 M. Yamashita, S. Hara, H. Matsunaga: Ultrafine polishing of tungsten and molybdenum mirrors for CO₂ laser. In: *Laser-Damage in Optical Materials: Collected Papers, 1969–1998*, Vol. 08 (SPIE, Bellingham 1999)
- 5.425 D.L. Hibbard: Electroless nickel for optical applications. In: *Advanced Materials for Optics and Precision Structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67, ed. by T.B. Parsonage M.A. Ealey, R.A. Paquin (SPIE, Bellingham 1997) p. 179
- 5.426 K. Parker, H. Shah: Residual stresses in electroless nickel plating, *Plating* **58**, 230 (1971)
- 5.427 K. Parker: Internal stress measurements of electroless nickel coatings by the rigid strip method. In: *Testing of Metallic and Inorganic Coatings*, Vol. STP 947, ed. by W.B. Hardig, G.D. di Ban (American Society for Testing and Materials, Philadelphia 1987) p. 111
- 5.428 AlumiplateTM is a registered trademark of Alumiplate, Inc., Coon Rapids, MN, USA for their aluminium plating process and coating
- 5.429 D. Vukobratovich, A. Gerzoff, M.K. Cho: Thermo-optic analysis of bi-metallic mirrors, *Optomechanical Design and Precision Instruments*, Proc. SPIE **3132**, 12–23 (1997)
- 5.430 C.J. Shih, A. Ezis: Application of hot-pressed silicon carbide to large high-precision optical structures, *Silicon Carbide Materials for Optics and Precision Structures*, Proc. SPIE **2543**, 24 (1995)
- 5.431 J.S. Goela, M.A. Pickering: Optics applications of chemical vapor deposited β -SiC. In: *Advanced Materials for Optics and Precision Structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67, ed. by T.B. Parsonage M.A. Ealey, R.A. Paquin (SPIE, Bellingham 1997) p. 71
- 5.432 V. Rehn, J.L. Stanford, A.D. Behr, V.O. Jones, W.J. Choyke: Total optical integrated scatter in the vacuum ultraviolet: Polished CVD SiC, *Appl. Opt.* **16**, 111 (1977)
- 5.433 V. Rehn, W.J. Choyke: Total optical integrated scatter in the vacuum ultraviolet: Polished CVD SiC, *Nucl. Instrum. Methods* **117**, 173 (1980)
- 5.434 E. Sein, F. Safa, D. Castel, P. Deny: Silicon carbide, a sound solution for space optics, Proc. 52Int'l Astronautical Congress Toulouse (2001) Paper IAF-01-1.1.06
- 5.435 M.A. Ealey, J.A. Wellman, G. Weaver: CERAFORM SiC: Roadmap to 2 meters and 2 kg/m² areal density. In: *Advanced Materials for Optics and Precision Structures*, Crit. Rev. Opt. Sci. Technol., Vol. 67, ed. by T.B. Parsonage M.A. Ealey, R.A. Paquin (SPIE, Bellingham 1997) p. 53
- 5.436 E. Tobin, M. Magida, S. Kishner, M. Krim: Design, fabrication, and Test of a meter-class reaction bonded sic mirror blank, *Silicon Carbide Materials for Optics and Precision Structures*, Proc. SPIE **2543**, 12 (1995)
- 5.437 R.A. Paquin, M.B. Magida: Low scatter surfaces on silicon carbide, *Laser Induced Damage in Optical Materials: 1989*, NIST Spec. Publ. **801**, 256 (1990)
- 5.438 M. Krödel, G.S. Kutter, M. Deyerler, N. Pailer: Short carbon-fiber reinforced ceramic – Cescic® – for optomechanical applications, *Optomechanical Design And Engineering 2002*, Proc. SPIE **4771**, 230 (2002)
- 5.439 R. Plummer, D. Bray: Guidelines for design of SuperSiC® silicon carbide mirror substrates, precision components, *Optomechanical Design and Engineering 2002*, Proc. SPIE **4771**, 265 (2002)
- 5.440 S. Musikant: *Optical Materials* (Dekker, New York 1985) p. 1985
- 5.441 Hoya Optical Glass Catalog
- 5.442 Ohara Optical Glass Catalog (Ohara, Brandsburg 1995)
- 5.443 Schott Optical Glass Catalog (Schott, Mainz 2001)